

Clinical Reasoning Manual

NBEO Part II PAM / TMOD. An emergency gate, condition cards, and all 350 items with a generated, auditable key.

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350	70	701	178
ITEMS, ALL OF THEM HERE	CASES OF 3 TO 6 ITEMS	DISTRACTOR EXPLANATIONS	PAGES, NO FILLER

WHAT CHANGED IN THIS EDITION

The previous key was 82% option A, and 95% in Session 2: guessing A scored 272 of 333 items with no stem read. Letters are now generated to a distribution and audited on page 4, options are length-matched, and every rationale explains the tempting wrong answer as well as the right one.

THIS BOOK, AND ITS PAIR

Two books hold the same complete bank in two orders. This one is arranged by clinical domain, each case followed by its key. The Exam Simulator holds the identical 350 items arranged the other way, on the same generated key, so a score from either is a score of the same paper.

Six passes

Reading this manual is not studying it. Retrieval under time pressure is what transfers to the exam room.

PASS	WORK	PROOF IT WORKED
1 • Systems	Learn the algorithm, the emergency gate, the condition cards, the drug anchors.	You can state recognition, danger, next test, treatment, contraindication, follow-up, and failure trigger without looking.
2 • Retrieval	Closed-book recall at roughly 1, 3, 7, 14, and 30 days, always from a changed prompt.	Successful recalls are logged. Hours are not.
3 • Session 1	175 items, 210 minutes, uninterrupted, confidence marked on every item.	Scored by item type as well as total, with pacing and error categories written down.
4 • Repair	High-confidence misses first. Each becomes one if-then rule.	Every false rule has a replacement you can recite.
5 • Session 2	Second 175 items under identical conditions.	Weak item types improve; late-session accuracy holds.
6 • Taper	Red flags, drugs, formulas, image algorithms, logistics.	No new content in the final 48 hours. Sleep protected.

Error log: seven fields, one line each

Concept	Short disease or decision label.
Answer contrast	What you chose against what was correct.
Error type	Knowledge, discrimination, image reading, management sequence, calculation, or misreading the stem.
False rule	The exact belief that produced the miss, written as a sentence.
New rule	One executable if-then replacement.
Next recall	Tomorrow for a high-confidence miss, three days for low-confidence, seven days after a success.
Transfer test	A changed patient or the competing diagnosis.

WHY SCORE BY ITEM TYPE

The blueprint weights treatment and management far above basic science, and TMOD is decided separately. A 74% total can hide a failing treatment score. Every item here is tagged, so scoring by tag takes two minutes and tells you what to repair.

INTERLEAVE, DO NOT BLOCK

Study competing pairs together: episcleritis against scleritis, optic neuritis against ischaemic neuropathy, AMD against central serous, HSV against zoster, preseptal against orbital. The exam tests the discriminator, never the textbook description.

What is checked, and what is not

A guide that labels its own uncertainty is more useful than one that asserts it was verified. Every factual claim carries one of three states.

Verification table

CLAIM	STATE	BASIS
350 items, presented as cases of 3-6 items	VERIFIED	NBEO exam page, fetched 15 Aug 2026
Two sessions of 175 items, 3.5 hours each	VERIFIED	NBEO exam page
Approximately 120 TMOD items, separate breakout decision	VERIFIED	NBEO exam page
Delivered at Pearson centres; tutorial being updated	VERIFIED	NBEO exam page
Per-domain item ranges, 17 domains	UNCONFIRMED	Transcribed from prior edition. Content Matrix PDF resolves but was not machine-read
Item-type percentage bands	UNCONFIRMED	As above
Deep link to the Pearson tutorial	REMOVED	URL did not resolve; reach it from the exam page

The defect this edition exists to fix

ANSWER KEY	SESSION 1	SESSION 2	BOTH
Single-answer items	161	172	333
Correct answer was A	67.1%	95.3%	81.7%
Longest run of one letter	30	140	-
Letters never correct	E	D, E	E

WHAT THAT MEANT IN PRACTICE

Answering A to every single-answer item scored 272/333, about 82%, without reading one stem. Session 2 alone scored 95%. The two sessions were also biased by different amounts, so a Session 1 score could not be compared with a Session 2 score, which broke the guide's own repair-and-confirm workflow.

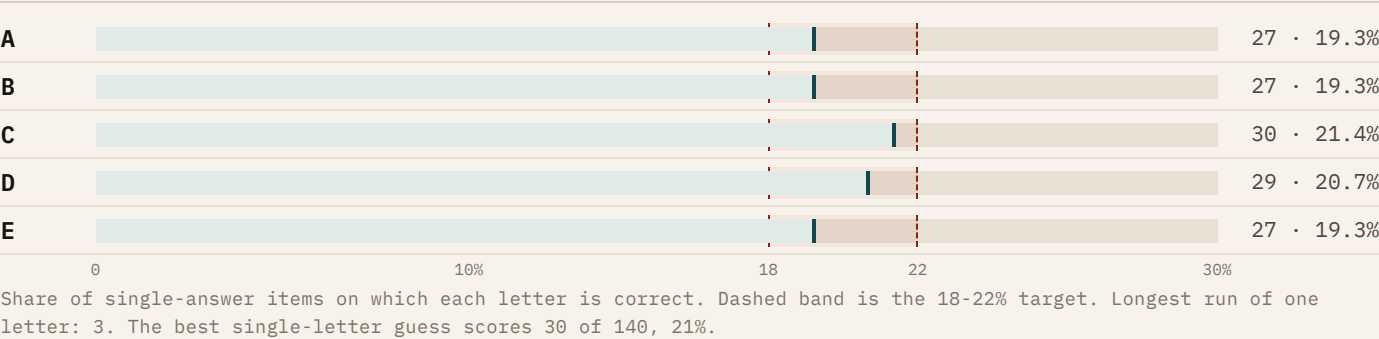
REBUILD RULE, AUDITABLE

Each letter A to E is correct on 18-22% of single-answer items in each session, no letter repeats more than three times consecutively, all five letters appear in both sessions, and option length is matched within each item. The build fails if any of that breaks. Re-run `tools/build.py --audit` after any edit to the bank.

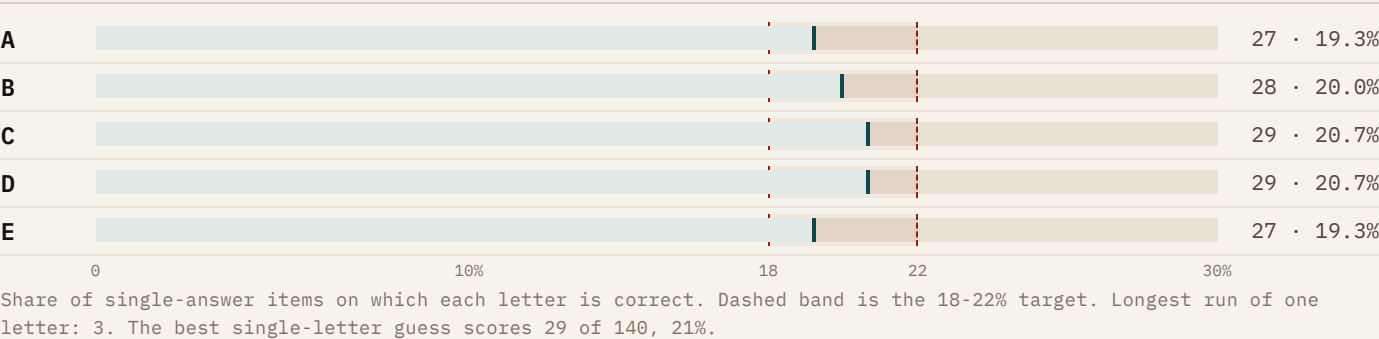
The key, audited

A biased key trains the wrong skill and then flatters you for it. This page is the evidence that the bank you are about to sit does not.

Session 1, 140 single-answer items



Session 2, 140 single-answer items



How the letters were chosen

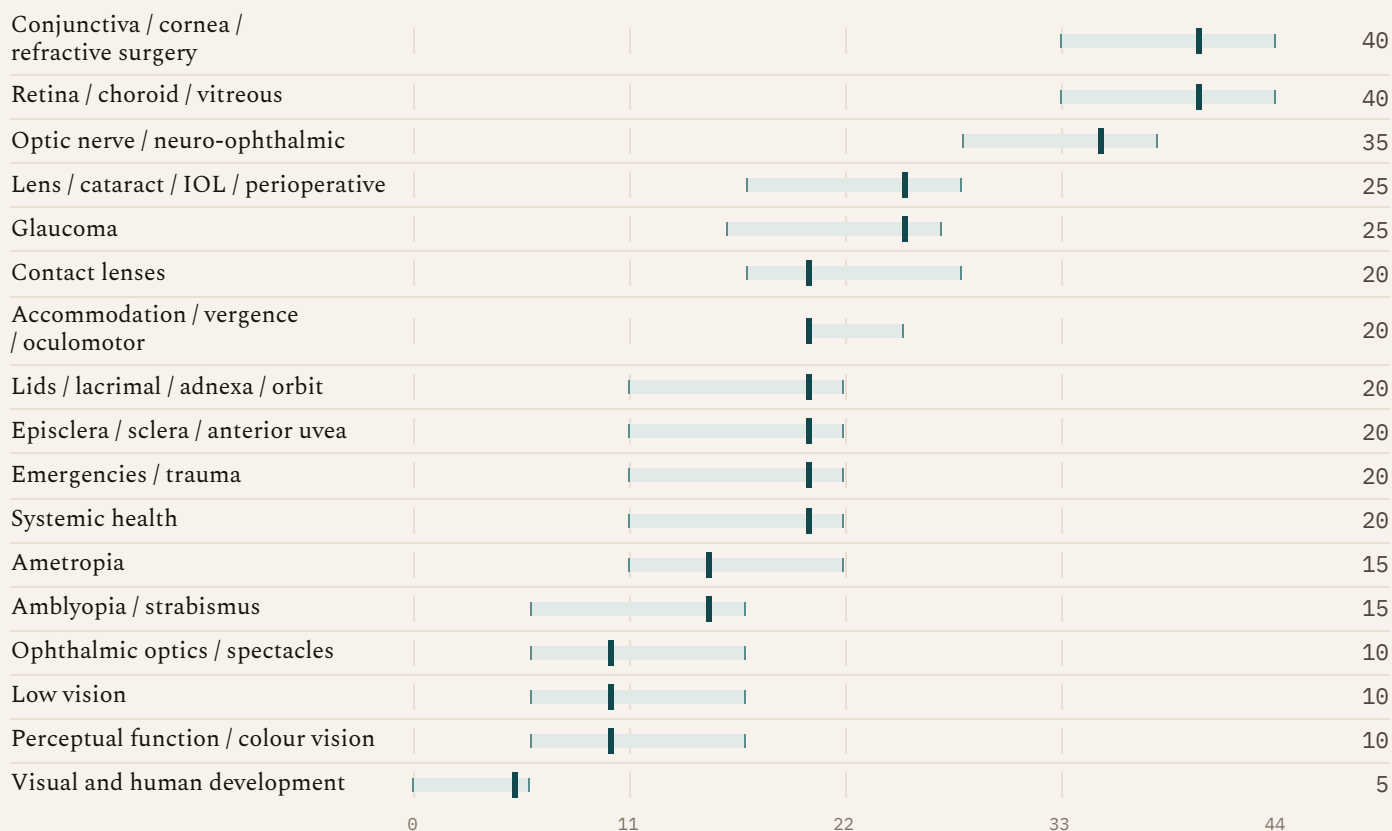
Not by the author	Item writers mark the correct option in place and never see a letter. Letters are assigned afterwards by a generator, so a habit of putting the right answer first cannot survive into print.
Even by construction	Each session's single-answer items are dealt an even multiset of A to E, so every letter lands on 19 to 21% before any adjustment.
No streaks	The sequence is rejected and redealt until no letter is correct more than three times in a row.
Order preserved	Options are rotated, not shuffled, so graded or paired options stay adjacent and a numeric ladder stays in order.
Length controlled	Within each item, the longest option is at most about twice the shortest, and across a session the correct option is the longest no more often than a third of the time.
Checked on build	The manual will not build if any of those conditions fails. The numbers above are printed from the same run that produced these pages.

TEST IT YOURSELF

Mark A for every single-answer item in either session and score it against the answer key in Part 10. You should land near a fifth. The previous edition scored 82% for exactly that, and 95% in Session 2, which is set out on page 3.

Know the target

Band shows the published range. Bar shows this manual's 350-item allocation. Scale runs 0 to 44 items.



READ IT THIS WAY

Cornea and retina alone can carry 88 items. Revision split evenly across 17 domains is misallocated by a factor of eight at the extremes. Development items usually arrive inside paediatric cases.

Case conventions worth memorising

BVA	Best achievable or best-corrected acuity. If it is reduced, pinhole improvement has already been accounted for, so do not spend an item hunting refractive causes.
Near acuity	Assume 16 inches unless the case says otherwise.
Fields	Side by side: OD right, OS left. Stacked: OD above OS.
Images	Fundus, OCT, fields, angiography, ultrasound, radiology, anterior segment. Describe before diagnosing.

245

ITEMS · NORMAL HEALTH, DISEASE, TRAUMA

105

ITEMS · REFRACTIVE, SENSORY, OCULOMOTOR

Where the marks are

Four item types, weighted very unevenly. Every item here is tagged, and the score sheets total by tag.

Item-type weighting and this manual's allocation

ITEM TYPE	BAND	HERE	WHAT IT ACTUALLY ASKS
Treatment and management	40-57%	158 · 45%	Choice, mechanism, dose, schedule, duration, follow-up, education, escalation.
Diagnosis	26-34%	116 · 33%	Most likely diagnosis, the supporting or missing datum, the next test.
Basic science, applied	11-17%	56 · 16%	Anatomy, physiology, microbiology, optics, pharmacology, tied to this patient.
Law, ethics, public health	4-10%	20 · 6%	Duty, consent, refusal, records, delegation, telehealth, screening arithmetic.

Bands transcribed, marked unconfirmed on page 3. The right-hand column is the real allocation across 350 items and 16 domains, counted at build time. 70 items are all-or-none, a format rather than a type.

Two item formats, two disciplines

Single answerONE OF FIVE	Select all that applyALL OR NONE
READ The question before the options. Decide what you think, then look. COMPARE Options in pairs. The pair differing on one variable shows what is tested. COMMIT Once. Changed management answers usually move to the more cautious wrong option.	COUNT Underline the requested number before reading the options. JUDGE Each option alone, true or false for this patient. PREFER Patient-specific over merely true. SCORE Every correct choice and no incorrect choice. A partial set earns zero.
TRAP Length is not a signal in this manual. Correct options are the longest in roughly a fifth of items, which is chance.	TRAP Findings that prove the diagnosis are not the same as findings that measure organ function. Read which the item wants.

What a complete management answer contains

Immediate action	What happens in the next minutes, before any prescription.
Agent and route	Drug or procedure, form, concentration, site of action.
Dose and duration	Loading and maintenance, frequency, taper, stop point.
Contraindication	The host factor in the stem that changes the choice.
Follow-up interval	A number, not as needed. Often the only difference between two options.
Measurable response	What you will remeasure to know it worked.
Escalation trigger	What failure looks like, and what you do then.

HOW THIS IS EXAMINED

Wrong options are right actions in the wrong order, right drugs at the wrong interval, or plans missing the contraindication the stem planted. An option with no interval and no failure trigger is incomplete.

Pacing, and the seven-step algorithm

175 items in 210 minutes is 72 seconds per item. The cost of running late is not the last case, it is the rereading you do in the middle.

Budget by case size

CASE	BUDGET	MOVE	SITUATION	DECISION
3 items	4 min	Read once, answer the direct items, flag one uncertainty.	Two options survive and you have spent 90 seconds	Choose the one that acts sooner on the more dangerous possibility, flag, move.
4 items	5 min	Commit before option-hunting for a rescue.	The stem contains a number you have not used	Reread. Numbers are planted to be used: pulse, weight, dose, pressure, age.
5 items	6 min	The typical unit. Do not reread the stem for each item.	An image is described but not shown	Answer from the described findings. Do not invent what is not stated.
6 items	7 min	Diagnosis, threat, test, treatment, follow-up, then the science.	You do not know the disease	Answer the management item anyway. Threat level is often decidable without the name.
Checkpoints: item 50 by 60 minutes, item 100 by 120, item 150 by 180. That leaves 30 minutes for the last 25 items and the flags. Behind schedule means cutting rereading, not standards.			15 minutes left, six flags	Spend it on flagged management items, not on flagged basic science.

Seven-step case algorithm

1 Frame	Age, laterality, tempo, pain, vision loss, systemic context, medications, risk behaviour.
2 Threat gate	Could delay cost life, the eye, or the fellow eye? Run the eighteen patterns in Part 3 before anything else.
3 Localise	Media, cornea and anterior segment, retina, optic nerve, post-chiasmal pathway, neuromuscular junction, orbit, binocular motor system.
4 Build three	Most likely, most dangerous plausible, best mimic. The mimic is usually one of the five options.
5 Discriminate	The one finding or test that most changes probability or management. Not a panel.
6 Treat in order	Stabilise, protect tissue and vision, treat the cause, prevent recurrence and fellow-eye injury.
7 Close the loop	Interval, measurable response, adverse effects, red flags, referral urgency, and what failure triggers.

INTERLEAVE, DO NOT BLOCK

Study competing pairs together: episcleritis against scleritis, optic neuritis against ischaemic neuropathy, AMD against central serous, HSV against zoster, preseptal against orbital, pupillary block against ciliochoroidal effusion. Part 7 sets them out side by side, because the exam tests the discriminator and never the textbook description.

Six patterns that cannot wait

Recognise, act, and know the specific thing that must not be done. The prohibition is examined as often as the treatment.

Chemical injury SURFACE	Open globe TRAUMA
RECOGNISE Exposure history. Do not wait for acuity or pH before acting.	RECOGNISE Full-thickness wound, Seidel, peaked pupil, uveal prolapse, low IOP with a mechanism.
DO FIRST Irrigate immediately and copiously. Evert lids, sweep fornices for retained particles, continue until pH is neutral and stays neutral after a pause. Then grade limbal ischaemia and epithelial loss.	DO FIRST Rigid shield, nil by mouth, antiemetic and analgesia, tetanus and systemic antibiotics per protocol, CT orbits when appropriate, immediate surgical referral.
NEVER Neutralise alkali with acid. Delay irrigation to find equipment.	NEVER Tonometry, gonioscopy, ultrasound, or any pressure on the globe.
Orbital cellulitis ORBIT	Microbial keratitis CORNEA
RECOGNISE Lid swelling plus any of: painful or restricted motility, proptosis, reduced acuity or colour, RAPD, raised IOP, fever, severe chemosis.	RECOGNISE Contact lens wear, epithelial defect over a stromal infiltrate, chamber reaction, pain, central location.
DO FIRST Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ophthalmology and ENT. Drain a subperiosteal abscess when vision or the apex is threatened.	DO FIRST Measure and photograph. Culture central, large, deep, atypical, post-surgical, or unresponsive ulcers. Intensive topical antimicrobial, review inside 24 hours.
NEVER Call it preseptal until orbital signs have been actively excluded.	NEVER Send a central ulcer home on conjunctivitis dosing, or patch it.
Acute angle closure GLAUCOMA	Giant cell arteritis SYSTEMIC
RECOGNISE Pain, halos, nausea, corneal oedema, shallow chamber, mid-dilated fixed pupil, very high IOP.	RECOGNISE Over 50 with new headache, jaw claudication, scalp tenderness, polymyalgia or constitutional symptoms, chalky pallid disc oedema.
DO FIRST Aqueous suppressants the patient can safely take, systemic acetazolamide if appropriate, control inflammation and nausea, then definitive iridotomy. Assess the fellow eye.	DO FIRST ESR, CRP, and platelets, and start high-dose systemic corticosteroid immediately. Temporal artery imaging or biopsy follows treatment.
NEVER Lead with pilocarpine at extreme pressure: an ischaemic sphincter will not respond.	NEVER Let a normal ESR reassure you, or wait for biopsy. The fellow eye is what is at stake.

CONTINUED ON PAGE 9

Retinal artery occlusion, tear and detachment, endophthalmitis, papilloedema, acute third-nerve palsy and painful Horner syndrome, then six more on page 10 that arrive looking mild.

And six more

Retinal artery occlusion VASCULAR RECOGNISE Sudden painless profound monocular loss, dense RAPD, retinal whitening or cherry-red spot. DO FIRST Treat as acute stroke: activate the emergency stroke pathway now, with carotid and cardiac source evaluation and a GCA screen by age. NEVER Spend the treatment window on ocular massage or other office manoeuvres.	Tear and detachment RETINA RECOGNISE New flashes, shower of floaters, pigment in the vitreous, vitreous haemorrhage, curtain or field loss. DO FIRST Dilated peripheral examination with indentation, B-scan if the view is blocked, urgent retina referral. Macula-on detachment is the most time-critical. NEVER Accept a normal central OCT as evidence against a peripheral break.
Endophthalmitis POSTOPERATIVE RECOGNISE Pain and severe acuity loss after surgery or injection, hypopyon, fibrin, vitritis, lost red reflex. DO FIRST Same-hour retina-level care: intraocular sampling then intravitreal antibiotics, with vitrectomy by severity. NEVER Treat it as routine postoperative inflammation, or wait for culture before treating.	Papilloedema NEURO RECOGNISE Bilateral disc oedema with headache, transient obscurations, pulsatile tinnitus, diplopia, possible sixth-nerve palsy. DO FIRST Urgent brain imaging including venous evaluation, then lumbar puncture with opening pressure when imaging makes it safe. NEVER Puncture before imaging, or dismiss true oedema as crowded discs without evidence.
Third-nerve palsy NEURO-VASCULAR RECOGNISE Ptosis with the eye down and out; pupil involvement, pain, or an incomplete palsy raises compression. DO FIRST Emergency aneurysm-focused CTA or MRA and neurovascular referral. Modern practice images acute isolated palsies. NEVER Treat pupil sparing as proof against aneurysm in an acute palsy.	Painful Horner syndrome NEURO-VASCULAR RECOGNISE Acute ptosis and miosis, anisocoria greater in darkness, dilation lag, with head or neck pain. DO FIRST Emergency head and neck vascular imaging for carotid dissection. NEVER Delay imaging to complete pharmacologic localisation.

TREATMENT GRAMMAR, EVERY TIME

Immediate action, drug or procedure, dose and form and schedule and duration, contraindication, patient education, follow-up interval, the measurable response, and the escalation trigger. An answer that names a drug but no interval or failure trigger is incomplete, and incomplete options are how these items are written.

Six that are missed because they look mild

These are the ones that arrive looking like something routine. The recognition line is the whole point.

Acute retinal necrosis VIRAL RETINA RECOGNISE Peripheral confluent whitening, occlusive arteritis, dense vitritis, rapid circumferential spread, often with pain and a red eye. DO FIRST Same-day retina referral. Systemic antiviral therapy starts on clinical suspicion, with intravitreal antiviral cover for posterior pole threat. NEVER Treat as idiopathic intermediate uveitis with steroid alone. Steroid without antiviral cover accelerates it.	Sympathetic ophthalmia UVEA RECOGNISE Bilateral granulomatous panuveitis weeks to years after penetrating injury or intraocular surgery to the other eye. The fellow eye is the eye at risk. DO FIRST Urgent high-dose systemic corticosteroid with immunomodulatory planning, and specialist co-management. NEVER Dismiss mild fellow-eye inflammation after trauma as sympathetic irritation.
Ciliochoroidal effusion angle closure GLAUCOMA RECOGNISE Bilateral acute myopic shift with uniformly shallow chambers, days after starting a sulfa-derived drug such as topiramate. DO FIRST Stop the drug with the prescriber, suppress aqueous, cycloplege to deepen the chamber, treat pressure medically. NEVER Treat it as pupillary block. Iridotomy does not help and miotics make the anterior rotation worse.	Non-accidental injury PAEDIATRIC RECOGNISE Multi-layered retinal haemorrhages extending to the periphery, retinoschisis, in an infant with a history that does not fit the findings. DO FIRST Admit through the paediatric safeguarding pathway the same day, document findings and images verbatim, notify per jurisdiction. NEVER Arrange outpatient follow-up and let the child leave, or record an opinion about who caused it.
Retinoblastoma PAEDIATRIC TUMOUR RECOGNISE Leukocoria, new strabismus, or unexplained poor fixation under three years. Any white reflex is abnormal until proven otherwise. DO FIRST Same-week referral to ocular oncology for examination under anaesthesia, with B-scan and MRI. Examine the parents and siblings. NEVER Biopsy an intraocular mass in a child, and never book routine review for leukocoria.	Pituitary apoplexy CHIASM RECOGNISE Sudden severe headache with bitemporal field loss, ophthalmoplegia, and reduced consciousness or hypotension. DO FIRST Emergency department now: urgent MRI, endocrine support for adrenal insufficiency, neurosurgical assessment. NEVER Book an outpatient field to confirm a chiasmal defect in an acutely unwell patient.

THE GATE IN ONE SENTENCE

Before you name the disease, decide whether delay costs life, the eye, or the fellow eye, because that decision is answerable from the stem even when the diagnosis is not.

Ocular surface and adnexa

Recall each card in one order: decisive pattern, confirming test, first action, trap. Then name the most dangerous mimic.

Preseptal cellulitis LIDS / ORBIT PATTERN Lid erythema and oedema with normal acuity, pupils, colour, motility, and globe position. CONFIRM Clinical. Image when the examination is limited, severe, atypical, or orbital spread is possible. ACT Oral antibiotic covering skin and sinus organisms, review inside 24-48 hours, written orbital red flags. TRAP In a febrile child with sinusitis this is a diagnosis of exclusion. Motility and colour are the gate.	Dacryocystitis LACRIMAL PATTERN Tender erythematous swelling below the medial canthal tendon, with reflux or tearing. CONFIRM Clinical, culture the drainage, assess for orbital extension. ACT Systemic antibiotics, warm compresses, drainage of an abscess, definitive lacrimal surgery once settled. TRAP Do not probe or irrigate through acutely infected tissue.
Recurrent chalazion LIDS PATTERN Firm lesion returning to the same site, madarosis, yellow tarsal thickening, unilateral chronic blepharoconjunctivitis. CONFIRM Full-thickness biopsy with conjunctival mapping, not a third curettage. ACT Urgent oculoplastic referral. Evert both lids, examine conjunctiva and regional nodes. TRAP Sebaceous carcinoma masquerades as chalazion and as blepharitis, and spreads pagetoid.	Thyroid eye disease ORBIT PATTERN Lid retraction, proptosis, exposure, restrictive diplopia, apical crowding. CONFIRM Thyroid function, colour and pupil testing, fields, exophthalmometry; imaging when atypical or neuropathy is suspected. ACT Surface protection, smoking cessation, urgent specialist care for optic neuropathy or corneal breakdown. TRAP Activity and severity are separate axes. Muscle belly enlargement with tendon sparing is the signature.
Adenoviral keratoconjunctivitis CONJUNCTIVA PATTERN Follicles, watery discharge, preauricular node, high contagion; subepithelial infiltrates may follow. CONFIRM Clinical; rapid antigen testing helps in selected settings. ACT Supportive care and strict hygiene. Corticosteroid only for visually significant infiltrates, monitored. TRAP Contagious period and instrument disinfection matter as much as the drop.	Episcleritis vs scleritis EPISCLERA / SCLERA PATTERN Sectoral mobile vessels that blanch with phenylephrine, against deep violaceous non-blanching injection with boring pain and globe tenderness. CONFIRM Phenylephrine, palpation, B-scan for posterior disease, phenotype-directed systemic workup. ACT Episcleritis: lubricant, cool compress, NSAID. Scleritis: systemic anti-inflammatory or immunomodulatory therapy, urgent specialist. TRAP Topical steroid alone is inadequate in scleritis and dangerous if the cause is infectious.

Cornea and refractive surgery

Recall each card in one order: decisive pattern, confirming test, first action, trap. Then name the most dangerous mimic.

Bacterial keratitis CORNEA	HSV keratitis CORNEA
PATTERN Epithelial defect over a dense stromal infiltrate, pain, chamber reaction, lens wear or trauma. CONFIRM Measure and photograph. Scrape for culture when central, large, deep, atypical, post-surgical or unresponsive. ACT Intensive topical antimicrobial with a loading phase, no lens wear, review inside 24 hours.	PATTERN Dendrite with terminal bulbs and reduced corneal sensation for epithelial disease; stromal haze or endothelial oedema with keratic precipitates for immune disease. CONFIRM Clinical, with sensation testing. PCR when atypical. ACT Oral or topical antiviral for epithelial disease. Add steroid only for immune stromal or endothelial disease, always under antiviral cover.
TRAP Never patch, and never treat a central ulcer on conjunctivitis dosing.	TRAP Steroid over an active dendrite converts a treatable ulcer into a geographic one.
Acanthamoeba keratitis CORNEA	Keratoconus ECTASIA
PATTERN Pain far out of proportion to signs, water exposure or poor lens hygiene, radial keratoneuritis, poor response to antibacterials and antivirals. CONFIRM Culture on non-nutrient agar with bacterial overlay, PCR, or confocal. Repeat if suspicion persists. ACT Prolonged intensive biguanide therapy, often with a diamidine, cornea specialist involved from the start.	PATTERN Rising irregular astigmatism, scissoring reflex, Vogt striae, Fleischer ring, inferior steepening and thinning on tomography. CONFIRM Serial tomography for progression: thinnest point, maximum keratometry, posterior elevation. ACT Cross-linking for documented progression, specialty lenses for vision. Treat the eye rubbing and the atopy.
TRAP A ring infiltrate is late and not required for the diagnosis. Steroid worsens it.	TRAP Cross-linking stabilises, it does not correct. Progression is a tomographic judgement, not a refraction change.
Post-LASIK DLK REFRACTIVE SURGERY	Neurotrophic keratopathy SURFACE
PATTERN Diffuse sterile interface haze in the first days, no epithelial defect, no chamber reaction, minimal pain. CONFIRM Clinical, staged by extent and central involvement. ACT Intensive topical steroid, close review. Flap lift and irrigation for stage three or four.	PATTERN Persistent epithelial defect with rolled edges, minimal pain, reduced sensation, after zoster, HSV, surgery or a fifth-nerve lesion. CONFIRM Corneal sensation testing before anaesthetic, staged by epithelial and stromal loss. ACT Preservative-free lubrication, punctal occlusion, autologous serum or nerve growth factor therapy, tarsorrhaphy or bandage lens for progression.
TRAP Focal infiltrate, pain, discharge or chamber reaction is infectious keratitis, where steroid is the wrong answer.	TRAP The absence of pain is why it presents late and perforates. Sensation is the test that names it.

Retina, choroid and vitreous

Neovascular AMD MACULA PATTERN New distortion or a central scotoma over days to weeks, subretinal or intraretinal fluid, haemorrhage, pigment epithelial elevation. CONFIRM OCT for fluid and its compartment, angiography or OCT angiography when the lesion type is unclear. ACT Urgent retina referral for anti-VEGF therapy, with a loading course and interval-based review. Amsler self-monitoring and fellow-eye counselling. TRAP Good acuity is a reason to treat quickly, not a reason to wait. Drusen do not explain new fluid.	Central serous chorioretinopathy MACULA PATTERN Younger patient, hyperopic shift, mild distortion, a shallow serous detachment with a thick choroid, steroid exposure or high stress. CONFIRM OCT for subretinal fluid and choroidal thickness. Angiography for chronic or atypical disease. ACT Remove exogenous corticosteroid where possible, observe acute episodes, treat persistent disease with photodynamic therapy or a mineralocorticoid antagonist per specialist. TRAP Anti-VEGF is not first-line here, and the corticosteroid in the stem is the answer to the management item.
Proliferative diabetic retinopathy RETINA PATTERN New vessels at the disc or elsewhere, vitreous or preretinal haemorrhage, with or without centre-involving oedema. CONFIRM Dilated examination, OCT for the macula, wide-field imaging or angiography for the periphery. ACT Panretinal photocoagulation or anti-VEGF per specialist, treat centre-involving oedema, and drive systemic control: glycaemia, blood pressure, lipids, renal review. TRAP Anti-VEGF without a reliable follow-up plan is unsafe in proliferative disease. Ask who is coming back.	Retinal vein occlusion VASCULAR PATTERN Sector or four-quadrant haemorrhage with dilated tortuous veins, cotton wool spots, macular oedema. CONFIRM OCT for oedema, RAPD and field for ischaemia, blood pressure, glucose, lipids, and a clotting or inflammatory screen in the young. ACT Anti-VEGF for centre-involving oedema, treat neovascular complications, and manage the vascular risk factors that caused it. TRAP An ischaemic central occlusion threatens the angle. Undilated review at 4 to 6 weeks is not enough.
Rhegmatogenous detachment VITREORETINAL PATTERN Flashes, a shower of floaters, a curtain, relative field loss, pigment or haemorrhage in the vitreous. CONFIRM Dilated peripheral examination with indentation. B-scan when the view is obscured. ACT Urgent retina referral. Macula-on disease is the most time-critical thing on this page after an artery occlusion. TRAP A normal central OCT says nothing about the periphery. Positioning advice is not treatment.	Hydroxychloroquine retinopathy TOXIC PATTERN Often asymptomatic early. Parafoveal outer retinal thinning, later a bull's-eye. Asian eyes may show a pericentral pattern. CONFIRM Baseline then annual screening after five years, earlier with risk: dose above roughly 5 mg/kg, renal impairment, tamoxifen. Fields plus OCT, with mfERG or FAF when equivocal. ACT Discuss stopping with the prescriber when toxicity is confirmed. Damage does not reverse. TRAP Fundus appearance is a late sign. Calculate the dose per real body weight and test the pericentral field in Asian patients.

Glaucoma, uvea and the optic nerve

Primary open angle glaucoma GLAUCOMA PATTERN Open angle, structural rim and RNFL loss matching a repeatable field defect, often asymmetric, usually asymptomatic. CONFIRM Gonioscopy, corrected pressure with central thickness in mind, disc photography, OCT, and at least two reliable fields before committing. ACT Set a target pressure from the damage, the baseline and life expectancy. First-line selective laser trabeculoplasty or a prostaglandin analogue, then reassess against the target. TRAP Pressure is a risk factor, not the diagnosis. Treating a number without structure and function is how normal-tension disease is missed.	Angle closure mechanisms GLAUCOMA PATTERN Pupillary block gives a convex iris that flattens after iridotomy. Plateau iris persists after iridotomy. Effusion gives bilateral myopic shift and uniform shallowing. CONFIRM Gonioscopy with indentation, anterior segment OCT or ultrasound, and reassessment after any iridotomy. ACT Iridotomy for block, iridoplasty or long-term miotic for plateau, drug withdrawal with cycloplegia for effusion. TRAP One treatment does not fit the three mechanisms. Naming the mechanism is the item.
Anterior uveitis UVEA PATTERN Pain, photophobia, circumlimbal injection, cells and flare, keratic precipitates, posterior synechiae, pressure high or low. CONFIRM Grade cells and flare, dilate to check the posterior segment, and phenotype before ordering: granulomatous, bilateral, recurrent, or systemic features. ACT Topical steroid at a frequency matched to the grade with a real taper, cycloplegia for pain and synechiae, monitor pressure, treat the cause. TRAP Stopping steroid when the eye looks comfortable causes the rebound. The taper is the treatment plan.	Optic neuritis OPTIC NERVE PATTERN Younger patient, subacute monocular loss over days, pain on eye movement, RAPD, dyschromatopsia, disc normal or swollen. CONFIRM MRI of brain and orbits with contrast, and antibody testing for MOG and aquaporin-4 when the picture is atypical or severe. ACT Neurology referral. Intravenous corticosteroid speeds recovery without changing final acuity, and is used for severe or bilateral loss and for MRI-defined risk. TRAP Oral prednisone at standard dose alone is the wrong answer. Painless loss over 50 with a pallid swollen disc is arteritic, not this.
Ischaemic optic neuropathy OPTIC NERVE PATTERN Sudden painless loss with an altitudinal defect and a swollen disc. Chalky pallid oedema, jaw claudication and constitutional symptoms mean arteritic. CONFIRM ESR, CRP and platelets immediately for any suspicion over 50; temporal artery imaging or biopsy after treatment starts. ACT High-dose systemic corticosteroid at once for arteritic disease. Non-arteritic disease gets vascular risk management and sleep apnoea screening. TRAP A normal ESR does not exclude arteritis. The fellow eye is what the treatment protects.	Compressive and chiasmal disease PATHWAY PATTERN Slowly progressive loss, disc pallor without prior swelling, bitemporal defects respecting the vertical midline, or a junctional defect. CONFIRM Formal fields both eyes, OCT for a band or bow-tie pattern, and MRI with attention to the sella and chiasm. ACT Neurosurgery and endocrinology referral by urgency, with acute presentations treated as apoplexy. TRAP A field defect that respects the vertical midline is never retinal. Pallor with no history of swelling means chronic compression.

Anchor, caveat, monitor

A memorised dose used in the wrong tissue or the wrong host is a wrong answer.

Antimicrobial anchors

INDICATION	ANCHOR	CAVEAT THAT GETS EXAMINED
Bacterial keratitis	Fluoroquinolone for small peripheral ulcers; fortified therapy for central, deep or resistant disease. Load every 5-15 minutes, then hourly.	Culture first when central, large, deep or unresponsive. Do not wait for results to treat.
HSV epithelial keratitis	Aciclovir 400 mg five times daily, or valaciclovir 500 mg three times daily, 7-10 days. Topical ganciclovir 0.15% is an alternative.	No steroid over an active epithelial ulcer. Steroid belongs to immune stromal or endothelial disease, with antiviral cover.
Herpes zoster ophthalmicus	Valaciclovir 1 g three times daily, or aciclovir 800 mg five times daily, 7-10 days, renally adjusted.	Ocular disease lags the rash. A quiet first examination does not end surveillance.
Fungal keratitis	Natamycin 5% hourly initially for filamentous organisms, prolonged course, cornea specialist involved.	Corticosteroid accelerates progression. Vegetative trauma and feathery margins are the clues.
Acanthamoeba keratitis	Intensive biguanide therapy, PHMB or chlorhexidine 0.02%, often with a diamidine, for months.	Pain out of proportion, water exposure. Early disease is mislabelled HSV.
Hyperacute conjunctivitis	Systemic ceftriaxone-based treatment per current CDC guidance, saline lavage, STI testing, partner care.	Copious purulence with corneal risk is an emergency, not a drop problem.

Contraindication anchors

Topical beta-blocker	CAUTION	Alpha-2 agonist	CAUTION
AVOID Bronchospastic asthma or COPD, bradycardia, heart block, decompensated heart failure.		AVOID Brimonidine in infants and very young children: CNS depression and apnoea.	
REDUCE Punctal occlusion and lid closure lower systemic absorption.		EXPECT Fatigue, dry mouth, late follicular allergy, hypotension.	
TRAP A resting pulse in the low 50s is in the stem for a reason.		TRAP Check interacting systemic drugs before prescribing.	
Miotic	CAUTION	Topical corticosteroid	CAUTION
AVOID Susceptible peripheral retina, active inflammation, ciliochoroidal effusion mechanisms.		AVOID Active HSV epithelial disease, fungal and Acanthamoeba keratitis, uncontrolled bacterial infection.	
EXPECT Brow ache, induced myopia, poor night vision.		MONITOR IOP, epithelial healing, cataract, and the taper.	
TRAP Useless with an ischaemic sphincter, harmful in topiramate-type effusion.		TRAP Name tissue, cause, dose, and monitoring, or the answer is incomplete.	

Lowering pressure

Pressure-lowering classes

CLASS	MECHANISM	ANCHOR, AND THE CAVEAT THAT GETS EXAMINED
Prostaglandin analogue	Increases uveoscleral outflow	Once nightly, the largest single-agent reduction. Hyperaemia, lash growth, iris and periocular pigmentation, orbital fat atrophy. Caution with active uveitis or a history of herpetic keratitis.
Beta-blocker	Reduces aqueous production	Twice daily, or once for gel-forming. Avoid in bronchospastic disease, bradycardia, heart block and decompensated failure. A resting pulse in the low 50s is in the stem for a reason.
Alpha-2 agonist	Reduces production, some outflow	Two or three times daily. Late follicular allergy is common. Contraindicated in infants and young children: CNS depression and apnoea.
Topical carbonic anhydrase inhibitor	Reduces production	Two or three times daily. Stinging, metallic taste. Caution with sulfonamide reactions and low endothelial reserve.
Systemic acetazolamide	Reduces production strongly	For acute pressure crises. Paraesthesia, malaise, metabolic acidosis, renal stones, hypokalaemia. Avoid in sickle disease and significant renal impairment.
Rho kinase inhibitor	Increases trabecular outflow	Once nightly, additive to other classes. Hyperaemia is very common, with subconjunctival haemorrhage and corneal verticillata.
Cholinergic agonist	Increases trabecular outflow	Useful in plateau iris and pseudophakic eyes. Brow ache, induced myopia, poor night vision. Useless against an ischaemic sphincter and harmful in ciliochoroidal effusion.

CHOOSING THE SECOND AGENT

Add a class with a different mechanism, not a second drug of the same one. Check adherence and instillation technique before escalating, and count the preservative load on a compromised surface. Every added bottle costs adherence, which is the commonest reason a target pressure is missed.

Controlling inflammation

Four steps, and the specific thing each one fails at. Naming the step is usually easier than naming where it stops working, which is what the item asks.

Anti-inflammatory ladder, and where each step fails

Topical NSAID	SURFACE AND MACULA	Topical steroid	ANTERIOR SEGMENT
USE	Postoperative pain and macular oedema prophylaxis, episcleritis, some allergic disease.	USE	Anterior uveitis, immune stromal keratitis with antiviral cover, significant postoperative inflammation.
FAILS	Not adequate for significant intraocular inflammation.	FAILS	Does not reach the posterior segment reliably, and does not control scleritis.
TRAP	Prolonged use on a compromised surface, particularly a neurotrophic or post-refractive cornea, causes melt.	TRAP	Contraindicated over active HSV epithelial disease, fungal and Acanthamoeba keratitis, and uncontrolled bacterial infection. Monitor pressure and the taper.
Periocular and intravitreal steroid	POSTERIOR	Systemic steroid and steroid sparing	HOST
USE	Intermediate and posterior uveitis, refractory macular oedema.	USE	Scleritis, arteritic ischaemic neuropathy, sympathetic ophthalmia, sight-threatening systemic disease.
FAILS	Not for undiagnosed infectious disease, and not a substitute for systemic control in systemic vasculitis.	FAILS	Long-term control. Immunomodulatory therapy is planned from the start, co-managed.
TRAP	Steroid response and cataract are expected, not surprises. Exclude infection first, every time.	TRAP	Glucose, bone, blood pressure, mood, infection risk and vaccination status are all part of the correct answer.

HOST FACTORS THAT CHANGE THE DRUG

Pregnancy and breastfeeding, age at either extreme, renal and hepatic function, asthma and cardiac conduction, sulfonamide reactions, sickle disease, immunosuppression, and the systemic drug list. When a stem gives you a weight, a pulse, a creatinine or a pregnancy, it is the discriminator, not decoration.

Arithmetic, fast

The fastest marks on the exam, and the easiest to lose to a sign error.

Formulas

NAME	FORM	USE AND SIGN DISCIPLINE
Spherical equivalent	sph + cyl/2	Halve the cylinder with its own sign. -6.00 -2.00 x 180 gives -7.00 D.
Transposition	sph + cyl, flip, 90°	New sphere is old sphere plus old cylinder, reverse the cylinder sign, rotate the axis by 90 degrees.
Prentice rule	$\Delta = \text{cm} \times \text{D}$	Decentration in centimetres. Base follows the thick edge for plus lenses, the thin edge for minus.
Vertex conversion	$F2 = F1/(1 - dF1)$	Distance in metres, signs preserved. Matters beyond about ± 4.00 D.
Kestenbaum estimate	1 / acuity fraction	Starting magnification only. Refine by task, working distance, contrast, field, endurance.
Equivalent viewing power	1 / metres	10 cm working distance is 10 D.
Hofstetter	$18.5 - 0.30 \text{ age}$	Average. Minimum is $15 - 0.25 \text{ age}$, maximum $25 - 0.40 \text{ age}$. A screening comparison, not a diagnosis.
Gradient AC/A	$\Delta \text{phoria} / \Delta \text{D}$	State the sign convention before you calculate.
Screening arithmetic	$\text{TP}/(\text{TP}+\text{FN})$	Sensitivity. Specificity is $\text{TN}/(\text{TN}+\text{FP})$. PPV rises with prevalence; NPV falls.

Drills: cover the right column

PROMPT	WORKING AND ANSWER
Transpose +2.00 -3.00 x 180	-1.00 +3.00 x 090
Spherical equivalent of -4.00 -2.00 x 090	-4.00 + (-1.00) = -5.00 D
4 mm decentration through +5.00 D	$0.4 \times 5 = 2\Delta$
-10.00 D spectacle at 12 mm, to the corneal plane	$-10/(1 - (0.012 \times -10)) = -8.93 \text{ D}$
Starting magnification for 20/160	$160/20 = 8\times$
Hofstetter average amplitude at age 20	$18.5 - 6.0 = 12.5 \text{ D}$
Hydroxychloroquine 400 mg daily at 60 kg	6.67 mg/kg, above the 5 mg/kg threshold
Soft toric rotated 10° left of a 180 axis	LARS: order 010
GP steepened 0.50 D, over-refraction -0.50 D, from -3.00 D	SAM: $-3.00 -0.50 -0.50 = -4.00 \text{ D}$
Anisometropia -2.00 OD, -5.00 OS, 10 mm below centre	2Δ and 5Δ base down: 3Δ imbalance

Marks are lost to sign errors in transposition and vertex, to forgetting that Prentice needs centimetres, and to quoting a magnification estimate as a prescription. Write one line of working every time.

Lenses, and the arithmetic that follows

Fitting decisions and screening arithmetic. Both are fast marks, and both are lost to sign errors rather than to ignorance.

Contact lens and refractive decisions that get examined

SITUATION	THE RULE, AND THE ARITHMETIC IF THERE IS ANY
Soft toric rotation	Left add, right subtract, from the lens marking as observed. Rotated 10 degrees left of a 180 axis means order axis 010.
Rigid lens fit change	Steeper add minus, flatter add plus. Steepening 0.50 D from -3.00 D with a -0.50 D over-refraction gives -4.00 D.
Lens-related red eye	Stop wear, exclude infiltrative keratitis, and decide sterile against infectious on pain, epithelial defect, chamber reaction and location.
Vertex distance	Convert beyond about four dioptres. $F2 = F1$ divided by $(1 \text{ minus } dF1)$, distance in metres, signs preserved.
Induced prism	Prentice: prism dioptres equal decentration in centimetres times power. Anisometropia through a reading position is the classic item.
Myopia control	Discuss the options with their evidence, expected effect size and rebound, and document the discussion, including the option of doing nothing.

Screening arithmetic, worked once

QUANTITY	FORMULA	WORKED EXAMPLE: 100 WITH DISEASE, 900 WITHOUT, 80 DETECTED, 90 FALSE POSITIVES
Sensitivity	$TP / (TP + FN)$	$80 / 100 = 80\%$
Specificity	$TN / (TN + FP)$	$810 / 900 = 90\%$
Positive predictive value	$TP / (TP + FP)$	$80 / 170 = 47\%$
Negative predictive value	$TN / (TN + FN)$	$810 / 830 = 98\%$

THE POINT OF THAT TABLE

Sensitivity and specificity are properties of the test. Predictive values are properties of the population. Halve the prevalence and the positive predictive value falls while the test does not change, which is exactly what the item is testing.

The cheapest marks on the paper

Fifteen or so items turn on duty, consent and documentation. They are the least prepared part of the exam, and the answers are almost always the same shape: do the clinically safe thing, say so plainly, write it down, and keep the patient in the system.

Law, ethics and public health

ITEM TYPE	WHAT THE CORRECT ANSWER LOOKS LIKE
Informed refusal	Explain the risk in plain terms, offer the alternative, document the refusal and the specific risks discussed, arrange the shortest safe review, and keep the door open. Do not discharge the patient for refusing.
Capacity and consent	Capacity is decision-specific. Assess understanding, retention, weighing and communication. Involve a surrogate only when capacity is genuinely absent.
Minors	Consent from a parent or guardian, assent from the child, and jurisdiction-specific exceptions. Safeguarding concerns override confidentiality.
Records	Contemporaneous, factual, timed and attributed. Amend by addendum, never overwrite. Record what you were told, what you found, what you advised and what was declined.
Delegation	Delegate tasks, never clinical judgement. The delegating clinician remains responsible for the decision.
Telehealth	Establish that the standard of care can be met remotely. If the decision needs a finding you cannot obtain, arrange in-person examination.
Driving and work	Advise against driving where the standard is not met, document the advice, and follow the local reporting duty exactly.
Notifiable and public health	Report as jurisdiction requires, treat the patient, and address contacts and partners where the disease demands it.

THE SHAPE OF A CORRECT ANSWER

Act on the clinical risk first. Explain in plain language. Document what you found, advised and were refused. Set the shortest safe review interval. Follow the local duty exactly, and never make the patient leave the system as a punishment for disagreeing with you.

Competing pairs

Each pair is two conditions that trade places under time pressure. The decider row is the single finding that separates them, which is the finding the item will turn on.

Orbital cellulitis		vs	Cavernous sinus thrombosis
LATERALITY	Unilateral, stays unilateral		Crosses to the other eye within hours to days
CRANIAL NERVES	Motility limited by swelling and pain		III, IV, VI and V1-V2 out of proportion to swelling
PUPIL	RAPD if the apex is crowded		RAPD plus a poorly reactive mid-dilated pupil
SYSTEMIC	Fever with sinus disease		Fever, rigors, obtunded, meningism
DECIDER	Bilateral signs, or multiple cranial neuropathies out of proportion to orbital swelling, mean cavernous sinus and neurosurgical admission rather than orbital drainage.		

Acanthamoeba keratitis		vs	Herpes simplex stromal keratitis
PAIN	Severe, out of proportion to the signs		Proportionate, often modest
INFILTRATE	Radial perineural, then an incomplete ring		Diffuse or disciform central haze
ON ANTIVIRALS	Unchanged or worse over weeks		Settles, then recurs on withdrawal
RISK FACTOR	Tap water, swimming, poor lens hygiene		Previous cold sore or previous dendrite
DECIDER	Radial perineural infiltrates or a ring infiltrate in a lens wearer exposed to water outweigh any pseudodendrite. Scrape and culture for amoeba before more steroid goes in.		

Scleritis		vs	Episcleritis
PAIN	Deep and boring, wakes the patient		Gritty or uncomfortable, sleeps through it
PHENYLEPHRINE	Deep injection does not blanch		Redness blanches within minutes
DAYLIGHT	Blue-violet hue with an oedematous sclera		Bright red vessels over white sclera
SYSTEMIC LINK	Rheumatoid disease, vasculitis, infection		Usually none, most often idiopathic
DECIDER	Redness that will not blanch with phenylephrine, in an eye tender to palpation, means scleritis, systemic work-up and systemic treatment rather than a topical prescription.		

Neovascular AMD		vs	Polypoidal choroidal vasculopathy
PATIENT	Older, soft drusen in both eyes		Often Asian ethnicity, fewer soft drusen
LESION	Grey-green subretinal membrane		Orange-red nodules at the posterior pole
HAEMORRHAGE	Small bleed beside the lesion		Large and recurrent, sometimes massive
IMAGING	Fluid on OCT, leakage on angiography		Polyps on indocyanine green angiography
DECIDER	Recurrent large haemorrhage or orange nodules justify indocyanine green angiography, because polypoidal disease may need photodynamic therapy as well as anti-VEGF.		

Competing pairs, continued

Central retinal vein occlusion		vs	Ocular ischaemic syndrome
ONSET	Abrupt, often noticed on waking		Gradual blur over weeks, with ache in some eyes
HAEMORRHAGE	Dense, four quadrants, up to the disc		Sparse, mid-peripheral, few at the posterior pole
ARTERIES	Normal calibre or mildly narrowed		Markedly narrowed, spontaneous pulsation on gentle pressure
PRESSURE	Normal or raised early		Often low, because ciliary perfusion has fallen
DECIDER	Sparse mid-peripheral haemorrhage with narrowed arteries and a soft eye points to carotid disease, which sends the patient for urgent vascular imaging rather than for injections alone.		

Central serous chorioretinopathy		vs	Neovascular age-related maculopathy
AGE AND MACULA	Thirties to fifties, no drusen		Usually past sixty, drusen in both maculae
OCT FLUID	Clear subretinal dome, thick choroid		Fluid with hyper-reflective tissue, thin choroid
BLOOD	Absent		Present in most eyes at some stage
REFRACTION	Hyperopic shift, micropsia		No consistent shift, metamorphopsia
DECIDER	Hyper-reflective material above the pigment epithelium, or any haemorrhage, means a membrane and same-week anti-VEGF. Its absence with a thick choroid buys three months of observation.		

Neovascular glaucoma		vs	Acute primary angle closure
ONSET	Days to weeks of increasing ache		Hours, severe, with nausea and vomiting
CENTRAL CHAMBER	Deep, until the membrane contracts		Very shallow in both eyes
IRIS AND ANGLE	Vessels at the pupil margin, membrane in the angle		No vessels, angle closed by apposition
FELLOW EYE	Normal depth, open angle		Narrow, at risk, needs iridotomy
DECIDER	A deep central chamber with vessels at the pupil margin means the closure is neovascular, so the treatment is anti-VEGF and retinal laser rather than iridotomy to either eye.		

Optic neuritis		vs	Non-arteritic ischaemic optic neuropathy
AGE AND ONSET	Twenties to forties, worse over days		Past fifty, noticed on waking, then static
PAIN	Ache on eye movement in most cases		Painless
DISC	Normal in retrobulbar cases		Swollen, often segmental, with splinter haemorrhage
FIELD	Central depression		Altitudinal, respecting the horizontal
DECIDER	A swollen disc with an altitudinal field in an older patient moves the work to vascular risk and the fellow eye. A flat disc with pain on movement sends the patient for MRI and neurology.		

Competing pairs, continued

Arteritic AION		vs	Non-arteritic AION
DISC	Chalky pallid swelling, cup often spared		Hyperaemic segmental swelling in a small crowded disc
ACUITY	Counting fingers or worse in most eyes		Often 20/60 or better at onset
SYSTEMIC	Jaw ache, scalp tenderness, weight loss		Nocturnal hypotension, sleep apnoea, diabetes
MARKERS	ESR and CRP raised, platelets high		ESR and CRP normal for age
DECIDER	Raised ESR and CRP with jaw claudication move the case to same-day steroid and biopsy, whatever the disc looks like. Normal markers with a crowded fellow disc do not.		

Horner syndrome		vs	Physiological anisocoria
ANISOCORIA	Larger in dim light, with dilation lag		Equal in light and dark, no lag
LIDS	Mild ptosis with the lower lid raised		Lid position normal or symmetrically lax
APRACLOnidine	Reverses the anisocoria		No change in either pupil
COURSE	New, with neck, chest or vascular symptoms		Long-standing, visible in old photographs
DECIDER	Anisocoria that grows in the dark with dilation lag is denervation and needs imaging of the whole sympathetic chain. Anisocoria equal in light and dark needs nothing.		

Acquired fourth nerve palsy		vs	Skew deviation
TORSION	Higher eye extorted, often symptomatic		Higher eye intorted, part of a tilt reaction
HEAD TILT	Away from the palsy, relieves the torsion		Towards the lower eye, with a body lean
THREE STEP	Fits one muscle throughout		Often fits no single muscle
COMPANY	Isolated, follows the trauma		Brainstem signs, nystagmus, gait ataxia
DECIDER	The direction of torsion separates them: extorsion of the higher eye is a superior oblique palsy, intorsion is a skew and needs posterior fossa imaging.		

Sterile contact lens infiltrate		vs	Microbial keratitis
PAIN	Gritty, eases within an hour of lens removal		Severe, out of proportion, persists after removal
SITE	Peripheral, with a clear zone to the limbus		Central or paracentral, no clear zone
EPITHELIUM	Intact, or a stain smaller than the infiltrate		Defect wider than the infiltrate beneath it
CHAMBER	Quiet, no cells		Cells, flare, sometimes a hypopyon
DECIDER	An epithelial defect larger than the infiltrate under it, or any anterior chamber reaction, moves the eye to culture and hourly therapy whatever the size of the lesion.		

Competing pairs, continued

Ethambutol optic neuropathy		vs	Nutritional optic neuropathy	
ONSET	Weeks to months into a dose above 15 mg/kg		Months to years of poor diet or alcohol	
FIELDS	Central depression, sometimes bitemporal		Central or centrocaecal scotomata	
DECIDING TEST	Dose per kg, with renal function		B12, folate, red cell thiamine	
COURSE	Improves over months once the drug goes		Improves with repletion, not withdrawal	
DECIDER	The weight-based dose read against renal function separates them, and it is the one thing that can be acted on the same day. Both need the blood tests, only one needs a drug stopped.			

Pseudomyopia		vs	Axial myopia	
CYCLOPLEGIC REFRACTION	Plano or plus: the minus disappears		The minus stays, within 0.50 D of the dry result	
NRA AND PRA	NRA low, PRA full		Both near the norms for age	
RETINOSCOPY	Fluctuates, with pupils at 3 mm		Steady reflex, pupils normal for the room	
ON MINUS LENSES	Symptoms worsen and the minus grows		Blur settles, and it stays settled	
DECIDER	A cycloplegic refraction that comes back plano or plus in a young patient who reads for ten hours a day is spasm, whatever the dry retinoscopy says. Do not dispense the minus.			

Flat rigid fit		vs	Steep rigid fit	
FLUORESCEIN CENTRE	Dark apical touch		A central pool of fluorescein	
EDGE	Wide standing clearance, bubbles		Sealed edge, minimal clearance	
POSITION AND MOVEMENT	Rides high, moves 3 mm or more		Rides central to low, barely moves	
VISION	Clears for a moment after a blink		Blurs after the blink, clears between	
DECIDER	Read the apex first. Bearing means the lens is flatter than the cornea, so steepen it and add minus. Central pooling means steeper, so flatten it and add plus.			

Rod-cone dystrophy		vs	Cone-rod dystrophy	
FIRST SYMPTOM	Night blindness, then field loss		Glare, poor acuity, colour confusion	
FIELD	Mid-peripheral ring, then tunnel		Central scotoma with a full periphery	
ELECTRORETINOGRAPH	Rod responses go first		Cone and flicker responses go first	
FUNDUS	Mid-peripheral spicules, waxy pallor		Macular atrophy, often a bull's eye	
DECIDER	Ask which failed first, the dark or the detail. Rods first means field and mobility work. Cones first means magnification, glare control and colour advice.			

Competing pairs, continued

Divergence insufficiency		vs	Bilateral sixth nerve palsy
COMITANCE	The same in right and left gaze		Grows in the field of the weak muscle
ABDUCTION	Full, with normal saccadic velocity		Limited, with slowed abducting saccades
DISTANCE AND NEAR	Eso at distance, near almost straight		Eso at both, worse at distance
COURSE	Months, stable, no other signs		Days to weeks, often with other signs
DECIDER	Comitance and saccadic velocity decide it. Full, equal abduction in an older patient means prism. Slow abduction or an incomitant angle means imaging.		

Retinoblastoma		vs	Coats disease
FUNDUS	Chalky elevated mass with vitreous seeds		Yellow lipid exudate with retinal telangiectasia
LATERALITY	Unilateral or bilateral, often multifocal		Almost always unilateral, usually male
ULTRASOUND	Calcification within the mass		No calcification, exudate and detachment only
AGE	Median 18 months, most under 2 years		Usually 4 to 8 years at presentation
DECIDER	Calcification with vitreous seeding means retinoblastoma and a same-day oncology referral. Exudate with telangiectasia and no calcium means Coats, urgent but not oncological.		

Infectious keratitis after LASIK		vs	Diffuse lamellar keratitis
ONSET	Day 2 onwards, worsening hour by hour		Day 1 to 2, then improving on steroid
SYMPTOMS	Increasing pain with photophobia		Mild or no pain, blur dominates
INFILTRATE	Focal and dense, crosses into stroma		Diffuse granular haze within the interface
EPITHELIUM, CHAMBER	Defect present, cells in the chamber		Epithelium intact, chamber quiet
DECIDER	A focal infiltrate with an epithelial break or chamber cells means lift, culture and treat as infection. Diffuse haze with intact epithelium means hourly steroid and same-day review.		

Acute bacterial endophthalmitis		vs	Toxic anterior segment syndrome
ONSET	Day 2 to 5, worsening by the hour		Within 24 hours of surgery, then static
SYMPTOMS	Pain with lid swelling and injection		Little or no pain, blur dominates
CORNEA	Clear or focally hazy		Limbus-to-limbus oedema
VITREOUS	Dense opacities on ultrasound		Clear, the reaction stays anterior
DECIDER	Vitreous involvement with pain means tap and inject within the hour. Limbus-to-limbus oedema on day 1 with a clear vitreous means intensive steroid and a look at the theatre process.		

Competing pairs, continued

Acute retinal necrosis		vs	Cytomegalovirus retinitis
HOST	Immunocompetent, often after shingles		Deeply immunosuppressed, low CD4 count
VITREOUS	Dense vitritis with an aching eye		Little or no vitreous reaction, painless
BORDER	Scalloped, spreads circumferentially in days		Granular edge, creeps over weeks
VESSELS	Occlusive arteritis with sheathing		Haemorrhage with perivascular whitening
DECIDER	A well patient with dense vitritis and arteritis needs induction antiviral today. A granular indolent border in an immunosuppressed patient points at cytomegalovirus and its own agents.		

Choroidal melanoma		vs	Choroidal naevus
THICKNESS	Over 2 mm, and growing between records		Under 2 mm and unchanged for years
SURFACE	Orange lipofuscin, no drusen		Drusen, often with a pigmented halo
FLUID	Subretinal fluid at the margin		No fluid, retina flat over it
ULTRASOUND	Low internal reflectivity, excavation		Too flat to measure, no excavation
DECIDER	Thickness over two millimetres, subretinal fluid, orange pigment or documented growth each turn observation into a same-week ocular oncology referral.		

Internuclear ophthalmoplegia		vs	Ocular myasthenia mimicking it
CONVERGENCE	Full, because it bypasses the fasciculus		Variable, and it fatigues with effort
NYSTAGMUS	Coarse in the abducting eye		Absent, though a lid twitch may be seen
LIDS	Level and symmetrical		Ptosis, worsening through the day
COURSE	Fixed over days, with other lesions		Fluctuates within a day, better after rest
DECIDER	Fatigue, fluctuation and ptosis make it myasthenia and move the test to antibodies. A fixed adduction deficit with abducting nystagmus and full convergence needs MRI of the brainstem.		

Drug-induced ciliochoroidal effusion		vs	Acute primary angle closure
LATERALITY	Both eyes within hours of each other		One eye now, the fellow eye at risk later
CHAMBER	Shallow centrally and peripherally		Shallow peripherally, central depth near normal
REFRACTION	Myopic shift of 2 to 4 D		No shift once the corneal oedema clears
IRIDOTOMY	Angle stays closed afterwards		Angle opens and the attack breaks
DECIDER	A myopic shift with uniform shallowing in both eyes means the ciliary body has rotated forward. Stop the drug, cycloplege and suppress aqueous, and do not cut an iridotomy for it.		

Competing pairs, continued

Plateau iris syndrome		vs	Residual pupil block
IRIDOTOMY	Patent, and the angle closes anyway		Small, sealed or absent on retroillumination
IRIS CONTOUR	Flat plane with a sudden peripheral rise		Bowed forward from the pupil outward
CENTRAL DEPTH	Normal for the eye		Shallow, and it deepens once the hole works
INDENTATION	Double hump over the ciliary processes		Angle opens smoothly and stays open
DECIDER	A patent iridotomy with a flat iris and normal central depth means the ciliary body holds the angle shut. The answer is iridoplasty or lens extraction, never a second hole.		

PART 8

Seventy cases, by domain

Every case in the bank, grouped by clinical domain, each one followed immediately by its teaching key. This is the order you learn in.

WHY DOMAIN ORDER

Cornea sits with cornea, retina with retina, the neuro-ophthalmic cases together. Competing conditions land within a few pages of each other, which is where the discriminators stick. Reading a whole domain in one sitting builds the comparison the exam actually tests.

WHEN YOU WANT IT COLD INSTEAD

Use the Exam Simulator. Same 350 items in exam order, cases only, every key at the back, so a session can be sat timed with nothing visible.

KEY ON THE FACING PAGE

Answer the case, then turn one page. This book is for learning, so the feedback is immediate and deliberate.

WORKING A SINGLE WEAKNESS

The domain index gives every case with its item-type tags. To repair treatment items, work the Tx column across domains rather than reading the book front to back.

Domain index

This manual prints in this order: by domain, then by case number, with each teaching key on the page after its case. Competing conditions land within a few pages of each other, which is where the discriminators stick.

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CASE S1-26 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

The print that slides together after twenty minutes

A 19-year-old student has four months of print that blurs and slides together after twenty minutes of reading, with frontal headache that eases when one eye is covered. BVA 20/20 each, refraction +0.25 DS each. Cover test orthophoric at distance, 12 Δ exophoria at near. Near point of convergence 14 cm break, 18 cm recovery. Positive fusional vergence at near: blur 8, break 12, recovery 4. Amplitude of accommodation 11 D each.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Convergence paresis from a dorsal midbrain lesion
- B Convergence insufficiency with a remote near point
- C Accommodative insufficiency with reduced amplitude
- D Basic exophoria equal at distance and at near
- E Divergence excess intermittent exotropia

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings support this rather than an accommodative cause?

TREATMENT • SELECT 3

- A Amplitude of accommodation of 11 D in each eye
- B Refraction of +0.25 DS with 20/20 in each eye
- C Frontal headache after twenty minutes of reading
- D Near point of convergence 14 cm with recovery at 18 cm
- E Exophoria of 12 Δ at near with orthophoria at distance
- F Positive fusional vergence at near breaking at 12 Δ

ANSWER _____ CONFIDENCE L • M • H

3 Which management has the best evidence as first-line here?

TREATMENT

- A Office-based vergence therapy with home reinforcement
- B Base-in prism at near, prescribed by Sheard's criterion
- C Plus addition at near of +1.00 D for reading tasks
- D Home pencil push-ups alone, five minutes twice daily
- E Referral for bilateral medial rectus resection surgery

ANSWER _____ CONFIDENCE L • M • H

4 Twelve weeks of therapy done well and symptoms persist. What next?

TREATMENT

- A Prescribe a +1.50 near addition and stop the therapy
- B Discharge with reassurance that the condition is benign
- C Reassess, then add base-in prism if the near point stays remote
- D Repeat the same twelve-week therapy programme without changes
- E Refer for surgery on the horizontal muscles at near

ANSWER _____ CONFIDENCE L • M • H

5 Her college asks for a report on her reading difficulty. What is required?

LAW AND ETHICS

- A Written authorisation from the patient before releasing a report
- B Verbal consent recorded in the notes at the time of the request
- C Nothing, since the request comes from her educational institution
- D Written authorisation from a parent, as she is still a student
- E A summary with the identifiers removed, sent without consent

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-26 • why

1 • B Convergence insufficiency

An exophoria that appears only at near, a near point of 14 cm and positive vergence breaking at 12 are the findings that define it. Accommodation measures 11 D, which is normal at 19.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Accommodative insufficiency	Blur at near after twenty minutes is the classic accommodative complaint.	Amplitude is 11 D, which is age-normal, and the exophoria is at near only.
D	Basic exophoria	There is an exophoria, and the near symptoms fit an exodeviation.	The distance cover test is orthophoric, so the deviation is not equal.

2 • D, E, F Remote near point, near exophoria, low vergence

Both conditions blur print, so the separation sits in the vergence data: a remote near point, an exodeviation confined to near, and a reduced positive fusional range. Amplitude is age-normal.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Amplitude 11 D	It is the accommodative measurement, so it looks decisive here.	By Hofstetter the minimum at 19 is about 10 D, so 11 D is normal.
B, C	Refraction, headache	Both are in the record and both feel relevant to near work.	Neither separates the two, because both occur with either mechanism.

3 • A Office-based vergence therapy

Supervised in-office therapy with home reinforcement outperformed pencil push-ups and placebo in the CITT trial. Prism and plus adds treat the symptom without restoring fusional range.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Push-ups alone	They are free, familiar and easy to instruct in a single visit.	In the trial push-ups alone performed no better than placebo therapy.
B	Base-in prism	It relieves symptoms immediately and is easy to dispense.	It leaves the vergence deficit untreated, so the demand returns.

4 • C Reassess, then prism if measures unchanged

Therapy is judged on the measures, not the calendar. If the near point and the fusional range have not moved, prism supports the deficit, whereas repeating a failed programme costs months.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Repeat unchanged	Persistence is often the answer with therapy that works slowly.	Twelve weeks done well with no change in the measures is a failed course.
B	Discharge	The condition is benign and no pathology has been found.	The symptoms are disabling for a student and they persist untreated.

5 • A Written patient authorisation first

She is an adult, so the record is hers to release. Authorisation is written, specific to the recipient, and filed with the record; the request itself creates no permission.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	College request alone	Educational requests feel administrative rather than clinical.	A third party gains no right of access by asking on its own letterhead.
D	Parental authorisation	Student status suggests a parent is still the responsible party.	At 19 she consents for herself, whoever pays the fees.

IF-THEN RULES FROM THIS CASE

If exophoria appears only at near with a remote near point, treat the vergence system before reaching for prism. If twelve weeks of therapy moves no measure, then add prism.

CASE S1-28 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

The head tilt that started after the bicycle fall

A 22-year-old man has three weeks of vertical double vision since a fall from his bicycle with brief loss of consciousness. Worst looking down, and he misjudges stairs. Head tilted to the left. Right hypertropia 8 Δ in primary position, 14 Δ in left gaze, 18 Δ on right head tilt, 4 Δ on left tilt. Depression in adduction limited OD. Excyclotorsion 8 ° on double Maddox rod. Vertical fusion range 4 Δ.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Decompensated congenital right fourth nerve palsy
- B Right skew deviation from a posterior fossa lesion
- C Right Brown syndrome limiting elevation in adduction
- D Thyroid eye disease restricting the right inferior rectus
- E Acquired right fourth nerve palsy after head trauma

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings favour an acquired palsy over a decompensated congenital one?

TREATMENT • SELECT 3

- A Right hypertropia increasing in left gaze
- B Symptomatic excyclotorsion of 8 ° on double Maddox rod
- C Vertical fusion range of only 4 Δ
- D Onset dated to a head injury three weeks ago
- E Right hypertropia increasing on right head tilt
- F Habitual head tilt towards the left shoulder

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management now?

TREATMENT

- A Surgery on the right inferior oblique within a month
- B Botulinum toxin to the right inferior oblique now
- C Neurosurgical referral for the head injury this week
- D Occlusion or Fresnel prism, reassess at three months
- E Ground-in vertical prism in new spectacles today

ANSWER _____ CONFIDENCE L • M • H

4 What does the head tilt test measure in this patient?

BASIC SCIENCE

- A The vertical fusional range available in each head position
- B Vestibulo-ocular gain during a slow head roll manoeuvre
- C The comitance of the deviation between the two lateral gazes
- D Right intorters compete on right tilt, so the superior rectus elevates
- E Right extorters compete on right tilt, so the inferior oblique elevates

ANSWER _____ CONFIDENCE L • M • H

5 Which finding would point to skew deviation instead?

DIAGNOSIS

- A Hypertropia that increases on head tilt towards the higher eye
- B Limited depression of the higher eye in adduction
- C Vertical deviation that is larger in contralateral gaze
- D Incyclotorsion of the higher eye with a contralateral head tilt
- E Excyclotorsion of the higher eye measured on Maddox rod

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-28 • why

1 • E Traumatic right fourth nerve palsy

The three-step pattern is a right hypertropia worse in left gaze and on right tilt, with limited depression in adduction. Symptomatic extorsion and a 4 Δ fusion range date it to the injury.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Congenital fourth palsy	Trauma often decompensates a long-standing palsy, and the pattern matches.	A congenital palsy carries a large vertical fusion range and no torsional symptoms.
B	Skew deviation	Head trauma with a vertical deviation does raise a central cause.	Skew gives intorsion of the higher eye and no dependence on head tilt.

2 • B, C, D Torsion, small fusion range, dated onset

The three-step results and the head tilt are shared by both forms, so they cannot separate them. Torsional symptoms, a small vertical fusion range and a dated onset belong to the acquired palsy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, E	Three-step results	They are how the palsy is identified, so they look like the answer.	Both forms give the same pattern, which names the muscle and nothing else.
F	Habitual head tilt	A tilt suggests a posture adopted over many years.	New palsies adopt a tilt within days, so a tilt dates nothing.

3 • D Temporary prism or occlusion, review at 3 months

Traumatic fourth nerve palsies improve for up to six months, so the measurement to operate on does not exist yet. Temporary relief keeps him functional while the angle changes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Ground-in prism now	It gives comfortable single vision and looks like definitive care.	The deviation is still changing, so the prescription will soon be wrong.
A	Early surgery	Three weeks of disabling diplopia feels like enough waiting.	Spontaneous recovery is common, so operate only on a stable measurement.

4 • D Intorters compete, superior rectus unopposed

Tilting to the right calls on the right intorters, superior oblique and superior rectus. With the oblique palsied the superior rectus elevates unopposed, so the hypertropia grows.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Extorters on right tilt	Naming the other torsional pair sounds equally plausible.	Extorters act on tilt to the other side, which is why the deviation falls there.
A	Vertical fusion range	Fusion range is measured in this workup and it matters here.	Range is measured with prism, not by tilting the head.

5 • D Incyclotorsion of the higher eye

Skew is a central prenuclear imbalance and produces an ocular tilt reaction: the higher eye intorts and the head tilts towards the lower eye. A palsied superior oblique lets the eye extort.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Excyclotorsion	Torsion is the discriminator, so any torsion looks like the answer.	Direction decides: extorsion of the higher eye is the fourth nerve palsy.
B	Limited depression	It is an underaction, and underactions occur in central disease too.	Depression failing in adduction is superior oblique weakness itself.

IF-THEN RULES FROM THIS CASE

If a vertical deviation follows the three-step pattern with symptomatic extorsion, it is an acquired fourth nerve palsy. If the torsion runs the other way, think skew and image.

CASE S2-01 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

Distance blur that arrived in the middle of exam term

A 19-year-old student has six weeks of variable distance blur after ten-hour revision days, with frontal headache and near print that stays sharp. Dry subjective -1.75 DS each eye, BVA 20/20. Cyclopentolate 1% retinoscopy gives +0.25 DS each eye. Distance retinoscopy fluctuates, pupils 3 mm in room light, push-up near point 6 cm, NRA +0.75 D, PRA -3.50 D, 6 Δ esophoria at 40 cm.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Uncorrected latent hyperopia at distance
- B Convergence excess with esophoria at near
- C Early keratoconus with variable retinoscopy
- D Axial myopia of -1.75 D, newly manifest
- E Pseudomyopia from accommodative spasm

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings separate accommodative spasm from axial myopia?

DIAGNOSIS • SELECT 3

- A NRA reduced to +0.75 D at 40 cm
- B Fluctuating distance retinoscopy with 3 mm pupils
- C Equal refraction in the right and left eye
- D Symptoms that began during ten-hour revision days
- E Cycloplegic refraction of +0.25 D in each eye
- F Distance acuity of 20/20 through -1.75 D

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management?

TREATMENT

- A Withhold the minus, prescribe a +1.00 D add and facility therapy
- B Dispense a partial -1.00 DS both eyes with a near break routine
- C Atropine 1% both eyes for two weeks with a full-time distance add
- D Cyclopentolate 1% nightly for a week, then repeat the dry refraction
- E Dispense -1.75 DS both eyes and review the refraction in six months

ANSWER _____ CONFIDENCE L • M • H

4 At eight weeks the dry refraction is +0.25 DS both eyes and NRA is +2.00 D. What now?

TREATMENT

- A Taper the +1.00 add over a month and keep facility work going
- B Stop the add and the exercises today, and discharge her
- C Continue the +1.00 add full time and review in a year
- D Prescribe +0.25 DS for distance so the plus is never lost
- E Repeat the cycloplegic refraction every three months and hold the add

ANSWER _____ CONFIDENCE L • M • H

5 Her near point is 6 cm. What amplitude is that, against the Hofstetter mean at 19?

BASIC SCIENCE

- A 6.0 D, well below the mean of 12.8 D
- B 10.25 D, the minimum expected at 19
- C 12.8 D, exactly the mean for her age
- D 16.7 D, above the mean of 12.8 D
- E 20.6 D, above the maximum expected at 19

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-01 • why

1 • E Pseudomyopia from accommodative spasm

Cyclopentolate leaves +0.25 D, so the -1.75 D is ciliary tone and not axial length. An NRA of +0.75 D, a 6 cm near point and a fluctuating reflex with 3 mm pupils are sustained overaction.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Axial myopia	Six weeks of blur in a student reads as the myopia everyone expects.	Axial myopia does not vanish under cycloplegia. The plus reading settles it.
A	Latent hyperopia	Cycloplegia does uncover plus, which is what happened here.	Latent hyperopia blurs near before distance, and her manifest was minus.

2 • A, B, E Cycloplegic plus, low NRA, fluctuating reflex

Only measurements that move with ciliary tone can separate the two. Plus under cycloplegia, a reduced NRA and a fluctuating reflex with miosis all move. Acuity and symmetry do not.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	20/20 through -1.75 D	The minus lens works, which is exactly why it gets dispensed.	Both conditions read 20/20 through the minus, so it separates nothing.
C, D	Symmetry, revision history	Both fit the story of a student who worked too long and too close.	History raises the suspicion, and symmetry is common to both.

3 • A No minus, a +1.00 add, facility therapy

There is no refractive error to correct, so the lenses treat the spasm: plus at near unloads accommodation while facility work restores relaxation. Dispensed minus feeds the spasm.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Dispense -1.75 DS	It gives instant clear distance vision and she leaves happy.	Minus lenses drive more accommodation, so the spasm deepens and recurs.
C	Atropine for two weeks	Paralysing the ciliary muscle does break a spasm, and quickly.	Two weeks of atropine costs all of her near vision in an exam term.

4 • A Taper the add, keep the facility work

The refraction and the NRA have normalised, so the add has done its work and comes off gradually while the exercises hold the gain. Abrupt withdrawal is when spasm returns.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Stop and discharge	The numbers are normal, so the episode looks finished.	Relapse in the same term is common. Taper, and keep a review.
C	Add full time for a year	It worked, and leaving it alone risks nothing in the short run.	A permanent add at 19 teaches the eye to stop relaxing at all.

5 • D 16.7 D, above the 12.8 D mean at 19

Amplitude is the reciprocal of the near point in metres: $1 / 0.06 = 16.7$ D. The Hofstetter mean is $18.5 - 0.3 \times 19 = 12.8$ D, so she over-accommodates rather than running short.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	6.0 D	It copies the 6 in the near point straight into dioptres.	Six centimetres is 0.06 m, and $1 / 0.06$ is 16.7 D, not 6.
C	12.8 D, the mean	It is the correct Hofstetter mean, so it looks like the answer.	The mean is the comparison, not her measurement of 16.7 D.

IF-THEN RULES FROM THIS CASE

If the minus disappears under cycloplegia, the ciliary muscle wrote the prescription. Do not dispense it: unload the near task and train relaxation.

CASE S2-07 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

Print that gives up twenty minutes into the evening

A 23-year-old graduate student has three months of blur and frontal ache after 20 minutes of reading, worse each evening and absent at distance. Cycloplegic refraction +0.75 DS each eye, BVA 20/20. Push-up amplitude 4.50 D OD and 4.75 D OS. Accommodative facility 3 cycles per minute with ± 2.00 flippers, failing on the minus. MEM lag +1.25 D. NPC 7 cm. Near phoria 4 Δ exophoria. No medication.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Accommodative insufficiency, both eyes
- B Ill-sustained accommodation with a normal amplitude
- C Convergence insufficiency with a near exophoria
- D Latent hyperopia, uncorrected at near
- E Accommodative infacility with a normal amplitude

ANSWER _____ CONFIDENCE L • M • H

2 Which further finding would make this a systemic problem rather than a functional one?

DIAGNOSIS

- A Symptoms that clear away completely over a two-week summer holiday
- B A tonic pupil with light-near dissociation and vermiform movement
- C A 4 Δ exophoria that rises to 8 Δ by the evening
- D A dynamic retinoscopy lag that measures +1.75 D
- E Facility of 2 cycles per minute, still failing on minus

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate treatment?

TREATMENT

- A 6 Δ base in split between the lenses at near
- B Pencil push-ups to a small target held 5 cm from the nose
- C Full time +0.75 DS, with review in six months
- D A +1.00 D near add with amplitude and facility therapy
- E A +2.50 D near add worn for all near work

ANSWER _____ CONFIDENCE L • M • H

4 At eight weeks the amplitude is 8.50 D and facility 9 cycles per minute. What now?

TREATMENT

- A Stop everything today and discharge her from care
- B Continue the +1.00 add unchanged for another year
- C Increase the add to +1.50 D to protect the gain
- D Refer for imaging, since 8.50 D is still below par
- E Taper the add and keep the facility exercises going

ANSWER _____ CONFIDENCE L • M • H

5 Which two statements about her amplitude are correct?

BASIC SCIENCE • SELECT 2

- A Push-up measurement over-reads the true amplitude
- B Minus lens amplitude reads higher than push-up
- C 4.50 D is within normal limits at her age
- D An amplitude of 4.50 D at 23 is early presbyopia
- E The Hofstetter minimum at 23 years is 9.25 D

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-07 • why

1 • A Accommodative insufficiency

An amplitude of 4.50 D at 23 is under half the expected floor, with a +1.25 D lag and facility failing on minus. Distance is clear and the NPC is 7 cm, so accommodation is the deficit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Ill-sustained accommodation	Symptoms build after 20 minutes, which is what fatigue looks like.	Ill-sustained accommodation starts from a normal amplitude. Hers is 4.50 D.
C	Convergence insufficiency	Near symptoms with an exophoria is the commonest pairing taught.	An NPC of 7 cm and 4 Δ of exophoria will not carry these symptoms.

2 • B A tonic pupil with light-near dissociation

Accommodation runs on the same parasympathetic pathway as the pupil, so a tonic pupil localises the lesion to the ciliary ganglion. The other findings only grade the same deficit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Exophoria by evening	Worsening through the day suggests fatigue, and so myasthenia.	A phoria drifting 4 Δ by evening is ordinary in a symptomatic reader.
E	Facility of 2 cpm	A worse number looks like worse disease, so it looks significant.	It measures the same accommodation more severely, not a new mechanism.

3 • D A +1.00 add with accommodative therapy

The add makes 40 cm reachable now, and therapy raises the amplitude so the add can come off later. A large add or prism treats neither the amplitude nor the facility.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	A +2.50 D add	More plus makes reading easier still, and she is symptomatic now.	It fixes her at 40 cm and removes the stimulus therapy depends on.
A	6 Δ base in at near	Base in relieves a near exophoria, which she does have.	4 Δ of exophoria is not the problem, and prism adds no accommodation.

4 • E Taper the add, continue facility work

8.50 D is close to the 9.25 D floor for her age and the facility is normal, so the add is withdrawn while the exercises consolidate. Stopping both at once is when symptoms return.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stop and discharge	The numbers are nearly normal and she feels well, so it looks done.	Gains fade if the exercises stop at the same moment as the add.
D	Imaging now	8.50 D is still under the Hofstetter floor of 9.25 D at 23 years.	An amplitude that answers to therapy is functional. Imaging adds nothing.

5 • A, E Minimum 9.25 D, and push-up over-reads

The Hofstetter minimum is $15 - 0.25 \times 23 = 9.25$ D, so 4.50 D is under half of it. Push-up over-reads, because blur is tolerated and the target grows as it comes closer.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Minus lens reads higher	Both tests measure amplitude, so the direction is a coin toss.	Minus lenses minify and remove proximal cues, so they read lower.
C, D	Normal, or presbyopia	One excuses the number and the other explains it away by age.	9.25 D is the floor at 23, and presbyopia at 23 needs a cause.

IF-THEN RULES FROM THIS CASE

If near blur comes with an amplitude below $15 - 0.25 \times \text{age}$, treat the accommodation and not the phoria. If it fell suddenly, look for a drug, a pupil or a systemic cause.

CASE S2-08 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

Print that doubles at twenty minutes and clears on looking up

A 29-year-old accountant has two months of frontal headache and print that doubles after 20 minutes, clearing when he looks up. Refraction +0.75 DS each eye, worn, BVA 20/20. PD 6.0 cm. Cover test 2 Δ esophoria at 6 m and 12 Δ esophoria at 40 cm. Through +1.00 D over the near correction the near deviation falls to 5 Δ . NPC 5 cm, amplitude 9.00 D, base in at 40 cm blur 6, break 10.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Convergence excess esophoria with a high AC/A
- B Basic esophoria, equal at distance and near
- C Divergence insufficiency of adult onset
- D Latent hyperopia driving convergence at near
- E A partly recovered right sixth nerve palsy

ANSWER _____ CONFIDENCE L • M • H

2 With a PD of 6.0 cm, what is the calculated AC/A ratio?

BASIC SCIENCE

- A 4 Δ /D
- B 7 Δ /D
- C 10 Δ /D
- D 12 Δ /D
- E 16 Δ /D

ANSWER _____ CONFIDENCE L • M • H

3 Which three belong in first line management?

TREATMENT • SELECT 3

- A Bilateral medial rectus recession within the year
- B Atropine 1% at weekends to relax accommodation
- C A +1.00 D near addition worked into his prescription
- D Base-in fusional vergence therapy at 40 cm
- E A longer working distance with a break every 20 minutes
- F 6 Δ base out split between the two lenses

ANSWER _____ CONFIDENCE L • M • H

4 Through the +1.00 add he is comfortable at 40 cm but blurred at his 65 cm screen. What now?

TREATMENT

- A Set an intermediate power for 65 cm and re-check the eso through it
- B Raise the add to +2.00 D so that both working distances are covered
- C Add 4 Δ base in for screen work only, keeping the add
- D Return to single vision +0.75 and treat with therapy alone
- E Keep the add and move the screen in to 40 cm to match it

ANSWER _____ CONFIDENCE L • M • H

5 Which finding would send him for imaging rather than more lenses?

DIAGNOSIS

- A A gradient AC/A that measures 7 Δ /D
- B Symptoms that disappear entirely at the weekend
- C Esophoria larger at 6 m than at 40 cm, with slow abduction
- D Near esophoria rising to 16 Δ by the end of the working day
- E A base-in break at 40 cm of only 8 Δ

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-08 • why

1 • A Convergence excess, high AC/A

The near deviation exceeds the distance deviation by 10 Δ, and a +1.00 D lens removes 7 Δ of it, which is accommodative convergence. A base-in break of 10 Δ explains the symptoms.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Basic esophoria	There is esophoria at both distances, so the label fits loosely.	Basic esophoria differs by under 5 Δ. His differs by 10 Δ.
C	Divergence insufficiency	Adult esophoria with diplopia is its classic presentation.	That is worse at distance. His deviation is far worse at 40 cm.

2 • C 10 Δ/D by calculation

AC/A = PD + m(near - distance), esophoria positive: $6 + 0.4(12 - 2) = 10 \Delta/D$. The gradient through the +1.00 lens is $12 - 5 = 7 \Delta/D$, lower because proximal drive is excluded.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	7 Δ/D	It is the gradient ratio, worked correctly from the +1.00 lens.	The item asks for the calculated ratio, which includes the PD term.
A	4 Δ/D	It is 0.4×10 , the near term of the formula on its own.	The PD in centimetres has to be added: $6 + 4$, not 4 alone.

3 • C, D, E A plus add, base-in therapy, task change

A high AC/A means plus at near buys vergence directly. Training the base-in range and cutting the near demand attack the same imbalance from the other side.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	6 Δ base out	Prism is standard for a symptomatic phoria, and 6 Δ looks apt.	Base out adds to an esophoria. Relieving prism here is base in.
A	Medial rectus recession	The near angle is large, so surgery looks like the definitive fix.	A comitant phoria answering to a +1.00 lens is not a surgical problem.

4 • A An intermediate power set for 65 cm

The add has to match the distance he works at, and the deviation is re-measured through whatever power is prescribed. Demand at 65 cm is 1.54 D, not the 2.50 D of 40 cm.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Raise the add to +2.00	More plus relieves more esophoria, and it worked at 40 cm.	It blurs 65 cm further, since the focus then lands near 50 cm.
E	Move the screen to 40 cm	It matches the add he already tolerates, at no extra cost at all.	It raises demand from 1.54 D to 2.50 D, which is what hurts.

5 • C Distance eso above near, with slow abduction

A sixth nerve reverses the pattern: the deviation grows at distance and abducting saccades slow. Everything else here is a comitant near excess that answers to plus.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Near eso rising to 16 Δ	A bigger number late in the day reads like decompensation.	It is the same near excess, fatigued. The distance to near pattern holds.
A	Gradient of 7 Δ/D	It differs from the calculated 10, so it looks like an inconsistency.	Gradient runs below calculated by design, since proximal drive is out.

IF-THEN RULES FROM THIS CASE

If the near deviation exceeds the distance deviation by more than 5 Δ, the AC/A is high: try plus at near before any prism. If distance eso exceeds near eso, think sixth nerve.

CASE S2-09 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

Double across the road, single across the table

A 71-year-old has four months of horizontal doubling that appears only in the distance, worse when tired and at night on the road, with no pain. Refraction +2.00 DS each eye, BVA 20/25 each. Cover test 10 Δ esophoria at 6 m and 2 Δ at 40 cm, the same in right and left gaze and on head tilt. Abduction is full with normal saccadic velocity. No ptosis, no fatigue on upgaze, no proptosis, pupils equal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A A bilateral sixth nerve palsy, partly recovered
- B Ocular myasthenia with a variable esodeviation
- C Thyroid eye disease with tight medial recti
- D Decompensated childhood esophoria
- E Age-related divergence insufficiency

ANSWER _____ CONFIDENCE L • M • H

2 Which imaging finding would confirm the mechanism?

DIAGNOSIS

- A Enlarged medial recti with tendon sparing throughout the coronal views
- B An empty sella with distended optic nerve sheaths
- C Thickened muscle tendons with orbital fat stranding
- D Inferior sag of the lateral rectus with a degenerated LR-SR band
- E Enhancement of the sixth nerve in the prepontine cistern

ANSWER _____ CONFIDENCE L • M • H

3 Distance eso is 10 Δ and near eso 2 Δ . What base-out prism should be ground in?

TREATMENT

- A 4 Δ in total
- B 6 Δ in total
- C 8 Δ in total
- D 10 Δ in total
- E 12 Δ in total

ANSWER _____ CONFIDENCE L • M • H

4 In the prism he reads with discomfort at 40 cm. What is the best solution?

TREATMENT

- A Botulinum toxin to both medial recti, repeated as needed
- B A separate distance pair carrying the prism, with plain readers
- C Reduce the prism until both working distances are equally tolerable
- D A 15 Δ Fresnel prism across the whole of each lens
- E Refer for bilateral medial rectus recession this year

ANSWER _____ CONFIDENCE L • M • H

5 He drives daily and asks whether he may carry on. Which three duties apply today?

LAW AND ETHICS • SELECT 3

- A Tell him not to drive until the prism controls the diplopia
- B Follow the state rule on reporting a driver with diplopia
- C Certify him as fit, since his near vision is single
- D Leave the driving question to his primary care doctor
- E Give no advice until imaging excludes a sixth nerve
- F Record the deviation, the prism prescribed and the advice given

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-09 • why

1 • E Divergence insufficiency, age related

A comitant esodeviation 8 Δ larger at distance than at near, with full abduction and normal saccadic velocity at 71, is a divergence problem rather than a nerve or muscle one.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Sixth nerve palsy	Distance esotropia with diplopia is exactly how one presents.	A palsy is incomitant, with slowed abduction. Both are normal here.
B	Ocular myasthenia	Doubling worse when tired is the phrase that names myasthenia.	No ptosis, no fatigue on upgaze, and the deviation does not vary.

2 • D A sagging lateral rectus, degenerated band

The band between lateral and superior rectus degenerates with age, so the lateral rectus sags and its pulling direction drops. That is the anatomy behind a comitant distance esodeviation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Sixth nerve enhancement	It would prove the diagnosis everyone worries about first.	That needs an incomitant deviation with slow abduction to look for.
A	Enlarged medial recti	Thyroid disease is a real cause of adult esodeviation.	There are no lid signs, no proptosis and no limitation of ductions.

3 • B 6 Δ base out, 3 Δ in each lens

Prescribe about two thirds of the deviation: $10 \Delta \times 0.6 = 6 \Delta$, split as 3 Δ base out each eye, then confirm that fusion holds at 40 cm, where the eso is only 2 Δ.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	10 Δ in total	Full correction of the measured angle looks like the honest answer.	It leaves 8 Δ of base out at near, where he deviates only 2 Δ.
A	4 Δ in total	Under-prescribing prism feels safer than over-prescribing it.	It leaves 6 Δ uncorrected at distance, which is where he doubles.

4 • B Prism in a distance-only pair

The deviation is 10 Δ at distance and 2 Δ at near, so one prism cannot serve both. Two pairs match the prism to the deviation at each working distance.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Reduce the prism	A compromise usually works when two distances disagree.	It gives him diplopia at distance and discomfort at near, together.
E	Medial rectus recession	Surgery for divergence insufficiency is real, and it is durable.	It follows prism failure or a large angle, not a first prescription.

5 • A, B, F Record it, advise stopping, report as required

Uncontrolled distance diplopia is the driving hazard, so the duty is to advise against driving until it is controlled, to document that advice, and to report where the state requires it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Certify him as fit	Single near vision and 20/25 read as a pass on the numbers.	Distance doubling is what matters on the road, and it is present.
E	Wait for imaging	Deferring until the cause is known feels cautious and correct.	The hazard exists today, whatever a scan shows next month.

IF-THEN RULES FROM THIS CASE

If a comitant esodeviation is larger at distance than at near in an older patient with full abduction, prism it. If it is incomitant, or abduction is slow, image the sixth nerve.

CASE S2-29 • ACCOMMODATION, VERGENCE, OCULOMOTOR • 5 ITEMS

The eye that will not follow its partner across

A 31-year-old woman has six days of horizontal double vision, worse looking to the right, and two weeks of tingling in the right hand. Three years ago she lost vision OS over days with pain on eye movement, and recovered. BVA 20/20 both, mild temporal pallor OS. On right gaze the left eye will not adduct past the midline and the right eye shows coarse abducting nystagmus. Convergence is full. Pupils equal, lids level.

1 Where must the lesion be?

DIAGNOSIS

- A The left medial rectus or its insertion
- B The right abducens nucleus in the pons
- C The left medial longitudinal fasciculus in the pons
- D The right medial longitudinal fasciculus in the pons
- E The left third nerve in its subarachnoid course

ANSWER _____ CONFIDENCE L • M • H

2 Which finding shows that the medial rectus itself is working?

DIAGNOSIS

- A Corrected acuity of 20/20 in each eye
- B Full convergence to a near target
- C Full elevation and depression in both eyes
- D Equal pupils with symmetrical lids
- E Coarse abducting nystagmus on the right

ANSWER _____ CONFIDENCE L • M • H

3 Which three features point to demyelination rather than a stroke?

DIAGNOSIS • SELECT 3

- A Painful loss of vision OS three years ago
- B Temporal pallor of the left optic disc
- C Diplopia that is worse on right gaze
- D Coarse nystagmus in the abducting eye
- E Equal pupils and symmetrical lids
- F Age 31 with no vascular risk factors

ANSWER _____ CONFIDENCE L • M • H

4 What is the most appropriate next step?

TREATMENT

- A Prism correction and review in six weeks
- B Intravenous steroid before any imaging
- C Occlusion of one eye at home until the diplopia settles
- D MRI of brain and cord, with neurology referral
- E CT of the head today with a vascular opinion

ANSWER _____ CONFIDENCE L • M • H

5 Which pathway explains the failure of conjugate gaze here?

BASIC SCIENCE

- A Direct supply of the medial rectus by the sixth nerve
- B Internuclear fibres from the abducens nucleus crossing over
- C Frontal eye field fibres descending to the pons
- D Vestibular input through the medial vestibular nucleus
- E The rostral interstitial nucleus for vertical saccades

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-29 • why

1 • C The left medial longitudinal fasciculus

Fibres run from the right abducens nucleus across to the left third nerve nucleus in the left fasciculus. Left adduction fails on right gaze while convergence, which bypasses it, is full.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Right fasciculus	The nystagmus is in the right eye, which drags the side across.	A right lesion would stop the right eye adducting on left gaze.
B	Right abducens nucleus	The fibres do start there, and rightward gaze is what fails.	That lesion abolishes all gaze to the right, abduction included.

2 • B Convergence is full

Convergence reaches the medial rectus by a supranuclear route that does not use the fasciculus. A muscle that adducts fully for a near target cannot be paretic for gaze.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Abducting nystagmus	It is the other half of the syndrome and impossible to miss.	It reports the yoked eye overshooting, not what the muscle can do.
C, D	Vertical gaze, pupils	Both are normal, which feels like proof the muscles are intact.	Neither tests adduction, and both can be normal in a palsy.

3 • A, B, F Young, previous optic neuritis, disc pallor

Youth without vascular risk, an earlier painful visual loss and pallor of that disc give lesions separated in time and place. Two weeks of hand tingling adds a third site.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Diplopia on right gaze	It is the symptom that brought her in, so it feels central.	It localises the lesion. It says nothing about the cause.
D, E	Nystagmus, pupils	Both belong to the examination that made the diagnosis.	Both are identical whether the cause is a plaque or an infarct.

4 • D MRI of brain and cord, neurology

The question is how many lesions there are and where, which needs MRI of brain and cord with contrast. Treatment and prognosis follow the scan, so imaging comes before steroid.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	CT and vascular review	Brainstem signs do raise stroke, and CT is the quickest scan.	CT cannot see a pontine plaque, and she has no vascular risk.
B	Steroid first	Steroid shortens a relapse, and she is symptomatic now.	Steroid before imaging blurs the diagnosis it means to treat.

5 • B Internuclear fibres from the abducens nucleus

Interneurons in the abducens nucleus cross the midline and ascend in the opposite fasciculus to the medial rectus subnucleus. Cutting that link uncouples the two eyes on gaze.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Frontal eye field	It initiates the saccade, so it sits upstream of the movement.	A frontal lesion moves both eyes together, without a split.
A	Sixth nerve to medial rectus	The abducens nucleus is involved, so its nerve looks implicated.	The sixth nerve supplies lateral rectus. Medial rectus is third.

IF-THEN RULES FROM THIS CASE

If adduction fails on gaze to one side while convergence stays full, the lesion is internuclear and in the pons. In a young adult, image brain and cord for demyelination.

CASE S1-27 • AMBLYOPIA AND STRABISMUS • 5 ITEMS

The eye that turns in over a picture book

A 4-year-old boy has six months of an inward turn of the right eye, noticed over picture books and worse when he is tired. Cycloplegic retinoscopy +5.00 DS OD, +4.75 DS OS. Uncorrected cover test 20 Δ esotropia at distance, 30 Δ at near. With the full plus in trial frames, 4 Δ esophoria at distance and 8 Δ esotropia at near. Crowded acuity 20/30 OD, 20/25 OS. Stereopsis 400 seconds of arc. Fundi normal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Partial right sixth nerve palsy with an incomitant deviation
- B Sensory esotropia from untreated unilateral amblyopia
- C Refractive accommodative esotropia, convergence excess
- D Non-accommodative convergence excess with a high AC/A ratio
- E Infantile esotropia that has presented late to care

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate first management?

TREATMENT

- A Full cycloplegic plus worn constantly, review in six weeks
- B Half the cycloplegic plus, to preserve some fusional drive
- C Full correction with a +2.50 bifocal add from the start
- D Part-time occlusion of the left eye for two hours daily
- E Referral for bilateral medial rectus recession surgery

ANSWER _____ CONFIDENCE L • M • H

3 Which three findings show the deviation is accommodative in origin?

DIAGNOSIS • SELECT 3

- A Onset noticed six months ago, worse when tired
- B Cycloplegic refraction of +5.00 DS in the right eye
- C Distance deviation falls to 4 Δ with the full plus
- D Esotropia larger at near than at distance untreated
- E Crowded acuity of 20/30 in the right eye
- F Stereopsis of 400 seconds of arc on testing

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism links the refractive error to the deviation?

BASIC SCIENCE

- A Bilateral lateral rectus underaction from a mild deficit
- B Sensory deprivation from the anisometropia between the two eyes
- C Accommodative effort drives convergence by the AC/A link
- D Reduced negative fusional vergence range at near fixation
- E Immature divergence tone in the developing vergence system

ANSWER _____ CONFIDENCE L • M • H

5 Six weeks on full plus: aligned at distance, 12 Δ esotropia at near. What next?

TREATMENT

- A Refer for surgery on the medial recti for the residual angle
- B Add a +2.50 flat-top bifocal with the segment splitting the pupil
- C Increase the distance plus by 1.00 D over the cycloplegic finding
- D Prescribe 8 Δ base-out prism divided between the two eyes
- E Start part-time occlusion of the left eye for the near deviation

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-27 • why

1 • C Refractive accommodative esotropia

Uncorrected hypermetropia of +5.00 D with 20 Δ at distance collapsing to 4 Δ esophoria under the full plus makes the deviation accommodative. The residual 8 Δ at near marks the excess.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Non-accommodative excess	There is a genuine near excess of 8 Δ after correction.	That entity carries little hypermetropia and does not respond to plus.
E	Infantile esotropia	A large esotropia in a small child fits the label loosely.	Onset near three and a half years, and the response to plus, exclude it.

2 • A Full cycloplegic plus, constant wear

The deviation is driven by accommodative effort, so removing the effort with the full plus is the treatment. Six weeks lets alignment and acuity respond before anything is added.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Half the plus	Undercorrecting a hypermetrope is a familiar habit from adult practice.	Any residual accommodative demand keeps driving the esotropia.
C	Bifocal from the start	There is a near excess, and a bifocal treats it directly.	Measure the residual near angle on full plus first, because it often settles.

3 • B, C, D High hypermetropia, plus response, near excess

The response to plus proves the mechanism, supported by significant hypermetropia and a deviation larger at near. Acuity and stereopsis measure the cost of the deviation, not its cause.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Acuity 20/30	Reduced acuity in the deviating eye looks like the key abnormality.	It measures amblyopia risk, not what drives the deviation.
A	Onset six months ago	Age of onset is how esotropias are usually classified.	Onset narrows the list but proves nothing about accommodation.

4 • C Accommodative convergence through AC/A

Each dioptre of accommodation carries a fixed quantity of convergence. To clear +5.00 D of hypermetropia the child accommodates hard, and the linked convergence overwhelms fusional divergence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Low negative vergence	Negative fusional vergence is what would hold an esodeviation latent.	That is the reserve that fails, not the drive that creates the deviation.
B	Sensory deprivation	There is 0.25 D of anisometropia and unequal acuity.	Deprivation esotropia needs poor vision in one eye, not 20/30.

5 • B +2.50 bifocal, segment splitting the pupil

A residual near esotropia on full distance correction is a high AC/A, treated by removing the near accommodative demand. The segment has to sit high enough to be used, at the pupil margin.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Extra 1.00 D of plus	More plus relaxes more accommodation, so it looks like the same logic.	It blurs distance vision without touching the near excess.
A	Surgery now	Twelve prism dioptres at near is a real deviation.	Optical treatment is incomplete, and a bifocal usually aligns the near angle.

IF-THEN RULES FROM THIS CASE

If an esotropia shrinks under the full cycloplegic plus, accommodative drive is the disease. If the near angle persists on full plus, add a bifocal rather than more distance plus.

CASE S2-10 • AMBLYOPIA AND STRABISMUS • 5 ITEMS

An eye that swings out in bright sun

A 5-year-old girl has had an outward drift of the left eye since she was three, seen when she is tired or unwell and in bright sun, where she screws up that eye. She never reports doubling. Cycloplegic refraction +0.75 DS each eye, acuity 20/25 each on a crowded test, PD 5.4 cm. Cover test: intermittent alternating exotropia 25 Δ at 6 m and 12 Δ at 40 cm, manifest about half the visit. Stereoacuity 40 seconds when aligned.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Intermittent exotropia, divergence excess pattern
- B Intermittent exotropia, convergence weakness pattern
- C Constant exotropia with alternating suppression
- D Congenital exotropia with a large stable angle
- E Sensory exotropia from poor acuity on the left

ANSWER _____ CONFIDENCE L • M • H

2 Which three belong in the first plan?

TREATMENT • SELECT 3

- A Full-time occlusion of the left eye until it straightens
- B Score control at each visit on a standard scale
- C Overminus lenses of -2.00 D to recruit convergence
- D Part-time occlusion of the preferred eye as a trial
- E 12 Δ base in ground into her spectacles full time
- F Bilateral lateral rectus recession within the month

ANSWER _____ CONFIDENCE L • M • H

3 Which change over the next year most supports referral for surgery?

TREATMENT

- A Squinting in sunlight has become noticeably more frequent
- B Manifest most of the day, with stereoacuity down to 200 seconds
- C The distance angle measures 30 Δ on two consecutive visits
- D Her parents report more drifting in the hour before bed
- E The near angle rises to 24 Δ after 30 minutes of patching

ANSWER _____ CONFIDENCE L • M • H

4 After 30 minutes of occlusion the near angle measures 24 Δ. What does that show?

DIAGNOSIS

- A A secondary convergence insufficiency has developed
- B Accommodative convergence has failed at near
- C Proximal fusion was masking the true near angle
- D A high AC/A ratio is driving the distance angle
- E The exotropia has decompensated to a constant one

ANSWER _____ CONFIDENCE L • M • H

5 How should the surgical numbers be set?

TREATMENT

- A On the near angle, with a medial rectus resection added
- B On the post-occlusion near angle of 24 Δ
- C On the distance angle of 25 Δ alone
- D On the office near angle of 12 Δ
- E On the average of the distance and near angles

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-10 • why

1 • A Intermittent exotropia, divergence excess

The deviation is 13 Δ larger at distance than at near, it alternates, and it is manifest half the visit with 40 seconds of stereoacuity when straight. Equal acuity excludes a sensory cause.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Convergence weakness	It is the other pattern of the same disease, so it feels like a toss-up.	That pattern is larger at near. Hers is 25 Δ far and 12 Δ near.
C	Constant exotropia	Manifest half the visit sounds close to constant to a parent.	She realigns with 40 seconds of stereopsis, which a constant XT loses.

2 • B, C, D Control score, overminus, part-time patching

Control, not the size of the angle, drives management in intermittent exotropia, so it is scored at every visit. Overminus and part-time occlusion are the levers while acuity is equal.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	12 Δ base in full time	Base in does relieve an exodeviation, and the size looks right.	Full-time relieving prism trains her to accept the angle, not to fuse.
A	Full-time occlusion	Patching is the treatment her parents have already heard about.	Acuity is equal, so full-time occlusion risks the fusion she still has.

3 • B Poor control with falling stereoacuity

Surgery follows loss of control and loss of binocular function, not the size of the angle. Manifest most of the day with stereoacuity dropping is the deterioration that counts.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	30 Δ at distance	A bigger angle looks like a stronger case for operating soon.	Angle size alone predicts neither deterioration nor the outcome.
E	24 Δ on patching	The number jumps, which looks like the disease progressing.	That unmask the true near angle. It is a measurement, not a change.

4 • C Tenacious proximal fusion, now unmasked

With a PD of 5.4 cm the AC/A falls from $5.4 + 0.4(25 - 12) = 10.6$ to $5.4 + 0.4(25 - 24) = 5.8 \Delta/D$. The distance excess was proximal fusion at near, not a high AC/A.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	A high AC/A	A distance angle twice the near angle is the classic hint for it.	Once the near angle is unmasked the AC/A is $5.8 \Delta/D$, which is normal.
E	Decompensation	A larger measured angle reads naturally as a squint getting worse.	The near angle was always 24 Δ. Occlusion only revealed it.

5 • B On the unmasked near angle of 24 Δ

Occlusion removes the fusional convergence that hid part of the deviation, so 24 Δ at near and 25 Δ at distance are the true, nearly equal, angles to operate on.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	The office near angle	It is the number measured in clinic without any manipulation.	It is 12 Δ short of the truth, so the surgery under-corrects her.
A	An added resection	It looks like a way to cover a distance to near difference.	Once unmasked, the difference is 1 Δ. There is nothing to cover.

IF-THEN RULES FROM THIS CASE

If the near angle in an exotropia is smaller than the distance angle, patch one eye for 30 minutes and measure again. Plan the surgery on the larger angle.

CASE S1-29 • AMETROPIA • 5 ITEMS

The prescription that has moved twice this year

A 9-year-old girl reports distance blur. Her spectacles from a year ago read -1.25 DS each; cycloplegic refraction today is -2.50 DS OD and -2.75 DS OS. Axial length 24.30 mm OD and 24.40 mm OS, up 0.45 mm in the year. Both parents are myopic. She spends under an hour outdoors and about four hours on near work daily. Corrected acuity 20/20 each, orthophoric, accommodative lag +0.75 D. Fundi normal.

1 Which feature best predicts continued progression here?

DIAGNOSIS

- A Two myopic parents in the immediate family history
- B Spherical equivalent of -2.50 D at the present visit
- C Four hours of daily near work with little time outdoors
- D Accommodative lag of +0.75 D at the near point
- E Axial elongation of 0.45 mm in the year at age nine

ANSWER _____ CONFIDENCE L • M • H

2 Which single change has been shown to slow axial elongation?

TREATMENT

- A Low-dose atropine 0.05% at night to both eyes
- B Undercorrection of the distance prescription by 0.75 D
- C Single vision spectacles worn at the full correction
- D A progressive addition lens for the accommodative lag
- E Vergence therapy to reduce the accommodative demand

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate plan at this visit?

TREATMENT

- A Full correction, a control option chosen, axial length at six months
- B Full correction now, with review of the refraction in twelve months
- C Atropine 1% at night with photochromic lenses for the glare
- D Undercorrect by 0.50 D and increase time outdoors to two hours daily
- E Orthokeratology fitted today without discussing the alternatives

ANSWER _____ CONFIDENCE L • M • H

4 At six months on atropine 0.05%, axial length is up 0.25 mm. What next?

TREATMENT

- A Switch to single vision spectacles and advise outdoor time
- B Check compliance, then consider adding an optical treatment
- C Stop treatment, since this elongation shows the drop has failed
- D Increase the atropine concentration to 1% each night
- E Continue unchanged and remeasure in a further twelve months

ANSWER _____ CONFIDENCE L • M • H

5 Which three duties apply when starting the atropine off-label?

LAW AND ETHICS • SELECT 3

- A Notify the state board of optometry of the prescription
- B Obtain a second practitioner's signature before prescribing
- C Ask the child to countersign the consent entry herself
- D Explain that the use is outside the licensed indication
- E Record the parent's informed consent in the clinical notes
- F Document the alternatives offered and the review interval

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-29 • why

1 • E 0.45 mm of axial elongation in a year

Progression is measured as axial growth, and 0.45 mm in a year at nine is roughly twice the expected rate. History and habits raise risk but they do not measure what this eye is doing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Two myopic parents	Parental myopia is the best-known risk factor for onset.	It predicts who becomes myopic, not how fast this eye is growing.
D	Accommodative lag	Lag theories of progression are still widely taught.	Lag has not predicted progression in trials, and 0.75 D is unremarkable.

2 • A Low-dose atropine 0.05% nightly

Only low-dose atropine here has slowed axial growth; orthokeratology and dual-focus soft lenses are the other proven options. Undercorrection accelerates progression and single vision is the comparator.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Undercorrection	Less minus feels like less stimulus for the eye to grow.	Trials showed faster progression with undercorrection, not slower.
D	Progressive addition lens	Reducing the near demand is a plausible mechanism and is well marketed.	The effect measured in trials is small and does not change management.

3 • A Correct fully, start control, remeasure at six months

She is progressing fast enough to treat, so the visit ends with a chosen intervention, advice on outdoor time, and a measurement interval short enough to judge the effect.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Review in a year	Annual review is the routine interval for a child in spectacles.	Another 0.45 mm of growth passes before anyone notices the failure.
C	Atropine 1%	The strongest concentration should give the largest effect.	It brings glare, near blur and marked rebound for little extra benefit.

4 • B Confirm compliance, then combine treatments

A quarter of a millimetre in six months is inadequate control, and the two commonest reasons are missed doses and too weak a stimulus. Combination therapy is the next step, not a stronger drop.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Atropine 1%	Escalating the dose is the reflex when a drug underperforms.	Side effects rise steeply and rebound is worse, so combine instead.
E	Remeasure in a year	Trends need time, and a single measurement can mislead.	The trend is already unfavourable, and twelve months loses the window.

5 • D, E, F Disclose off-label use, consent, document alternatives

Off-label prescribing is lawful where it is clinically justified. The duty is disclosure of licensing status and alternatives, consent from the person with parental responsibility, and a record of both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Notify the board	Regulatory notification sounds like the safe and compliant step.	No such duty exists for prescribing within scope of practice.
C	Child countersigns	Involving the child respects her developing autonomy.	At nine she assents, while consent is the parent's to give and record.

IF-THEN RULES FROM THIS CASE

If axial length rises more than 0.20 mm in six months, treat rather than watch. If the drop is used off-label, record the licensing status, the alternatives and the consent.

CASE S2-02 • AMETROPIA • 5 ITEMS

Print in one eye that looks smaller and further away

A 34-year-old teacher collects new spectacles, OD -1.00 DS and OS -5.50 DS at a 12 mm vertex, 20/20 each eye. Her old pair was OD -1.00 DS, OS -3.00 DS. Since the change she has frontal headache after 20 minutes of marking, print in the left eye that looks smaller and set back, and she misjudges the bottom step. Cover test 2 Δ exophoria at 40 cm, motility full, both fundi normal.

1 What is the most likely explanation for her symptoms?

DIAGNOSIS

- A An incorrect cylinder axis in the left lens
- B A vertex distance error in the left lens
- C A decentred left lens inducing unwanted prism
- D Aniseikonia from correcting 4.50 D of anisometropia
- E Decompensated exophoria from the increased minus

ANSWER _____ CONFIDENCE L • M • H

2 At about 1.5% of image size per dioptre, what aniseikonia does 4.50 D predict?

BASIC SCIENCE

- A About 1.1%
- B About 2.3%
- C About 4.5%
- D About 6.8%
- E About 13.5%

ANSWER _____ CONFIDENCE L • M • H

3 Which three changes reduce the size difference?

TREATMENT • SELECT 3

- A Under-correct the left eye by 1.50 D of the minus
- B Split 2 Δ base in between the two lenses
- C Fit contact lenses, moving the correction to the cornea
- D Cut the vertex distance of the left lens to 8 mm
- E Order the left lens on a steeper base curve, thicker at the centre
- F Increase the vertex distance of the left lens to 16 mm

ANSWER _____ CONFIDENCE L • M • H

4 She wants to try contact lenses. What is the most appropriate plan?

TREATMENT

- A Fit the left eye alone and leave the right eye uncorrected
- B Fit the right eye alone and leave the left lens in the frame
- C Order iseikonic spectacles first, before any contact lens trial is arranged
- D Keep the spectacles and add a 1.5% size lens on the left
- E Fit both eyes in contact lenses and keep the spectacles for backup

ANSWER _____ CONFIDENCE L • M • H

5 Which test most directly measures the size difference she describes?

DIAGNOSIS

- A Maddox rod with a graded prism bar at 6 m
- B Worth four dot at 33 cm in dim illumination
- C Titmus stereoacuity, graded circles at 40 cm
- D Base-in to blur, break and recovery at near
- E Direct comparison with an aniseikonia plate set

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-02 • why

1 • D Aniseikonia across 4.50 D of anisometropia

Full correction of a 4.50 D difference at 12 mm makes the left image markedly smaller. This is a size error, not a focus error, which is why acuity is 20/20 and the step is misjudged.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Decompensated exophoria	2 Δ of exophoria with headache after 20 minutes fits the pattern.	2 Δ at near is physiological, and prism would not shrink one image.
B, C	Vertex error, decentration	Both are real dispensing faults that follow a big power change.	Either would blur or double the image, not scale one eye down.

2 • D About 6.8%, past the usual 3% limit

The spectacle plane costs roughly 1.5% per dioptre: $4.50 \times 1.5 = 6.8\%$. Most patients tolerate about 3%, so 6.8% predicts the symptoms she reports at 20 minutes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	About 4.5%	It reads the dioptres straight off as percent, which is close.	1% per dioptre understates the spectacle plane by about a third.
E	About 13.5%	It doubles the rate to 3% per dioptre, quoted for thick lenses.	The working figure at the spectacle plane is about 1.5% per dioptre.

3 • C, D, E Contacts, a shorter vertex, a steeper base OS

Spectacle magnification is a shape factor and a power factor. A minus lens closer to the eye, or on the cornea, minifies less, and a steeper, thicker front surface magnifies more.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	A longer vertex	Vertex distance is the right variable, and one direction must work.	Further out, a minus lens minifies more, so the gap widens.
A	Under-correct by 1.50 D	Cutting the power difference does cut the size difference.	It buys 2% of size at the cost of 1.50 D of blur in the worse eye.

4 • E Contact lenses in both eyes, spectacles kept

At the corneal plane the predicted size difference falls to roughly a quarter of the spectacle figure, near 1.5%. Both eyes have to move together, or the difference is rebuilt.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Left eye only	Only the left eye needs the large correction, so it looks efficient.	The right eye still needs -1.00 D, and she is left with unequal blur.
C	Iseikonic spectacles first	It is the textbook answer to a measured size difference.	Iseikonic lenses are thick, costly and slower than a lens trial.

5 • E Direct comparison aniseikonia plates

Aniseikonia is a difference in perceived size, so it is measured by showing a half field to each eye and asking which is larger. Phoria, fusion and stereo tests measure alignment.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stereoacuity	Stereopsis does fall as aniseikonia rises, so the number moves.	It records a consequence, and it moves for a dozen other reasons.
A, B	Maddox rod, Worth four dot	Both are quick binocular tests she would pass or fail today.	One measures phoria and one measures fusion. Neither compares size.

IF-THEN RULES FROM THIS CASE

If a new prescription corrects more than 2 D of anisometropia, predict about 1.5% of aniseikonia per dioptre, and move the correction to the cornea before rebuilding the lens.

CASE S1-02 • CONTACT LENSES • 5 ITEMS

The pain that woke a lens wearer at four in the morning

A 24-year-old woman wears monthly silicone hydrogel lenses, sleeps in them twice a week, and tops up the solution in her case rather than discarding it. Pain, photophobia and green discharge for 36 hours OS. BVA 20/25 OD, 20/200 OS. A 2.2 mm paracentral epithelial defect over a dense stromal infiltrate, 1.2 mm hypopyon, circumlimbal injection, stromal oedema. IOP 14 OD, 11 OS.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Marginal staphylococcal hypersensitivity keratitis
- B Contact lens induced sterile peripheral ulcer
- C Pseudomonas keratitis related to overnight wear
- D Adenoviral keratitis with subepithelial infiltrates
- E Acanthamoeba keratitis in a topping-up lens user

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the first hour of management?

TREATMENT • SELECT 3

- A Topical dexamethasone four times daily to limit scarring
- B Topical antifungal cover pending microscopy results
- C Pressure patch after cycloplegia for pain relief
- D Corneal scrape for Gram stain and culture before drops start
- E Hourly topical fluoroquinolone monotherapy, day and night
- F Send the lens, case and solution for culture as well

ANSWER _____ CONFIDENCE L • M • H

3 She is reviewed at 48 hours. Which finding most justifies changing therapy?

TREATMENT

- A The infiltrate is denser with 0.4 mm more stromal thinning
- B Culture grows Pseudomonas aeruginosa fully sensitive to ciprofloxacin
- C Pain is worse although the epithelial defect has shrunk by half
- D The hypopyon has settled and the anterior chamber is quieter
- E Injection persists and she complains of stinging on the drops

ANSWER _____ CONFIDENCE L • M • H

4 Which addition best manages her pain and anterior chamber reaction?

TREATMENT

- A Oral prednisolone 40 mg daily for three days
- B Topical cyclopentolate 1% twice daily with oral analgesia
- C Topical ketorolac four times daily with oral analgesia
- D Subconjunctival gentamicin with topical anaesthetic
- E Topical atropine 1% twice daily and a bandage lens

ANSWER _____ CONFIDENCE L • M • H

5 Which organism factor best explains the speed of stromal destruction?

BASIC SCIENCE

- A Endotoxin release driving neutrophil chemotaxis into the stroma
- B Adhesion to compromised epithelium through pili and flagella
- C Cyst formation that resists disinfection by the solution
- D Secreted proteases and elastase that digest stromal collagen
- E Biofilm formation on the lens surface and in the storage case

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-02 • why

1 • C Pseudomonas keratitis, overnight wear

A large epithelial defect over a dense suppurative infiltrate, with hypopyon and mucopurulent discharge inside 36 hours, is bacterial. Sleeping in lenses and topped-up solution point to Pseudomonas.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Sterile CLPU	Also lens related, also painful, and commoner than infection in this group.	CLPU is small, peripheral, without hypopyon, and the epithelium stays intact.
E	Acanthamoeba keratitis	Poor solution hygiene is the classic risk factor quoted for it.	Onset in 36 hours with hypopyon and pus is too fast and too suppurative.

2 • D, E, F Scrape and culture, hourly quinolone, lens culture

Culture is taken before the first drop, because therapy sterilises the plate within hours. Hourly fluoroquinolone covers Pseudomonas, and lens, case and solution often grow the same organism.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Topical steroid now	Scarring is the long-term risk, and steroid does reduce stromal infiltrate.	Steroid before the organism is known and controlled invites rapid melt.
B, C	Antifungal, patching	Both feel cautious: broad cover, and comfort for a very painful eye.	No fungal risk factor here, and patching an infected cornea incubates it.

3 • A Progressive thinning despite hourly cover

Response is judged on infiltrate density, defect size and thinning, not on symptoms or the sensitivity report. Progressive thinning on adequate cover means fortified drops and a perforation plan.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Sensitive on culture	It confirms the drug choice, which reads like the decision point.	Sensitivity in a dish says nothing about the cornea in front of you.
C	Worse pain, smaller defect	Pain is what she reports, so it feels like the outcome measure.	Pain lags healing. A defect halving on hourly drops is progress.

4 • B Cycloplegia plus oral analgesia

Cycloplegia relieves ciliary spasm and limits synechiae without touching epithelial healing or host defence. Analgesia is oral, because topical anaesthetic delays healing and masks progression.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Subconjunctival gentamicin	It puts potent Gram-negative cover straight at the site.	Hourly drops already give high corneal levels, and this adds needle risk.
C	Topical NSAID	It treats pain and inflammation without steroid, which sounds safe.	Topical NSAIDs are linked to melt in a cornea that is already thinning.

5 • D Secreted proteases digest collagen

Pseudomonas secretes proteases and elastase that cleave stromal collagen directly, which is why melting is measured in hours. Biofilm and adhesion explain how it arrived, not the speed.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Biofilm on lens and case	Biofilm is why topping up solution fails, and it is genuinely present.	It explains inoculation and persistence, not the rate of stromal loss.
C	Cyst formation	Resistance to disinfection fits the hygiene history exactly.	Cysts are an amoebic feature, and cysts do not lyse collagen.

IF-THEN RULES FROM THIS CASE

If a lens wearer has a stromal infiltrate under an epithelial defect, scrape and culture before the first drop. If thinning progresses on hourly cover, fortify and plan for perforation.

CASE S1-32 • CONTACT LENSES • 5 ITEMS

Gritty red eye on waking in a lens wearer

A 22-year-old soft lens wearer slept in her monthly lenses for two nights and woke with a gritty red right eye and mild watering. Discomfort eased within an hour of removing the lens. VA 20/20. A single 0.6 mm grey-white infiltrate lies 2 mm inside the temporal limbus with faint punctate stain over it, the cornea between it and the limbus is clear, injection is sectoral, the chamber is quiet and lid margins are clean.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Staphylococcal marginal keratitis from lid disease
- B Herpes simplex stromal keratitis, first episode
- C Early Acanthamoeba keratitis from tap water rinsing
- D Early bacterial keratitis of the peripheral cornea
- E Contact lens peripheral ulcer, sterile

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings argue for a sterile infiltrate rather than microbial keratitis?

DIAGNOSIS • SELECT 3

- A Symptoms that began overnight in the lens
- B Sectoral rather than diffuse conjunctival injection
- C Best corrected acuity of 20/20 in that eye
- D Clear cornea between the infiltrate and the limbus
- E An epithelial stain smaller than the infiltrate beneath it
- F An anterior chamber without cells or flare

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management today?

TREATMENT

- A Scrape for culture now and start hourly fortified antibiotics through the night
- B Topical dexamethasone four times daily and refitting in a daily disposable
- C Cyclopentolate, a pressure pad overnight, review in 48 hours
- D Stop lens wear, broad-spectrum antibiotic four times daily, review in 24 hours
- E Stop lens wear, unpreserved lubricants only, review in one week

ANSWER _____ CONFIDENCE L • M • H

4 At review the next day, which change most requires culture and intensive therapy?

TREATMENT

- A Symptoms are worse but the infiltrate is fainter and the stain has gone
- B A second 0.4 mm infiltrate has appeared in the same peripheral quadrant
- C Acuity has dropped to 20/30 from lid oedema and tear film debris
- D Injection has spread to all four quadrants, the infiltrate unchanged at 0.6 mm
- E Infiltrate now 1.8 mm and denser, epithelial defect wider than it, 1+ cells

ANSWER _____ CONFIDENCE L • M • H

5 She asks for her lens prescription today, to reorder online. What is the right response?

LAW AND ETHICS

- A Record that lens wear is unsafe until the cornea heals, and defer release
- B Release it with a written caution not to wear lenses until the eye settles
- C Decline any further lens supply and move her to spectacles permanently
- D Send her to another practitioner to decide about the prescription
- E Release it, since the fitting was verified and completed last year

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-32 • why

1 • E Sterile contact lens peripheral ulcer

A small peripheral infiltrate, clear cornea between it and the limbus, a quiet chamber, normal acuity and relief on lens removal is a hypersensitivity response, not an invading organism.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Bacterial keratitis	Any infiltrate in a wearer who slept in lenses could be microbial.	Microbial ulcers are painful, central or paracentral, and the chamber reacts.
A	Marginal keratitis	The lesion sits peripherally, where staph hypersensitivity lands.	Lid margins are clean and there is no blepharitis to drive the reaction.

2 • D, E, F Clear limbal zone, small stain, quiet chamber

A sterile infiltrate is an immune response to lens-borne antigen: the epithelium stays largely intact, the stroma reacts locally, and the chamber stays quiet. Microbial invasion breaks all three.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Overnight onset	Sleeping in lenses precedes both sterile and microbial disease.	Timing names the risk factor, not which of the two followed it.
C	Acuity 20/20	Preserved acuity feels reassuring and argues against infection.	A peripheral microbial ulcer can also spare the visual axis on day one.

3 • D Stop lenses, antibiotic cover, review at 24 hours

Removing the antigen source is the treatment. Cover is prophylactic while a small peripheral infiltrate is watched, and the 24-hour review is what separates a settling sterile lesion from an ulcer.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Lubricants, one week	Sterile infiltrates do settle on lens removal alone.	A week is too long to wait to find out the first diagnosis was wrong.
B	Steroid and a refit	Suppressing an immune infiltrate treats the mechanism directly.	Steroid without cover, and a lens back on a cornea not yet proven sterile.

4 • E Bigger, denser, defect wider than infiltrate, cells

Size, density, an epithelial break larger than the infiltrate beneath it, and any chamber reaction are the microbial markers. Progression on cover is the other indication to scrape.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	A second small infiltrate	New lesions sound like spread, and satellites suggest fungus.	Multiple small peripheral infiltrates are the usual sterile pattern.
D	Injection now diffuse	The eye looks angrier, which is what the patient reports.	Injection tracks irritation. The infiltrate and the chamber have not moved.

5 • A Defer release, record that wear is unsafe

The duty to hand over a lens prescription runs from a completed fitting on a healthy eye. An active corneal infiltrate makes that lens the wrong device today, and the record has to say so.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Release with a caution	It respects the release duty while passing on the clinical advice.	The supplier dispenses the power, not the caveat, and lenses arrive in days.
E	Release as requested	A verified fitting on file reads like a settled entitlement.	The entitlement is to a valid prescription, and validity turns on the eye today.

IF-THEN RULES FROM THIS CASE

If an infiltrate is small, peripheral, with a clear limbal zone and a quiet chamber, treat it as sterile and review at 24 hours. If it grows or the chamber reacts, culture it.

CASE S2-04 • CONTACT LENSES • 5 ITEMS

Vision that clears for a second after every blink

A 27-year-old is fitted with a monthly toric soft lens OD. Spectacle refraction -3.25 -1.75 x 170, BVA 20/20. Through the lens acuity is 20/40, clearing to 20/25 for a second after each blink. The lens centres well, moves 0.3 mm and is comfortable. Its 6 o'clock scribe mark settles 20° to the practitioner's right after every blink, back to the same place within 3 seconds. Over-refraction +0.25 -0.75 x 25.

1 What is the most likely cause of the reduced acuity?

DIAGNOSIS

- A Cylinder power ordered 0.50 D too low
- B Consistent rotational mislocation of the lens cylinder
- C Residual astigmatism from the lens front surface
- D Lens flexure over a flat fit with excessive lens movement
- E A lipid film that breaks up between blinks

ANSWER _____ CONFIDENCE L • M • H

2 Applying LARS, which axis should be ordered?

TREATMENT

- A -3.25 -1.75 x 10
- B -3.25 -1.75 x 30
- C -3.25 -1.75 x 130
- D -3.25 -1.75 x 150
- E -3.25 -1.75 x 170

ANSWER _____ CONFIDENCE L • M • H

3 Which three steps suit a lens that mislocates the same way after every blink?

TREATMENT • SELECT 3

- A Re-order with the axis compensated by LARS
- B Try a different stabilisation design, such as peri-ballast
- C Re-check the fit and steepen the base curve if it rides loose
- D Raise the cylinder power to overcome the residual astigmatism
- E Drop to the spherical equivalent of -4.00 DS
- F Add a lubricant every two hours for the fluctuation

ANSWER _____ CONFIDENCE L • M • H

4 About how much residual cylinder does a 20° mislocation of a -1.75 D cylinder leave?

BASIC SCIENCE

- A 0.30 D
- B 0.60 D
- C 0.90 D
- D 1.20 D
- E 1.75 D

ANSWER _____ CONFIDENCE L • M • H

5 At aftercare the mark sits 20° right on one blink and 10° left on the next. What now?

TREATMENT

- A Change to another stabilisation design, or fit rigid
- B Compensate the axis by the average of the two positions
- C Order both axes and let her alternate between them
- D Add -0.50 D of sphere to sharpen the average image
- E Continue in the same lens and review in three months

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-04 • why

1 • B Rotational mislocation of 20 degrees

A well centred lens moving 0.3 mm, with a mark that returns to the same 20° after every blink, is stable but mislocated. A cross cylinder over-refraction is the signature of a rotated cylinder.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	A lipid film	Vision that clears with a blink usually means the tear film.	A film clears after the blink then blurs slowly, and leaves no cylinder.
D	Flexure over a flat fit	Flexure is a real cause of residual cylinder in a soft toric.	Movement is 0.3 mm with good centration, so the lens is not loose.

2 • D -3.25 -1.75 x 150 ordered

Left add, right subtract. The mark sits 20° to the right, so $170 - 20 = 150$. Ordering 150 lets the lens rotate 20° right on the eye and land its cylinder on the 170 meridian.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Axis 170	It repeats the spectacle axis, where the cylinder belongs on the eye.	It is the axis already failing, because the lens does not sit at 170.
A	Axis 10	It adds 20° to 170, the right arithmetic in the wrong direction.	Adding is for rotation to the left. This mark sits to the right.

3 • A, B, C LARS axis, new stabilisation, re-check the fit

Repeatable mislocation can be compensated in the order, and a change of design or fit stops it recurring. Extra cylinder and lubricants do nothing about where the cylinder sits.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	More cylinder power	The over-refraction shows cylinder, so more cylinder looks logical.	A misplaced axis needs rotating, not strengthening. This adds error.
E	Spherical equivalent	It removes the axis problem, and some patients do accept it.	It leaves 1.75 D of cylinder uncorrected, so 20/40 stays 20/40.

4 • B About 0.60 D, at an oblique axis

Residual cylinder is $2C \sin(\theta/2)$: $2 \times 1.75 \times \sin 10^\circ = 0.61$ D, lying obliquely. That is the -0.75 D cross cylinder the over-refraction picked up at 25.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	1.75 D	It assumes a rotated cylinder counts as an uncorrected one.	At 20° most of the cylinder still works. Only the cross part is left.
D	1.20 D	It scales by 20/30, a proportion that sounds reasonable enough.	The relation is $2C \sin(\theta/2)$, not a linear share of the axis error.

5 • A Change the stabilisation, or fit rigid

LARS only works on a lens that mislocates by the same amount every time. Variable rotation means the design is not holding, so the design changes rather than the axis on the order.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Average the two positions	Averaging is how every other unstable measurement is handled.	It leaves the cylinder wrong in both positions, by 15° either way.
C	Two lenses, alternating	It covers both positions, so one lens is always right.	She cannot tell which is in, and rotation varies within a single day.

IF-THEN RULES FROM THIS CASE

If a toric lens mislocates by the same amount after every blink, compensate the ordered axis: left add, right subtract. If the mislocation varies, change the stabilisation, not the axis.

CASE S2-05 • CONTACT LENSES • 5 ITEMS

A rigid lens that rides high and drops on down gaze

A 31-year-old is fitted with a rigid lens OD. Keratometry 42.50 @ 180 and 43.50 @ 090, spectacle refraction -4.25 -1.00 x 180 at 12 mm, BVA 20/20. The diagnostic lens is 8.20 mm base curve, 9.4 mm diameter, -2.75 D, and over-refraction is plano at 20/25. It rides 2 mm high, moves 3 mm on a blink, and fluorescein shows central bearing, mid-peripheral pooling and standing edge clearance.

1 What does this fluorescein pattern and lens behaviour indicate?

DIAGNOSIS

- A A steep fit, with central pooling and a sealed, immobile edge
- B An alignment fit, with even clearance throughout
- C A lens decentred by a low riding upper lid
- D A warped lens with an irregular front surface
- E A flat fit, with apical bearing and excess edge clearance

ANSWER _____ CONFIDENCE L • M • H

2 The lens is refitted 0.25 mm steeper, at 7.95 mm. What power should be ordered?

TREATMENT

- A -2.75 D
- B -3.25 D
- C -4.00 D
- D -4.75 D
- E -5.50 D

ANSWER _____ CONFIDENCE L • M • H

3 How does a rigid lens correct her 1.00 D of corneal astigmatism?

BASIC SCIENCE

- A The lens front surface carries a compensating cylinder
- B Aberration in the tear layer cancels the corneal cylinder
- C The tear lens fills the flatter meridian, leaving a spherical surface
- D The lens flexes on to the cornea and takes up its toricity
- E The rigid back surface flattens the steep corneal meridian over hours of wear

ANSWER _____ CONFIDENCE L • M • H

4 At six weeks there is grade 2 staining at 3 and 9 o'clock, with an incomplete blink. What now?

TREATMENT

- A Increase the diameter and reduce edge lift, with blink training
- B Reduce the diameter so the lens clears the limbus with more room to move
- C Flatten the base curve to lift the lens off the apex
- D Add a topical antibiotic and a bandage lens for a week
- E Move to overnight wear so the surface stays covered

ANSWER _____ CONFIDENCE L • M • H

5 After six hours out of lenses the refraction is -3.00 -1.75 x 015 with distorted mires. Which two findings favour warpage?

DIAGNOSIS • SELECT 2

- A Corneal sensation is reduced in that eye
- B The axis has rotated 15° where the lens bore on the apex
- C The change reverses over two weeks out of lenses
- D Pachymetry reads 452 µm with the thin point displaced
- E Retinoscopy shows a scissors reflex centrally

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-05 • why

1 • E A flat fit with apical bearing

At 8.20 mm the lens is $337.5 / 8.20 = 41.16$ D, which is 1.34 D flatter than the flat meridian. It bears on the apex, pools mid-peripherally, and the lid catches the lifted edge and carries it up.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	A steep fit	Central pooling with apical clearance is the pattern learnt first.	The centre bears here, the edge stands off, and the lens rides high.
B	An alignment fit	Comfort and 20/25 suggest the lens is doing its job well enough.	An aligned lens moves 1 to 2 mm and shows no apical touch at all.

2 • C -4.00 D on a 7.95 mm base curve

$337.5 / 8.20 = 41.16$ D and $337.5 / 7.95 = 42.45$ D, so the new lens is 1.29 D steeper. Steeper add minus: $-2.75 - 1.25 = -4.00$ D.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	-2.75 D	It keeps the power that gave a plano over-refraction on the eye.	A steeper lens adds a plus tear lens, so the power has to go minus.
E	-5.50 D	It doubles the compensation, which feels safe with a steeper lens.	The base curve moved 1.29 D, so the power moves 1.25 D, not 2.75 D.

3 • C The tear lens neutralises corneal cylinder

Tears fill the space between a spherical rigid back surface and a toric cornea, replacing the corneal surface optically. The 1.00 D of corneal cylinder disappears with no cylinder in the lens.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Lens flexure	Flexure is real, and it is why soft lenses fail to mask cylinder.	A rigid lens resists flexure. That is what makes the tear lens work.
E	Corneal flattening	Rigid lenses can mould a cornea, as orthokeratology shows.	Moulding is a complication of the fit, not the optical mechanism.

4 • A Larger diameter, less edge lift, blink training

Staining at 3 and 9 o'clock is desiccation where the lid never covers the surface. A larger lens with a lower edge, plus a complete blink, restores the tear layer over the exposed cornea.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	A smaller diameter	Less lens on the eye sounds like less mechanical insult to it.	A smaller lens leaves more cornea exposed, so the staining worsens.
E	Overnight wear	A covered surface cannot dry out while she is asleep.	Closed eye wear in a rigid lens risks oedema and infection instead.

5 • B, C Axis rotation at the bearing site, then reversal

Warping follows the mechanics: the axis shifts where the lens bore, and the cornea returns to baseline out of lenses. Thinning with a displaced apex belongs to ectasia instead.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Thin point 452 μ m	Thinning is measurable, and it is the thing everyone fears here.	A displaced thin point at 452 μ m argues for ectasia, not moulding.
A, E	Scissors reflex, sensation	Both look like fine print from an irregular cornea workup.	One suggests ectasia and the other suggests herpetic disease.

IF-THEN RULES FROM THIS CASE

If a rigid lens bears on the apex and rides high, steepen it and add minus, quarter for quarter. If the refraction and the mires change out of lenses, suspect warpage.

CASE S1-03 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

The eye that has been red twice before

A 41-year-old man has four days of watering, mild photophobia and blur OD, and two similar episodes in the past two years. BVA 20/40 OD. A branching epithelial lesion with terminal bulbs stains with fluorescein, its bed stains faintly with rose bengal, corneal sensation is reduced OD, and two small anterior stromal opacities remain from an earlier episode. Chamber quiet, IOP 16 both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Thygeson superficial punctate keratitis
- B Herpes zoster pseudodendritic keratopathy
- C Herpes simplex epithelial keratitis, recurrent
- D Healing abrasion with a dendritic epithelial ridge
- E Acanthamoeba radial keratoneuritis, early

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate treatment now?

TREATMENT

- A Topical chloramphenicol with a bandage contact lens
- B Topical ganciclovir 0.15% gel five times daily
- C Topical ganciclovir gel with dexamethasone twice daily
- D Oral aciclovir 400 mg twice daily as sole therapy
- E Preservative-free lubricants and review in one week

ANSWER _____ CONFIDENCE L • M • H

3 At two weeks the epithelium is intact but acuity is 20/60 with central stromal haze and cells, and no keratic precipitates. What has happened?

DIAGNOSIS

- A Herpetic stromal keratitis without ulceration
- B Neurotrophic keratopathy from lost sensation
- C Herpetic endotheliitis with secondary oedema
- D Bacterial superinfection of a healed dendrite
- E Antiviral epithelial toxicity with pseudodendrites

ANSWER _____ CONFIDENCE L • M • H

4 Which regimen is most appropriate for this stage?

TREATMENT

- A Topical ciclosporin 0.05% twice daily as sole therapy
- B Topical prednisolone with antiviral cover, slow taper
- C Topical prednisolone hourly without antiviral cover
- D Oral aciclovir 400 mg five times daily as sole therapy
- E Topical ganciclovir gel five times daily as sole therapy

ANSWER _____ CONFIDENCE L • M • H

5 Which two statements about latency and recurrence are correct?

BASIC SCIENCE • SELECT 2

- A Recurrence reflects fresh exogenous reinfection
- B Corneal sensation recovers fully between episodes
- C Latency is maintained in corneal stromal keratocytes
- D Latent virus resides in the trigeminal ganglion
- E Recurrence follows reactivation and axonal transport

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-03 • why

1 • C Recurrent HSV epithelial keratitis

A branching lesion with terminal bulbs, an ulcer bed that stains with rose bengal, reduced sensation and old anterior stromal scars from earlier episodes make recurrent herpes simplex the answer.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Zoster pseudodendrite	Zoster also gives a raised branching lesion with reduced sensation.	Pseudodendrites are heaped mucus with no terminal bulbs and no ulcer bed.
D	Healing abrasion	Healing epithelium forms ridges that can look dendritic on staining.	A healing ridge has no terminal bulbs, and it does not recur three times.

2 • B Topical ganciclovir gel, five times daily

Live virus in the epithelium needs an antiviral at treatment dose until the lesion heals. Steroid on a live dendrite drives geographic ulceration, and lubricants do nothing to the virus.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Antiviral plus steroid	Steroid limits the scarring his earlier episodes plainly caused.	Steroid belongs to stromal disease. On a dendrite it enlarges the ulcer.
D	Oral aciclovir alone	Oral dosing avoids drop toxicity and reaches the ganglion too.	400 mg twice daily is the prophylactic dose, not a treatment dose.

3 • A Herpetic stromal keratitis, non-necrotising

Stromal haze with chamber cells appearing after the epithelium has healed is an immune stromal reaction, not live epithelial virus. No defect and no precipitates place the disease in the stroma.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Endotheliitis	Stromal thickening with a chamber reaction is close to its picture.	Endotheliitis needs keratic precipitates under the oedema. There are none.
E	Antiviral toxicity	Two weeks of gel five times daily does cause punctate toxicity.	Toxicity is epithelial and stains. This haze sits under an intact surface.

4 • B Steroid under antiviral cover, slow taper

Immune stromal disease needs steroid, and steroid needs antiviral cover to stop epithelial recurrence. Rapid withdrawal produces the rebound that scars the visual axis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Steroid, no cover	It treats the immune stroma, which is what is costing him vision.	Unopposed steroid reactivates epithelial virus and risks geographic ulcer.
E	Antiviral alone	It cleared the dendrite, so continuing it feels consistent.	Antiviral does not touch the immune infiltrate driving the stromal haze.

5 • D, E Ganglionic latency, axonal reactivation

Herpes simplex persists in the trigeminal ganglion and travels down ophthalmic axons when reactivated. Recurrence is endogenous, and sensory loss accumulates with each episode.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Exogenous reinfection	Repeat episodes suggest repeat exposure, as with conjunctivitis.	Recurrence is reactivation of the patient's own latent virus.
B, C	Sensation, keratocytes	Both sound reasonable: nerves recover, keratocytes host stromal disease.	Sensory loss is cumulative, and latency is ganglionic rather than corneal.

IF-THEN RULES FROM THIS CASE

If a branching epithelial lesion has terminal bulbs, treat live virus and hold the steroid. If stromal haze appears after the epithelium heals, add steroid only under antiviral cover.

CASE S1-04 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

Pain out of proportion to the slit lamp findings

A 27-year-old man rinses his monthly lenses under the tap and swims in them. Three weeks of worsening OS pain, now severe and disturbing sleep, with photophobia and blur. Two courses of topical antiviral and one of a fluoroquinolone have not helped. BVA 20/80 OS. Patchy anterior stromal infiltrates in an incomplete ring, radial perineural infiltrates, epithelial pseudodendrites, limbitis. IOP 18 both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Acanthamoeba keratitis from tap water exposure
- B Fungal keratitis with feathery stromal extension
- C Atypical mycobacterial keratitis after lens trauma
- D Pseudomonas keratitis, partially treated
- E Herpes simplex stromal keratitis, unresponsive

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings most strongly separate this from herpes simplex?

DIAGNOSIS • SELECT 3

- A A unilateral red eye with photophobia
- B Radial perineural stromal infiltrates
- C Pain far in excess of the slit lamp signs
- D An incomplete ring infiltrate in the stroma
- E Reduced corneal sensation in the affected eye
- F Epithelial irregularity with pseudodendrites

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate initial treatment?

TREATMENT

- A Therapeutic penetrating keratoplasty within the fortnight
- B Oral itraconazole with topical natamycin drops
- C Topical polyhexanide and propamidine, hourly at first
- D Hourly fortified vancomycin and amikacin drops
- E Topical prednisolone with the antiviral continued

ANSWER _____ CONFIDENCE L • M • H

4 At six weeks the infiltrate has cleared but a sterile ring inflammation persists with acuity 20/100. What now?

TREATMENT

- A Add topical steroid, continuing the biguanide
- B Stop the biguanide and start topical steroid
- C Add oral voriconazole to the biguanide
- D Restart hourly propamidine for two weeks
- E Proceed to therapeutic keratoplasty this month

ANSWER _____ CONFIDENCE L • M • H

5 Which property of the organism explains why treatment runs for months?

BASIC SCIENCE

- A It forms biofilm on the corneal surface
- B It survives inside keratocytes as a latent form
- C It seeds the trigeminal nerve and reactivates later
- D The cyst wall resists drugs and disinfectants
- E Trophozoites divide faster than epithelium heals

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-04 • why

1 • A Acanthamoeba keratitis

Pain far beyond the visible signs for three weeks, radial perineural infiltrates, a partial ring infiltrate and failure of antiviral and antibacterial therapy in a tap-rinsing swimmer name the amoeba.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	HSV stromal keratitis	Pseudodendrites and a chronic course took two clinicians there already.	Simplex gives neither radial perineural infiltrates nor a ring in weeks.
B	Fungal keratitis	Also indolent, also unmoved by antibacterials, also lens related.	Water exposure without vegetable trauma, and no feathery satellite edges.

2 • B, C, D Perineural infiltrates, pain, ring infiltrate

Radial keratoneuritis, a ring infiltrate and pain out of proportion are amoebic. Reduced sensation, pseudodendrites and unilateral photophobia occur in both and separate nothing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Reduced sensation	Sensory loss is the herpetic sign every candidate learns first.	Both diseases blunt sensation, so it cannot tell them apart.
A, F	Pseudodendrites, redness	Pseudodendrites are why the antiviral was started twice already.	Amoebic epithelium mimics dendrites, and redness is common to both.

3 • C Hourly biguanide with a diamidine

A biguanide is the agent that kills cysts, and a diamidine added to it covers trophozoites. Intensive hourly dosing in the first days decides whether the ring resolves.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Steroid plus antiviral	It was the working diagnosis, and steroid does settle the limbitis.	Steroid raises amoebic load and delays cure. Treat the organism first.
A	Early keratoplasty	Medical therapy is long and the eye is deteriorating week by week.	Grafting an active amoebic cornea recurs in the graft. Treat first.

4 • A Steroid added, biguanide continued

Inflammation persisting after the organism is controlled is immune, and steroid settles it. The biguanide continues because steroid can wake residual cysts.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Steroid, cover stopped	The infiltrate has cleared, so the anti-amoebic looks finished.	Cysts survive months. Steroid without cover produces relapse.
E	Keratoplasty now	20/100 with a scarred axis looks like a surgical problem.	Graft an inflamed eye and it fails. Wait for quiescence.

5 • D The cyst wall resists drugs

The double-walled cyst is dormant and poorly permeable, so it outlives short courses and multipurpose solutions. Cure means outlasting cysts, not merely clearing trophozoites.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Intracellular latency	Chronicity and relapse feel like an intracellular hiding place.	The amoeba is extracellular. It hides in a cyst, not in a cell.
C	Nerve seeding	Radial perineural infiltrates make the nerves look like a reservoir.	That infiltrate is inflammation around nerves, not neural latency.

IF-THEN RULES FROM THIS CASE

If a lens wearer has pain out of proportion to the signs and no response to antiviral or antibacterial therapy, scrape for amoeba. If steroid is added later, keep the biguanide running.

CASE S1-05 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

The teenager whose glasses change at every visit

A 17-year-old boy with atopic eczema rubs his eyes daily. Over 14 months OD refraction moved from -2.75 -1.50 x 175 to -4.00 -3.25 x 160, BVA 20/25. Tomography OD: maximum keratometry 51.8 D, up 1.9 D over those 14 months, thinnest point 442 μm , inferior steepening with a displaced thin point. OS Kmax 46.2 D, unchanged. No scarring, no oedema, endothelium normal.

1 Which finding establishes progression?

DIAGNOSIS

- A Maximum keratometry has risen 1.9 D in 14 months
- B The thinnest point measures 442 μm
- C Inferior steepening with a displaced thin point
- D Best corrected acuity is still 20/25 in that eye
- E Cylinder has increased by 1.75 D with axis rotation

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate management?

TREATMENT

- A Cross-linking of both eyes at the same session
- B Repeat tomography in 12 months, then decide
- C Intrastromal ring segments OD with spectacles
- D Epithelium-off cross-linking OD, monitor OS
- E Rigid gas permeable fitting OD, no cross-linking

ANSWER _____ CONFIDENCE L • M • H

3 Which two factors would defer standard epithelium-off cross-linking?

TREATMENT • SELECT 2

- A Active atopic keratoconjunctivitis with lid disease
- B Maximum keratometry above 50 D
- C Age 17 rather than over 18
- D Apical scarring visible on the slit lamp
- E Corrected acuity better than 20/30
- F Stromal thickness under 400 μm after debridement

ANSWER _____ CONFIDENCE L • M • H

4 Three months after treatment acuity is 20/60 with central stromal haze and no epithelial defect. What is the most likely cause?

DIAGNOSIS

- A Endothelial decompensation from irradiation
- B Sterile infiltrate from the bandage lens
- C Microbial keratitis after epithelial removal
- D Expected post-cross-linking stromal haze
- E Continued progression despite treatment

ANSWER _____ CONFIDENCE L • M • H

5 How does the treatment stiffen the cornea?

BASIC SCIENCE

- A Riboflavin absorbs UVA and new collagen bonds form
- B UVA denatures collagen and shrinks the stroma thermally
- C Riboflavin scavenges radicals and prevents enzymatic loss
- D UVA kills keratocytes so the stroma stops remodelling
- E Riboflavin blocks UVA from reaching the endothelium

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-05 • why

1 • A Kmax up 1.9 D over 14 months

Progression is a documented change in corneal shape, most reliably a rise in maximum keratometry with thinning. Refraction varies with fit and effort, and one thin reading is not a change.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Thinnest point 442 µm	A thin cornea is the hallmark of the disease and looks like severity.	One measurement is a state, not a change. Progression needs two dates.
D	Acuity still 20/25	Preserved vision reads as reassurance that nothing has moved.	Acuity holds until the axis distorts. It lags shape by many months.

2 • D Cross-link the progressing eye, watch the other

Cross-linking is indicated by documented progression, and this eye has it at 17 with 442 µm of tissue. The fellow eye is stable, so it is monitored rather than treated.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Rigid lens only	A rigid lens restores the acuity he actually came about.	A lens corrects the shape, it does not stop the shape changing.
B	Review in 12 months	Watchful waiting sounds proportionate in a young seeing eye.	He is already progressing. Another year of steepening costs tissue.

3 • A, F Stroma under 400 µm, active atopic disease

The standard protocol needs 400 µm of stroma after debridement to protect the endothelium, and an inflamed atopic surface heals badly. A high Kmax and young age argue for treating, not waiting.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Age 17	Consent and cooperation make young age feel like a reason to wait.	Young eyes progress fastest. Age brings the treatment forward.
B, E	Kmax 50 D, good acuity	Both read like thresholds: too steep to treat, too good to risk.	Neither defers anything. Documented progression is the indication.

4 • D Expected cross-linking haze

Haze at the demarcation line peaks in the first months and clears over a year as acuity recovers. An intact epithelium with no pain and no cells argues against infection and against decompensation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Microbial keratitis	Debridement plus a bandage lens is a real route to infection.	Infection brings pain, a defect and cells. None of the three is present.
A	Endothelial failure	The stroma was 442 µm thick, close to the safety limit.	Decompensation gives oedema and folds, not isolated stromal haze.

5 • A Riboflavin plus UVA cross-links collagen

UVA excites riboflavin, which releases reactive oxygen species and forms covalent bonds between collagen fibrils and proteoglycans. The stiffened stroma resists further steepening.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Riboflavin as a shield	It does shield deeper tissue, which is why the 400 µm rule exists.	Shielding is a safety effect. Stiffening comes from bond formation.
B	Thermal shrinkage	Older thermokeratoplasty worked that way, so it sounds familiar.	These UVA doses are non-thermal. Collagen is bonded, not shrunk.

IF-THEN RULES FROM THIS CASE

If maximum keratometry rises with thinning across two dates, cross-link rather than refit. If the stroma measures under 400 µm after debridement, change the protocol, not the indication.

CASE S2-16 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

Haze under the flap on the third day

A 34-year-old woman had bilateral LASIK 3 days ago and saw 20/20 with a clear interface on day 1. The right eye now has 24 hours of increasing pain, photophobia and blur, on prednisolone four times daily. BVA 20/60 OD. A 1.2 mm dense white infiltrate sits at the inferior flap edge and extends into the stroma beneath the flap, with an overlying epithelial defect, 2+ anterior chamber cells and dense sectoral injection.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Central toxic keratopathy after LASIK
- B Epithelial ingrowth beneath the flap
- C Pressure-induced stromal keratopathy
- D Infection at the flap interface
- E Diffuse lamellar keratitis of grade 3

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings separate this from diffuse lamellar keratitis?

DIAGNOSIS • SELECT 3

- A Topical steroid four times daily since surgery
- B A focal dense infiltrate with an epithelial defect
- C Pain and photophobia worsening on day 3
- D 2+ anterior chamber cells with sectoral injection
- E Haze visible at the lamellar interface
- F Best corrected acuity reduced to 20/60

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Increase prednisolone to hourly and review in 24 hours
- B Hourly moxifloxacin without lifting the flap
- C Amputate the flap and treat the bed with drops
- D Oral doxycycline, lubricants, review in 48 hours
- E Lift the flap, scrape for culture, fortified drops

ANSWER _____ CONFIDENCE L • M • H

4 Which empirical cover fits an onset on day 3, before cultures return?

TREATMENT

- A Moxifloxacin four times daily with lubricants
- B Fortified vancomycin with a fluoroquinolone
- C Topical amikacin with oral clarithromycin
- D Topical natamycin with oral voriconazole
- E Hourly polyhexanide with propamidine drops

ANSWER _____ CONFIDENCE L • M • H

5 Which property of the flap interface most affects treatment?

BASIC SCIENCE

- A Stromal lamellae there stay more hydrated
- B The flap is denervated, so surface healing is slower
- C Bowman layer blocks antibody entry to it
- D It is a space sealed from tears and limbal vessels
- E Keratocytes there do not express MHC class II

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-16 • why

1 • D Infectious keratitis at the interface

A focal dense infiltrate with an epithelial defect, pain worsening on day 3, chamber cells and sectoral injection is infection. Sterile interface disease is diffuse, comfortable and quiet.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Diffuse lamellar keratitis	Interface haze after LASIK on steroid is the pattern taught first.	It is diffuse granular haze, peaks on day 1 or 2, epithelium intact.
A	Central toxic keratopathy	It also appears in the first week and drops acuity sharply.	It is central and non-inflammatory, with striae, no infiltrate, no cells.

2 • B, C, D Focal infiltrate, rising pain, chamber cells

Focality with a break in the epithelium, pain that worsens after day 2, and an anterior chamber reaction are infective. Interface haze, reduced acuity and steroid use belong to both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E, F	Haze, acuity	They are the findings that make the eye look abnormal at the lamp.	Both occur in sterile lamellar keratitis, so neither separates them.
A	Steroid since surgery	Steroid is the treatment for DLK, so its use hints at that diagnosis.	Every LASIK eye is on steroid. It is routine, not a discriminator.

3 • E Lift the flap, culture, fortified antibiotics

The organism sits in a space that drops reach poorly, so the flap is lifted to sample, irrigate and treat directly. Steroid comes down while the infection is uncontrolled.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Hourly steroid	It is the correct treatment for the sterile disease this mimics.	Hourly steroid over an untreated infection is how interfaces melt.
B	Drops, flap left down	Fluoroquinolone cover looks adequate and avoids a procedure.	Penetration under the flap is poor and the culture is never taken.

4 • B Fortified vancomycin and a fluoroquinolone

Infection in the first week after LASIK is usually staphylococcal or streptococcal, so gram-positive cover with broad support starts at once. Atypical and fungal cover belongs to late onset.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Amikacin, clarithromycin	It is the classic regimen for the atypical mycobacteria of this surgery.	Those organisms present at 2 to 8 weeks, not on the third day.
A	Drops four times daily	Moxifloxacin covers the likely organisms and is already prescribed.	Four times daily does not treat an established stromal infiltrate.

5 • D A sealed space with poor drug access

The interface has no vessels and is shielded from the tear film, so drug levels there stay low and host defence arrives late. Access is why the flap is lifted rather than treated through.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	MHC expression	An immune privilege explanation sounds like the mechanistic answer.	Drug and cell access, not antigen presentation, is what fails here.
B	Denervation	LASIK does cut nerves, and that does slow epithelial healing.	Denervation does not explain why an interface infiltrate resists drops.

IF-THEN RULES FROM THIS CASE

If a post-LASIK interface is focally infiltrated, or the epithelium is broken, or the chamber reacts, lift and culture. If haze is diffuse with intact epithelium, treat as sterile with steroid.

CASE S2-18 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

A large epithelial defect that does not hurt

A 63-year-old man had herpes zoster ophthalmicus in the left V1 a year ago and uses preserved lubricants with a topical NSAID for discomfort. He reports 3 weeks of blur and no pain. BVA 20/80 OS. A 4 by 3 mm oval epithelial defect sits below centre with heaped grey edges, stromal thinning of about 30%, no infiltrate, no chamber cells. Corneal sensation absent in all quadrants. Closure complete, Bell present, Schirmer 12 mm.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Herpes simplex geographic ulcer
- B Exposure keratopathy from poor closure
- C Toxic keratopathy from preserved drops
- D Neurotrophic keratopathy after zoster
- E Microbial ulcer with a rolled edge

ANSWER _____ CONFIDENCE L • M • H

2 Which mechanism explains why the epithelium fails?

BASIC SCIENCE

- A Limbal stem cell loss from the original vesicles
- B Persistent viral replication in the epithelium
- C Loss of sensory neuropeptides that drive repair
- D Reduced aqueous tear volume from lacrimal scarring
- E Antibody deposition in the epithelial basement membrane

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate first change to treatment?

TREATMENT

- A Add a topical antibiotic four times daily
- B Start hourly preserved lubricant drops
- C Stop the NSAID and the preserved drops
- D Add a topical steroid to settle the heaped edge
- E Debride the edges and pad the eye for 48 hours

ANSWER _____ CONFIDENCE L • M • H

4 Which three measures are appropriate now?

TREATMENT • SELECT 3

- A Topical steroid to reduce the edge reaction
- B Repeated debridement of the heaped epithelium
- C Topical NSAID continued for the discomfort
- D Preservative-free lubricants through the day
- E A therapeutic contact lens with antibiotic cover
- F Autologous serum or recombinant nerve growth factor

ANSWER _____ CONFIDENCE L • M • H

5 Three weeks on, thinning is 70% with a clear central descemetocoele. What now?

TREATMENT

- A Continue lubricants and review in 4 weeks
- B Penetrating keratoplasty within the week
- C Topical steroid to reduce the melting
- D Bandage lens with hourly antibiotic alone
- E Amniotic membrane with temporary tarsorrhaphy, now

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-18 • why

1 • D Neurotrophic keratopathy

A painless persistent defect with heaped edges and thinning, in a cornea with no sensation a year after zoster, is failure of trophic supply. No infiltrate and no cells argue against infection.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Microbial ulcer	Any persistent defect with thinning raises infection first.	There is no infiltrate, no chamber reaction, and no pain in 3 weeks.
B	Exposure keratopathy	Post-zoster lids scar, and exposure also gives a lower defect.	Closure is complete with a Bell phenomenon, and tear volume is adequate.

2 • C Loss of trophic sensory neuropeptides

Sensory nerves release substance P and CGRP, which support epithelial mitosis, migration and adhesion. Without them the surface neither closes nor sticks, and blinking is not prompted.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Aqueous deficiency	Dryness is the usual reason a surface fails after zoster.	Schirmer 12 mm is adequate, and dryness gives punctate staining.
B	Live virus	The trigger was viral, so ongoing infection sounds plausible.	This is a year on, with no dendrites and no infiltrate.

3 • C Stop the NSAID and preserved drops

Topical NSAIDs impair epithelial healing and can drive melt in an anaesthetic cornea, and preservative adds surface toxicity. Removing both costs nothing and may close the defect.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Add steroid	The heaped edge looks inflamed and steroid flattens it.	Steroid slows healing and thins a cornea already down by 30%.
A	Add antibiotic	An open surface invites infection, so cover feels prudent.	Preserved cover adds toxicity without treating why it stays open.

4 • D, E, F Unpreserved tears, bandage lens, serum or NGF

Treatment replaces the missing trophic support and protects the surface: unpreserved lubrication, a lens that shields the epithelium under cover, and serum or growth factor to restore healing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, C	Steroid, NSAID	Both quieten an eye that looks irritated at the slit lamp.	Both slow epithelial closure and both invite melt in this cornea.
B	Repeated debridement	Heaped edges do not adhere, so removing them feels constructive.	Repeated debridement of an anaesthetic cornea enlarges the defect.

5 • E Amniotic membrane and tarsorrhaphy today

A descemetocele is hours from perforation, so the surface is covered and the lids are closed to take the mechanical load off it. Grafting an anaesthetic cornea repeats the same failure.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Keratoplasty this week	The cornea is nearly perforated, and a graft replaces it.	A graft on an anaesthetic surface melts. Cover and close first.
A	Lubricants, 4 weeks	Conservative care has been the theme of the whole illness.	At 70% thinning, the next month is when the cornea perforates.

IF-THEN RULES FROM THIS CASE

If a corneal defect is painless in an anaesthetic cornea, stop every toxic drop and support the surface. If thinning passes half depth, cover the cornea and close the lids.

CASE S2-24 • CORNEA AND REFRACTIVE SURGERY • 5 ITEMS

A crescent of cornea melting at the edge

A 61-year-old woman with 12 years of seropositive rheumatoid arthritis, on methotrexate 20 mg weekly, has eight days of pain and watering OD. Her joints have been swollen for two months. ESR 68, CRP 42. BVA 20/50 OD. A crescentic peripheral stromal ulcer from 2 to 5 o'clock, 1.2 mm wide and 60% deep, with an epithelial defect, an infiltrated leading edge and limbal injection. No scleral oedema, no hypopyon.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Terrien marginal degeneration with peripheral thinning
- B Staphylococcal marginal hypersensitivity keratitis
- C Infectious peripheral keratitis from *Staphylococcus aureus*
- D Peripheral ulcerative keratitis from rheumatoid vasculitis
- E Mooren ulcer in an otherwise healthy cornea

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate systemic treatment?

TREATMENT

- A Oral doxycycline with preservative-free lubricants
- B A full-dose oral NSAID with gastric protection
- C High-dose oral steroid with escalation of immunosuppression
- D Methotrexate increased to 25 mg weekly, review in a month
- E Hourly topical prednisolone with a bandage lens

ANSWER _____ CONFIDENCE L • M • H

3 Which three local measures belong alongside systemic treatment?

TREATMENT • SELECT 3

- A Topical antibiotic cover over the epithelial defect
- B Hourly topical prednisolone to the ulcer bed
- C A topical NSAID four times daily for the pain
- D A bandage lens worn continuously for a month
- E Preservative-free lubricants through the day
- F Oral doxycycline as an anticollagenase

ANSWER _____ CONFIDENCE L • M • H

4 At day 10 a 0.8 mm perforation appears in the gutter. What now?

TREATMENT

- A A large penetrating keratoplasty as the first procedure
- B A bandage lens with hourly topical steroid alone
- C Aqueous suppression with a pressure patch for a week
- D Surgery deferred until the joint disease is quiet
- E Tissue adhesive or a patch graft, immunosuppression continued

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism explains the benefit of doxycycline here?

BASIC SCIENCE

- A It clears commensal bacteria from the lid margin
- B It thickens the lipid layer of the tear film
- C It inhibits matrix metalloproteinases in the stroma
- D It suppresses the B cells making rheumatoid factor
- E It blocks prostaglandin synthesis in the epithelium

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-24 • why

1 • D Rheumatoid peripheral ulcerative keratitis

A crescentic peripheral ulcer with an epithelial defect and an infiltrated advancing edge, in seropositive disease with swollen joints and an ESR of 68, is corneal vasculitis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Mooren ulcer	It looks identical at the slit lamp and melts the periphery too.	Mooren is idiopathic, diagnosed once vasculitis is excluded.
A	Terrien degeneration	It thins the periphery and is easy to reach for in an older eye.	Terrien is painless, epithelialised, and slow over years.

2 • C Oral steroid plus stronger immunosuppression

Corneal melt in rheumatoid disease is active systemic vasculitis and it destroys tissue in days. Steroid buys control while a stronger agent is started with the rheumatologists.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	More methotrexate	She is on it already, so dose escalation looks like the next step.	It takes weeks to act, and it has failed her at 20 mg weekly.
E	Topical steroid only	Steroid is the anti-inflammatory reflex for a melting cornea.	Topical steroid thins the stroma further and misses the cause.

3 • A, E, F Lubricants, doxycycline, antibiotic cover

The local aims are surface protection, less proteolysis and no secondary infection over a bare bed. Steroid drops and NSAID drops both thin a gutter that is already 60% deep.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Hourly prednisolone	It settles the visible inflammation at the leading edge.	It suppresses stromal repair, and the ulcer deepens instead.
C, D	NSAID drops, bandage lens	One eases severe pain, the other covers a bare stromal bed.	NSAIDs are linked with melt, and a lens hides progression.

4 • E Seal it, keep the immunosuppression running

A small perforation is sealed with the least tissue that will hold, because a large graft in vasculitic cornea melts at its own edge. Systemic control continues through the repair.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Large keratoplasty	A definitive graft seems to solve cornea and perforation at once.	The host bed is inflamed, so the new rim melts in the same way.
D	Defer surgery	Operating on active vasculitis is genuinely a poor bet.	The eye is open now. A leaking cornea cannot wait for joints.

5 • C Metalloproteinase inhibition

Tetracyclines inhibit the matrix metalloproteinases released by neutrophils and keratocytes, which is the enzymatic step digesting collagen at the base of the gutter.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Lid margin bacteria	Doxycycline is prescribed for blepharitis for exactly that.	This is a sterile vasculitic melt, not a lid-driven ulcer.
D	B cell suppression	Rheumatoid factor is central, and B cell therapy does help.	That is rituximab. Doxycycline does not touch B cells.

IF-THEN RULES FROM THIS CASE

If rheumatoid disease brings a peripheral gutter with an epithelial defect, treat it as systemic vasculitis and escalate systemic therapy. If it perforates, seal it and keep treating.

CASE S1-14 • EMERGENCIES AND TRAUMA • 5 ITEMS

Everything went black in one eye at breakfast

A 74-year-old man lost the vision of his right eye painlessly 90 minutes ago. No headache, no scalp tenderness, no pain on chewing. Atrial fibrillation, on no anticoagulant. Blood pressure 176/92, pulse irregular at 96. BVA hand movements OD, 20/25 OS. Dense RAPD OD. OD: pale oedematous posterior retina, a cherry red spot, segmented columns in the arterioles, no plaque at the disc. IOP 16 both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Non-arteritic anterior ischaemic optic neuropathy
- B Non-arteritic central retinal artery occlusion
- C Ophthalmic artery occlusion with choroidal ischaemia
- D Arteritic retinal occlusion in giant cell arteritis
- E Ischaemic central retinal vein occlusion

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate immediate action?

TREATMENT

- A Carotid duplex in clinic today, cardiology review next week
- B High dose oral prednisolone now, pending inflammatory markers
- C Intravenous acetazolamide with rebreathing to raise carbon dioxide
- D Ocular massage and paracentesis, with stroke referral tomorrow
- E Immediate transfer to a stroke unit for assessment and imaging

ANSWER _____ CONFIDENCE L • M • H

3 Which three belong in the same-day workup?

TREATMENT • SELECT 3

- A Electroretinography to confirm inner retinal ischaemia
- B Thrombophilia screen before any anticoagulation is started
- C Erythrocyte sedimentation rate and C-reactive protein today
- D Blood pressure, rhythm assessment and an electrocardiogram
- E Carotid imaging arranged within the coming week
- F Fluorescein angiography to time the arterial filling delay

ANSWER _____ CONFIDENCE L • M • H

4 Which new finding would move this towards an arteritic cause?

DIAGNOSIS

- A Amaurosis fugax in the same eye last month
- B An irregular pulse at 96 with no anticoagulation
- C Scalp tenderness with pain in the jaw on chewing
- D Segmented columns of blood within the arterioles
- E A calcific plaque visible at the disc margin

ANSWER _____ CONFIDENCE L • M • H

5 Which anatomic feature explains the cherry red spot?

BASIC SCIENCE

- A The fovea is thin and takes its oxygen from the perfused choroid
- B A cilioretinal artery supplies the fovea in most eyes and spares it
- C Foveal pigment epithelium is denser, so it reads red against pale retina
- D Xanthophyll at the fovea absorbs light scattered by swollen retina
- E Ganglion cells crowd the foveal centre and tolerate ischaemia longer

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-14 • why

1 • B Non-arteritic central retinal artery occlusion

Sudden painless loss with a pale inner retina, a cherry red spot and segmented arteriolar flow is inner retinal infarction. Untreated fibrillation supplies the embolic source.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Ophthalmic artery occlusion	Hand movements vision is severe enough to suggest the whole supply has gone.	A cherry red spot needs an intact choroid under the fovea, so the choroid is perfused.
D	Arteritic occlusion	At 74 the arteritic form must be excluded, and it changes treatment completely.	No headache, scalp tenderness or jaw pain, and the embolic source is already known.

2 • E Immediate transfer to a stroke unit

Retinal artery occlusion is an acute ischaemic stroke of the eye, and at 90 minutes the patient is inside the window where brain imaging and prevention change outcome.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Massage, then refer	These manoeuvres are the traditional bedside response and cost nothing to try.	Neither has proven benefit, and the overnight delay wastes the stroke window.
A	Duplex today, cardiology later	It investigates the source and looks appropriately urgent.	Investigation without a stroke pathway leaves the brain unprotected today.

3 • C, D, E Inflammatory markers, ECG, carotid imaging

The workup asks where the embolus came from and whether arteritis is hiding. Markers, rhythm and carotid imaging answer that; the ocular tests only restate what the fundus already shows.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Fluorescein angiography	It documents the occlusion objectively and looks like the definitive test.	The diagnosis is already clinical. Angiography delays the stroke referral.
B	Thrombophilia screen	It sounds like due diligence before committing to anticoagulation.	At 74 with fibrillation the source is known, and the screen changes nothing.

4 • C Scalp tenderness with jaw claudication

Arteritis is diagnosed on systemic features and markers, not on the fundus, because the retinal picture is identical. Jaw claudication is the most specific symptom in the set.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Calcific plaque	A visible plaque is dramatic and it proves the artery is diseased.	A plaque names an embolic source, which argues against arteritis.
D	Segmented columns	It is the sign of stagnant arterial flow, which fits an inflamed vessel.	Segmentation follows any severe occlusion, whatever closed the artery.

5 • A Thin fovea over a perfused choroid

Infarcted inner retina turns opaque and pale, but the fovea has almost no inner retina and is nourished by the choroid, so the intact red choroidal bed shows through it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Cilioretinal artery	A cilioretinal artery genuinely spares central vision when one is present.	It exists in a minority of eyes and spares a wedge of retina, not the red spot.
E	Ganglion cell tolerance	Survival of the central retina would explain a normal looking centre.	Ganglion cells are absent at the foveal centre and die fastest elsewhere.

IF-THEN RULES FROM THIS CASE

If retinal arterial occlusion is diagnosed inside the stroke window, the destination is the stroke unit, not the eye clinic.
If systemic arteritic symptoms are present, treat before biopsy.

CASE S1-35 • EMERGENCIES AND TRAUMA • 5 ITEMS

The after-hours call about flashes and floaters

An after-hours call. A 61-year-old, -8.00 D myopic and pseudophakic four months, describes six hours of new floaters and arcing flashes in the right eye, with no shadow and no pain. She declines to be seen before a family wedding in four days. A practice audit of 200 such callers found 30 with a tear or detachment; the phone protocol flagged 27 of the 30 and 27 of the other 170.

1 What should be arranged on this call?

TREATMENT

- A Reassurance that this follows surgery, with routine review
- B Bed rest with the head still, and a call back in 48 hours
- C A dilated fundus examination with indentation today
- D A dilated examination at the practice within two weeks
- E Emergency department attendance tonight for a B-scan

ANSWER _____ CONFIDENCE L • M • H

2 From the audit, what is the chance that a flagged caller has a tear or detachment?

BASIC SCIENCE

- A 10%
- B 15%
- C 50%
- D 84%
- E 90%

ANSWER _____ CONFIDENCE L • M • H

3 Which three features make her decision to wait an informed refusal?

LAW AND ETHICS • SELECT 3

- A The specific risk of a detachment was named, not implied
- B She was told what change should make her call back at once
- C A signed waiver of liability is held on file
- D Her daughter agreed that waiting was reasonable
- E The notes were written up the following morning
- F She can restate in her own words what delay may cost

ANSWER _____ CONFIDENCE L • M • H

4 She still declines. What must the record show?

LAW AND ETHICS

- A A note to her family doctor recording that the call was made
- B The risks named, her refusal, the earliest appointment offered, and return criteria
- C That the patient declined advice, with the date and time of the call
- D A letter discharging her from the practice for refusing urgent care
- E Her signature on a refusal form, to be collected at the next visit

ANSWER _____ CONFIDENCE L • M • H

5 She calls on day three with a shadow in the upper nasal field. What now?

TREATMENT

- A Dilated examination at the practice the next morning
- B Reassurance that a shadow follows a large floater load
- C Bed rest with the eye positioned, and referral in 48 hours
- D Repeat the phone protocol and act on whether it flags
- E Same-day vitreoretinal referral, treating it as a detachment

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-35 • why

1 • C Dilated examination with indentation today

Photopsia with new floaters is a symptomatic vitreous detachment, and axial myopia with recent surgery raises the odds of a tear. Only a dilated peripheral view with indentation settles it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Within two weeks	It books a proper examination, and most of these are uncomplicated.	A tear is treatable while the retina is attached, and that window is days.
E	Emergency department	It matches the urgency and gets her seen out of hours.	What she needs is dilated indentation, not an ultrasound scan.

2 • C 50%, one flagged call in two

Flagged calls are 27 true and 27 false, so $27/54 = 50\%$. Sensitivity $27/30 = 90\%$ and specificity $143/170 = 84\%$ describe the protocol; what a flag means also depends on prevalence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	90%	It is the sensitivity, the figure the protocol is sold on.	Sensitivity is the hit rate among those with disease, not among those flagged.
B	15%	It is the prevalence, which sounds like the base rate answer.	Prevalence is the chance before the protocol ran, not after a flag.

3 • A, B, F Understanding, the named risk, return criteria

A refusal is informed when the patient has capacity, hears the specific consequence, and knows what would change the plan. A signature and a relative's view add nothing to any of the three.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Signed waiver	A signature looks like the document that settles the question.	A waiver does not transfer the consequences of inadequate advice.
E	Notes written next day	The file is what the episode is judged on, so its timing feels central.	When the note was typed says nothing about what she was told.

4 • B Risks named, refusal, offer, return criteria

The entry has to show that the choice was informed and that the door stays open: what was disclosed, what she decided, the soonest slot offered, and the symptoms that mean she calls at once.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Patient declined	It is true, brief, and what busy after-hours notes usually say.	It records the outcome without the disclosure that makes it defensible.
D	Discharge letter	Refusing urgent advice feels like a breakdown of the relationship.	Abandoning a patient mid-episode creates the liability it tries to avoid.

5 • E Same-day vitreoretinal referral

A field defect after days of photopsia means retina is off until proved otherwise, and macula-on repair is time critical. A superior nasal shadow places the break inferiorly and temporally.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Practice, next morning	It gets her the dilated examination she should have had on day one.	The examination is no longer the question. A field defect needs a surgeon.
D	Repeat the protocol	The protocol is the practice's agreed triage tool for these calls.	Triage tools sort symptoms. A field defect has already answered it.

IF-THEN RULES FROM THIS CASE

If photopsia comes with new floaters, examine dilated with indentation within a day. If a refusal is reasoned, record the risk named, the offer made, and what should bring the patient back.

CASE S2-15 • EMERGENCIES AND TRAUMA • 5 ITEMS

The eye that will not look up after football

A 14-year-old took an elbow to the left orbit 5 hours ago at football, watched by his coach. Little bruising and no proptosis. Elevation OS stops 10° above the horizontal with pain on the attempt, vertical diplopia, and four episodes of vomiting. Pulse 48 and regular. Left cheek and upper gum numb. BVA 20/20 each eye, pupils equal, IOP 16 both, fundi normal. CT: small inferior wall defect with tissue drawn through it.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Large blowout fracture with enophthalmos
- B Inferior rectus contusion without entrapment
- C Retrobulbar haemorrhage limiting movement
- D Traumatic fourth nerve palsy on the left
- E Trapdoor floor fracture with entrapment

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A Ice, nasal decongestant and review in 48 hours
- B Press-on prism for the diplopia, review in 2 weeks
- C Refer now for surgical release within 24 hours
- D Oral steroid and review in the fracture clinic next week
- E Wait 10 to 14 days for swelling to settle, then repair

ANSWER _____ CONFIDENCE L • M • H

3 Which action best addresses the pulse of 48 and the vomiting?

TREATMENT

- A Nil by mouth, urgent theatre, cardiac monitoring
- B Intravenous atropine, then review in 48 hours
- C Antiemetic and overnight observation on the ward
- D ECG and a cardiology opinion before orbital repair
- E Reassure: bradycardia follows any orbital blow

ANSWER _____ CONFIDENCE L • M • H

4 Which three duties apply before the child leaves the department?

LAW AND ETHICS • SELECT 3

- A Refer the same day by speaking to the operating team
- B Record the mechanism in the words given, and who attended
- C Note whether the injury fits the account, and escalate if not
- D Delay the referral until a parent signs a consent form
- E Notify safeguarding about every orbital injury in a minor
- F Send photographs of the face to the child's school

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism explains so little bruising with such tight restriction?

BASIC SCIENCE

- A The elastic paediatric floor springs back on tissue
- B Bone fragments transect the inferior rectus belly
- C Haemorrhage inside the muscle limits it mechanically
- D Orbital fat necrosis tethers the globe inferiorly
- E Infraorbital nerve injury weakens elevation

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-15 • why

1 • E Trapdoor floor fracture, tissue entrapped

A white eye with severe painful limitation of elevation, infraorbital numbness, vomiting and bradycardia after blunt injury in a teenager is tissue gripped by a floor that has sprung shut.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Large blowout fracture	It is the adult pattern after this mechanism and also gives diplopia.	A large defect gives bruising and enophthalmos, not tight painful tethering.
B	Rectus contusion	Contusion explains pain and limitation without invoking surgery.	Contusion does not drive repeated vomiting with a pulse of 48.

2 • C Surgical release today, within 24 hours

Entrapped muscle is ischaemic, and in children fibrosis with permanent restriction follows within days. The interval that suits an adult blowout is the interval in which this eye loses movement.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Wait 10 to 14 days	It is the correct rule for an adult blowout fracture with diplopia.	Paediatric trapdoor entrapment is released within a day or two.
D	Steroid, clinic next week	Steroid does reduce orbital oedema, and the eye still sees 20/20.	The problem is mechanical trapping, and a week of it scars the muscle.

3 • A Nil by mouth and urgent theatre

Bradycardia with vomiting is the oculocardiac reflex from traction on trapped tissue, and it settles when the tissue is freed. It marks urgency and warns the anaesthetist.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Atropine, review later	Atropine does treat reflex bradycardia during orbital handling.	It masks the sign while the muscle stays trapped and ischaemic.
D	Cardiology first	A pulse of 48 in a teenager looks like a cardiac problem.	The cause is orbital. Waiting for cardiology delays the release.

4 • A, B, C Same-day referral, record account, check fit

The duties are a same-day spoken referral, a contemporaneous record of the account and of who gave it, and a judgement on whether the injury fits that account. Escalation follows a discrepancy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Wait for consent	Surgery on a minor needs consent, so waiting looks correct.	Consent is arranged alongside the referral. It does not delay it.
E, F	Blanket notify, photographs	Both feel like caution about a child with a facial injury.	Notification follows concern, and school sharing has no lawful basis.

5 • A Elastic bone snaps back and grips tissue

A child's floor is greenstick and returns to position, so the defect stays small, blood is contained and muscle or its sheath is held. That is why the eye looks well and moves badly.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Muscle transected	Total loss of elevation suggests the muscle has been divided.	A divided muscle is slack, not tight. Forced duction would be free.
E	Nerve injury	Cheek numbness proves the infraorbital nerve is injured.	That nerve is sensory. It has no part in elevating the eye.

IF-THEN RULES FROM THIS CASE

If a child cannot elevate after blunt orbital injury, treat entrapment as a same-day operation, and more urgently still with nausea or bradycardia. The adult 2-week rule does not apply.

CASE S2-17 • EMERGENCIES AND TRAUMA • 5 ITEMS

Wet cement, and a cornea that has gone grey

A 41-year-old builder splashed wet cement into the right eye 30 minutes ago and rinsed with bottled water for about a minute on site. Severe pain, lids swollen, counting fingers OD. Conjunctiva blanched from 5 to 8 o'clock with perilimbal ischaemia over 4 clock hours, cornea hazy with the whole epithelium gone, iris detail faint. Grey particles in the upper fornix. Fornix pH 10.5.

1 What is the first action?

TREATMENT

- A Irrigate with saline, sweep the fornices, recheck pH
- B Measure acuity and pressure, then start irrigation
- C Irrigate for 15 minutes by the clock, then examine
- D Instil anaesthetic and antibiotic, then refer for irrigation
- E Neutralise with a dilute acid buffer, then irrigate

ANSWER _____ CONFIDENCE L • M • H

2 Which three belong in the first hour?

TREATMENT • SELECT 3

- A Topical anaesthetic so irrigation can be tolerated
- B Double eversion and sweeping of both fornices
- C Recheck of fornix pH 5 minutes after stopping
- D Applanation pressure before any irrigation
- E Pad the eye and transfer for irrigation there
- F Topical steroid before the surface is clean

ANSWER _____ CONFIDENCE L • M • H

3 Which finding most determines the visual prognosis?

DIAGNOSIS

- A Loss of the whole corneal epithelium
- B Acuity of counting fingers on arrival
- C A fornix pH of 10.5 at presentation
- D Corneal haze obscuring iris detail
- E Perilimbal ischaemia over 4 clock hours

ANSWER _____ CONFIDENCE L • M • H

4 Why does alkali keep damaging after the surface is rinsed?

BASIC SCIENCE

- A Heat of hydration burns the deeper stroma
- B It is held in the epithelium and goes no deeper
- C It binds corneal collagen and stays on the surface
- D Hydroxyl ions saponify membranes and pass inwards
- E It coagulates surface protein into a barrier layer

ANSWER _____ CONFIDENCE L • M • H

5 The pH is normal and the surface is clean. Which plan fits the next 2 weeks?

TREATMENT

- A Amniotic membrane graft with no topical steroid
- B Tears, steroid, cycloplegic, ascorbate and doxycycline
- C Topical steroid alone, tapered over the first 3 days only
- D Bandage lens with a topical antibiotic only
- E Antibiotic with pressure padding for a week

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-17 • why

1 • A Irrigate to a normal pH, fornices swept

Alkali burns for as long as it remains, and retained cement keeps releasing it, so irrigation runs to a pH endpoint with the particles removed. Measurement and referral wait their turn.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Acuity and IOP first	Recording vision before treatment is the habit of every clinic.	Every minute of contact deepens the burn. Numbers can be taken later.
C	Fifteen minutes, then stop	A fixed interval is easy to remember and feels thorough.	Time is not the endpoint. Cement holds the pH up far beyond 15 minutes.

2 • A, B, C Anaesthetic, fornix sweep, delayed pH recheck

Anaesthesia makes irrigation effective, sweeping removes the reservoir of cement, and a delayed recheck catches the pH climbing again as trapped particles keep dissolving.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D, E	Pressure, transfer	Both are standard trauma care and both feel responsible.	Neither removes alkali, and both spend the minutes that matter.
F	Early steroid	Inflammation is intense and steroid is used in these eyes.	It belongs after decontamination, not while alkali is still present.

3 • E Extent of perilimbal ischaemia

Limbal vessels mark the stem cell reserve, and their loss predicts failure to re-epithelialise, conjunctivalisation and scarring. Epithelial loss and haze at 30 minutes say little.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, B	Epithelium, acuity	They are the most dramatic findings in front of you.	Both recover fully if the limbus survives, and both mislead early.
C	A pH of 10.5	It grades the insult and drives the irrigation endpoint.	It measures the surface now, not the reserve the eye heals from.

4 • D Hydroxyl ions saponify lipid and penetrate

Alkali dissolves cell membranes and travels through stroma into the chamber, so injury continues after the surface looks clean. Acid coagulates protein and is more often held at the surface.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C, E	Barrier formation	Coagulation limiting depth is true, but it is true of acid.	Alkali forms no barrier. It saponifies its way inwards.
A	Heat of hydration	Cement does generate heat, and the eye is intensely painful.	Depth here comes from hydroxyl ions, not from temperature.

5 • B Tears, steroid, cycloplegic, ascorbate, doxycycline

Steroid controls inflammation while the epithelium is watched, cycloplegia settles the iris, and ascorbate with doxycycline supports collagen and limits melt. Lubrication carries the surface.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Steroid, 3-day taper	Steroid is the mainstay, and a short course avoids stromal melt.	Three days is too short for this burn and misses the melting phase.
A	Membrane, no steroid	Amniotic membrane does help severe burns re-epithelialise.	It is an adjunct. Withholding steroid leaves the inflammation untreated.

IF-THEN RULES FROM THIS CASE

If the injury is alkali, irrigate to a pH endpoint with the fornices swept before anything is measured. If the limbus is ischaemic, the prognosis follows the clock hours lost.

CASE S1-08 • EPISCLERA, SCLERA, ANTERIOR UVEA • 5 ITEMS

A red aching eye that cannot bear the sunlight

A 29-year-old man has three days of aching OD with photophobia and blur, and six months of early morning low back stiffness that eases as he moves. BVA 20/40 OD. Circumlimbal injection, 3+ cells with 1+ flare, fine keratic precipitates inferiorly, a strand of fibrin, posterior synechiae at 7 o'clock. IOP 11 OD, 15 OS. No vitreous cells, fundus normal. A similar episode affected OS last year.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Intermediate uveitis with anterior spillover
- B Toxoplasma retinochoroiditis with iritis
- C Herpetic anterior uveitis with trabeculitis
- D Acute anterior uveitis, HLA-B27 associated
- E Fuchs uveitis syndrome, first presentation

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate initial treatment?

TREATMENT

- A Oral prednisolone 40 mg daily with a topical antibiotic
- B Topical ketorolac four times daily with cyclopentolate
- C Sub-Tenon triamcinolone injection at this visit
- D Hourly topical prednisolone 1% with cyclopentolate
- E Topical prednisolone four times daily with lubricants

ANSWER _____ CONFIDENCE L • M • H

3 Which two features point to a systemic association worth investigating?

DIAGNOSIS • SELECT 2

- A Inflammatory back pain for six months
- B Recurrence alternating between the eyes
- C Fine keratic precipitates inferiorly
- D Intraocular pressure of 11 mmHg
- E Photophobia with circumlimbal injection

ANSWER _____ CONFIDENCE L • M • H

4 Which referral and test plan fits him best?

TREATMENT

- A Rheumatology referral, HLA-B27 and sacroiliac imaging
- B Rheumatology referral with a full autoimmune screen
- C Chest imaging with serum ACE and calcium
- D Syphilis and tuberculosis testing before any steroid
- E No investigation, since first-line therapy is unchanged

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism best explains the low pressure in the inflamed eye?

BASIC SCIENCE

- A Reduced episcleral venous pressure from injection
- B Reduced aqueous production by an inflamed ciliary body
- C Increased outflow through an inflamed trabecular meshwork
- D Fibrin plugging the trabecular meshwork
- E Posterior synechiae obstructing the pupil

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-08 • why

1 • D HLA-B27 acute anterior uveitis

Sudden painful unilateral non-granulomatous inflammation with fibrin, a low pressure and recurrence in the other eye, in a young man with inflammatory back pain, is the B27 pattern.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Herpetic uveitis	It recurs unilaterally too, with fibrin and synechiae of its own.	Pressure is low, not high, with no sectoral iris atrophy or scar.
A	Intermediate uveitis	Recurrent uveitis at this age raises it, and spillover is real.	No vitreous cells and no snowbanks. The reaction is purely anterior.

2 • D Hourly topical steroid with cycloplegia

Three plus cells with fibrin needs intensive topical steroid rather than a four times daily dose, and cycloplegia both eases spasm and breaks the synechiae that are forming.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Steroid four times daily	It is the right drug at a familiar starting frequency.	Underdosing fibrinous uveitis leaves synechiae behind. Start hourly.
C	Sub-Tenon steroid	One injection secures delivery and removes any compliance worry.	Kept for resistant or posterior disease. Drops reach the front.

3 • A, B Inflammatory back pain, alternating recurrence

Morning stiffness that eases with movement suggests axial spondyloarthritis, and recurrent alternating acute anterior uveitis is its ocular signature. The rest occurs in any acute iritis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Fine precipitates	Precipitate character does separate granulomatous disease.	Fine precipitates are non-specific and appear in any acute iritis.
D, E	Low pressure, glare	Both are abnormal and both belong to the picture in front of you.	Ciliary hyposecretion and photophobia occur whatever the cause.

4 • A Rheumatology, HLA-B27, sacroiliac imaging

The history points at one disease, so the tests are aimed there rather than scattered. Naming axial disease changes his systemic treatment even though it does not change today's drops.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Full autoimmune screen	Casting wide feels thorough once a systemic cause is suspected.	Unaimed panels generate false positives and delay the right referral.
E	No investigation	Treatment today is identical whichever result comes back.	The result changes his spinal care and his relapse risk, not his drops.

5 • B Ciliary hyposecretion in acute inflammation

Inflamed ciliary epithelium secretes less aqueous, so pressure falls despite debris in the meshwork. When pressure rises instead, suspect trabeculitis or a herpetic cause.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Fibrin in the meshwork	Fibrin is visible in this chamber and does obstruct outflow.	Obstruction raises pressure. This eye is soft, so inflow has fallen.
E	Posterior synechiae	Pupil block is a familiar pressure mechanism in uveitis.	Seclusion raises pressure through iris bombe. It does not lower it.

IF-THEN RULES FROM THIS CASE

If acute anterior uveitis recurs and alternates in a young adult, ask about inflammatory back pain and refer. If fibrin is present, dose the steroid hourly and dilate the pupil.

CASE S1-09 • EPISCLERA, SCLERA, ANTERIOR UVEA • 5 ITEMS

A boring pain that wakes her at three in the morning

A 54-year-old woman with nine years of seropositive rheumatoid arthritis has 10 days of deep boring OS pain that wakes her, with tenderness on palpation of the globe. BVA 20/30 OS. A violaceous patch of deep injection nasally that does not blanch with topical phenylephrine, an oedematous sclera in daylight, and a 1 mm peripheral corneal gutter with intact epithelium. IOP 14 both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Nodular episcleritis with adjacent corneal drying
- B Diffuse anterior scleritis with a corneal gutter
- C Peripheral ulcerative keratitis without scleritis
- D Necrotising scleritis with established thinning
- E Sclerosing keratitis from long-standing dry eye

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings separate this from episcleritis?

DIAGNOSIS • SELECT 3

- A Watering with mild photophobia
- B Symptoms present for 10 days
- C Injection that phenylephrine does not blanch
- D Pain that wakes her from sleep
- E Tenderness of the globe on palpation
- F Sectoral rather than diffuse redness

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate initial treatment?

TREATMENT

- A Oral NSAID with gastric cover, rheumatology review
- B Topical prednisolone four times daily with lubricants
- C Oral prednisolone 1 mg per kg starting today
- D Topical NSAID with a bandage contact lens
- E Subconjunctival triamcinolone to the affected quadrant

ANSWER _____ CONFIDENCE L • M • H

4 Three weeks later the pain is worse, the gutter is deeper, and an avascular scleral patch has appeared. What now?

TREATMENT

- A Increase the oral NSAID to the maximum dose
- B Topical ciclosporin with punctal occlusion
- C Scleral patch graft before any medical change
- D Oral prednisolone alone, tapered over four weeks
- E Urgent systemic immunosuppression, shared care

ANSWER _____ CONFIDENCE L • M • H

5 Why does the cornea thin at the periphery in this disease?

BASIC SCIENCE

- A Peripheral stroma holds fewer keratocytes for repair
- B Limbal vessels deliver immune complexes and proteases
- C Tear film breaks up faster over the peripheral cornea
- D Peripheral cornea is thinner than the centre to begin with
- E Limbal stem cells fail and the epithelium breaks down

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-09 • why

1 • B Diffuse anterior scleritis with a gutter

Pain that wakes her, tenderness on palpation, violaceous injection that phenylephrine does not blanch, and an oedematous sclera in daylight define scleritis. The gutter marks corneal involvement.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Nodular episcleritis	Both are red and sectoral, and episcleritis is far commoner.	Episcleral vessels blanch with phenylephrine and rarely wake anyone.
D	Necrotising scleritis	Rheumatoid disease is exactly where necrotising scleritis appears.	No avascular patch and no translucent thinning. The sclera is swollen.

2 • C, D, E No blanching, night pain, globe tenderness

Deep vessels do not blanch, and only scleral inflammation gives boring pain that wakes a patient and tenderness felt through the lid. Sector redness, watering and duration occur in both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Sectoral redness	Nodular episcleritis is sectoral, so a sector feels diagnostic.	Both diseases can be sectoral. The plane of the vessels decides.
A, B	Watering, 10 days	Symptoms and duration read like markers of severity.	Episcleritis waters as well, and it can run for several weeks.

3 • A Oral NSAID with gastric cover, refer

Non-necrotising scleritis usually answers a full-dose oral NSAID, and topical steroid does not reach the sclera. The corneal gutter and her rheumatoid disease bring in the physician.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Topical steroid alone	Steroid is the reflex anti-inflammatory for a painful red eye.	Drops do not penetrate to the sclera. This treatment is systemic.
E	Subconjunctival steroid	It places drug exactly where the inflammation can be seen.	Avoided where sclera may thin. The perforation risk is real.

4 • E Urgent systemic immunosuppression

An avascular patch with a deepening gutter is necrotising disease, which threatens the globe and signals active systemic vasculitis. Steroid alone relapses, so a steroid-sparing agent starts now.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	More NSAID	It is the class that seemed to be controlling her at first.	Necrotising disease has outrun NSAIDs. Change class, not dose.
C	Patch graft first	Thinning sclera looks like a structural problem to repair.	Grafts melt in inflamed sclera. Control the disease, then rebuild.

5 • B Limbal vessels bring complexes and proteases

The peripheral cornea abuts limbal vessels, so immune complexes deposit there and recruit neutrophils whose proteases digest stroma. The avascular centre is spared.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Peripheral thinness	Anatomy makes it sound as though the thinnest tissue goes first.	The peripheral cornea is thicker than the centre, not thinner.
C	Tear break-up	Rheumatoid patients are dry, and dryness does harm epithelium.	Dryness gives punctate staining, not a vascularised immune gutter.

IF-THEN RULES FROM THIS CASE

If a red eye is tender to palpation and the redness does not blanch with phenylephrine, treat it systemically and ask why. If a scleral patch turns avascular, escalate immunosuppression.

CASE S1-17 • GLAUCOMA • 5 ITEMS

Pressures that crept up between two visits

A 63-year-old woman has no visual symptoms. Her mother went blind from glaucoma. Asthma controlled on an inhaler, no other medication, resting pulse 54. BVA 20/20 both eyes. Goldmann IOP 26 OD, 28 OS. Central corneal thickness 545 and 548 μm . Gonioscopy: open to the ciliary band, no pigment band, no pseudoexfoliation. Cup-disc 0.75 OD and 0.8 OS with inferior rim thinning. Repeatable 24-2: superior nasal step OS, full OD.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Secondary glaucoma from pseudoexfoliation
- B Primary open angle glaucoma, worse in the left eye
- C Ocular hypertension with physiologically large cups
- D Normal tension glaucoma with asymmetric discs
- E Pigmentary glaucoma at an early stage

ANSWER _____ CONFIDENCE L • M • H

2 Which first-line topical agent fits this patient?

TREATMENT

- A Brinzolamide 1% three times daily in both eyes
- B Brimonidine 0.2% twice daily in both eyes
- C Pilocarpine 2% four times daily in both eyes
- D Timolol 0.5% twice daily in both eyes, with punctal occlusion taught
- E A prostaglandin analogue at night in both eyes

ANSWER _____ CONFIDENCE L • M • H

3 Which three make up an adequate baseline before treatment starts?

TREATMENT • SELECT 3

- A Gonioscopy of both eyes with the angle graded
- B Central corneal thickness in both eyes
- C Anterior segment imaging used instead of gonioscopy
- D Corneal endothelial cell count in both eyes
- E Fundus autofluorescence of both maculae
- F Two repeatable threshold visual field examinations

ANSWER _____ CONFIDENCE L • M • H

4 Three months on the prostaglandin, IOP is 21 OD and 23 OS. What next?

TREATMENT

- A Switch to the lowest available concentration of a beta-blocker
- B Refer for trabeculectomy in the left eye on the next available list
- C Repeat the visual fields before changing any treatment
- D Continue the same drop and review again in six months
- E Add a topical carbonic anhydrase inhibitor and review in six weeks

ANSWER _____ CONFIDENCE L • M • H

5 Which finding decides that this is glaucoma rather than ocular hypertension?

DIAGNOSIS

- A An intraocular pressure of 28 in the left eye by Goldmann
- B A cup-disc ratio of 0.8 in the left eye
- C A mother blinded by glaucoma in later life
- D Central corneal thickness of 548 μm
- E Inferior rim thinning matched by a nasal field defect

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-17 • why

1 • B Primary open angle glaucoma, worse OS

Open angles, pressures in the mid twenties, inferior rim thinning and a matching superior nasal defect on repeatable fields complete the diagnosis. Gonioscopy excludes the secondary causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Ocular hypertension	Acuity is normal and she has no symptoms, so nothing seems lost yet.	A repeatable nasal step with matching rim thinning is established damage.
D	Normal tension glaucoma	Asymmetric discs with a family history is the usual normal tension story.	Pressures of 26 and 28 on normal corneas are not normal tension.

2 • E Prostaglandin analogue at night

A pulse of 54 with asthma rules out a beta-blocker whatever the drop technique. A prostaglandin gives the largest single-agent reduction on once nightly dosing, which also protects adherence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Timolol with occlusion	Punctal occlusion is taught precisely to limit systemic absorption.	It reduces but does not remove absorption. Asthma and bradycardia are absolute.
A	Brinzolamide	It avoids the respiratory and cardiac risk entirely, which looks safest.	It lowers pressure less and dosing three times daily costs adherence.

3 • A, B, F Repeat fields, gonioscopy, corneal thickness

Treatment is measured against a baseline, so the field must be repeatable, the angle mechanism must be known, and pressure must be interpreted against corneal thickness.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Imaging over gonioscopy	Angle imaging is objective, repeatable and comfortable for the patient.	It cannot grade pigment, synechiae or recession, and it does not replace the lens.
D, E	Cell count, autofluorescence	Both are quantitative tests that fill out a thorough looking record.	Neither informs glaucoma diagnosis, target setting or progression.

4 • E Add a second agent, review at six weeks

A 25% reduction from 26 and 28 sets targets near 19 and 21, and the left eye with the field defect has not reached its. Adding a second agent is the next step, checked at six weeks.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Continue, review later	The pressure has fallen and a single drop is easier to sustain.	The worse eye misses its target, and six months lets damage accumulate.
C	Repeat fields first	Progression is judged on fields, so more data before acting seems careful.	Fields cannot change at twelve weeks. Pressure is the actionable number.

5 • E Rim thinning matched by a field defect

Ocular hypertension is raised pressure without damage. Structure and function agreeing in the same hemifield, the inferior rim and the superior field, is the damage that names the disease.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Cup-disc 0.8	A large cup is the image everyone associates with the diagnosis.	Large discs carry large cups. Ratio alone cannot separate the two.
C	Family history	It raises risk severalfold and rightly changes how closely she is watched.	Risk factors decide surveillance, not diagnosis.

IF-THEN RULES FROM THIS CASE

If the pulse is under 60 or the chest is asthmatic, no beta-blocker at any concentration. If structural loss and field loss agree in the same hemifield, treat it as glaucoma.

CASE S1-18 • GLAUCOMA • 5 ITEMS

A red eye, a headache, and vomiting since evening

A 66-year-old woman, +3.25 D in both eyes, has six hours of severe right brow ache with blurred vision and haloes around lights. She has vomited twice and took a cold remedy yesterday. BVA counting fingers OD, 20/25 OS. OD: injected conjunctiva, hazy oedematous cornea, mid-dilated oval unreactive pupil, very shallow chamber, IOP 54. No cells, no keratic precipitates, no lens opacity flecks. OS: IOP 18, narrow angle, no synechiae.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Phacomorphic closure from an intumescent lens
- B Plateau iris syndrome after pharmacological dilation
- C Acute primary angle closure with pupil block
- D Neovascular glaucoma with an angle membrane
- E Acute anterior uveitis with a raised pressure

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate immediate treatment?

TREATMENT

- A Laser peripheral iridotomy at once, through the oedematous cornea, with no drops
- B Pilocarpine 2% now, repeated every five minutes for three doses
- C Intravenous mannitol as the first step, before any topical agent
- D Urgent lens extraction on the emergency list this evening
- E Topical pressure lowering with oral acetazolamide, supine, review in an hour

ANSWER _____ CONFIDENCE L • M • H

3 Which two steps belong in the definitive plan once the pressure falls?

TREATMENT • SELECT 2

- A Laser peripheral iridotomy to the affected eye
- B Prophylactic iridotomy to the fellow eye
- C Long-term prostaglandin drops in both eyes
- D Selective laser trabeculoplasty to the angle
- E Iridoplasty repeated at monthly intervals

ANSWER _____ CONFIDENCE L • M • H

4 One week after iridotomy, IOP is 24 with synechiae over 20% of the angle. What next?

TREATMENT

- A Observe without treatment and review in three months
- B Start a topical pressure lowering agent and monitor the angle
- C Repeat the iridotomy at a second site in the same eye
- D Add pilocarpine long term to hold the angle open
- E Refer for trabeculectomy with an antimetabolite

ANSWER _____ CONFIDENCE L • M • H

5 Which finding shows that pupil block, not a plateau configuration, drives the closure?

DIAGNOSIS

- A A mid-dilated oval pupil that does not react to light
- B Corneal oedema at a pressure of 54 in that eye
- C A narrow angle in the fellow eye with no synechiae and normal pressure
- D Onset in the evening after a dose of cold remedy
- E A convex iris with a very shallow central chamber in a hyperopic eye

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-18 • why

1 • C Acute primary angle closure

A hyperopic eye with a very shallow chamber, a pressure of 54, corneal oedema and a fixed mid-dilated pupil is acute closure. The quiet chamber and clear lens exclude the secondary causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Acute anterior uveitis	Pain, redness and blurred vision are the uveitis triad, and pressure can rise.	No cells and no keratic precipitates, and uveitis does not shallow the chamber.
B	Plateau iris syndrome	The cold remedy gives a dilating trigger, and plateau eyes close on dilation.	Plateau eyes have a chamber of normal central depth and a flat iris plane.

2 • E Medical pressure reduction, then reassess

At 54 the iris sphincter is ischaemic and pilocarpine works poorly, and laser is unreliable through a hazy cornea. Lowering pressure medically clears the cornea and makes iridotomy possible.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Intensive pilocarpine	Constricting the pupil is the mechanism that should reopen the angle.	A sphincter ischaemic at 54 will not respond, and repeated doses add anterior rotation.
A	Immediate iridotomy	Iridotomy is the definitive treatment, so doing it now sounds decisive.	The oedematous cornea will not transmit the beam reliably or safely.

3 • A, B Iridotomy to both eyes

Iridotomy removes the pressure gradient that bows the iris forward. The fellow eye is narrow with the same anatomy and roughly half of such eyes close within five years, so it is treated too.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Long-term prostaglandin	Drops lower pressure and feel like the standard glaucoma answer.	Drops do not address the mechanism, and the angle can still close on them.
D	Laser trabeculoplasty	It is laser to the angle, which sounds like the right target.	It needs a visible open angle to treat, and it does not relieve pupil block.

4 • B Topical agent with angle monitoring

A patent iridotomy has relieved the block, so the residual pressure comes from synechial and trabecular damage. That is treated medically first, with gonioscopy to track any further closure.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Second iridotomy	If pressure remains high the first opening may look inadequate.	A patent iridotomy needs no second site. The problem is now the angle itself.
A	Observe three months	The crisis has passed and 24 is far from the pressure she presented with.	Synechiae progress silently, and this eye has already lost angle.

5 • E Convex iris, shallow central chamber

Pupil block bows the whole iris forward and shallows the chamber centrally as well as peripherally. Plateau eyes keep a normal central depth with a flat iris and a forward ciliary body.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Fixed mid-dilated pupil	It is the signature sign of the attack and it is genuinely present.	It reflects sphincter ischaemia at high pressure, whatever closed the angle.
D	Cold remedy trigger	A dilating drug precipitates both mechanisms, so it feels diagnostic.	Plateau eyes close on dilation too. The trigger does not name the mechanism.

IF-THEN RULES FROM THIS CASE

If the cornea is oedematous at a very high pressure, lower the pressure medically before attempting laser. If one eye has closed, the fellow narrow eye gets a prophylactic iridotomy.

CASE S1-19 • GLAUCOMA • 5 ITEMS

Ache and redness in the eye that failed in spring

A 71-year-old man had an ischaemic central retinal vein occlusion in the right eye four months ago and now has three weeks of ache, redness and haloes. BVA counting fingers OD, 20/30 OS. OD: oedematous cornea, IOP 44, deep central chamber, fine vessels crossing the pupil margin onto the iris, gonioscopy shows fibrovascular tissue bridging the angle with synechiae over 180°. Disc pale, old haemorrhages. OS: IOP 15, angle open.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Steroid response glaucoma after treatment of the occlusion
- B Ghost cell glaucoma from degenerate red blood cells
- C Uveitic glaucoma with trabeculitis and iris vessels
- D Neovascular glaucoma with a fibrovascular angle membrane
- E Acute angle closure in an eye with an old occlusion

ANSWER _____ CONFIDENCE L • M • H

2 Which two findings establish that the angle is closed by membrane, not apposition?

DIAGNOSIS • SELECT 2

- A Fibrovascular tissue bridging the angle on gonioscopy
- B Peripheral anterior synechiae over 180° of the angle
- C Fine vessels crossing the pupil margin onto the iris
- D Corneal oedema at a pressure of 44 in that eye
- E Three weeks of increasing ache, redness and haloes

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management now?

TREATMENT

- A Trabeculectomy with mitomycin C on the next available list
- B Cyclodiode laser now, with no retinal treatment planned
- C Retrobulbar alcohol for pain, since the eye sees very little
- D Intravitreal anti-VEGF with topical pressure lowering, panretinal laser to follow
- E Panretinal photocoagulation alone now, with pressure drops and review in four weeks

ANSWER _____ CONFIDENCE L • M • H

4 Pressure stays 38 on maximal drops, the eye is painful and sees hand movements. What next?

TREATMENT

- A Glaucoma drainage device with a tube to the chamber
- B Enucleation, since the eye has no useful vision left
- C Retrobulbar chlorpromazine with the drops continued
- D Cyclodiode laser to lower the pressure and settle the pain
- E Trabeculectomy with mitomycin C and a releasable suture

ANSWER _____ CONFIDENCE L • M • H

5 Which mediator most directly drives the angle change?

BASIC SCIENCE

- A Prostaglandin released from a chronically inflamed uvea
- B Transforming growth factor beta from lens epithelium
- C Complement activated on the trabecular endothelium
- D Ferritin deposited from degraded intraocular blood
- E Vascular endothelial growth factor from ischaemic retina

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-19 • why

1 • D Neovascular glaucoma

Ischaemic retina four months ago, new vessels at the pupil margin and fibrovascular tissue bridging the angle with broad synechiae is neovascular glaucoma at the synechial stage.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Acute angle closure	Ache, haloes, corneal oedema and a pressure of 44 read like an acute attack.	The central chamber is deep, and the closure is by membrane over three weeks.
B	Ghost cell glaucoma	There has been intraocular blood, and degenerate cells do block outflow.	Ghost cells need vitreous haemorrhage and give no iris or angle vessels.

2 • A, B Angle membrane with broad synechiae

Appositional closure is reversible on indentation and leaves the angle anatomy intact. Tissue bridging the angle with permanent adhesions is mechanical closure that no drop can reopen.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Vessels at pupil margin	Rubeosis is the hallmark of the disease, so it feels like the answer.	It proves neovascularisation, not that the angle is already sealed.
D, E	Oedema, three weeks of ache	Both show how high and how sustained the pressure has been.	Neither examines the angle, and both occur with appositional closure.

3 • D Anti-VEGF and drops, laser to follow

Anti-VEGF regresses the angle and iris vessels within days while drops carry the pressure, and panretinal laser then removes the ischaemic drive permanently. The order matters, not the choice.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Panretinal laser alone	It treats the underlying ischaemia, which is the root of the problem.	It takes weeks to work and the cornea is too hazy to laser adequately now.
A	Trabeculectomy	Pressure is 44 with a sealed angle, which sounds like a drainage problem.	Filtration fails in an actively neovascular eye. Regress the vessels first.

4 • D Cyclodiode laser for pressure and pain

With minimal vision, a painful eye and a neovascular angle, reducing aqueous production is the lowest risk route to comfort. Filtering procedures scar closed and carry a real risk of loss.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Drainage device	Tubes are the recognised choice in neovascular glaucoma with vision.	Reserved for eyes with useful vision. This eye needs comfort, not a tube.
B	Enucleation	It is definitive for a blind painful eye and ends the problem.	Pain has not yet been treated. Enucleation is a last resort, not a next step.

5 • E VEGF from the ischaemic retina

Hypoxic retina secretes vascular endothelial growth factor, which diffuses forward into the aqueous and drives new vessels on the iris and in the angle. That pathway is why anti-VEGF works within days.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Uveal prostaglandin	The eye is red and aching, so an inflammatory mediator fits the picture.	Prostaglandins cause inflammation and lower pressure. They grow no vessels.
D	Ferritin from old blood	There was retinal haemorrhage, so iron products are genuinely present.	Iron toxicity is a late siderotic process and grows no fibrovascular tissue.

IF-THEN RULES FROM THIS CASE

If an ischaemic eye develops a high pressure with iris vessels, treat the ischaemia and the pressure together. If the eye is painful with no useful vision, cyclodestruction before filtration.

CASE S2-32 • GLAUCOMA • 5 ITEMS

The eye that took a squash ball twenty years ago

A 44-year-old man was struck in the right eye by a squash ball at 24 and told nothing was broken. He has no symptoms now. IOP 30 OD, 15 OS, CCT 545 μm both eyes. Gonioscopy OD: a wide ciliary band over four clock hours, torn iris processes, focal meshwork pigment, angle open elsewhere. OS unremarkable. Sphincter tears and a focal lens opacity OD. C/D 0.8 with an inferior notch OD, 0.4 OS. Superior arcuate defect OD.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Pigmentary glaucoma from iris chafing on the lens
- B Angle recession glaucoma in the injured eye
- C Pseudoexfoliative glaucoma of unilateral onset
- D Ghost cell glaucoma from an old vitreous bleed
- E Primary open angle glaucoma, markedly asymmetrical

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings date the disease to the old injury rather than to primary glaucoma?

DIAGNOSIS • SELECT 3

- A Intraocular pressure of 30 mmHg in that eye
- B Cup-disc ratio 0.8 with an inferior notch OD
- C Central corneal thickness of 545 μm
- D A band of ciliary body visible over four clock hours
- E Torn iris processes at the recessed angle
- F Sphincter tears with a focal lens opacity OD

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate first-line treatment for the pressure OD?

TREATMENT

- A Trabeculectomy with mitomycin C in the right eye now
- B Oral acetazolamide 250 mg four times daily, long term
- C Topical prostaglandin analogue OD, review at four weeks
- D Selective laser trabeculoplasty to the recessed angle OD
- E Topical pilocarpine 2% four times daily to that eye

ANSWER _____ CONFIDENCE L • M • H

4 Which finding at review would most justify escalating treatment?

TREATMENT

- A IOP 24 mmHg with the disc and field unchanged over a year
- B IOP 19 mmHg with conjunctival hyperaemia and lash growth
- C IOP 30 mmHg recorded once on a day he missed the drop
- D A pressure difference of 9 mmHg between the two eyes
- E IOP 24 mmHg with nerve fibre layer thinning progressing OD

ANSWER _____ CONFIDENCE L • M • H

5 He drives to work. What is the correct position on his licence?

LAW AND ETHICS

- A He may drive while his treated pressure stays below 21 mmHg
- B Driving is his own affair, so nothing is advised or recorded
- C Assessment rests on the binocular field, so he drives on for now
- D Any glaucoma diagnosis must be notified to the authority at once
- E A monocular arcuate defect disqualifies him until pressure is controlled

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-32 • why

1 • B Angle recession glaucoma OD

A wide band of ciliary body over four clock hours with torn iris processes is a recessed angle, and the sphincter tears and focal lens opacity date it to the injury. The fellow eye is normal.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Asymmetrical POAG	Open angles with a notched disc and arcuate loss is its usual picture.	Primary disease does not recess the ciliary band or tear iris processes.
C	Pseudoexfoliation	It is the commonest cause of a genuinely unilateral high pressure.	No flakes on the pupil margin or capsule, and the pigment is focal.

2 • D, E, F Recessed band, torn processes, sphincter tears

Only the structural scars carry a date: a recessed ciliary band, torn processes, and a torn sphincter beside a focal lens opacity. The other findings measure severity, not mechanism.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, B	Pressure and cupping	They are the abnormal numbers, and they are what drives the treatment.	Both occur in primary disease, so neither dates the damage to 20 years ago.
C	Corneal thickness	Thickness corrects the pressure reading, which makes it feel diagnostic.	545 μm is average, and it says nothing about why this angle is damaged.

3 • C Topical prostaglandin OD, review at four weeks

A notched disc with a matching arcuate defect needs the pressure down, and a prostaglandin gives the largest single-agent reduction. Medical therapy comes before laser or a filtering operation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Laser trabeculoplasty	It is a standard first line where the meshwork is open and visible.	A traumatised meshwork answers laser poorly and unpredictably.
A	Trabeculectomy now	Established field loss at 30 mmHg looks like a surgical problem.	He has had no medical trial. Surgery is not the opening move here.

4 • E Progressive thinning at 24 mmHg

Escalation follows progression rather than a number. Structural loss continuing at 24 mmHg means the target is too high for this nerve, while a stable disc at the same pressure means it is not.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stable at 24 mmHg	24 sits above the usual target, so treating the number feels safer.	A nerve holding steady for a year says the current target is working.
C	Spike after a missed dose	A reading of 30 on treatment reads as outright failure of the drop.	An unmedicated pressure measures adherence, not the regimen.

5 • C Binocular field decides, advice recorded

Fitness turns on the binocular field, and a full fellow field covers this arcuate defect. He is told the position, formally field tested if the defect spreads, and the advice is recorded.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Notify at once	The word glaucoma sounds notifiable in and of itself.	One eye affected with a full fellow field does not meet the test.
A	Pressure under 21 mmHg	21 is the number patients remember, so it feels like the threshold.	Licensing is decided on field and acuity, never on a pressure reading.

IF-THEN RULES FROM THIS CASE

If one eye alone runs a high pressure, gonioscope both angles before naming the disease. If a wide ciliary band is seen, expect a poor answer to trabeculoplasty and monitor for life.

CASE S2-33 • GLAUCOMA • 5 ITEMS

Both eyes blurred a week after a new tablet

A 34-year-old woman began topiramate for migraine 11 days ago, raised to 50 mg twice daily four days ago. Both eyes blurred over 36 hours, with brow ache. Refraction has moved from -1.00 to -4.50 D in each eye. BVA 20/40 each. IOP 42 OD, 40 OS. Corneas hazy, chambers shallow centrally and peripherally, iris plane flat with no bombe, angles closed on gonioscopy in both eyes. Lenses clear. B-scan shows annular choroidal elevation.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Acute primary angle closure in both eyes at once
- B Topiramate effusion with secondary angle closure
- C Plateau iris syndrome presenting acutely in both eyes
- D Phacomorphic closure from an intumescent lens
- E Posner-Schlossman syndrome with bilateral spikes

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the first 24 hours?

TREATMENT • SELECT 3

- A Topical pilocarpine 2% hourly to both eyes
- B Argon laser peripheral iridoplasty to both eyes
- C Stop the topiramate today and tell the prescriber
- D Topical aqueous suppressants with atropine 1%
- E Topical prednisolone to settle the ciliary effusion
- F Bilateral laser peripheral iridotomy at this session

ANSWER _____ CONFIDENCE L • M • H

3 The pressure is 34 mmHg at six hours. What is the next step?

TREATMENT

- A Add oral acetazolamide and a hyperosmotic, keeping the cycloplegic
- B Perform laser peripheral iridotomy in both eyes now
- C Add pilocarpine 2% hourly to both eyes and review
- D Arrange urgent lens extraction for the eye with the higher pressure
- E Paracentesis of both anterior chambers at the slit lamp

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism explains the myopic shift?

BASIC SCIENCE

- A Corneal oedema steepening the anterior corneal curvature
- B Ciliary muscle spasm driving an accommodative response
- C Axial elongation under a sustained high pressure
- D Anterior rotation of the ciliary body and lens
- E Osmotic swelling of the lens raising its refractive index

ANSWER _____ CONFIDENCE L • M • H

5 Which single finding most reliably separates this closure from primary angle closure?

DIAGNOSIS

- A Intraocular pressure above 40 mmHg in each eye
- B Corneal oedema with acuity reduced to 20/40
- C A closed angle on gonioscopy in both eyes
- D Brow ache with blurring over the last 36 hours
- E A central chamber as shallow as the periphery

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-33 • why

1 • B Drug-induced ciliochoroidal effusion

Simultaneous bilateral closure, a 3.50 D myopic shift, shallowing at the centre as well as the periphery, and annular choroidal elevation place the block behind the iris. The drug started 11 days ago.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Primary angle closure	The pressures, the haze and the closed angles are identical to it.	Primary closure takes one eye at a time and shifts no refraction.
D	Phacomorphic closure	A lens moving forward does shallow a chamber and close an angle.	Both lenses are clear, and both eyes closed together in 36 hours.

2 • C, D, E Stop the drug, aqueous suppression, atropine, steroid

The block sits behind the iris, so the drug comes out, the ciliary body is relaxed and aqueous is suppressed while the effusion settles. The diaphragm falls back over days and the angles reopen.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Laser iridotomy	Closed angles at 40 mmHg is the textbook indication for a hole.	There is no pupil block to relieve, so the angle stays shut afterwards.
A, B	Pilocarpine, iridoplasty	Both pull peripheral iris out of the angle in other closures.	Miotics rotate the ciliary body further forward and worsen this one.

3 • A Acetazolamide and a hyperosmotic, atropine kept

Medical control buys the days the effusion needs once the drug is withdrawn. Nothing here calls for a hole in the iris or the loss of a clear lens, and both act on a mechanism that is absent.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Iridotomy now	A pressure of 34 on maximal drops reads as failure needing laser.	The angle is shut by forward rotation, which an iridotomy leaves intact.
D	Lens extraction	Removing the lens does deepen a crowded anterior segment for good.	These lenses are clear and the mechanism resolves in days regardless.

4 • D Forward lens movement on a rotated ciliary body

Fluid in the supraciliary space rotates the ciliary body forward and slackens the zonule, so the lens moves anteriorly and thickens. The eye turns myopic and the chamber shallows across its width.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Lens swelling	Lens hydration is a real cause of acute myopia in hyperglycaemia.	These lenses are clear, and swelling would not lift the choroid.
B	Ciliary spasm	Spasm gives myopia and brow ache, and both are present here.	Spasm closes no angle and produces no annular choroidal elevation.

5 • E Central as well as peripheral shallowing

Pupil block bows the peripheral iris and leaves central depth near normal. Uniform shallowing with a myopic shift means the whole lens-iris diaphragm has come forward, which is a different disease.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Pressure above 40	The height of the pressure feels like a measure of the mechanism.	Both mechanisms reach the forties, so the number separates nothing.
B, C	Oedema, closed angle	These are the findings that name an angle closure in the first place.	Shared by both, and neither says what pushed the iris forward.

IF-THEN RULES FROM THIS CASE

If both angles close at once with a myopic shift, stop the new drug and look behind the iris. If the chamber is shallow centrally as well as peripherally, an iridotomy will not open it.

CASE S2-34 • GLAUCOMA • 5 ITEMS

Pressure that climbed again after a laser hole

A 42-year-old woman had bilateral laser iridotomies 14 months ago for appositional closure. Refraction +0.75 both eyes, BVA 20/20 each. IOP 15 OD, 30 OS. Both iridotomies are patent on retroillumination, the iris plane flat, central chamber depth normal. Indentation gonioscopy OS shows a double hump, angle closed to the meshwork over three quadrants. OD closes on dark provocation. Ciliary processes sit anteriorly on ultrasound.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Plateau iris syndrome persisting after iridotomy
- B Malignant glaucoma with aqueous misdirection
- C Chronic angle closure with peripheral synechiae
- D Pigment dispersion from chafing at the laser edge
- E Incomplete iridotomy with residual pupil block

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings show that pupil block is not the mechanism?

DIAGNOSIS • SELECT 3

- A Refraction of +0.75 in both eyes
- B Both iridotomies are patent on retroillumination
- C Normal central chamber depth with a flat iris
- D A double hump at the angle on indentation
- E Intraocular pressure of 30 mmHg in the left eye
- F The angle closes on dark provocation OD

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate treatment for the closure OS?

TREATMENT

- A Argon laser iridoplasty OS with a pressure-lowering drop
- B Repeat the iridotomy OS with a larger opening
- C Trabeculectomy with mitomycin C in the left eye
- D Clear lens extraction in the left eye this month
- E Observation with gonioscopy and imaging repeated in six months

ANSWER _____ CONFIDENCE L • M • H

4 How should the right eye be managed?

TREATMENT

- A Discharge to routine care, with no further gonioscopy
- B Regular gonioscopy, treating on a pressure rise or synechiae
- C Prophylactic iridoplasty OD at the same session as the left eye
- D Long-term topical pilocarpine 1% to the right eye
- E Repeat the iridotomy OD, since the angle still closes

ANSWER _____ CONFIDENCE L • M • H

5 She is offered a systemic anticholinergic for bladder symptoms. What is correct?

TREATMENT

- A Tell the prescriber that dilation can close her angles
- B No precaution is needed now that the iridotomies are patent
- C Refuse the tablet until a trabeculectomy has been done
- D Add pilocarpine while she takes the tablet, then stop it
- E Reassure her, since only eye drops dilate enough to matter

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-34 • why

1 • A Plateau iris syndrome after iridotomy

Appositional closure with a rising pressure despite two patent iridotomies, a flat iris with normal central depth, a double hump on indentation, and anteriorly placed ciliary processes define plateau iris.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Residual pupil block	Closure after a laser suggests the hole is too small or has sealed.	Both holes transilluminate and the central chamber depth is normal.
B	Malignant glaucoma	It too closes an angle in an eye that already has a patent iridotomy.	It follows filtering surgery and shallows the chamber centrally.

2 • B, C, D Patent holes, flat iris, double hump

A patent hole removes block, and its signature, a bowed iris over a shallow centre, is absent. The double hump shows ciliary processes holding the peripheral iris against the meshwork instead.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, E	Pressure and refraction	One is the abnormal number, the other the classic closure risk factor.	Neither locates the block, and +0.75 argues against a crowded eye.
F	Closure in the dark	It proves the angle does close, which is the worry in front of you.	It shows that closure happens, not what pushes the iris forward.

3 • A Iridoplasty OS with a pressure-lowering drop

Iridoplasty contracts peripheral iris stroma and pulls it out of the angle, which is the only laser that treats this configuration. A drop holds the pressure while the angle is reopened.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	A second iridotomy	Another hole feels logical when the first one did not hold the angle.	The hole is patent, so a second changes nothing about the ciliary body.
C	Trabeculectomy	30 mmHg after a laser that failed looks like a surgical eye.	Filtration is for uncontrolled damage, not an angle that can be opened.

4 • B Gonioscopy on a schedule, treat on change

This eye holds 15 mmHg with no synechiae, so it is followed with the test that shows apposition. Treatment is triggered by a pressure rise or new adhesions, not by the configuration alone.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Iridoplasty both eyes	The configuration is bilateral, so treating both looks consistent.	No pressure rise and no synechiae here. There is nothing yet to treat.
A	Discharge	15 mmHg with 20/20 and a patent iridotomy reads as a normal eye.	An angle that closes in the dark needs gonioscopy, not discharge.

5 • A Warn the prescriber, dilation can close the angle

A patent iridotomy removes pupil block but not plateau configuration, so a dilated pupil still crowds the angle. The prescriber needs that before the first tablet, not after the attack.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	No precaution needed	The iridotomy is the treatment everyone associates with angle closure.	It prevents block alone, and this angle closes without any block.
E	Only drops matter	Topical drugs reach the iris directly, which sounds like the greater risk.	Systemic anticholinergics dilate the pupil as effectively as a drop.

IF-THEN RULES FROM THIS CASE

If an angle still closes with a patent iridotomy, a flat iris and normal central depth, the block sits at the iris root. Treat the configuration with iridoplasty, and warn about dilation.

CASE S2-35 • GLAUCOMA • 5 ITEMS

The screening result he does not want to act on

A community programme screened 1000 adults over 60 for glaucoma, in a population where the prevalence is 4%. The instrument has a sensitivity of 90% and a specificity of 85%. A 63-year-old man screens positive: IOP 25 OD, 24 OS, cup-disc 0.7 both eyes, a brother with glaucoma. He has capacity, declines referral and further testing today, and asks for a copy of his record to take elsewhere.

1 What is the positive predictive value of a positive screen in this population?

BASIC SCIENCE

- A 4%, since that is the prevalence in the group
- B 20%, so four in five positive screens are false
- C 90%, the share of true cases the test detects
- D 85%, the share of healthy people it clears
- E About 99%, since very few cases are missed

ANSWER _____ CONFIDENCE L · M · H

2 The same test is used in a clinic where prevalence is 20%. Which two statements are correct?

BASIC SCIENCE • SELECT 2

- A The positive predictive value rises to about 60%
- B Sensitivity and specificity are unchanged
- C The negative predictive value rises to about 99%
- D False positives per 1000 screened rise to 200
- E Specificity rises because fewer positives are false
- F The positive predictive value stays at about 20%

ANSWER _____ CONFIDENCE L · M · H

3 He declines referral and further testing. What must be done today?

LAW AND ETHICS

- A Record what was explained, the risks accepted, and his decision
- B Record that he refused and close the episode as declined
- C Ask him to sign a waiver releasing the practice from liability
- D Refer him anyway, since a positive screen obliges a referral
- E Involve his brother, who already understands the family risk

ANSWER _____ CONFIDENCE L · M · H

4 He asks for a copy of his record to take to another practice. What is correct?

LAW AND ETHICS

- A Refuse until he accepts the referral he has declined
- B Give him a copy within the statutory period
- C Send it to the other practice only, and not to him
- D Withhold the screening images as practice property
- E Charge for the copy and release it once it is paid

ANSWER _____ CONFIDENCE L · M · H

5 What is the most appropriate clinical plan at this visit?

TREATMENT

- A Repeat the screening test in twelve months and do nothing now
- B Discharge him with advice to return if his vision changes
- C Write to his physician asking for the referral to be made
- D Offer fields and disc imaging here, with a written plan
- E Start a prostaglandin analogue today on the screening result

ANSWER _____ CONFIDENCE L · M · H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-35 • why

1 • B 20%

Of 1000 screened, 40 have disease and 36 are detected, while 144 of the 960 healthy screen positive. That makes 180 positives, of which 36 are true, so 36 divided by 180 gives 20%.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	90%, the sensitivity	Sensitivity is the figure quoted for the test, so it reads as its accuracy.	Sensitivity looks forward from disease, not back from a positive result.
A	4%, the prevalence	It is the other number in the stem, and the disease is genuinely rare.	A positive result moves his probability from 4% up to 20%.

2 • A, B PPV rises to 60%, the test itself unchanged

At 20% prevalence, 200 of 1000 have disease and 180 are detected, while 120 of the 800 healthy screen positive. So 180 of 300 positives are true. The instrument has not changed at all.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	NPV rises to 99%	A better positive value at higher prevalence sounds like it lifts both.	More disease means more missed cases, so the negative value falls to 97%.
D, E	200 false positives	Both follow from treating prevalence as a property of the instrument.	With 800 healthy people, false positives fall to 120, and 85% is fixed.

3 • A Document the explanation, the risk and the refusal

A refusal is valid when a person with capacity has been told what the test is for and what untreated disease costs. What makes it informed is the record of that exchange, and an open door back.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Signed waiver	A signature looks like the strongest protection the practice can hold.	Liability cannot be signed away, and a form is not a conversation.
D	Refer him anyway	The clinical risk is real, and referral feels like the safe default.	Acting against the refusal of a competent adult is the greater wrong.

4 • B A copy to him, inside the statutory period

The file belongs to the practice and the information in it belongs to him, so a person with capacity may have a copy on request. The original, including the refusal entry, stays on file.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Clinician to clinician	A transfer between professionals feels tidier and more secure.	He is entitled to his own record, and he decides where it goes next.
A	Refuse until he agrees	It reads as leverage that might yet protect his sight.	Access to a record is not conditional on accepting treatment.

5 • D Definitive testing offered here, then review

A positive screen is not a diagnosis, and four in five are false, so fields and imaging come before any drop. Offering the tests he will accept keeps him in view and dates the next decision.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Start a prostaglandin	25 mmHg with a 0.7 cup looks like enough to justify treating now.	Treating on a screen commits him to therapy for an unmade diagnosis.
A	Rescreen in a year	A year is a reasonable interval for following a glaucoma suspect.	He holds an untested positive result and a family history, not a clearance.

IF-THEN RULES FROM THIS CASE

If a screen is positive, work out the positive predictive value before acting on it. If a person with capacity refuses, record what was explained, what was refused, and the review offered.

CASE S1-07 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

Vision that faded again two years after surgery

A 71-year-old man had uncomplicated phacoemulsification OS 26 months ago and reached 20/20. He now reports 20/50 OS, glare in sunlight and difficulty with print. Refraction is unchanged from his one-month postoperative result. The implant is centred in the bag, with a wrinkled pearly membrane behind it and pearls at its edge. Cornea clear, IOP 16, no cells, macula normal on OCT.

1 What is the most likely cause of the drop in vision?

DIAGNOSIS

- A Refractive change from posterior corneal astigmatism
- B Pseudophakic cystoid macular oedema
- C Posterior capsule opacification behind the implant
- D Late in-the-bag intraocular lens dislocation
- E Fuchs endothelial dystrophy, decompensating

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate treatment?

TREATMENT

- A Intravitreal triamcinolone for the membrane
- B New spectacles with an anti-glare coating
- C Lens exchange with polishing of the capsule
- D Nd:YAG laser posterior capsulotomy in clinic
- E Surgical membrane peel by a pars plana approach

ANSWER _____ CONFIDENCE L • M • H

3 Two days after the laser he reports new floaters and a shadow in the upper field. What is the priority?

DIAGNOSIS

- A Repeat the laser to enlarge the opening
- B Dilated peripheral retinal examination today
- C Reassurance, since floaters are expected after laser
- D Measure the pressure and review in one month
- E Macular OCT to look for postoperative oedema

ANSWER _____ CONFIDENCE L • M • H

4 Which pressure finding one hour after the laser needs treatment before he leaves?

TREATMENT

- A Pressure risen from 16 to 21 mmHg
- B Pressure unchanged with pigment in the chamber
- C Pressure fallen to 12 with a shallow chamber
- D Pressure 18 with a few red cells behind the iris
- E Pressure risen from 16 to 31 mmHg

ANSWER _____ CONFIDENCE L • M • H

5 Which two duties apply when consenting him for the laser?

LAW AND ETHICS • SELECT 2

- A Record that alternatives, including no treatment, were discussed
- B Obtain consent from a relative as well as from him
- C Warn him that the implant will be replaced
- D Guarantee the acuity he had at one month
- E Disclose the raised risk of retinal detachment

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-07 • why

1 • C Posterior capsule opacification

A wrinkled pearly membrane with Elschnig pearls behind a centred implant accounts for glare and reduced acuity years after surgery. A normal OCT and stable refraction exclude the alternatives.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Pseudophakic CMO	It is the classic late cause of blur after cataract surgery.	OCT shows a normal macula, and CMO peaks in months, not two years.
A	Refractive change	New spectacles are the simplest explanation for gradual blur.	Refraction is unchanged from one month after his surgery.

2 • D Nd:YAG posterior capsulotomy

A YAG capsulotomy clears the axis in minutes without entering the eye, and his symptoms justify it. Surgery is kept for a dense fibrotic plate the laser cannot open.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Lens exchange	It removes membrane and pearls together in one definitive operation.	It trades a two-minute laser for intraocular surgery and its risks.
B	Anti-glare spectacles	Glare is his main complaint, and coatings genuinely reduce it.	The scatter is inside the eye. No coating reaches behind the implant.

3 • B Same-day dilated peripheral examination

A field shadow with new floaters after capsulotomy is a retinal detachment until proven otherwise, and the laser raises that risk. Only a dilated peripheral view answers the question.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Reassure about floaters	Debris and floaters are genuinely common in the first days.	A field defect is not debris. The periphery has to be seen today.
E	Macular OCT	Post-laser oedema is real and OCT is the test that finds it.	OCT misses the peripheral break causing an upper field shadow.

4 • E A spike to 31 mmHg before discharge

Pressure peaks in the first hours after capsulotomy as pigment and debris load the outflow. A rise into the thirties is treated and rechecked before he goes home.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Rise to 21 mmHg	It is a rise, and any rise after laser feels like a complication.	A few points of transient rise is expected and settles untreated.
B	Pigment, same pressure	Visible pigment is exactly the mechanism of the spike.	Pigment without a pressure rise needs recheck, not treatment.

5 • A, E Disclose detachment risk, record alternatives

Consent covers the material risks a patient in his position would want, detachment and pressure rise among them, and the alternatives including doing nothing. Outcomes are explained, not promised.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Relative consents too	Involving family feels safer for an older patient.	He has capacity. The consent is his alone to give.
D	Guarantee 20/20	He reached 20/20 before, so promising it sounds honest.	A guaranteed outcome is not consent. State the expected range.

IF-THEN RULES FROM THIS CASE

If vision falls years after cataract surgery with a clear macula and stable refraction, look behind the implant. If a field shadow follows capsulotomy, dilate and examine the periphery that day.

CASE S2-19 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

Pain and a lost view on the third day after surgery

A 74-year-old with type 2 diabetes had uneventful phacoemulsification 3 days ago and read 20/25 on day 1. Pain and rapid loss began 12 hours ago. BVA hand movements. Lids swollen, conjunctiva intensely injected, wound Seidel negative, 2 mm hypopyon, fibrin across the intraocular lens, IOP 28. No fundus view. B-scan: dense vitreous opacities with the retina attached.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Fibrinous uveitis after surgery
- B Retained lens matter with inflammation
- C Uveitis-glaucoma-hyphaema syndrome
- D Acute bacterial endophthalmitis
- E Toxic anterior segment syndrome

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A Intravitreal antibiotic after the cultures are reported
- B Increase the topical steroid and review tomorrow
- C Vitreous tap and intravitreal vancomycin with ceftazidime
- D Hourly fortified topical antibiotics with review in 12 hours
- E Subconjunctival gentamicin and oral moxifloxacin

ANSWER _____ CONFIDENCE L • M • H

3 Which three belong in the first hour?

TREATMENT • SELECT 3

- A Intravitreal amphotericin as routine cover
- B Systemic ciprofloxacin as the sole therapy
- C Removal of the intraocular lens at once
- D Aqueous and vitreous samples before injecting
- E Intravitreal vancomycin with ceftazidime
- F Vitrectomy today if vision is light perception

ANSWER _____ CONFIDENCE L • M • H

4 At 48 hours the eye is worse despite injection. What now?

TREATMENT

- A Add oral prednisolone and continue observation
- B Continue the current treatment and review in 48 hours
- C Change to intravitreal amphotericin for fungus
- D Proceed to enucleation to control the infection
- E Repeat intravitreal antibiotics and refer for vitrectomy

ANSWER _____ CONFIDENCE L • M • H

5 Which source and organism best fit this onset?

BASIC SCIENCE

- A Surface flora through the wound, mostly staphylococcal
- B Haematogenous streptococci from a distant focus
- C Propionibacterium sequestered in the capsular bag
- D Contaminated irrigating fluid causing sterile inflammation
- E Fungal spores carried in the viscoelastic device

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-19 • why

1 • D Acute bacterial endophthalmitis

Pain with loss to hand movements on day 3, a hypopyon, fibrin and dense vitreous opacities is infection inside the eye. Vitreous involvement is what separates it from anterior causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Toxic anterior segment	It also gives fibrin and a hypopyon after routine surgery.	It starts within 24 hours, spares the vitreous, and is not painful.
B	Retained lens matter	Retained cortex does inflame an eye during the first week.	It does not produce a hypopyon with dense vitreous opacity in hours.

2 • C Tap and inject vancomycin with ceftazidime

Bacteria multiply in vitreous within hours and only intravitreal drug reaches it, so the sample and the injection happen at one sitting, today. Topical and oral routes do not reach vitreous.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Topical, review in 12 hours	Intensive drops treat corneal infection and feel proportionate.	Drops do not reach the vitreous, and 12 hours is the retina's margin.
A	Wait for cultures	Targeted therapy after a result is sound antimicrobial practice.	Therapy here is empirical. Sampling and injection go together.

3 • D, E, F Sample, inject, vitrectomy if light perception

Samples are taken before drug is given, cover is broad and intravitreal, and presenting vision decides surgery: light perception only is the threshold at which vitrectomy does better.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Amphotericin routinely	Fungal endophthalmitis is devastating, so cover looks safer.	Fungus is rare on day 3 and amphotericin is toxic to the retina.
B, C	Oral only, lens out	Both look aggressive in roughly the right direction.	Oral drug underdoses the vitreous, and the lens stays for now.

4 • E Reinject and refer for vitrectomy

Failure to improve means the organism load has not been cleared, so drug is given again and the vitreous is removed with it. Steroid alone and further delay both cost retina.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Oral steroid	Inflammation is doing the visible damage inside the eye.	Steroid with uncontrolled infection accelerates the loss.
C	Switch to an antifungal	Failure of antibacterial therapy suggests the wrong organism.	Reinjection and vitrectomy come first. Fungal onset is later and slower.

5 • A Ocular surface flora through the wound

Most acute cases come from the patient's own lid and conjunctival flora entering at surgery, which is why antisepsis matters more than the choice of antibiotic. Coagulase-negative staphylococci lead.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Propionibacterium	It is the organism every postoperative list carries.	It presents weeks to months later, indolently, with a capsular plaque.
D	Sterile, from fluid	Contaminated fluid does cause clusters after cataract surgery.	Sterile syndromes appear on day 1 and spare the vitreous.

IF-THEN RULES FROM THIS CASE

If an eye loses vision with pain in the first week after surgery and the vitreous is involved, tap and inject the same hour.
If vision is light perception, arrange vitrectomy.

CASE S2-20 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

A cloudy cornea the morning after routine surgery

A 68-year-old woman is seen the morning after uneventful phacoemulsification. Painless blur, BVA 20/200. Limbus-to-limbus corneal oedema, 1.5 mm of fibrin in the chamber, a fixed mid-dilated pupil, IOP 32, lids barely swollen, no vitreous cells on examination or on ultrasound. Two other patients from the same operating list have similar findings this morning.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Acute bacterial endophthalmitis
- B Descemet membrane detachment
- C Retained lens fragment reaction
- D Corneal decompensation from surgical trauma
- E Toxic anterior segment syndrome

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings favour a toxic cause over infection?

DIAGNOSIS • SELECT 3

- A Fibrin layered in the anterior chamber
- B Intraocular pressure of 32 mmHg
- C Best acuity reduced to 20/200
- D Onset within 24 hours of surgery
- E Limbus-to-limbus corneal oedema
- F Two others from the same list affected

ANSWER _____ CONFIDENCE L • M • H

3 How should this eye be managed today?

TREATMENT

- A Urgent vitrectomy with intravitreal cover
- B Topical antibiotic hourly, review in 48 hours
- C Oral acetazolamide alone, review in a week
- D Intensive topical steroid with pressure control
- E Tap and intravitreal antibiotics within the hour

ANSWER _____ CONFIDENCE L • M • H

4 Which action addresses the cause for the rest of the list?

TREATMENT

- A Switch to a different intraocular lens model
- B Send the remaining viscoelastic for sterility testing only
- C Continue the list with extra topical antibiotic
- D Sequester the instruments and fluids, audit reprocessing
- E Change the antibiotic prophylaxis for future lists

ANSWER _____ CONFIDENCE L • M • H

5 Which exposure most commonly causes this syndrome?

BASIC SCIENCE

- A Preservative-free lidocaine given into the chamber
- B Povidone-iodine 5% applied to the lid skin
- C A hydrophobic acrylic lens in the capsular bag
- D Topical NSAID started on the first day
- E Residue left on reprocessed surgical instruments

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-20 • why

1 • E Toxic anterior segment syndrome

Onset within 24 hours, limbus-to-limbus oedema with a fixed dilated pupil, no vitreous involvement and three eyes from one list is a toxic insult delivered into the anterior chamber.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Endophthalmitis	Fibrin with a raised pressure after surgery reads as infection.	Infection is painful, starts later, and takes the vitreous with it.
D	Surgical decompensation	A hard cataract can leave a swollen cornea on day 1.	Trauma gives focal oedema, not a fixed pupil in three patients.

2 • D, E, F Day 1 onset, whole-cornea oedema, a cluster

Toxic injury is immediate, spread across the whole cornea because the insult was inside the chamber, and shared by whoever shared the exposure. Fibrin, pressure and acuity occur in both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Fibrin in the chamber	A fibrinous chamber is the sign that dominates the examination.	Both diagnoses produce fibrin, so it never decides the question.
B, C	Pressure, acuity	Both are abnormal and both look like markers of severity.	Severity is shared. Timing and distribution are not.

3 • D Intensive steroid, pressure control, same-day review

The injury is a sterile chemical one, so treatment is anti-inflammatory with pressure control and a recheck the same day. If vitreous cells appear or pain begins, treat the eye as infected.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Tap and inject	Missing endophthalmitis is the costlier error, so cover it.	Day 1 onset, clear vitreous, no pain, and three eyes on one list.
B	Antibiotic, 48 hours	It hedges towards infection and avoids an injection.	It treats nothing here and leaves the eye unseen for 2 days.

4 • D Quarantine instruments and fluids, audit reprocessing

A cluster points at the process rather than the patient, so shared instruments, solutions and reprocessing steps are held and examined. Nothing else stops the next eye being injured.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Change prophylaxis	Adjusting antibiotics feels like the standard response to a cluster.	The insult is toxic, not infective. Prophylaxis changes nothing.
B	Test viscoelastic only	Viscoelastic is a real culprit, so testing it looks targeted.	Testing one item leaves detergent, autoclave and solution errors in use.

5 • E Residue on reprocessed instruments

Poor cleaning of phaco tips and cannulas leaves detergent, denatured viscoelastic and endotoxin to enter the chamber. Preserved intracameral drugs and solutions of wrong pH do the same.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Preservative-free lidocaine	Intracameral drug is an established cause of this syndrome.	It is preserved drug that injures. Unpreserved lidocaine is standard.
B, C	Iodine, lens	Both touch the eye, and iodine is toxic to endothelium at strength.	Antisepsis at 5% on skin and an acrylic lens give no diffuse oedema.

IF-THEN RULES FROM THIS CASE

If the anterior segment inflames within 24 hours with limbus-to-limbus oedema and a clear vitreous, treat as toxic and examine the list. If the vitreous is involved, treat as infection.

CASE S2-21 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

The eye that bleeds a little every few months

A 68-year-old man had phacoemulsification OD 14 months ago, complicated by a capsule tear, and a single-piece acrylic implant was placed in the sulcus. Four episodes since of ache and red haze OD. He takes aspirin 75 mg daily after a coronary stent. BVA 20/40 OD. A 1 mm layered hyphaema, 2+ cells, pigment on the endothelium, a superonasal iris transillumination defect, an open angle heavy with pigment. IOP 32 OD, 15 OS.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Chronic pseudophakic endophthalmitis from *Cutibacterium acnes*
- B Uveitis-glaucoma-hyphaema syndrome from a sulcus implant
- C Herpetic anterior uveitis with trabeculitis and atrophy
- D Pigmentary glaucoma with spontaneous microhyphaema
- E Neovascular glaucoma with rubeosis and recurrent bleeding

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the initial management of this eye?

TREATMENT • SELECT 3

- A Laser trabeculoplasty to the pigmented angle
- B Intravitreal anti-VEGF for the recurrent bleeding
- C Ultrasound biomicroscopy to locate the haptics
- D Topical steroid with an aqueous suppressant
- E Referral for repositioning or exchange of the implant
- F A prostaglandin analogue as first-line pressure control

ANSWER _____ CONFIDENCE L • M • H

3 The pressure is 32 mmHg with a 1 mm hyphaema. Which pressure treatment fits today?

TREATMENT

- A Oral acetazolamide 250 mg four times daily for a week
- B Selective laser trabeculoplasty at this visit
- C Topical pilocarpine 2% with the steroid four times daily
- D Topical timolol with dorzolamide, review in 48 hours
- E Topical latanoprost at night, review in one week

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism best explains the raised pressure in this eye?

BASIC SCIENCE

- A A neovascular membrane contracting across the angle
- B Steroid response raising trabecular outflow resistance
- C Pigment and red cells loading the trabecular meshwork
- D Peripheral anterior synechiae closing the angle
- E Ciliary block with forward rotation of the ciliary body

ANSWER _____ CONFIDENCE L • M • H

5 He asks about his aspirin. Which plan is most appropriate?

TREATMENT

- A Ask cardiology for a shorter-acting substitute
- B Continue the aspirin and treat the chafing implant
- C Stop the aspirin until the hyphaema has cleared
- D Stop the aspirin and start tranexamic acid instead
- E Continue the aspirin and observe for six months

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-21 • why

1 • B UGH syndrome from a sulcus implant

A single-piece haptic in the sulcus chafes the back of the iris: a transillumination defect, dispersed pigment, recurrent bleeding, cells and 32 mmHg in that eye alone, with a normal fellow eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Cutibacterium endophthalmitis	It is also a late, low-grade, relapsing uveitis in a pseudophakic eye.	No capsular plaque, and blood with an iris defect is mechanical.
D	Pigmentary glaucoma	Heavy angle pigment and iris transillumination are its signature.	That is bilateral, in young myopes, and it does not bleed.

2 • C, D, E Image the haptics, settle the eye, refer

The eye is settled medically while the cause is confirmed and the implant dealt with. Chafing is mechanical, so no drop cures it, and laser and anti-VEGF treat mechanisms not at work here.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Prostaglandin first line	It is the strongest single agent for a pressure of 32 mmHg.	It is avoided while the chamber is inflamed and bleeding.
A, B	Trabeculoplasty, anti-VEGF	One targets the pigmented angle, the other the bleeding vessels.	The angle is loaded by chafe, and there are no new vessels.

3 • D Aqueous suppression with a beta-blocker and a CAI

Aqueous suppression lowers pressure without stirring the eye, and 48 hours is short enough to catch a further rise. Prostaglandins and miotics worsen inflammation, and laser adds pigment load.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Latanoprost at night	It is the usual first choice and the most effective single drop.	It aggravates active uveitis, which is what is driving this eye.
C	Pilocarpine with steroid	Miosis pulls the iris off the haptic and outflow improves.	Miotics increase inflammation and drag iris across the optic.

4 • C Pigment and blood clog the meshwork

Gonioscopy shows an open angle heavy with pigment, and the chamber holds red cells. Both load the meshwork, which is why pressure rises with each bleed and falls between them.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Angle closure by synechiae	Chronic uveitis does scar the angle shut and raise pressure.	Gonioscopy shows the angle open along its whole circumference.
B	Steroid response	He is likely to be given steroid, and steroid does raise pressure.	The pressure was already 32 mmHg before any steroid was given.

5 • B Keep the aspirin, deal with the implant

The bleeding is mechanical, from a haptic rubbing iris vessels, so withdrawing antiplatelet cover risks stent thrombosis without stopping the bleed. The implant is the treatable part.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stop the aspirin	Antiplatelet drugs do prolong bleeding, and this eye keeps bleeding.	A coronary stent makes that a cardiac risk for no ocular gain.
E	Aspirin, then observe	Bleeds have settled by themselves four times already.	Pressure is 32 with each episode. Waiting costs optic nerve.

IF-THEN RULES FROM THIS CASE

If a pseudophakic eye bleeds again with pigment dispersion and an iris transillumination defect, image the haptics. If pressure and bleeding recur, move the implant.

CASE S2-22 • LENS, CATARACT, IOL, PERIOPERATIVE • 5 ITEMS

A wobble in the vision nine years after surgery

A 79-year-old man had phacoemulsification OS nine years ago and read 20/20 afterwards. Six months of blur that changes with head position, with a monocular double image. BVA 20/70 OS, and 20/25 with +2.25 D added to his old prescription. Pseudoexfoliative material on both pupil margins. OS: the implant and a contracted capsule sit low and tilted, the optic edge crossing the pupil, moving on eye movement. IOP 22 OS, 19 OD.

1 What is the most likely cause of the change in vision?

DIAGNOSIS

- A Posterior capsule opacification behind a centred implant
- B Late in-the-bag dislocation of the implant and capsule
- C Out-of-the-bag dislocation with a broken haptic
- D Capsular phimosis without loss of zonular support
- E Pseudophakic macular oedema causing fluctuation

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings show that the zonule, not the capsule alone, has failed?

DIAGNOSIS • SELECT 3

- A Nine years since an uncomplicated operation
- B Phacodonesis of implant and capsule together
- C The complex sitting low and tilted in the pupil
- D A hyperopic shift of +2.25 D since surgery
- E Pseudoexfoliative material on both pupil margins
- F An intraocular pressure of 22 mmHg OS

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Referral for planned surgical refixation of the complex
- B Emergency same-day vitrectomy and implant removal
- C Nd:YAG laser to open the contracted anterior capsule
- D Observation three-monthly while acuity stays at 20/70
- E New spectacles with the +2.25 D change, review in six months

ANSWER _____ CONFIDENCE L • M • H

4 Which change would make this a same-day referral?

TREATMENT

- A Acuity falling a further two lines over three months
- B Pressure rising to 26 mmHg on no treatment
- C The double image becoming constant in all positions
- D The optic edge crossing the visual axis at rest
- E Free dislocation into the vitreous, pressure 42 mmHg

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism best explains the late loss of support here?

BASIC SCIENCE

- A Vitreous traction pulling on the implant haptics
- B Enzymatic digestion of the zonule by chronic uveitis
- C A capsule tear made during the original operation
- D Pseudoexfoliative fibrilopathy weakening the zonule
- E Residual lens epithelium contracting the capsule

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-22 • why

1 • B Late in-the-bag dislocation

Implant and contracted capsule move together, sit low and tilted, and the eye has gained +2.25 D, so the whole complex has fallen back on a failing zonule. Pseudoexfoliation explains why.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Capsule opacification	It is the usual reason vision fades years after cataract surgery.	A membrane does not tilt the optic, wobble, or shift refraction.
C	Out-of-the-bag dislocation	Both leave a displaced implant in an eye operated years ago.	The capsule has come with it, so the haptics are still in the bag.

2 • B, C, D Wobble, tilt and a hyperopic shift

Movement of implant and capsule as one unit, a tilted low position and a change in refraction all measure lost support. A capsule that has only contracted stays where the zonule holds it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Pseudoexfoliation	It is the reason the zonule failed, so it reads as the evidence.	It predicts failure. It does not show that anything has moved.
A, F	IOP 22, nine years	One is abnormal, the other the interval that invites suspicion.	Pressure reflects the meshwork, and time alone measures nothing.

3 • A Planned surgical refixation

The complex is mobile and the optic edge crosses the pupil, so acuity will not hold and full dislocation is likely. It is a planned list operation, not an emergency and not a laser.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Emergency vitrectomy	A mobile implant in a nine-year-old eye feels like an emergency.	It is still in place with a flat retina. The list can be planned.
C	YAG to the phimosis	The capsule has contracted, and laser does open a tight rim.	Cutting a supporting rim on a loose complex hastens the drop.

4 • E Free dislocation with a pressure of 42

An implant loose in the vitreous with a pressure of 42 threatens optic nerve and retina today. The other changes are reasons to bring the planned operation forward, not to admit him.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Two lines over months	Falling acuity is what the operation is for in the first place.	A slow drop moves him up the list. It is not an emergency.
D	Edge across the axis	It explains the double image and the poor acuity he reports.	It is the same disease at the same speed, with the eye intact.

5 • D Pseudoexfoliative zonular weakness

Pseudoexfoliative material on both pupil margins marks a fibrillopathy that degrades the zonule over years, and it is the commonest reason a bagged implant descends late.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Capsule contraction	The capsule here has plainly contracted around the optic.	Contraction pulls the rim in. It does not detach the zonule.
C	Old capsule tear	A tear at surgery is the classic route to a displaced implant.	That implant falls early, and his operation was uneventful.

IF-THEN RULES FROM THIS CASE

If pseudophakic vision changes with head position, look for movement of the whole complex rather than a membrane. If it is mobile but still in place, plan surgery rather than admit.

CASE S1-01 • LIDS, LACRIMAL, ADNEXA, ORBIT • 5 ITEMS

The swollen eyelid that hurts to move

A 13-year-old boy has three days of worsening left eyelid swelling and frontal headache, after a week of purulent nasal discharge. Temperature 38.4 °C. BVA 20/20 OD, 20/40 OS. Mild left RAPD, red saturation 60% OS, 3 mm proptosis, tense chemosis, painful restriction of elevation and abduction. IOP 17 OD, 27 OS. Cornea clear, chamber quiet, fundus normal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Cavernous sinus thrombosis complicating sinusitis
- B Orbital cellulitis of ethmoid sinus origin
- C Orbital rhabdomyosarcoma with secondary inflammation
- D Preseptal cellulitis with reactive chemosis
- E Idiopathic orbital inflammatory syndrome

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings measure optic nerve function, rather than supporting the infective diagnosis?

DIAGNOSIS • SELECT 3

- A Red saturation reduced to 60% OS
- B Purulent nasal discharge for one week
- C Intraocular pressure 27 mmHg OS
- D Best corrected acuity 20/40 OS
- E Temperature 38.4 °C
- F Left relative afferent pupillary defect

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Outpatient MRI within 48 hours, then decide about admission
- B Intravenous antibiotics booked for the morning, oral cover overnight
- C Oral corticosteroid with a topical antibiotic to settle the swelling
- D Oral amoxicillin-clavulanate, nasal decongestant, review in 24 hours
- E Same-day referral for contrast CT, intravenous antibiotics, ophthalmology and ENT

ANSWER _____ CONFIDENCE L • M • H

4 Which anatomic route most commonly explains this presentation?

BASIC SCIENCE

- A Direct spread across the lamina papyracea from the ethmoid sinus
- B Retrograde thrombophlebitis through valveless facial and orbital veins
- C Extension through the optic canal from the sphenoid sinus
- D Ascending infection along the nasolacrimal duct from the inferior meatus
- E Haematogenous seeding of the orbit from an occult bacteraemia

ANSWER _____ CONFIDENCE L • M • H

5 He is admitted on intravenous antibiotics. Which change over 18 hours most supports drainage?

TREATMENT

- A Acuity falls to 20/80 with worse colour and an abscess crowding the apex
- B Blood cultures grow an organism fully sensitive to current therapy
- C Chemosis increases while pupil, colour and acuity stay unchanged
- D Temperature settles after two doses while the lid oedema persists
- E CT shows a 4 mm medial subperiosteal collection, acuity improving

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-01 • why

1 • B Orbital cellulitis, ethmoid origin

Proptosis, painful restricted motility, RAPD, reduced colour and acuity, and raised IOP all place the disease behind the septum. Fever with a week of purulent rhinorrhoea names the source.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Cavernous sinus thrombosis	The correct escalation of this infection, and it also gives proptosis and chemosis.	Signs stay strictly unilateral, with no multiple cranial neuropathy and no V1-V2 loss.
D	Preseptal cellulitis	The visible finding is a swollen lid, and preseptal disease is commoner at this age.	Preseptal disease spares acuity, pupil, colour, motility and globe position.
C	Rhabdomyosarcoma	Rapid proptosis in a child is the classic teaching phrase for it.	Fever and purulent sinus disease over three days fit infection, not tumour.

2 • A, D, F RAPD, red desaturation, reduced acuity

The item asks for afferent function, not for evidence of infection. An RAPD, dyschromatopsia and reduced best acuity each measure optic nerve conduction directly.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	IOP 27 mmHg	It is genuinely abnormal, and orbital pressure is what threatens the nerve.	It measures orbital pressure, not conduction. Normalising it proves nothing.
B, E	Fever, purulent discharge	They confirm infection and drive the diagnosis.	Neither measures the eye. A febrile child can have a normal afferent system.

3 • E Same-day referral, contrast CT, IV antibiotics

Orbital cellulitis with afferent dysfunction threatens the eye and, through intracranial extension, life. Imaging, intravenous therapy and surgical availability are arranged together, today.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	IV antibiotics tomorrow	It sees that intravenous therapy is needed and feels like a fair compromise.	The overnight delay is where vision goes. The decision point is the RAPD, now.
A	Outpatient MRI in 48 hours	MRI is the better soft-tissue study, so it reads as the more thorough answer.	Contrast CT answers today's questions faster: sinus, collection, apex, bone.

4 • A Lamina papyracea, from the ethmoid

The medial orbital wall is paper-thin and perforated by vessels, which makes contiguous ethmoid spread the usual route and explains why abduction and elevation fail first.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Valveless venous spread	A real mechanism, and how mid-face infection reaches the cavernous sinus.	Not the usual route from sinusitis, and it does not localise medially.
C	Optic canal, sphenoid	Sphenoid disease is dangerous and sits against the orbital apex.	Rare in children, and it would show apical signs before medial restriction.

5 • A Falling function with an apical abscess

Drainage is driven by function and location, not by the presence of a collection. Falling afferent function with apical involvement is the indication, as is failure to improve on adequate therapy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Abscess, function improving	An abscess on imaging feels like a surgical finding by definition.	Small medial collections often resolve on antibiotics. Imaging does not decide.
C	Chemosis increasing	It looks like deterioration in front of you.	Surface swelling with unchanged afferent function is no reason to open an orbit.

IF-THEN RULES FROM THIS CASE

If lid swelling comes with any change in acuity, colour, pupil, motility or globe position, treat it as orbital until imaging says otherwise. If function falls, drain.

CASE S2-13 • LIDS, LACRIMAL, ADNEXA, ORBIT • 5 ITEMS

The lump that keeps coming back in the same place

A 71-year-old woman has a third firm nodule at the same site in the left upper lid over 14 months. Two incision and curettage procedures, no specimen sent either time. Over the lesion the lid margin has lost its lashes and its posterior ridge, the tarsus is yellow and thickened, and the eye has stayed red through lid hygiene and 6 weeks of topical antibiotic. Left preauricular node 1 cm, firm and mobile.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Nodular basal cell carcinoma of the lid
- B Squamous cell carcinoma of the lid margin
- C Chronic staphylococcal blepharoconjunctivitis
- D Sebaceous carcinoma of the tarsal glands
- E Recurrent chalazion with granulation tissue

ANSWER _____ CONFIDENCE L • M • H

2 Which single feature argues most strongly against a benign chalazion?

DIAGNOSIS

- A Lash loss with destruction of the posterior lid margin
- B Recurrence at the same site after curettage
- C Yellow colour of the thickened tarsal tissue
- D A red eye that has not settled on treatment
- E Firmness of the nodule on palpation

ANSWER _____ CONFIDENCE L • M • H

3 Which procedure should be arranged to obtain the diagnosis?

TREATMENT

- A Repeat incision and curettage, curettings to histology
- B Shave biopsy of the lid margin lesion under anaesthesia
- C Excision with 2 mm margins as the first procedure
- D Fine needle aspiration of the nodule and the node
- E Full-thickness lid biopsy plus conjunctival map biopsies

ANSWER _____ CONFIDENCE L • M • H

4 Which three features of this tumour explain why it is missed?

DIAGNOSIS • SELECT 3

- A It metastasises before the lid lesion is visible
- B It is confined to the lower lid in most cases
- C Pagetoid spread within conjunctival epithelium
- D Multicentric foci with skip lesions in the tarsus
- E Lipid lost in processing unless fresh tissue is used
- F It arises from the eccrine glands of the lid

ANSWER _____ CONFIDENCE L • M • H

5 Histology confirms it. Which plan is appropriate?

TREATMENT

- A Wide excision with margin control, plus node imaging
- B Cryotherapy to the lid margin with topical mitomycin
- C Primary radiotherapy to the lid and the orbit
- D Exenteration as the first definitive procedure
- E Excision alone, with review of the node in 3 months

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-13 • why

1 • D Sebaceous carcinoma of the tarsus

Recurrence at one site, lash loss with destruction of the posterior margin, yellow tarsal thickening and a unilateral chronic conjunctivitis are destructive disease. The node raises the stakes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Recurrent chalazion	Two surgeons treated it as one, and chalazia genuinely do recur.	A chalazion does not destroy lash follicles or the posterior lid margin.
C	Chronic blepharitis	Unilateral chronic conjunctivitis with lid disease reads as blepharitis.	Blepharitis is bilateral and builds no firm nodule inside the tarsus.

2 • A Madarosis with loss of margin architecture

Loss of lash follicles and of the posterior margin means the lesion is replacing tissue rather than distending it. Recurrence, colour and chronic redness each have benign explanations.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Recurrence after surgery	Three episodes in 14 months is what prompted the referral.	Incompletely drained chalazia recur too. Recurrence is not destruction.
C, E	Yellow, firm nodule	Yellow lipid and firmness feel like the signature of this tumour.	Old granulation tissue is firm and yellow as well.

3 • E Full-thickness biopsy with map biopsies

The tumour sits within the tarsus and spreads inside conjunctival epithelium, so a superficial sample misses it. Mapping defines pagetoid extent, which decides how much tissue must go.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Curettage with histology	It samples the lesion and repeats a familiar, quick procedure.	Curettings miss tarsal depth and pagetoid spread, and blur the margins.
C	Excision with 2 mm margins	Excising it looks decisive and treats the lesion in one visit.	Margins are planned after histology and mapping, not before them.

4 • C, D, E Pagetoid spread, skip lesions, lipid lost

Intraepithelial spread imitates chronic blepharoconjunctivitis, skip lesions make one negative sample falsely reassuring, and lipid dissolves in routine processing so the cells look bland.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Eccrine origin	Naming a gland of the lid sounds like the pathological detail.	It arises in meibomian and Zeis glands, which are sebaceous.
A, B	Early spread, lower lid	Both sound like facts that would explain a missed tumour.	The upper lid is the commoner site, and nodes follow visible disease.

5 • A Wide excision with margins, node assessed

Clearance is judged on controlled margins, and pagetoid disease needs conjunctival treatment as well. A palpable node is imaged and sampled because this tumour spreads through lymphatics.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Cryotherapy, mitomycin	Topical treatment spares the lid and is used for conjunctival spread.	They are adjuncts to excision, not treatment for an invasive tarsal mass.
E	Excision, node later	The node is small and mobile, so watching it seems reasonable.	A 1 cm node with an invasive tumour is imaged and sampled now.

IF-THEN RULES FROM THIS CASE

If a chalazion recurs at one site, or takes lashes or the posterior lid margin, send full-thickness tissue. If a lid carcinoma is confirmed, examine and image the nodes.

CASE S2-14 • LIDS, LACRIMAL, ADNEXA, ORBIT • 5 ITEMS

Watering since the first weeks of life

A 9-month-old boy has watered from the right eye since the age of 3 weeks, with crusted lashes each morning that clean away easily. No photophobia and no lid squeezing. Corneas clear, horizontal diameters 10.5 mm each. The right tear lake is high, and fluorescein clears from it in 8 minutes against 2 minutes on the left. Mucoid reflux on pressure over the sac. No sac swelling, no redness, no fever.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Epiblepharon with lower lash contact
- B Congenital obstruction at the valve of Hasner
- C Primary congenital glaucoma with epiphora
- D Punctal agenesis of the lower canaliculus
- E Congenital dacryocystocele with mucus reflux

ANSWER _____ CONFIDENCE L • M • H

2 Which finding most reliably excludes primary congenital glaucoma?

DIAGNOSIS

- A Mucoid reflux on pressure over the sac
- B Dye disappearance delayed to 8 minutes
- C Symptoms present since the age of only 3 weeks
- D Corneal diameters of 10.5 mm, corneas clear
- E Absence of photophobia and of lid squeezing

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management now?

TREATMENT

- A Teach sac massage and hygiene, review at 12 months
- B Probing and irrigation under anaesthesia this month
- C Continuous topical antibiotic four times daily
- D Silicone intubation of the duct within 6 weeks
- E Refer for dacryocystorhinostomy before school age

ANSWER _____ CONFIDENCE L • M • H

4 Which three situations make earlier referral necessary?

TREATMENT • SELECT 3

- A Mucoid reflux on pressure over the lacrimal sac
- B Repeated episodes of acute dacryocystitis
- C A tense bluish swelling below the medial canthus
- D Symptoms unchanged beyond 12 to 15 months of age
- E Crusted lashes needing cleaning every morning
- F Watering that worsens in cold wind outdoors

ANSWER _____ CONFIDENCE L • M • H

5 At 13 months the symptoms are unchanged. What now?

TREATMENT

- A Balloon dacryoplasty as the first procedure
- B Dacryocystorhinostomy with silicone tubes
- C Topical antibiotic for a further 3 months
- D Refer for probing and irrigation under anaesthesia
- E Continue massage and review again at the age of 24 months

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-14 • why

1 • B Congenital nasolacrimal duct obstruction

Watering from the first weeks with morning crusting, a high tear lake, delayed dye disappearance and mucoid reflux from the sac places the block low in a patent, comfortable eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Congenital glaucoma	Epiphora since infancy is the presentation the two conditions share.	Corneal diameters of 10.5 mm, clear corneas, no photophobia or squeezing.
E	Dacryocystocele	Mucoid reflux comes from a sac that is holding its contents.	A dacryocystocele presents in the neonate as a tense bluish swelling.

2 • D Normal corneal diameter with a clear cornea

Raised pressure in infancy stretches and clouds the cornea, so normal diameters with clear corneas are hard evidence against it. The other findings show a blocked duct but exclude nothing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	No photophobia	Photophobia and squeezing are the taught triad with epiphora.	Early congenital glaucoma can water without obvious light aversion.
A	Mucoid reflux	Reflux proves the duct is blocked, which feels like enough.	A blocked duct and a raised pressure can sit in the same eye.

3 • A Massage and hygiene, review at 12 months

Most obstructions at Hasner open during the first year, so conservative care with a review date is the default at 9 months. Antibiotic is for purulent episodes, not as a standing course.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Probing now	It cures the problem in one visit after 8 months of watering.	At 9 months most open without anaesthesia. Probing can wait.
C	Continuous antibiotic	Morning crusting looks infective and drops settle it briefly.	It treats discharge, not obstruction, and selects resistant organisms.

4 • B, C, D Dacryocystitis, tense sac, persistence

Infection of the sac, a distended sac in a neonate, which can also obstruct the nose, and persistence past the age at which spontaneous opening stops all move the child to surgical care.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E, F	Crusting, wind	They are the symptoms the parents most want stopped.	Both are typical of simple obstruction that still resolves on its own.
A	Mucoid reflux	Reflux from the sac sounds like a sac that needs an operation.	It is expected in ordinary obstruction and does not change the timing.

5 • D Probing and irrigation under anaesthesia

Spontaneous opening becomes unlikely after the first year, and probing at this age succeeds as a single procedure in most children. Intubation and bypass surgery are for failures.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Wait until 24 months	Conservative care was the right answer for the whole first year.	The resolution curve flattens after 12 months. Waiting buys nothing.
B	Primary bypass surgery	It is the definitive operation, so it looks like the thorough answer.	It is reserved for failed probing and intubation at this age.

IF-THEN RULES FROM THIS CASE

If an infant waters with clear corneas of normal diameter, treat as duct obstruction and review at 12 months. If the sac becomes infected or tensely distended, refer at once.

CASE S1-33 • LOW VISION • 5 ITEMS

The post she can no longer read

A 79-year-old retired teacher with geographic atrophy reads headlines but not her post. Best corrected acuity is 20/125 each eye, stable over a year. She reads 1.6 M print at 25 cm slowly and cannot hold a line of text. Amsler shows a dense scotoma just to the right of fixation in the better eye. Both hands have a mild resting tremor. She still drives locally in daylight. Her kitchen has one ceiling light.

1 By Kestenbaum's rule, which add starts her on 1 M print, and at what working distance?

BASIC SCIENCE

- A +4.00 D at 25 cm
- B +5.00 D at 20 cm
- C +6.25 D at 16 cm
- D +8.00 D at 12.5 cm
- E +10.00 D at 10 cm

ANSWER _____ CONFIDENCE L • M • H

2 Which finding best explains why she cannot hold a line of print?

DIAGNOSIS

- A Illumination at the page is far below the level she needs
- B Her acuity reserve for 1 M print is smaller than 2 to 1
- C Uncorrected astigmatism blurs the smaller letters most
- D The scotoma sits to the right of fixation, over the text ahead
- E Contrast sensitivity is reduced across all spatial frequencies

ANSWER _____ CONFIDENCE L • M • H

3 Which two measures will do most for sustained reading of 1 M print?

TREATMENT • SELECT 2

- A Bring a directable task lamp close to the page
- B Train eccentric viewing to move the scotoma off the text
- C Add a yellow tint to raise apparent contrast
- D Fit a 3x hand magnifier held at arm's length
- E Supply an anti-reflection coating on her spectacles

ANSWER _____ CONFIDENCE L • M • H

4 Which device suits reading her post at the kitchen table?

TREATMENT

- A A closed-circuit magnifier sited in the sitting room
- B An illuminated stand magnifier at a fixed focal height
- C An illuminated hand magnifier held at its focal distance
- D A +6.25 D half-eye pair used at her habitual 25 cm
- E A 6x monocular telescope braced against the brow

ANSWER _____ CONFIDENCE L • M • H

5 She mentions that she still drives locally. What is the duty here?

LAW AND ETHICS

- A Tell her she is below the acuity standard, record it, and cite her duty to notify
- B Notify the licensing authority the same day without discussing it with her
- C Advise her to limit driving to daylight and familiar routes, and review in a year
- D Leave the question to her family doctor, who holds the wider record
- E Note the acuity in the record and raise driving at the next visit

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-33 • why

1 • C +6.25 D add, working at 16 cm

Kestenbaum: the starting add is the reciprocal of the acuity fraction, so $125/20 = 6.25$ D. Its focal distance is $1/6.25 = 0.16$ m, and the page has to be held there.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	+10.00 D at 10 cm	It is the familiar answer, but for 20/200 rather than 20/125.	The reciprocal of 125/20 is 6.25, not 10.
A	+4.00 D at 25 cm	It matches the 25 cm she already uses and feels comfortable.	+4.00 magnifies too little to bring 1.6 M performance down to 1 M.

2 • D Scotoma to the right of fixation

Letter acuity measures one target at a time. A scotoma on the side the eye is moving into hides the next word, so saccades land in the blind area and the line is lost.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Acuity reserve	Too small a reserve does slow reading, and hers is modest.	Reserve limits speed and endurance, not the ability to track a line.
A	Poor illumination	One ceiling light is genuinely inadequate for atrophy.	Light raises contrast and comfort, but it cannot fill a scotoma.

3 • A, B Task lighting at the page, eccentric viewing

Atrophy needs light on the page, and a scotoma over the incoming text needs a new preferred locus. Both act on what limits her, unlike a coating, a tint, or a lens held out of focus.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Yellow tint	Tints are popular and can cut glare on a bright page.	They also cut light reaching a retina that needs more of it.
D	3x hand magnifier	3x is close to the magnification the arithmetic asks for.	Held at arm's length it is out of focus, and her tremor moves it.

4 • B Illuminated stand magnifier

A stand magnifier holds the lens at its focal distance for her, which a resting tremor cannot do with a hand-held lens, and its own light supplies what one ceiling fitting does not.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Hand magnifier	It is portable, cheap, and the usual first device offered.	A resting tremor moves the lens off its focal distance continuously.
D	+6.25 D at 25 cm	The add is the one the arithmetic gives, and the habit is hers.	A +6.25 D lens focuses at 16 cm. At 25 cm the print is still blurred.

5 • A State the standard, document, name the duty to notify

The consultation is where this is settled: a plain statement that she is below the standard, the advice not to drive, her response, and who must notify. The record of that talk is the evidence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Notify without telling her	Reporting duties exist and other road users are genuinely at risk.	Notification never precedes telling the patient, whoever has to do it.
C	Daylight, familiar routes	It sounds proportionate and preserves her independence.	An acuity below the standard is below it in daylight on a known road.

IF-THEN RULES FROM THIS CASE

If letter acuity is better than reading performance, look for the scotoma and the lighting before adding magnification.
If acuity is below the driving standard, say so in the consultation.

CASE S2-06 • LOW VISION • 5 ITEMS

Doorways he cannot find once the light goes

A 42-year-old carpenter has trouble finding doorways after dusk, going back to his teens, and now catches furniture indoors. BVA 20/60 OD, 20/70 OS with -2.50 DS. Goldmann fields measure 12° horizontally in each eye. Mid-peripheral bone spicule pigment, waxy disc pallor, narrowed arterioles, early posterior subcapsular lens change. Rod responses are non-detectable, cone flicker reduced. He reads 2 M print at 25 cm.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Cone-rod dystrophy with peripheral pigment
- B Choroideremia with mid-peripheral atrophy
- C Gyrate atrophy with raised plasma ornithine
- D Vitamin A deficiency retinopathy after bowel resection surgery
- E Rod-cone dystrophy of the retinitis pigmentosa spectrum

ANSWER _____ CONFIDENCE L • M • H

2 He reads 2 M at 25 cm. What equivalent power puts 1 M print within reach?

TREATMENT

- A +2.00 D at 50 cm
- B +4.00 D at 25 cm
- C +5.00 D at 20 cm
- D +6.00 D at 16.7 cm
- E +8.00 D at 12.5 cm

ANSWER _____ CONFIDENCE L • M • H

3 Which three belong in the first low vision plan?

TREATMENT • SELECT 3

- A Genetic testing with counselling for him and his family
- B A 4x spectacle telescope as his main reading device
- C 40 Δ Fresnel prisms on both lenses to widen the field
- D A reversed 3x telescope worn full time for mobility
- E Orientation and mobility referral for the 12° field
- F A +8.00 D add with strong task lighting for 1 M print

ANSWER _____ CONFIDENCE L • M • H

4 Why did his night vision fail years before his acuity?

BASIC SCIENCE

- A Bipolar cells fail to pass the rod signal onward
- B Rod loss begins in the mid periphery, where rods are densest
- C Cone loss at the fovea spreads outward to the rods
- D Pigment epithelium pump failure detaches the rod outer segments
- E Cone photopigment regenerates too slowly in dim light

ANSWER _____ CONFIDENCE L • M • H

5 His state requires 20/40 and 110° of field to drive, and he drives daily. What is the duty?

LAW AND ETHICS

- A Ask him to raise it himself at licence renewal, and take no action today
- B Refer for on-road assessment before saying anything about the licence
- C Restrict him to daylight driving close to home, noted in the record
- D Withhold the field result from the record until the retinal specialist confirms the diagnosis
- E Record the field, tell him he is below the standard, advise stopping, and report as required

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-06 • why

1 • E Rod-cone dystrophy, RP spectrum

Night blindness from his teens, a 12° field, non-detectable rod responses with cones still working, and mid-peripheral spicules with waxy pallor are the rod-cone sequence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Cone-rod dystrophy	Acuity is already 20/60, so the cones are plainly involved too.	Cone-rod disease takes acuity and colour first, and spares the field.
B	Choroideremia	It gives night blindness and mid-peripheral change in a man.	Its atrophy bares choroidal vessels and white sclera, not spicules.

2 • E +8.00 D, reading at 12.5 cm

He needs twice the magnification 25 cm gives: $1 / 0.25 = 4.00$ D, and $4.00 \times 2 = 8.00$ D. The working distance follows, since $1 / 8 = 0.125$ m, so the page comes to 12.5 cm.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	+4.00 D at 25 cm	It is the dioptric equivalent of the distance he already uses.	It reproduces 2 M print. Halving print size needs double the power.
D	+6.00 D at 16.7 cm	It is more power than he has now, so the print does get smaller.	1.5 times the power reaches 1.3 M print. The figure needed is 8.00 D.

3 • A, E, F Mobility, a +8.00 add with light, genetics

The field costs him mobility, the acuity costs him print, and the diagnosis carries inheritance. Each is met directly: training, an add with light, and testing that gates future therapy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	40 Δ Fresnel prisms	Peripheral prisms genuinely help field loss, in hemianopia.	They shift a field rather than widen a 12° ring, and they blur it.
D	Reversed 3x telescope	Minification does put more of the room inside a 12° field.	It cuts 20/60 to 20/180. It is a spotting device, not worn full time.

4 • B Rods die first, and rods are peripheral

In rod-cone disease the rods degenerate first, and rod density peaks about 20° from fixation. Scotopic vision and the mid-peripheral field go long before the cone-rich fovea.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Bipolar transmission	It is the mechanism of congenital stationary night blindness.	That is stationary from birth, with a normal fundus and full field.
E	Slow cone regeneration	Dark adaptation is slow, so a pigment cycle fault fits loosely.	Cones do not carry scotopic vision, and his cone responses remain.

5 • E Measure, tell him, advise stopping, report

A 12° field is far under 110°, so the finding is not borderline. The duty is to measure, to tell him plainly, to advise against driving, and to record and report as the state requires.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	On-road assessment first	Functional testing sounds fairer than a number on a chart.	No road test rewrites a 12° field against a 110° standard.
C	A daylight restriction	It keeps some independence, and dusk is when he struggles most.	The field fails in daylight too, and only the state sets restrictions.

IF-THEN RULES FROM THIS CASE

If night blindness comes years before the acuity falls, test the field and the rods, not just the chart. If the field is under the driving standard, record it, say so, and report as required.

CASE S1-30 • OPHTHALMIC OPTICS AND SPECTACLES • 5 ITEMS

The new spectacles that double the print

A 58-year-old woman cannot tolerate new single vision spectacles: R +5.00 DS, L +1.00 DS, optical centres set at her distance interpupillary distance. She reads with her eyes 8 mm below the optical centres. Print doubles vertically after a minute of reading. Corrected acuity 20/20 each. Cover test orthophoric at distance, 2 Δ exophoria at near. Vertical fusion range 2 Δ. Frame well adjusted, vertex distance equal.

1 By Prentice's rule, what vertical imbalance does the reading position induce?

BASIC SCIENCE

- A 0.8 Δ, base up before the right eye
- B 1.6 Δ, base up before the right eye
- C 3.2 Δ, base up before the right eye
- D 4.0 Δ, base up before the right eye
- E 4.8 Δ, base up before the right eye

ANSWER _____ CONFIDENCE L • M • H

2 What explains the diplopia at the reading position?

DIAGNOSIS

- A Incorrect interpupillary distance creating a horizontal imbalance
- B Aniseikonia from the difference in spectacle magnification
- C Induced vertical imbalance exceeding her vertical fusion range
- D A decompensating vertical phoria unmasked by new spectacles
- E A mild right superior oblique palsy revealed by near work

ANSWER _____ CONFIDENCE L • M • H

3 Which three changes would reduce the imbalance at the reading position?

TREATMENT • SELECT 3

- A Contact lens correction of the anisometropia for near work
- B A larger frame with a higher fitting cross position
- C Reducing the vertex distance by 2 mm at both eyes
- D Polycarbonate lenses to reduce the lens thickness
- E A separate near pair with centres set at the reading position
- F Slab-off on the more plus lens if a bifocal is dispensed

ANSWER _____ CONFIDENCE L • M • H

4 Which single solution is most appropriate for her reading?

TREATMENT

- A Add 3 Δ base in at near, split between the lenses
- B A smaller frame with a higher fitting cross for both eyes
- C A dedicated near pair centred at the reading position
- D Grind 3 Δ base down into the right distance lens
- E Reduce the right lens to +4.00 DS to balance the prisms

ANSWER _____ CONFIDENCE L • M • H

5 Which dispensing check would have caught this before collection?

TREATMENT

- A Confirm the interpupillary distance to 1 mm
- B Check the base curve matches the previous pair
- C Measure the pantoscopic tilt and adjust the pads
- D Compute the induced prism at the near gaze position
- E Verify the back vertex distance at both eyes

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-30 • why

1 • C 3.2 Δ net, base up before the right eye

Prism equals decentration in centimetres times lens power. Right 0.8 cm x 5.00 D gives 4.0 Δ base up, left 0.8 cm x 1.00 D gives 0.8 Δ base up, so the net imbalance is 3.2 Δ.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	4.0 Δ	It is the prism induced by the right lens, correctly calculated.	The left lens induces 0.8 Δ the same way, and only the difference counts.
E	4.8 Δ	Adding the two prisms feels right when both eyes are affected.	Both bases are up, so the two prisms subtract rather than add.

2 • C Induced imbalance beats the fusion range

Vertical fusional amplitude is small, 2 Δ here, and the reading position generates 3.2 Δ. The deviation appears only in downgaze, which is why her distance vision is comfortable.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Decompensating phoria	New spectacles often unmask a phoria, and the timing fits exactly.	The distance cover test is orthophoric and the symptom is position dependent.
E	Superior oblique palsy	Vertical diplopia worst on reading is the palsy's textbook symptom.	No torsion, no gaze or tilt pattern, and it began with the new lenses.

3 • A, E, F Near pair, slab-off, contact lenses

The imbalance is decentration multiplied by unequal power, so each fix removes one of the two: centre the lenses where she looks, cancel the difference, or move the correction to the cornea.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Higher fitting cross	Adjusting the fitting height does sound like the lever to pull.	Raising the centres increases the drop in downgaze and the imbalance with it.
D	Polycarbonate lenses	Thinner lighter lenses do improve tolerance of an anisometropic Rx.	Material changes cosmetics and weight, never the induced prism.

4 • C Dedicated near pair, centres lowered

Centring each lens where she looks removes the decentration, and with it the induced prism, without touching her distance correction or her acuity in either eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Reduce the right power	Equalising the powers does remove the imbalance arithmetically.	It leaves the right eye 1.00 D under-corrected and blurred at all distances.
D	Base-down in the right	Prescribing the opposing prism looks like a direct correction.	It corrects downgaze and breaks fusion at distance, where she is comfortable.

5 • D Calculate induced prism at the near position

With 4.00 D of anisometropia, any vertical drop below the optical centres creates imbalance, so the calculation belongs at the design stage, before the lenses are cut and glazed.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Vertex distance	Vertex distance matters at these powers and is easy to check.	It changes effective power, not the prism induced by decentration.
A	IPD to 1 mm	Centration error is the problem, so a PD check feels right.	Horizontal centration is correct here, and the drop is vertical.

IF-THEN RULES FROM THIS CASE

If lens powers differ by more than 1.00 D, compute the induced prism at the reading position before dispensing. If it exceeds the vertical fusion range, centre a separate near pair.

CASE S1-31 • OPHTHALMIC OPTICS AND SPECTACLES • 5 ITEMS

Clearer when the frame slides down her nose

A 24-year-old returns three days after collecting new spectacles: distance vision is less crisp than in her old pair and a dull brow ache builds through the afternoon. Subjective refraction two weeks earlier, in a trial frame at a 14 mm vertex, was -12.00 DS each eye giving 20/20. The dispensed frame sits at 9 mm. Acuity is now 20/25 each eye, and clearer when she slides the frame down her nose. Corneae, motility and fundi are normal.

1 What best explains the reduced acuity and the brow ache?

DIAGNOSIS

- A Accommodative insufficiency unmasked by a full distance correction
- B Aniseikonia between the two high-powered lenses
- C Over-minus at the shorter vertex distance of the dispensed frame
- D Excess pantoscopic tilt introducing unwanted oblique astigmatism
- E Base-in prism from optical centres set wider than her pupils

ANSWER _____ CONFIDENCE L • M • H

2 Which power at the 9 mm plane focuses this eye as -12.00 DS did at 14 mm?

BASIC SCIENCE

- A -13.00 DS
- B -12.50 DS
- C -12.00 DS
- D -11.25 DS
- E -10.25 DS

ANSWER _____ CONFIDENCE L • M • H

3 Which two features separate over-minus from an under-corrected refraction here?

DIAGNOSIS • SELECT 2

- A Best acuity of 20/25 in each eye
- B Both eyes carry the same spherical power
- C The old spectacles were more comfortable
- D Acuity improves as the frame slides away from the eye
- E Brow ache that builds with sustained near and distance work
- F The refraction was performed at a 14 mm vertex

ANSWER _____ CONFIDENCE L • M • H

4 What is the most appropriate action?

TREATMENT

- A Order aspheric lenses of the same power to cut edge distortion
- B Prescribe a +0.75 D reading addition for the brow ache
- C Reassure her that she will adapt, and review in four weeks
- D Remake both lenses at -11.25 DS for the 9 mm vertex
- E Adjust the pads to hold the frame at the 14 mm refracting vertex

ANSWER _____ CONFIDENCE L • M • H

5 She asks about soft contact lenses. Which trial power fits this refraction?

TREATMENT

- A -12.00 D
- B -11.25 D
- C -10.75 D
- D -10.25 D
- E -9.50 D

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-31 • why

1 • C Over-minus from a shorter dispensed vertex

A minus lens moved closer to the eye gains effective minus power, so -12.00 specified at 14 mm is too strong at 9 mm. Sliding the frame out restores the refracting vertex and the acuity.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Pantoscopic tilt	Tilt does induce cylinder, and it grows fast with lens power.	Tilt-induced blur does not clear when the frame slides away from the eye.
B	Aniseikonia	Magnification effects are real at -12.00 and cause asthenopia.	Both eyes carry the same power, so the two retinal images match in size.

2 • D -11.25 DS at the 9 mm vertex

The lens sits 5 mm closer, so it needs less minus: $F = -12.00 / (1 + 0.005 \times 12.00) = -11.32$ D, which is -11.25 to the nearest quarter dioptre.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	-13.00 DS	It assumes a shorter vertex needs more minus, the commoner reflex.	Closer to the eye a minus lens gains effective power, so it needs less.
E	-10.25 DS	It is the right arithmetic for the corneal plane, a contact lens power.	The lens plane here is 9 mm, not 0 mm, so only 5 mm of the shift applies.

3 • D, E Clearer with the frame out, plus accommodative ache

Under-correction does not improve as the lens moves out, and it does not force accommodation. Both features here are the signature of too much minus in front of an eye working to clear it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	14 mm refracting vertex	It is where the error starts, so it feels like the key finding.	It gives the mechanism, not the direction the error runs in.
A	Acuity 20/25	Reduced acuity is the presenting complaint and is genuinely abnormal.	Over-minus and under-correction both blur distance by a similar amount.

4 • D Remake at -11.25 DS for the fitted vertex

The frame was measured correctly, the power was not. Respecifying the power for the plane it will be worn at keeps the short vertex, which a high myope wants for field and magnification.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Hold the frame at 14 mm	It restores the conditions the refraction was measured under.	It gives away the field and the reduced minification a short vertex buys.
B	Reading addition	The ache is accommodative, and an add relieves accommodative strain.	It leaves the distance power wrong and asks a 24-year-old to stop accommodating.

5 • D -10.25 D at the corneal plane

At the cornea the whole 14 mm applies: $F = -12.00 / (1 + 0.014 \times 12.00) = -10.27$ D. Order -10.25 and check the over-refraction on the eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	-12.00 D	Copying the spectacle power is quick and works at low powers.	At -12.00 it over-minuses by about 1.75 D at the corneal plane.
B	-11.25 D	It is the right compensation, but for the spectacle plane at 9 mm.	A contact lens sits at the cornea, so the full 14 mm shift applies.

IF-THEN RULES FROM THIS CASE

If a high power is dispensed at a vertex other than the one it was refracted at, respecify the power for the plane it will be worn at. If the lens moves closer, minus needs less minus.

CASE S2-03 • OPHTHALMIC OPTICS AND SPECTACLES • 5 ITEMS

A line of print that separates in her new lenses

A 58-year-old librarian has ten days in her first progressive lenses: OD +1.25 DS, OS +3.25 DS, add +2.25, at a 14 mm vertex. Distance is comfortable, but at 40 cm the line of print separates vertically after a minute. Covering either eye leaves the print single. Reading gaze falls 10 mm below the distance optical centres. Cover test 1 Δ hyperphoria at 6 m, 3 Δ at 40 cm, motility full.

1 What is the most likely cause of the vertical separation at near?

DIAGNOSIS

- A A decompensating right hyperphoria in her fifties
- B Fitting crosses set 3 mm too low in both lenses
- C A progressive corridor too narrow for her task
- D The addition over-plussed by +0.50 D for 40 cm
- E Induced vertical prism from 2.00 D of anisometropia

ANSWER _____ CONFIDENCE L • M • H

2 By Prentice rule, what vertical imbalance does reading 10 mm below the centres induce?

BASIC SCIENCE

- A 0.5 Δ
- B 1.25 Δ
- C 2.0 Δ
- D 3.25 Δ
- E 4.5 Δ

ANSWER _____ CONFIDENCE L • M • H

3 Which three changes reduce the imbalance without altering her distance correction?

TREATMENT • SELECT 3

- A Split 2 Δ base out between the two lenses
- B Conventional slab-off on the right lens, the less plus of the pair
- C Single vision readers with the centres at her reading position
- D A shorter corridor design, cutting the drop to the near zone
- E A deeper frame, giving more lens below the optical centres
- F Reduce the addition from +2.25 to +1.75 D

ANSWER _____ CONFIDENCE L • M • H

4 What should be checked before any lens is remade?

TREATMENT

- A Lens material and coatings against the original order
- B Monocular centration and fitting heights in the frame as worn
- C Corneal topography of both eyes for irregular astigmatism
- D The addition, by re-measuring her amplitude of accommodation
- E Stereoacuity and fusion ranges through the finished lenses

ANSWER _____ CONFIDENCE L • M • H

5 She asks for lined bifocals instead. Which segment choice limits the imbalance?

TREATMENT

- A A round top before the right eye and a flat top before the left
- B Flat top 35 segments in both lenses, set as low as the frame allows
- C A round top before the left eye and a flat top before the right
- D Flat top 28 segments in both lenses, set 2 mm higher on the left
- E Executive segments in both lenses, with the line at the centres

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-03 • why

1 • E Vertical imbalance from anisometropia

Reading 10 mm below the optical centres of lenses that differ by 2.00 D induces unequal base-up prism. That is why the two lines separate at near while distance stays comfortable.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Decompensating hyperphoria	There is a measured 3 Δ hyperphoria at near to point at.	It appears in these lenses at the reading point, and not without them.
C	A narrow corridor	Corridor width is the usual complaint in a first progressive.	A narrow corridor blurs and swims. It does not split print vertically.

2 • C 2.0 Δ of vertical imbalance at 40 cm

Prentice: prism equals decentration in cm times power. Right $1.0 \times 1.25 = 1.25 \Delta$ base up, left $1.0 \times 3.25 = 3.25 \Delta$ base up, so the imbalance is 2.0 Δ.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	3.25 Δ	It is the left eye's prism, the larger of the two figures.	The symptom comes from the difference between the eyes, not one lens.
E	4.5 Δ	It adds the two prisms instead of subtracting them.	Both are base up, so they partly cancel: $3.25 - 1.25 = 2.0 \Delta$.

3 • B, C, D Slab-off right, separate readers, short corridor

The imbalance is proportional to the power difference and to how far below the centres she reads. Slab-off cancels it in the reading zone, readers move the centres, a short corridor cuts the drop.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	A deeper frame	More lens below the centre sounds like more room to read in.	Reading further below the centres multiplies the prism, not reduces it.
A	2 Δ base out	The magnitude is right, and prism is the right family of answer.	The imbalance is vertical. Horizontal prism does nothing to it.

4 • B Centration and heights, in the frame as worn

A height error of 3 mm changes the induced imbalance by 0.6 Δ, since $0.3 \text{ cm} \times 2.00 \text{ D} = 0.6 \Delta$. The fit is verified as worn before a slab-off is ordered on the wrong lens.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Re-measure the addition	The add is the other new number in a first progressive, and it is high.	A wrong add blurs at 40 cm. It cannot separate print vertically.
C	Corneal topography	Irregular astigmatism does cause monocular doubling of print.	Her doubling is binocular: it clears when either eye is covered.

5 • C Dissimilar segments, round top on the left

Reading above the optical centre of a round top adds base-down prism in front of the more plus left eye, offsetting its excess base up. A flat top centre sits near the reading point and adds little.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Round top on the right	It is the right technique, applied to the other eye of the pair.	Base down is needed where the base up is largest, in the left lens.
D	Matched flat tops, raised	Raising one segment does move where she reads in that lens.	The distance powers still differ, so the imbalance survives the change.

IF-THEN RULES FROM THIS CASE

If anisometropia passes 1.50 D, work out the Prentice prism at the reading point before dispensing. If the imbalance passes about 1.5 Δ, plan slab-off or separate readers.

CASE S1-20 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Ache behind the eye, then the colours went flat

A 29-year-old woman has four days of worsening blur in the left eye with ache on eye movement, and says colours look washed out on that side. No headache. Two years ago she had a fortnight of numbness in the right leg that resolved without investigation. BVA 20/20 OD, 20/80 OS. RAPD OS. Ishihara 14 of 14 OD, 4 of 14 OS. Central depression on 24-2 OS. Both discs flat and normal in colour. IOP 14 both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Leber hereditary optic neuropathy, first eye
- B Posterior ischaemic optic neuropathy after hypotension
- C Demyelinating retrobulbar optic neuritis
- D Non-arteritic anterior ischaemic optic neuropathy
- E Compressive optic neuropathy at the orbital apex

ANSWER _____ CONFIDENCE L • M • H

2 Which feature best separates this from an ischaemic optic neuropathy?

DIAGNOSIS

- A Pain on eye movement with a normal disc at 29
- B A relative afferent pupillary defect on that side
- C Acuity of 20/80 with a central field depression
- D Four of fourteen Ishihara plates read correctly
- E An intraocular pressure of 14 in both eyes

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management?

TREATMENT

- A Oral prednisolone 1 mg/kg for eleven days with a taper
- B Observation alone, since most eyes recover their acuity
- C Same-day inflammatory markers and a temporal artery biopsy
- D Contrast MRI of brain and orbits, with neurology referral
- E Intravenous methylprednisolone started today, imaging later

ANSWER _____ CONFIDENCE L • M • H

4 Which two statements belong in the counselling?

TREATMENT • SELECT 2

- A Colour vision returns fully in every case by six weeks
- B Recovery is unlikely once acuity has fallen to 20/80
- C Most eyes recover useful acuity over two to three months
- D Steroids speed recovery but do not change the final acuity
- E Low dose oral steroid prevents a second attack later

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism explains the ache on eye movement?

BASIC SCIENCE

- A Ischaemia of the short posterior ciliary arteries at the disc
- B Stretch of the ciliary nerves crossing the posterior sclera
- C Raised orbital pressure from oedema within the muscle cone
- D Meningeal irritation transmitted along the optic canal
- E Traction on the inflamed nerve sheath by the muscle origins

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-20 • why

1 • C Demyelinating retrobulbar optic neuritis

A young woman with subacute unilateral loss, pain on movement, dyschromatopsia, an RAPD and a normal disc has retrobulbar inflammation. The earlier leg symptom points to demyelination.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Anterior ischaemic	It is the commonest optic neuropathy and shares the RAPD and colour loss.	It needs a swollen disc and it does not hurt on eye movement.
E	Apical compression	Compression also spares the disc early and causes progressive loss.	Compression progresses over months, without pain on movement.

2 • A Pain on movement with a normal disc

An RAPD, reduced acuity and dyschromatopsia occur in both, so neither separates them. Pain on movement with a flat disc in a woman of 29 is specific to inflammation of the nerve behind the globe.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Afferent pupil defect	It is the sign that proves the nerve is at fault, so it feels decisive.	Any unilateral optic neuropathy gives it. It localises without discriminating.
D	Ishihara 4 of 14	Colour loss is taught as the marker of optic neuritis.	Ischaemic neuropathy desaturates colour in proportion to acuity too.

3 • D Contrast MRI, then neurology

Imaging counts the lesions that set the risk of multiple sclerosis and excludes compression, and that result drives disease modifying treatment. Steroids do not change the final acuity.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Steroids first	Treating the inflammation immediately feels like the active choice.	Steroids only shorten the attack, and imaging must come before immunotherapy.
A	Oral prednisolone	It avoids admission and appears to give the same anti-inflammatory effect.	Standard oral dosing alone was associated with more recurrent attacks.

4 • C, D Recovery expected, steroids only speed it

The natural history is good, with most eyes back near normal acuity within three months, and steroid benefit is confined to the speed of that recovery. Residual colour and contrast loss is common.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Colour fully recovers	Acuity recovers so well that colour is assumed to follow it.	Colour and contrast deficits persist in many eyes with 20/20 acuity.
B	Poor acuity, poor outcome	Presenting acuity predicts outcome in most other optic neuropathies.	Even eyes at counting fingers usually recover substantially here.

5 • E Traction on the inflamed nerve sheath

The superior and medial rectus origins share the dural sheath of the nerve at the apex, so every movement pulls on an inflamed, swollen sheath. This is why the pain is worse on adduction and elevation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Ciliary artery ischaemia	It is the mechanism of the disease this case must be separated from.	Ischaemic neuropathy is painless, and the disc here is flat and normal.
C	Raised orbital pressure	Orbital congestion genuinely causes ache worsened by movement.	There is no proptosis, no chemosis and no restriction of movement.

IF-THEN RULES FROM THIS CASE

If a young adult loses vision in one eye with pain on movement and a normal disc, image the brain and orbits before reaching for steroids. If the disc is swollen instead, rethink the diagnosis.

CASE S1-21 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Vision gone on waking, and the scalp hurts to comb

A 74-year-old woman woke with painless loss of vision in the right eye. Six weeks of scalp tenderness on combing, jaw ache after chewing, weight loss and night sweats. BVA counting fingers OD, 20/25 OS. Right RAPD. Chalky pallid disc oedema OD with two flame haemorrhages, cup 0.2 OS. Macula and retinal vessels normal OD. ESR 78 mm/h, CRP 42 mg/L, platelets 486. Right temporal artery firm and non-pulsatile.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Non-arteritic anterior ischaemic optic neuropathy of the right optic nerve
- B Arteritic anterior ischaemic optic neuropathy, giant cell arteritis
- C Central retinal artery occlusion with secondary disc swelling
- D Demyelinating optic neuritis of the right optic nerve
- E Diabetic papillopathy with segmental oedema of the disc

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings separate an arteritic cause from a non-arteritic one?

DIAGNOSIS • SELECT 3

- A Right relative afferent pupillary defect
- B Painless onset noticed on waking
- C Jaw ache after chewing for six weeks
- D Chalky pallid oedema of the swollen disc
- E ESR 78 mm/h with CRP 42 mg/L
- F Cup-to-disc ratio 0.2 in the fellow eye

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate immediate management?

TREATMENT

- A Repeat ESR and CRP in 48 hours, then decide about steroid
- B High-dose systemic corticosteroid today, biopsy within two weeks
- C Withhold systemic steroid until the temporal artery biopsy is reported
- D Oral prednisolone 20 mg daily with rheumatology review next week
- E Aspirin 300 mg daily and rheumatology review within 48 hours

ANSWER _____ CONFIDENCE L • M • H

4 Which vascular event best explains the appearance of this disc?

BASIC SCIENCE

- A Stenosis of the internal carotid artery reducing perfusion
- B Occlusion of the short posterior ciliary arteries
- C Thrombosis of the central retinal vein at the lamina
- D Embolic occlusion of a cilioretinal artery at the disc
- E Vasculitis of the pial vessels of the canalicular nerve

ANSWER _____ CONFIDENCE L • M • H

5 Ten days into high-dose steroid, which change most needs escalation?

TREATMENT

- A Insomnia, tremor and low mood since the steroid started
- B ESR fallen to 40 mm/h with the jaw ache still present
- C Right acuity unchanged at counting fingers since day one
- D Transient greying of the left eye with CRP risen to 38
- E Fasting glucose 9.2 mmol/L with nocturia since starting steroid

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-21 • why

1 • B Arteritic AION from giant cell arteritis

Chalky pallid oedema, vision at counting fingers and an RAPD point to arteritic infarction rather than a crowded-disc event. Jaw ache, scalp tenderness, ESR 78 and CRP 42 name the vasculitis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Non-arteritic AION	Commoner, and it also gives sudden painless loss with disc oedema.	Oedema here is chalky and pallid, with systemic symptoms and an ESR of 78.
C	Retinal artery occlusion	Sudden profound loss in an elderly arteritic patient fits an arterial event.	No retinal opacification and no cherry-red spot; the lesion is a pallid swollen disc.

2 • C, D, E Jaw claudication, pallid oedema, raised markers

Both entities give sudden painless loss with an RAPD and disc oedema. Jaw claudication, chalky pallor of the swelling and raised ESR and CRP are the arteritic markers.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Small cup in fellow eye	A crowded disc is a real risk marker, so it reads as supporting evidence.	A small cup argues for the non-arteritic form, not against it.
A, B	RAPD, painless onset	Both are present and both are abnormal, so they read as diagnostic.	Both occur in either form, so neither discriminates.

3 • B Steroid now, biopsy within two weeks

Sight in the fellow eye is what is at risk and it can go within days. Steroid starts on clinical suspicion with an ESR of 78, and biopsy stays positive for one to two weeks after treatment begins.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Wait for biopsy	Confirming before committing an elderly patient to steroid feels careful.	The fellow eye can infarct while waiting, and biopsy yield survives early steroid.
D	Prednisolone 20 mg	It treats, and a lower dose limits harm in a 74-year-old.	Established visual loss needs high dose; 20 mg does not protect the other eye.

4 • B Short posterior ciliary artery occlusion

The laminar and prelaminar nerve is fed by the short posterior ciliary arteries. Granulomatous inflammation of these medium-sized vessels infarcts that watershed and gives pallid swelling.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Central retinal vein	Vein occlusion also swells the disc and drops vision suddenly.	No dilated tortuous veins and no scattered retinal haemorrhages.
A	Carotid stenosis	Ocular ischaemic syndrome is real and she has vascular risk.	It gives retinal signs and gradual change, not chalky disc pallor.

5 • D Fellow eye symptoms with rising CRP

New obscurations in the uninvolved eye with a rising CRP mean active vasculitis and a second eye at risk. Steroid hyperglycaemia is managed alongside, not by cutting the dose.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Glucose 9.2	A real steroid effect that needs treatment and looks like a reason to taper.	It calls for glycaemic control, not less steroid while sight is at risk.
B	ESR 40, jaw ache	A falling ESR reads as control, so the residual ache seems minor.	Symptoms hold the taper where it is, but no visual threat means no escalation.

IF-THEN RULES FROM THIS CASE

If disc oedema is pallid, or comes with jaw claudication or scalp tenderness, start steroid before the biopsy. If the fellow eye reports obscurations, escalate the same day.

CASE S1-22 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Headaches worst on waking, with a whoosh in both ears

A 28-year-old woman has three months of daily headache, worst on waking and on coughing, with brief greyouts of vision on standing and a whooshing noise in both ears. Horizontal diplopia at distance. BMI 38. Doxycycline for four months. BVA 20/20 each, no RAPD. Bilateral disc oedema with obscured crossing vessels, no spontaneous venous pulsation. Abduction limited OS. Perimetry: enlarged blind spots, nasal step OS. BP 128/78.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Malignant hypertension with bilateral disc swelling and exudates
- B Bilateral non-arteritic ischaemic optic neuropathy
- C Papilloedema from raised intracranial pressure
- D Bilateral optic neuritis of demyelinating origin
- E Buried optic disc drusen with pseudo-oedema

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the initial workup?

TREATMENT • SELECT 3

- A Dispense base-out prism for the distance diplopia
- B MRI of the brain with venography to exclude venous thrombosis
- C Lumbar puncture with an opening pressure, after imaging
- D Stop the doxycycline and review the medication list
- E Start an oral corticosteroid taper for the disc oedema
- F Order carotid duplex ultrasound of both sides

ANSWER _____ CONFIDENCE L • M • H

3 Which mechanism explains the abduction deficit?

BASIC SCIENCE

- A Stretch of the sixth nerve over the petrous ridge
- B Infarction of the sixth nerve nucleus in the dorsal pons
- C Restriction of the left medial rectus by an infiltrative myopathy
- D Compression of the nerve at the orbital apex by swollen muscle
- E Demyelinating plaque in the paramedian pontine reticular formation

ANSWER _____ CONFIDENCE L • M • H

4 Which change would move care to urgent surgical decompression?

TREATMENT

- A Weight unchanged after three months of dietary advice
- B Acuity falling to 20/60 with the field closing in ten days
- C Headache still present after two weeks of acetazolamide
- D Discs unchanged at six weeks with stable fields and acuity
- E Pulsatile tinnitus louder while acuity stays at 20/20

ANSWER _____ CONFIDENCE L • M • H

5 Which finding would indicate buried disc drusen rather than true oedema?

DIAGNOSIS

- A Peripapillary haemorrhage at the inferior disc margin
- B Transient greyouts of vision lasting a few seconds
- C Lumpy margin with autofluorescence and clear crossing vessels
- D Absent spontaneous venous pulsation on careful observation
- E Small crowded disc with no physiological cup and blurred margins

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-22 • why

1 • C Papilloedema, raised intracranial pressure

Positional headache, pulsatile tinnitus, obscurations and bilateral oedema with no venous pulsation are pressure signs. Preserved acuity with a sixth nerve deficit fits pressure, not neuritis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Bilateral optic neuritis	A young woman with swollen discs and headache is the neuritis stereotype.	Acuity is 20/20 with no RAPD and no pain on eye movement.
A	Malignant hypertension	It swells both discs and is the emergency nobody wants to miss.	BP is 128/78, with no arteriolar narrowing and no exudates.

2 • B, C, D MRI with venography, LP after imaging, stop the drug

Pressure has to be measured and secondary causes excluded before it is called idiopathic. Venous sinus thrombosis is the dangerous mimic, and tetracyclines are a recognised trigger.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Steroid taper	Steroid does lower pressure briefly and the discs look inflamed.	It masks the picture, rebounds on withdrawal, and the weight gain worsens the disease.
A	Prism for diplopia	The diplopia is real and prism is the optometric reflex.	The sixth nerve deficit resolves with the pressure, so prism chases a moving target.

3 • A Sixth nerve stretch from raised pressure

A raised pressure gradient displaces the brainstem downward and tethers the sixth nerve where it crosses the petrous ridge. That makes it a false localising sign rather than a second disease.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Sixth nerve nucleus	A nuclear lesion is the neat anatomical answer for an abduction deficit.	Nuclear lesions give a gaze palsy, and there is no gaze or facial sign here.
C	Medial rectus restriction	Thyroid myopathy limits abduction and is common in young women.	Nothing suggests restriction, and swollen discs with obscurations point intracranially.

4 • B Falling acuity with progressive field loss

Function decides. Loss of acuity and field over days is fulminant disease and needs shunting or optic nerve sheath fenestration, whereas headache and tinnitus alone do not.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Headache at two weeks	The symptom that brought her in has not improved, so treatment looks failed.	Headache lags and has other causes, so it is not the surgical trigger.
D	Discs unchanged	Persistent oedema on examination reads as uncontrolled pressure.	Oedema resolves slowly. Stable fields and acuity mean the nerve is coping.

5 • C Autofluorescent lumpy margin, vessels clear

Drusen elevate the disc without swelling the nerve fibre layer, so crossing vessels stay visible and the calcified bodies autofluoresce. Absent venous pulsation and a small cup occur in both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	No venous pulsation	Its absence is taught as the sign of raised pressure.	It is absent in a fifth of normal eyes, so it cannot separate the two.
A	Peripapillary bleed	Haemorrhage looks like the active and dangerous option.	Drusen bleed too, from the same crowded disc margin.

IF-THEN RULES FROM THIS CASE

If both discs are swollen with normal acuity, measure the pressure and image the venous sinuses before calling it idiopathic. If acuity or field falls over days, refer for decompression.

CASE S1-23 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

The lid that dropped with pain behind the eye

A 58-year-old man has six hours of a drooping right lid, vertical and horizontal double vision, and severe pain behind the right eye. Smoker, hypertensive. Right ptosis 5 mm. The right eye rests down and out, with limited elevation, depression and adduction, and full abduction. Right pupil 6 mm, sluggish to light; left 3.5 mm, brisk. No RAPD. Deficit unchanged after 60 seconds of sustained upgaze. BP 168/96, glucose 6.1.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Pupil-involving third nerve palsy from an aneurysm
- B Microvascular ischaemic third nerve palsy sparing the pupil
- C Ocular myasthenia gravis with variable ptosis and diplopia
- D Cavernous sinus lesion involving the third and fourth nerves
- E Giant cell arteritis presenting as an ocular motor palsy

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A ESR, CRP and high-dose steroid started today
- B Fresnel prism now with review in two weeks
- C Same-day CT angiography with neurosurgical referral
- D MRI of the brain booked within the next week
- E Observation for four weeks for spontaneous recovery

ANSWER _____ CONFIDENCE L • M • H

3 Which three findings localise this to the nerve rather than muscle or junction?

DIAGNOSIS • SELECT 3

- A Down and out rest position with full abduction
- B Deficit constant through 60 seconds of upgaze
- C Ptosis of 5 mm on the affected side
- D Vertical and horizontal double vision
- E Severe pain behind the affected eye
- F Dilated pupil of 6 mm, sluggish to light

ANSWER _____ CONFIDENCE L • M • H

4 Why is the pupil involved in a compressive lesion of this nerve?

BASIC SCIENCE

- A Ischaemia of the vasa nervorum strikes the outer fibres first
- B Sympathetic fibres from the carotid plexus join the nerve trunk
- C Pupillomotor fibres run superficially and are compressed first
- D The ciliary ganglion is carried inside the nerve sheath here
- E Parasympathetic axons are the most metabolically active fibres

ANSWER _____ CONFIDENCE L • M • H

5 Angiography is negative. Which plan is most appropriate now?

TREATMENT

- A Strabismus surgery once the deviation is measurable at six weeks
- B High-dose oral steroid for presumed inflammatory neuropathy
- C Botulinum toxin to the right lateral rectus this week
- D Vascular risk review, occlusion or prism, reassess at six weeks
- E Repeat vascular imaging every week until recovery begins

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-23 • why

1 • A Aneurysmal pupil-involving third nerve palsy

A complete external palsy with a 6 mm sluggish pupil and abrupt severe pain is compression until imaging proves otherwise. A posterior communicating artery aneurysm is what must be excluded today.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Microvascular palsy	He is 58 and hypertensive, which is the vasculopathic profile exactly.	The pupil is 6 mm and sluggish, and the glucose is 6.1.
C	Ocular myasthenia	Ptosis with diplopia and no visual loss is the myasthenic pattern.	The pupil is involved and the deficit does not fatigue on upgaze.

2 • C Same-day CT angiography, neurosurgery

A pupil-involving palsy with pain is a warning leak until proven otherwise, and the rupture risk is measured in hours. CT angiography is fast, available and sensitive for aneurysms of this size.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	MRI within a week	MRI is the better parenchymal study and it does get imaging done.	A week is longer than the interval in which these aneurysms rupture.
E	Observe four weeks	Microvascular palsies do recover in four to twelve weeks.	Observation belongs to pupil-sparing palsies, and only after imaging.

3 • A, B, F Pupil involvement, down and out, no fatigue

Myasthenia never touches the pupil and it fatigues, while myopathy spares the pupil and ignores nerve territory. A fixed down and out eye with a sluggish pupil is a nerve trunk lesion.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Ptosis 5 mm	Ptosis is the headline sign and it is severe here.	Levator weakness occurs in nerve, junction and muscle disease alike.
E	Pain behind the eye	Pain suggests compression, so it feels like a localiser.	Pain tells you the cause is urgent, not where the lesion sits.

4 • C Superficial pupillomotor fibres

The parasympathetic fibres travel on the dorsomedial surface of the nerve, fed by pial vessels. Extrinsic compression reaches them first, while microvascular disease infarcts the core.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Vasa nervorum ischaemia	It is the real mechanism of the vasculopathic palsy that spares the pupil.	Ischaemia hits the central fibres, which is why the pupil is spared.
D	Ciliary ganglion inside	The ganglion does sit close to the inferior division route.	It lies in the orbit behind the globe, not within the nerve trunk.

5 • D Risk review, prism or occlusion, six-week check

With imaging clear, a vasculopathic palsy is the working diagnosis and most recover by twelve weeks. Relief of diplopia is temporary while the angle changes, and the blood pressure is the treatable part.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Surgery at six weeks	The deviation is disabling and surgery is definitive.	Wait at least six months for a stable measurement; most of this recovers.
B	High-dose steroid	Steroid is the reflex when a palsy fails to recover quickly.	No inflammatory markers, no orbital signs, and it worsens his risk profile.

IF-THEN RULES FROM THIS CASE

If ptosis and an ocular motor deficit come with a dilated sluggish pupil, image the vessels the same day. If the pupil is spared in a vasculopath, imaging still comes before observation.

CASE S1-24 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

The smaller pupil that hides in the dark

A 52-year-old smoker has three weeks of a drooping right upper lid, aching right-sided neck pain and a hoarse voice, with a cough for two months. BVA 20/20 each. Right upper lid sits 2 mm lower, right lower lid 1 mm higher. In dim light pupils are 2.5 mm OD and 4.0 mm OS; in bright light 3.0 mm and 4.2 mm. Right pupil takes over five seconds to dilate. Motility full, no proptosis, no RAPD.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Physiological anisocoria with an involutional ptosis
- B Right tonic pupil with segmental iris palsy
- C Partial right third nerve palsy sparing motility
- D Pharmacological miosis from a topical agent
- E Right oculosympathetic paresis, preganglionic

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings place the lesion in the sympathetic pathway?

BASIC SCIENCE • SELECT 3

- A Right pupil takes over five seconds to dilate
- B Right lower lid sits 1 mm higher than the left
- C Right upper lid sits 2 mm lower than the left
- D Best corrected acuity of 20/20 in each eye
- E Full motility with no proptosis on either side
- F Anisocoria larger in dim light than in bright light

ANSWER _____ CONFIDENCE L • M • H

3 Which pharmacological test best confirms the diagnosis at this visit?

TREATMENT

- A Apraclonidine 0.5% to both eyes, looking for reversal of anisocoria
- B Pilocarpine 0.125% to both eyes, looking for a supersensitive response
- C Phenylephrine 2.5% to both eyes, comparing the dilation achieved
- D Hydroxyamphetamine 1% to both eyes, comparing dilation of the two
- E Tropicamide 1% to both eyes, comparing the speed of dilation

ANSWER _____ CONFIDENCE L • M • H

4 Which imaging plan is most appropriate?

TREATMENT

- A Chest radiograph now, with review in three months if normal
- B No imaging now, review in three months to see if the ptosis changes
- C CT of the neck and chest apex with carotid imaging, this week
- D MRI of the brain alone, arranged as an outpatient in one month
- E Carotid duplex ultrasound alone, arranged within one week

ANSWER _____ CONFIDENCE L • M • H

5 Which change would make carotid dissection the leading concern?

DIAGNOSIS

- A Dilation lag of eight seconds instead of five in the dark
- B Weight loss of 4 kg with a cough productive of blood
- C Abrupt onset today with severe neck pain and pulsatile tinnitus
- D Gradual onset over three weeks with the voice unchanged throughout
- E Ipsilateral facial anhidrosis with a normal chest radiograph

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-24 • why

1 • E Right Horner syndrome, preganglionic

Ptosis with a raised lower lid, a smaller pupil in the dark and dilation lag are sympathetic denervation. Neck pain, hoarseness and cough put the lesion on the preganglionic run through the chest.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Physiological anisocoria	A fifth of people have it, and lid laxity at 52 is ordinary.	Physiological anisocoria is equal in light and dark and has no dilation lag.
C	Partial third palsy	Ptosis plus anisocoria is the third nerve pattern.	The wrong pupil is small here, and motility is full.

2 • A, B, F Dark anisocoria, dilation lag, raised lower lid

Sympathetic denervation weakens the dilator and the Muller muscle, so the deficit shows in the dark, dilation lags and the lower lid rises. Upper lid ptosis on its own has many causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Upper lid 2 mm lower	It is the sign that brought him in and it is on the affected side.	Aponeurotic and mechanical ptosis do the same thing without pupil signs.
D, E	20/20, full motility	Normal findings feel reassuring and therefore relevant.	They exclude other disease but they localise nothing.

3 • A Apraclonidine 0.5%, reversal confirms

Denervation supersensitivity of the dilator lets a weak alpha agonist reverse the anisocoria: the small pupil dilates and the ptosis lifts. Hydroxyamphetamine localises, but only after confirmation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Hydroxyamphetamine	It is the classic Horner drop and it does separate the levels.	It answers where, not whether, and it cannot be used on the same day.
B	Pilocarpine 0.125%	Dilute pilocarpine is a supersensitivity test, so the logic feels right.	That test is for a tonic pupil, where the sphincter is denervated.

4 • C Neck and chest apex imaging with carotid study

The sympathetic chain runs from hypothalamus to orbit, and a hoarse voice with neck pain in a smoker points at the apex and the carotid. Imaging has to cover the whole preganglionic route.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Brain MRI alone	Central lesions do cause Horner syndrome and MRI is the neuro study.	The clues sit in the neck and chest, and a brain study misses both.
B	No imaging	Isolated Horner can be benign, and the signs are unchanged for weeks.	Hoarseness, neck pain and a smoking history make observation unsafe.

5 • C Abrupt onset with neck pain and tinnitus

Dissection presents with sudden neck or facial pain, Horner syndrome within hours and pulsatile tinnitus, and it can embolise within days. Hoarseness and weight loss point instead to the apex.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Weight loss, haemoptysis	It is the most alarming option and it does signal serious disease.	It points at an apical tumour, which is urgent but is not a dissection.
A	Longer dilation lag	A bigger number looks like a worse lesion.	Lag grades the denervation, not the cause or the level.

IF-THEN RULES FROM THIS CASE

If anisocoria is greater in the dark and the pupil dilates late, call it sympathetic and confirm with apraclonidine. If the voice or the neck is involved, image the chest apex.

CASE S2-11 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Three months into treatment the colours went flat

A 58-year-old man on isoniazid, rifampicin, pyrazinamide and ethambutol for 3 months reports 6 weeks of dimming vision and washed-out colour. Weight 52 kg, ethambutol 1200 mg daily, eGFR 42. BVA 20/60 OD, 20/70 OS. Pupils equal and brisk, no RAPD. Ishihara 2 of 14 plates each eye. Symmetric central scotomas on both fields. Discs flat with normal colour. IOP 15 both eyes.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Bilateral non-arteritic ischaemic optic neuropathy
- B Compressive lesion at the optic chiasm
- C Leber hereditary optic neuropathy
- D Isoniazid-related nutritional optic neuropathy
- E Ethambutol toxic optic neuropathy

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings place this at the optic nerves as a symmetric process?

DIAGNOSIS • SELECT 3

- A Dense red-green loss, 2 of 14 plates each eye
- B Central scotomas of matched size in both fields
- C No RAPD despite acuity of 20/60 or worse
- D Onset while on four-drug antituberculous therapy
- E Ethambutol at 23 mg/kg with an eGFR of 42
- F Best corrected acuity reduced in both eyes

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Add high-dose pyridoxine and continue current doses
- B Start oral prednisolone and repeat fields in 4 weeks
- C Contact the TB team today to stop ethambutol
- D Reduce ethambutol to 15 mg/kg and review in 3 months
- E Stop all four antituberculous drugs immediately

ANSWER _____ CONFIDENCE L • M • H

4 He asks what to expect. Which advice and follow-up plan is correct?

TREATMENT

- A Discharge now, since stopping the drug settles the matter
- B Hydroxocobalamin injections will restore the colour vision
- C Vision may fall for weeks after stopping, review at 6 weeks
- D Vision should return to normal within 2 weeks of stopping the drug
- E Restart ethambutol at a lower dose once acuity improves

ANSWER _____ CONFIDENCE L • M • H

5 Which mechanism best explains the pattern of loss?

BASIC SCIENCE

- A Infarction in the short posterior ciliary artery territory
- B Drug deposition in the retinal pigment epithelium
- C Excitotoxic loss of the magnocellular ganglion cells
- D Metal chelation impairing mitochondrial complex IV
- E Immune demyelination of the retrobulbar optic nerve

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-11 • why

1 • E Ethambutol toxic optic neuropathy

Symmetric painless loss over weeks, dense red-green failure and matched central scotomas, on 23 mg/kg with an eGFR of 42, is dose-related toxicity. Flat discs place it behind the globe.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Bilateral NAION	Age and vascular risk make ischaemia the reflex answer in an older man.	Ischaemia is sudden, usually sequential, and swells the disc it damages.
B	Chiasmal compression	Bilateral field loss with normal discs does point towards a chiasm.	Compression gives temporal loss and asymmetry, not matched central scotomas.

2 • A, B, C Colour loss, matched scotomas, no RAPD

Dyschromatopsia, symmetric central scotomas and an absent RAPD in eyes that both see 20/60 or worse localise to the optic nerves and say the two nerves are equally affected.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D, E	Drug and dose	They name the cause and are the reason the drug has to stop.	Exposure and dose explain why the nerves are failing, not where.
F	Acuity down in both eyes	It is the abnormality the patient actually came with.	Reduced acuity grades severity and localises nothing at all.

3 • C Stop ethambutol today, through the TB team

Toxicity is dose and duration dependent, and recovery depends on how early the drug stops. The TB service substitutes cover the same day so the regimen is not left short.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Dose reduction	15 mg/kg is the low-risk dose, so it reads as the safe compromise.	Risk is low at that dose in intact nerves. This nerve is already failing.
E	Stop all four drugs	It removes every possible culprit in one decision.	Only ethambutol needs to go. Losing cover invites resistant disease.

4 • C Warn of further decline, review at 6 weeks

Loss from ethambutol often progresses for 1 to 3 months after withdrawal, and recovery, when it comes, takes 6 months or longer. Early review records the floor and catches other causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Restart at a lower dose	The drug is useful and the dose was high, so a smaller dose feels safe.	Rechallenge after toxicity risks permanent loss. Substitute instead.
D	Normal within 2 weeks	Removing a cause usually stops a disease, so quick recovery sounds right.	Decline for weeks after stopping is typical, and recovery is slow.

5 • D Mitochondrial failure in papillomacular axons

Ethambutol chelates copper and zinc and impairs complex IV. The small, heavily loaded axons of the papillomacular bundle fail first, which produces central scotomas and red-green loss.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Ciliary artery infarction	It is the mechanism behind the ischaemic neuropathy on this list.	Infarction is abrupt, usually unilateral, and swells the disc.
C	Magnocellular loss	Naming a ganglion cell population sounds precise and mechanistic.	Toxicity hits small parvocellular axons, which is why colour goes first.

IF-THEN RULES FROM THIS CASE

If colour and central vision fall in both eyes on ethambutol, stop the drug the same day. If the loss is asymmetric or a disc swells, look for a second cause.

CASE S2-28 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

Two years of headache and a field she never noticed

A 38-year-old woman has 14 months of amenorrhoea with milky nipple discharge, and two years of dull headache. BVA 20/25 OD, 20/30 OS. Pupils equal and briskly reactive, with no afferent defect. Fields: OD has lost the upper and lower temporal quadrants, stopping at the vertical midline, and OS has lost the same temporal quadrants; the nasal half of each field is full. Mild temporal pallor of both discs.

1 Where must the lesion be?

DIAGNOSIS

- A Both optic nerves, in the orbit
- B The left occipital cortex, below the calcarine
- C The optic chiasm, at the crossing fibres
- D The left optic nerve, close to the chiasm
- E The right optic tract, behind the chiasm

ANSWER _____ CONFIDENCE L • M • H

2 Which finding would place the lesion in front of the chiasm instead?

DIAGNOSIS

- A Enlargement of both blind spots
- B An afferent pupillary defect on one side
- C Reduced red saturation in the worse eye only
- D Temporal pallor of both optic discs
- E Best acuity of 20/30 in the worse eye

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Fields repeated in three months to confirm the pattern
- B Routine neurology referral in six weeks
- C MRI of the pituitary and chiasm, with endocrine bloods
- D Urgent unenhanced CT of the head today
- E Prolactin measured alone, then a decision

ANSWER _____ CONFIDENCE L • M • H

4 Prolactin is 4200 mU/L on a macroadenoma. Which treatment comes first?

TREATMENT

- A Observation with six-monthly field tests
- B Hydrocortisone cover, then delayed surgery
- C A dopamine agonist, with the fields monitored
- D Transsphenoidal decompression within the week
- E Fractionated radiotherapy to the sella

ANSWER _____ CONFIDENCE L • M • H

5 Which two statements about the crossing fibres are correct?

BASIC SCIENCE • SELECT 2

- A Temporal retinal fibres cross to the other side
- B Macular fibres never reach the chiasm
- C Crossing fibres carry the nasal half of the field
- D Nasal retinal fibres cross in the chiasm
- E Inferonasal fibres cross low and anteriorly

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-28 • why

1 • C The optic chiasm

Loss of both temporal fields, stopping at the vertical midline, can only come from nasal retinal fibres, and they cross in one place. Amenorrhoea with galactorrhoea names what sits below.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Left optic nerve	One eye is a line worse, which invites a lesion on that side.	A nerve lesion takes the whole field of one eye, not two halves.
E	Right optic tract	A tract lesion is retrochiasmal and splits at the midline too.	That gives a left homonymous defect, not both temporal halves.

2 • B A one-sided afferent pupillary defect

Asymmetry of pupillary input means one optic nerve conducts worse than the other, which is prechiasmal. Colour, acuity and disc pallor all fall with chiasmal compression as well.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Bilateral temporal pallor	Pallor points at the optic nerve, so it reads as prechiasmal.	Chronic chiasmal compression atrophies crossing fibres in both.
C, E	Colour, acuity	Both measure afferent function and both are abnormal here.	Chiasmal compression reduces both. Neither localises in front.

3 • C MRI of the sella with endocrine bloods

MRI shows the mass, its extent and the chiasm, and hormones decide the treatment, so the two are ordered together. Compression that has grown for two years is imaged within days.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Unenhanced CT	It is quick, available, and excludes a large mass or a bleed.	It underestimates soft tissue at the sella and misses the chiasm.
A	Repeat the fields	Confirming an unfamiliar field defect feels careful.	The pattern is unambiguous, and delay costs crossing fibres.

4 • C A dopamine agonist first

A prolactin of 4200 mU/L makes this a prolactinoma, and these shrink on a dopamine agonist, often with fields recovering in weeks. Surgery is for failure or sudden visual collapse.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Surgery first	The chiasm is compressed, and decompression sounds mechanical.	A prolactinoma shrinks medically. Surgery is the second line.
A	Observation	Acuity is nearly normal, so watching looks proportionate.	Both temporal fields have gone already. It is progressing.

5 • D, E Nasal fibres cross, inferonasal ones lowest

Nasal retina, which sees the temporal field, crosses in the chiasm, and the inferonasal bundle crosses lowest and furthest forward. That is why superior temporal loss often comes first.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Temporal fibres cross	Half the fibres cross, so which half is easy to invert.	Temporal retina stays uncrossed. Nasal retina is what crosses.
C	Nasal half of the field	Nasal fibres and nasal field sound like the same thing.	Nasal retina sees the temporal field. Retina and field are flipped.

IF-THEN RULES FROM THIS CASE

If both temporal fields are lost at the vertical midline, localise to the chiasm and image within days. If one pupil shows an afferent defect, the disease is in front of it.

CASE S2-30 • OPTIC NERVE AND NEURO-OPHTHALMIC • 5 ITEMS

The same half of the world gone in both eyes

A 71-year-old man with atrial fibrillation, on no anticoagulant, noticed three days ago while gardening that he bumps into things on his left and misses the left of the page. Blood pressure 176/98. BVA 20/25 both. Fields: each eye has lost the whole left half, stopping at the vertical midline, with a spared island of about 4° around fixation. Pupils brisk, no afferent defect, discs normal.

1 Where must the lesion be?

DIAGNOSIS

- A The left occipital cortex, above the calcarine fissure
- B The right temporal lobe, in Meyer loop
- C The right parietal lobe, above the ventricle
- D The right occipital cortex, both calcarine banks
- E The right optic tract, just behind the chiasm

ANSWER _____ CONFIDENCE L • M • H

2 Which feature best separates a cortical lesion from an optic tract lesion?

DIAGNOSIS

- A Loss in each eye that respects the vertical midline
- B Normal pupil responses in both eyes
- C Sudden onset in atrial fibrillation
- D Normal optic discs three days after onset
- E Congruous halves with a spared central island

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management today?

TREATMENT

- A CT within a week, then a stroke clinic review
- B Aspirin started, with clinic review in a month
- C Field retraining and prisms arranged first
- D Routine neurology referral in six weeks
- E Same-day referral to a stroke service for imaging

ANSWER _____ CONFIDENCE L • M • H

4 Which three belong in his care over the next month?

TREATMENT • SELECT 3

- A Occlusion of one eye to settle confusion
- B Weekly field tests to track recovery
- C Steroid to reduce cortical swelling
- D Anticoagulation decided by the stroke team
- E Blood pressure brought to target
- F Low vision assessment with reading strategies

ANSWER _____ CONFIDENCE L • M • H

5 He asks about driving. What is the correct advice?

LAW AND ETHICS

- A He may drive by day with a passenger beside him
- B No restriction follows loss on one side
- C He must stop driving and notify the authority
- D He may drive once the field stops changing
- E He may drive while acuity stays at 20/25

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-30 • why

1 • D The right occipital cortex

The left half of both fields has gone at the vertical midline, so the lesion is retrochiasmal on the right. A complete, congruous defect with a spared central island is occipital.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Meyer loop	A temporal lesion is retrochiasmal and gives homonymous loss.	Meyer loop takes the upper quadrant, not a whole half field.
A	Left occipital cortex	Occipital is the right place, and sides are easy to invert.	The left cortex serves the right half of both fields.

2 • E Congruity with a spared central island

Fibres from the two eyes lie together in the cortex, so cortical defects match closely and often spare a central island. Tract lesions are incongruous and may give a pupil defect.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Respects the midline	The midline rule is the first thing taught about these fields.	Everything behind the chiasm respects it, the tract included.
B	Normal pupils	Normal pupils feel like evidence the lesion is far back.	Pupils are often normal in tract lesions too. It rarely helps.

3 • E Same-day stroke service referral

Three days from onset, in atrial fibrillation with no anticoagulation, he is at high risk of a second embolus. The pathway that investigates and prevents that starts at the stroke service.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Aspirin and a month	Antiplatelet therapy is part of secondary prevention.	Fibrillation needs anticoagulation, decided after imaging.
C	Prisms and retraining	Rehabilitation is what will help the field he has lost.	It treats the deficit and ignores the artery about to close.

4 • D, E, F Anticoagulation, blood pressure, low vision help

The month belongs to preventing the next infarct, so rhythm and a pressure of 176/98 are treated, while reading strategies restore what a left-sided field defect takes away.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Occlude one eye	Occlusion helps double vision, so it sounds like a comfort measure.	Both eyes lose the same half. Covering one removes nothing.
B, C	Weekly fields, steroid	One tracks recovery, the other treats swelling around an infarct.	Fields are repeated at months, and steroid has no role here.

5 • C Stop driving, notify the authority

A homonymous hemianopia fails the visual field standard whatever the acuity. He stops driving now, the duty to declare is his, and any return is decided on a formal field test.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Acuity is enough	The acuity standard is the one most drivers know about.	Field and acuity are separate standards. This one fails on field.
D	Wait until it is stable	Some cortical field loss does recover in the first months.	Recovery is judged by the authority, not by how he feels.

IF-THEN RULES FROM THIS CASE

If both eyes lose the same half of the field at the vertical midline, the lesion is behind the chiasm and it is a stroke referral. Congruity with a spared centre means occipital.

CASE S1-34 • PERCEPTUAL FUNCTION AND COLOUR VISION • 5 ITEMS

Both eyes, dimmer colours and harder print

A 58-year-old man in month four of treatment for pulmonary tuberculosis reports six weeks of dimmer colours and harder print. He weighs 60 kg and takes ethambutol 1500 mg daily with rifampicin, isoniazid and pyrazinamide; eGFR is 42. Best corrected acuity 20/40 OD and 20/50 OS. Ishihara 2 of 17 plates each eye. No RAPD. Discs and maculae look normal. Fields show symmetric central depression.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Ethambutol toxic optic neuropathy
- B Bilateral simultaneous demyelinating optic neuritis
- C Nutritional optic neuropathy of B12 deficiency
- D Autosomal dominant optic atrophy presenting late
- E Occult macular dystrophy with central loss

ANSWER _____ CONFIDENCE L • M • H

2 By Köllner's rule, which colour loss pairs with disease at this site?

BASIC SCIENCE

- A Red-green loss, because optic nerve disease affects it first
- B Blue-yellow loss, because outer retinal disease affects it first
- C Blue-yellow loss, because the S cones are the sparsest type
- D Total loss of hue, because all three cone classes fail together
- E Red-green loss, because the L and M genes lie on one chromosome

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Reduce ethambutol to 15 mg/kg and reassess in three months
- B Start high-dose oral steroid and repeat the fields in a month
- C Start hydroxocobalamin and a B-complex, and review in six weeks
- D Repeat colour vision in four weeks before contacting anyone
- E Speak to the treating physician today to stop ethambutol

ANSWER _____ CONFIDENCE L • M • H

4 Which two actions belong in the plan once the drug is stopped?

TREATMENT • SELECT 2

- A Genetic testing for OPA1 before accepting the diagnosis
- B Monthly full-field electroretinography until recovery
- C A course of oral hydroxocobalamin while the nerve recovers
- D Written communication to the prescriber with the dose per kg and eGFR
- E Recorded acuity, colour and threshold fields now and at six weeks

ANSWER _____ CONFIDENCE L • M • H

5 Which further finding would move the diagnosis away from drug toxicity?

DIAGNOSIS

- A Temporal disc pallor appearing at three months
- B Ishihara errors on every plate in both eyes
- C A further fall in eGFR from 42 to 35
- D A dense RAPD with pain on movement of one eye
- E Bitemporal wedge defects on threshold fields

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-34 • why

1 • A Ethambutol toxic optic neuropathy

1500 mg for 60 kg is 25 mg/kg, and an eGFR of 42 slows clearance. Symmetric central depression, colour loss out of proportion to acuity, and no RAPD fit a dose-related toxic insult.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	B12 deficiency	It produces the same painless, symmetric, central colour loss.	The dose here is 25 mg/kg with reduced clearance, which explains it.
B	Optic neuritis	Acquired colour loss with reduced acuity is the classic teaching set.	Neuritis is painful, usually unilateral, and gives an RAPD.

2 • A Red-green loss, from optic nerve disease

Köllner's rule pairs outer retinal disease with blue-yellow loss and optic nerve disease with red-green loss. Glaucoma and dominant optic atrophy are the standing exceptions.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Blue-yellow, outer retina	It is the other half of the rule, stated correctly.	The maculae are normal and the field loss is central and symmetric.
E	Red-green, gene linkage	It reaches the right answer through a real piece of genetics.	Gene arrangement explains inherited defects, not acquired ones.

3 • E Stop ethambutol today, through the prescriber

Toxicity is dose and duration dependent, and recovery turns on how fast the drug goes. Stopping is the physician's decision, so the call happens today, with the dose per kg and the eGFR.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Reduce to 15 mg/kg	Toxicity is dose related, so a lower dose looks like the fix.	No dose is safe once the nerve is affected, and three months is too long.
D	Recheck in four weeks	Confirming progression before acting feels careful.	Four more weeks of the drug is four more weeks of loss.

4 • D, E Write to the prescriber, and fix a measured baseline

The two jobs are to close the loop with the prescriber in writing and to fix a measurable baseline. Recovery is judged against recorded acuity, colour and fields, not against recall.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Hydroxocobalamin	Vitamin repletion is standard in nutritional optic neuropathy.	This is a drug effect. There is no deficiency here to replace.
B	Monthly ERG	It sounds like objective monitoring of the visual pathway.	The ERG measures retina. The lesion here is in the optic nerve.

5 • D Unilateral RAPD with pain on eye movement

Toxic neuropathy is symmetric and painless, so an RAPD means the two nerves are unequally affected and pain points at inflammation. Pallor, dense colour loss and worse renal function all fit toxicity.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Temporal disc pallor	A new abnormal sign after months reads like a different disease.	Pallor is the late appearance of the toxic insult already described.
E	Bitemporal wedges	Bitemporal loss usually means compression at the chiasm.	Ethambutol has a known chiasmal pattern with bitemporal defects.

IF-THEN RULES FROM THIS CASE

If colour and central field fall symmetrically with no RAPD, check every drug against weight and renal function. If a dose exceeds 15 mg/kg, the drug is the diagnosis until it is stopped.

CASE S1-10 • RETINA, CHOROID, VITREOUS • 5 ITEMS

Door frames that started to bend six weeks ago

A 78-year-old woman reports six weeks of distortion OD, with door frames bowed and print she managed in June now hard to read. She smokes 15 cigarettes a day. BVA was 20/25 OD in June and is 20/60 today. Amsler shows central metamorphopsia OD. Fundus OD: soft confluent drusen, a grey-green subretinal lesion and a small subretinal haemorrhage temporal to fixation. OS drusen only, 20/25.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Chronic central serous chorioretinopathy
- B Retinal macroaneurysm with macular exudation
- C Vitreomacular traction with a lamellar defect
- D Geographic atrophy involving the foveal centre
- E Neovascular age-related macular degeneration

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A Refer for focal laser photocoagulation of the lesion
- B Refer for macular OCT and anti-VEGF within one week
- C Refer for macular OCT within one week, then observe
- D Routine referral for assessment in six to eight weeks
- E Start high-dose antioxidant supplements and review

ANSWER _____ CONFIDENCE L • M • H

3 Which three belong in her care alongside the injections?

TREATMENT • SELECT 3

- A Smoking cessation support offered at this visit
- B Amsler self-monitoring of the fellow eye
- C Low vision assessment and reading aids
- D A supplement containing beta-carotene
- E Surrender of her driving licence this week
- F Laser prophylaxis to drusen in the fellow eye

ANSWER _____ CONFIDENCE L • M • H

4 After three monthly injections OD is 20/40 with a little persistent subretinal fluid. What now?

TREATMENT

- A Stop injections and monitor with monthly OCT
- B Switch to a different anti-VEGF agent today
- C Add photodynamic therapy to the next injection
- D Extend to four-monthly injections immediately
- E Continue injections and set the treatment interval

ANSWER _____ CONFIDENCE L • M • H

5 Which process most directly produces the leakage in this eye?

BASIC SCIENCE

- A VEGF driven vessel growth through Bruch membrane
- B Lipofuscin accumulation in the pigment epithelium
- C Choriocapillaris dropout with photoreceptor loss
- D Vitreous traction on the internal limiting membrane
- E Complement activation forming drusen deposits

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-10 • why

1 • E Neovascular AMD, right eye

Six weeks of metamorphopsia with acuity falling from 20/25 to 20/60, a grey-green subretinal lesion, subretinal blood and confluent soft drusen is choroidal neovascularisation on an AMD background.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Geographic atrophy	Same disease and same drusen, and it also destroys reading vision.	Atrophy is slow, grey-white with visible choroidal vessels, and dry.
A	Chronic CSC	Subretinal fluid with distortion belongs to its picture too.	Her age, the drusen and the subretinal blood do not fit CSC.

2 • B Urgent referral for OCT and anti-VEGF

Vision lost to active neovascularisation returns only if injections start within days to weeks. Imaging and treatment are arranged on one pathway rather than one after the other.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Routine referral	Six weeks feels prompt measured against usual waiting times.	Fibrosis is permanent. Each week of delay costs recoverable acuity.
E	Antioxidant supplements	Supplements do have trial evidence in age-related disease.	They lower risk in the fellow eye. They do not treat active disease.

3 • A, B, C Stop smoking, Amsler the fellow eye, low vision aids

Smoking is her one modifiable risk, the fellow eye needs a home test that catches conversion early, and reading aids restore function while treatment runs.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Beta-carotene formula	It resembles the supplement trial she may have read about.	Beta-carotene raises lung cancer risk in smokers. Use the later formula.
E	Surrender the licence	Central loss in one eye reads like the end of driving.	OS is 20/25. She is assessed against the standard, not disqualified.

4 • E Continue injections, then titrate the interval

Acuity improved and fluid is reducing, which is a response. Treatment continues while the interval is titrated to the eye, because stopping after loading invites recurrence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stop and monitor	Three injections is the loading course, so it sounds complete.	Loading starts therapy. Fluid remains, so injections continue.
B	Switch the agent	Persistent fluid reads as failure of the drug in use.	This eye improved. Switching is for non-response, not partial response.

5 • A VEGF driven growth through Bruch membrane

Hypoxic, inflamed outer retina raises VEGF, which drives choroidal vessels through breaks in Bruch membrane. Those new vessels are permeable, so fluid and blood collect under the retina.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Complement and drusen	Complement is central to the disease and drusen are visible here.	It explains the background, not the leaking vessel in front of you.
B	Lipofuscin build-up	It underlies the pigment epithelial ageing that starts all this.	Lipofuscin drives atrophy over years, not acute subretinal fluid.

IF-THEN RULES FROM THIS CASE

If distortion is new and acuity has fallen with a grey-green lesion or subretinal blood, refer for injection within days. If fluid persists while vision improves, continue and titrate.

CASE S1-11 • RETINA, CHOROID, VITREOUS • 5 ITEMS

Reading vision that faded over two months

A 52-year-old man with type 2 diabetes of 18 years, HbA1c 9.8%, reports two months of blur in the right eye. BVA 20/60 OD, 20/25 OS. OD: flat new vessel fronds at the disc and on the superior arcade, faint vitreous haze, dot-blot haemorrhages in four quadrants, venous beading. OCT OD shows cystoid spaces through the fovea, central subfield 445 μm . IOP 16 both eyes, angles open, iris without vessels.

1 Which classification best fits the right eye?

DIAGNOSIS

- A Moderate non-proliferative retinopathy, oedema off the centre
- B Proliferative retinopathy with vitreous haemorrhage, no oedema
- C Ocular ischaemic syndrome on a background of retinopathy
- D Severe non-proliferative retinopathy with centre-involving oedema
- E Proliferative retinopathy with centre-involving macular oedema

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate initial treatment of the right eye?

TREATMENT

- A Vitrectomy with endolaser and anti-VEGF given at the end of surgery
- B Panretinal photocoagulation at this visit, macular review in three months
- C Intravitreal anti-VEGF now, panretinal laser once the centre is drier
- D Focal macular laser to the leaking points, then panretinal laser
- E Dexamethasone implant now, with panretinal laser deferred six months

ANSWER _____ CONFIDENCE L • M • H

3 Which three steps belong in the plan alongside intravitreal treatment?

TREATMENT • SELECT 3

- A OCT of both maculae at each treatment visit to guide retreatment
- B Dilated assessment of the left eye for proliferation and its own need
- C Stopping aspirin because the vitreous is already showing some blood
- D Deferring the next dilated examination for six months of injections
- E Advising that all reading be done with a stronger near addition
- F Same-week diabetes team review of glycaemic control and blood pressure

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism most directly explains the thickening at the fovea?

BASIC SCIENCE

- A Breakdown of the inner blood-retinal barrier at capillary endothelium
- B Failure of the pigment epithelial pump at the outer blood-retinal barrier
- C Traction on the fovea by a taut thickened posterior hyaloid membrane
- D Loss of choriocapillaris perfusion beneath the central macula
- E Osmotic swelling of Muller cells driven by hyperglycaemia alone

ANSWER _____ CONFIDENCE L • M • H

5 After three monthly injections the central subfield is 402 μm and acuity 20/60. What next?

TREATMENT

- A Keep monthly injections and add a dexamethasone implant now
- B Stop injections and complete panretinal photocoagulation instead
- C Refer for vitrectomy with peeling of the internal limiting membrane
- D Continue the same agent monthly for three further doses
- E Switch to a different anti-VEGF agent and reassess after three doses

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-11 • why

1 • E PDR with centre-involving oedema

Fronds at the disc and on the arcade are new vessels, which places the eye past non-proliferative disease. Cystoid thickening through the fovea at 445 µm makes the oedema centre-involving.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Severe NPDR	Haemorrhage in four quadrants and venous beading do meet the severe count.	New vessels are present, and their presence alone ends the non-proliferative grade.
B	PDR without oedema	The fronds and the vitreous haze dominate, so the macula gets read second.	Cystoid spaces through the fovea at 445 µm are oedema at the centre.

2 • C Anti-VEGF first, panretinal laser after

Anti-VEGF answers both problems at once: new vessels regress and the fovea dries. Panretinal laser applied over a thick centre worsens acuity first, so it follows rather than leads.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Panretinal laser first	Laser is the definitive treatment for proliferation and this eye needs it.	Scatter laser over centre-involving oedema drives thickness and acuity the wrong way.
A	Vitrectomy	Vitreous haze reads as bleeding, and surgery clears the media in one step.	The haze is faint and the view supports laser and injection, so surgery waits.

3 • A, B, F Systemic review, serial OCT, fellow eye

An HbA1c of 9.8% and an untreated fellow eye drive the eye disease as hard as any injection does. Retreatment is decided on measured thickness, so OCT is repeated at each visit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Stopping aspirin	Antiplatelet therapy with blood in the vitreous feels like an obvious risk.	Aspirin does not worsen diabetic haemorrhage, and stopping it costs cardiac cover.
D	Six-month dilated review	Monthly injection visits give the impression the eye is being watched.	Injection visits do not examine the periphery, where proliferation progresses.

4 • A Inner blood-retinal barrier breakdown

Pericyte loss and damaged endothelial tight junctions let plasma into the retina, which collects in the outer plexiform and inner nuclear layers as the cystoid spaces seen here.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Outer barrier pump failure	The pigment epithelium does clear subretinal fluid, so its failure floods the macula.	That mechanism gives serous elevation, not intraretinal cysts.
C	Taut posterior hyaloid	A real diabetic pattern, and traction genuinely thickens the fovea.	No hyaloid deformation of the fovea is described, only cystoid spaces.

5 • E Switch anti-VEGF agent, then reassess

Thickness fell under 10% across a full loading course and acuity has not moved, which defines a poor response rather than a slow one. Changing agent is the next step before changing class.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Same agent, three more	Some eyes are slow responders, and persistence is often rewarded.	A 43 µm fall over three doses is non-response, not slow response.
C	Vitrectomy with peel	Surgery is the recognised answer to oedema that will not dry.	Peeling is for tractional oedema, and no traction is present here.

IF-THEN RULES FROM THIS CASE

If new vessels and centre-involving oedema sit together, dry the centre with anti-VEGF before laying panretinal laser.
If thickness barely moves across loading, change the drug.

CASE S1-12 • RETINA, CHOROID, VITREOUS • 5 ITEMS

Vision clouded over in one eye on waking

A 68-year-old man noticed painless blur in the left eye on waking three days ago. Treated hypertension, home readings 158/96. No diabetes. BVA 20/25 OD, 20/200 OS. Trace RAPD OS. IOP 15 both eyes, angles open, iris without vessels. OS: veins dilated and tortuous in four quadrants, flame and dot haemorrhages to the periphery, several cotton wool spots, blurred hyperaemic disc, central subfield 620 μm .

1 What is the most likely diagnosis?

DIAGNOSIS

- A Undiagnosed severe non-proliferative diabetic retinopathy
- B Central retinal vein occlusion with macular oedema
- C Hemiretinal vein occlusion with macular oedema
- D Ocular ischaemic syndrome from carotid disease
- E Papilloedema with bilateral venous stasis

ANSWER _____ CONFIDENCE L • M • H

2 Which three findings raise the concern that this occlusion is ischaemic?

DIAGNOSIS • SELECT 3

- A Best corrected acuity of 20/200 in the affected eye
- B Several cotton wool spots at the posterior pole
- C Veins dilated and tortuous in all four quadrants
- D Central subfield thickness measured at 620 μm
- E Blurred disc margin with a hyperaemic disc
- F Relative afferent pupillary defect in the affected eye

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management of the macular oedema?

TREATMENT

- A Observation for three months, since many occlusions settle alone
- B Intravitreal dexamethasone implant as the first-line agent
- C Oral acetazolamide with aspirin and review at six weeks
- D Macular grid laser to the thickened area, review at three months
- E Intravitreal anti-VEGF now, with monthly acuity and OCT review

ANSWER _____ CONFIDENCE L • M • H

4 Which systemic step matters most at this visit?

TREATMENT

- A Erythrocyte sedimentation rate and C-reactive protein today
- B Same-week review of blood pressure and glucose in general practice
- C Carotid duplex within the week to exclude a stenosis on the affected side
- D Thrombophilia screen, given the inherited risk of thrombosis
- E Immediate stroke unit referral for antiplatelet loading today

ANSWER _____ CONFIDENCE L • M • H

5 Where does the obstruction sit anatomically?

BASIC SCIENCE

- A In the superior ophthalmic vein behind the apex, draining intracranially
- B At the choriocapillaris, where venules collect into the vortex veins
- C In the short posterior ciliary vessels around the optic nerve head
- D At an arteriovenous crossing under a stiffened arteriole, superotemporally
- E In the vein at the lamina cribrosa, sharing a sheath with the artery

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-12 • why

1 • B Central retinal vein occlusion

Haemorrhage and venous dilatation occupy all four quadrants of one eye, which places the obstruction at the central trunk. Unilateral disease with a normal fellow eye excludes the systemic patterns.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Hemiretinal occlusion	It produces the same signs and is easy to accept without mapping the quadrants.	Hemiretinal disease respects the horizontal midline. All four quadrants are involved.
A	Severe NPDR	Dot haemorrhages and cotton wool spots are the diabetic vocabulary.	Onset overnight in one eye only, and the fellow eye carries no retinopathy.

2 • A, B, F RAPD, 20/200, cotton wool spots

Perfusion is judged on afferent function and capillary closure, not on how much blood or fluid is visible. An RAPD, acuity at 20/200 and cotton wool spots each point at closure.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Thickness 620 µm	It is the most abnormal number on the page and it does explain the blur.	Thickness measures leakage. A very thick macula can sit on a perfused bed.
C, E	Tortuous veins, full disc	Both look dramatic and both belong to the occlusion.	Both appear in perfused occlusion as well, so neither separates the two.

3 • E Anti-VEGF now, monthly review

Anti-VEGF is first line for oedema after venous occlusion and it also suppresses the drive to anterior segment new vessels. Monthly review catches both non-response and rubeosis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Observation	Spontaneous improvement is real, and the eye may recover without needles.	Acuity is 20/200 with 620 µm of thickening. Waiting loses recoverable vision.
D	Macular grid laser	Grid laser is the older standard and it sounds definitive for leakage.	Grid laser gave no acuity benefit in central occlusion and is not used for it.

4 • B Same-week blood pressure and glucose

Home readings of 158/96 name the modifiable driver, and undetected diabetes is the other. Treating them protects the fellow eye, which is the outcome that matters most here.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Carotid duplex	It is the right investigation when the retinal problem is arterial.	Carotid disease causes arterial events, not venous thrombosis at the disc.
D	Thrombophilia screen	Thrombosis invites a clotting workup and it feels thorough.	At 68 with hypertension the yield is negligible. Reserve it for young patients.

5 • E Vein at the lamina cribrosa

Artery and vein cross the lamina cribrosa inside one adventitial sheath, so arteriolar stiffening compresses the vein there and the resulting turbulence seeds thrombus.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Arteriovenous crossing	It is the correct site for the branch occlusion the signs otherwise resemble.	A crossing lesion drains one sector. Four quadrants means the central trunk.
A	Superior ophthalmic vein	It is the next vessel downstream, so obstruction there sounds plausible.	Orbital venous obstruction gives proptosis and dilated lid veins, absent here.

IF-THEN RULES FROM THIS CASE

If haemorrhage and dilated veins cross all four quadrants, the lesion is at the lamina cribrosa. If acuity, pupil or cotton wool spots are bad, treat the eye as ischaemic and watch the angle.

CASE S1-13 • RETINA, CHOROID, VITREOUS • 5 ITEMS

Flashes on Friday, and spots ever since

A 58-year-old woman, -5.50 D in both eyes, has four days of new floaters and arcs of light in the right eye, worse in dim rooms. No shadow in the field. BVA 20/20 both eyes, IOP 14. OD: pigment granules in the anterior vitreous, a Weiss ring, no cells, settled blood inferiorly. With indentation, a horseshoe tear at 11 o'clock with a vitreous strand on the flap, retina flat, no subretinal fluid. OS: inferotemporal lattice, no breaks.

1 Which feature carries the highest risk of progression to detachment?

DIAGNOSIS

- A Axial myopia of -5.50 D in both eyes
- B Pigment granules dispersed in the anterior vitreous
- C A horseshoe tear with traction on its flap
- D Settled vitreous blood inferior to the break
- E Lattice degeneration in the inferotemporal fellow eye

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate management of the right eye?

TREATMENT

- A Vitrectomy with endolaser and a gas tamponade
- B Laser retinopexy surrounding the tear within 24 hours
- C Cryotherapy to the tear and laser to the fellow lattice
- D Review in two weeks with a drawing of the periphery
- E Scleral buckle to relieve the traction on the flap

ANSWER _____ CONFIDENCE L • M • H

3 Which two symptoms after retinopexy require same-day return?

TREATMENT • SELECT 2

- A A shadow spreading from the periphery towards the centre
- B A sudden fresh increase in floaters with new flashes
- C Mild ache around the brow for two days after laser
- D Faint glare in bright sunlight lasting about a week
- E Slight morning blur that clears after some blinking

ANSWER _____ CONFIDENCE L • M • H

4 Why do tears of this kind form where they do?

BASIC SCIENCE

- A Vitreous collagen inserts firmly at the posterior vitreous base, and separation tears there
- B The internal limiting membrane is thinnest at the equator, so it splits first under traction
- C Pigment epithelial adhesion is weakest superiorly, so the retina lifts there first
- D Choroidal vessels tether the retina at the equator and resist vitreous movement
- E The subretinal space is widest at the equator, so fluid enters most easily there

ANSWER _____ CONFIDENCE L • M • H

5 What does the standard of care require of the practitioner at this visit?

LAW AND ETHICS

- A A dilated record made now, with the patient asked to telephone in a week
- B Referral held until the vitreous blood clears enough to grade the break
- C A referral letter posted today, with review arranged in one month
- D Same-day documented referral, the break located, warning symptoms in writing
- E Advice to attend an emergency department if the vision changes

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-13 • why

1 • C Horseshoe tear under traction

A flap tear held open by a vitreous strand keeps pulling with every eye movement, so fluid can enter continuously. That combination carries the highest conversion rate of any break.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Fellow eye lattice	Lattice is the best known predisposing lesion and it is genuinely present.	Lattice without breaks in a symptom-free eye is low risk and needs no treatment.
B, D	Pigment and blood	Both are the classic markers that a break exists somewhere.	They point to the tear. They do not measure the traction that opens it.

2 • B Laser retinopexy within 24 hours

A symptomatic flap tear with the retina still flat is sealed by surrounding retinopexy, which is quick, safe in clinic and highly effective. Delay is what turns the break into a detachment.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Review in two weeks	Vision is 20/20 with no field loss, so the eye looks stable enough to watch.	Acuity stays normal until the fovea goes. Traction is active now.
A	Vitrectomy with gas	Surgery removes the traction itself rather than walling off the break.	The retina is attached. Intraocular surgery adds risk that laser does not carry.

3 • A, B Spreading shadow, or new floaters and flashes

Retinopexy seals one break but does not stop the vitreous separating further. A spreading shadow means fluid under the retina, and a fresh shower of floaters means a new break.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Brow ache	Pain after any laser feels like the sign that something has gone wrong.	Brief ache follows the contact lens and the burns. It carries no retinal meaning.
D, E	Glare, morning blur	Both are new visual symptoms in an eye that has just been treated.	Neither maps to retinal separation. Both settle without any intervention.

4 • A Firm adhesion at the posterior vitreous base

Syncretic vitreous collapses forward and pulls hardest on its firmest attachment, the posterior margin of the vitreous base. Traction concentrated at that line tears full thickness retina.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Thin limiting membrane	Something must be weakest where the retina breaks, so thickness sounds right.	Break site follows where vitreous is attached, not where retina is thin.
C	Weak superior adhesion	Superior tears really are commoner, so a superior weakness sounds plausible.	Superior tears reflect gravity acting on the collapsing gel, not local adhesion.

5 • D Same-day referral, located and documented

A symptomatic flap tear is treated within days, so the referral is made the same day. The record carries the location and the written symptoms that oblige the patient to return at once.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Posted letter, one month	It refers and it documents, which covers the two obvious duties.	Postal timing does not match a break that can detach within days.
B	Wait for the blood	A clearer view would let the whole periphery be graded properly.	The break is already seen and the retina is flat. Waiting only loses the window.

IF-THEN RULES FROM THIS CASE

If floaters and flashes are new, indent and search the whole periphery. If a flap tear is found with the retina still flat, seal it within a day rather than review it.

CASE S1-15 • RETINA, CHOROID, VITREOUS • 5 ITEMS

A grey smudge in the middle of the page

A 41-year-old man reports six weeks of a central grey patch in the right eye, with objects on that side looking smaller and further away. He finished a course of oral corticosteroid for back pain three weeks ago and works night shifts. BVA 20/40 OD, improving to 20/30 with +1.00, 20/20 OS. OD: shallow serous elevation at the macula, no blood, no drusen, fine pigment flecks. OCT: subfoveal fluid, choroid 470 μm .

1 What is the most likely diagnosis?

DIAGNOSIS

- A Pachychoroid neovascularopathy with a type 1 membrane
- B Acute central serous chorioretinopathy
- C Optic disc pit maculopathy with subretinal fluid
- D Inflammatory choroidal neovascular membrane
- E Cystoid macular oedema of uncertain cause

ANSWER _____ CONFIDENCE L • M • H

2 Which two features argue against a neovascular membrane?

DIAGNOSIS • SELECT 2

- A No drusen in either macula at the age of 41
- B Micropsia with distortion on the Amsler chart
- C A hyperopic shift of +1.00 in the affected eye
- D Choroidal thickness measured at 470 μm
- E Subretinal fluid with no hyper-reflective material

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management now?

TREATMENT

- A Focal laser to the leakage point identified on angiography
- B Observation to three months, with steroid avoidance and sleep advice
- C Half-dose photodynamic therapy to the leaking area within two weeks
- D Intravitreal anti-VEGF monthly for three doses, then reassessment
- E Oral eplerenone for three months with electrolyte monitoring

ANSWER _____ CONFIDENCE L • M • H

4 Four months on, fluid persists and acuity has fallen to 20/50. What next?

TREATMENT

- A Monthly intravitreal anti-VEGF for three doses
- B Oral eplerenone with two-weekly electrolytes
- C Full-fluence photodynamic therapy to the leak
- D Half-dose photodynamic therapy guided by angiography
- E Continued observation for a further three months

ANSWER _____ CONFIDENCE L • M • H

5 Which structure is the primary site of the disease?

BASIC SCIENCE

- A Bruch membrane, through defects of which new vessels grow forward
- B Outer plexiform layer, where Muller cells swell and form cysts
- C Vitreoretinal interface, where traction lifts the photoreceptors
- D Choroid, whose hyperpermeable vessels overwhelm the pigment epithelium
- E Retinal capillary endothelium, with inner barrier breakdown at the fovea

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-15 • why

1 • B Acute central serous chorioretinopathy

A shallow serous detachment with a thickened choroid, micropsia and a hyperopic shift in a man of 41 weeks after corticosteroid exposure is the classic presentation. No blood and no drusen.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Pachychoroid neovascular	It shares the thick choroid and the same age group, and it also leaks fluid.	No subretinal hyper-reflective material and no haemorrhage on OCT or fundus.
E	Cystoid macular oedema	Central blur with distortion is the everyday description of macular oedema.	The fluid is under the retina, not in cystoid spaces within it.

2 • A, E No hyper-reflective material, no drusen

A membrane shows as hyper-reflective tissue above the pigment epithelium, usually with blood, and usually on a drusen-bearing macula. Their absence is what argues against it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Thick choroid	A thick choroid is the signature of serous disease, so it feels exclusive to it.	Pachychoroid neovascular membranes grow on exactly this thick choroid.
B, C	Micropsia, hyperopic shift	Both are textbook symptoms of the serous condition being considered.	Both follow from any fluid lifting the fovea, membrane or not.

3 • B Observe to three months, remove steroid

Most acute serous detachments resolve without treatment inside three months, and the recent corticosteroid course is a removable driver. Treatment is held for persistence or recurrence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Photodynamic therapy now	It is the most effective treatment, so offering it early looks like good care.	At six weeks the natural course is still resolution. Early treatment adds risk.
D	Anti-VEGF	Subretinal fluid at the macula is the situation anti-VEGF usually treats.	Without new vessels there is no target, and trials show no benefit.

4 • D Half-dose photodynamic therapy

Fluid past three to four months threatens the photoreceptors, and reduced-dose photodynamic therapy has the best evidence for resolving it while sparing the pigment epithelium.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Observe three months more	The condition is known for resolving on its own, so patience feels safe.	Past four months the outer retina thins irreversibly. The window is closing.
C	Full-fluence therapy	It uses the right modality and full dose sounds more likely to work.	Full fluence carries atrophy and ischaemia risk that the half dose avoids.

5 • D Hyperpermeable choroid, overwhelmed RPE

Choroidal congestion and hyperpermeability drive fluid at the pigment epithelium faster than its pump can clear it, so fluid collects under the retina. The measured choroidal thickness is the clue.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Inner barrier breakdown	It is the mechanism behind most macular fluid a candidate sees.	Inner barrier failure gives intraretinal cysts, not a smooth serous dome.
A	Bruch membrane defects	The pigment epithelium is involved, so its basement complex sounds central.	That is the route for neovascular disease, which this eye does not show.

IF-THEN RULES FROM THIS CASE

If a young adult has serous macular fluid, look for steroid exposure and a thick choroid, and observe to three months. If fluid persists past four months, treat rather than watch.

CASE S2-12 • RETINA, CHOROID, VITREOUS • 5 ITEMS

The white reflex in the holiday photograph

A 14-month-old girl is brought in after a family photograph showed a white reflex in the right eye. Six weeks of intermittent right esotropia. Born at term, no supplemental oxygen, no family history of eye disease. She fixes and follows with the left eye only. No red reflex OD, white pupillary reflex, quiet conjunctiva, clear lens, normal corneal diameters. Dilated view OD: elevated chalky retinal mass with vitreous seeds.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Stage 5 retinopathy of prematurity
- B Retinoblastoma of the right eye
- C Coats disease with exudative detachment
- D Persistent fetal vasculature
- E Ocular toxocariasis with a granuloma

ANSWER _____ CONFIDENCE L • M • H

2 What is the most appropriate next step?

TREATMENT

- A Arrange a B-scan and review the child in 2 weeks
- B Fine needle biopsy of the mass to confirm the diagnosis
- C Routine referral to paediatric ophthalmology in 6 weeks
- D Topical steroid for presumed intraocular inflammation
- E Same-day referral to a retinoblastoma service

ANSWER _____ CONFIDENCE L • M • H

3 Which three steps belong in the specialist assessment of this child?

DIAGNOSIS • SELECT 3

- A RB1 testing for the child and both parents
- B CT of the head to demonstrate calcification
- C Transvitreal biopsy of the mass for histology
- D Bone marrow aspirate before any treatment
- E Examination under anaesthesia with indentation of both eyes
- F MRI of the orbits and the brain

ANSWER _____ CONFIDENCE L • M • H

4 Which action best meets the duty owed to this child?

LAW AND ETHICS

- A Refer today and close the episode once the letter is sent
- B Wait for the parents to choose a date that suits them
- C Contact the safeguarding team before making any referral
- D Refer today, escalate to safeguarding if she is not brought
- E Give the parents a letter to hand to their family doctor this week

ANSWER _____ CONFIDENCE L • M • H

5 Which genetic point should shape the counselling offered?

BASIC SCIENCE

- A Around 15% of unilateral cases carry a germline mutation
- B Unilateral disease excludes a germline mutation
- C Germline cases follow autosomal recessive inheritance
- D Pineoblastoma risk arises only with bilateral tumours
- E Sporadic tumours need one somatic hit at the RB1 locus

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-12 • why

1 • B Retinoblastoma, right eye

Leukocoria with an elevated chalky retinal mass and vitreous seeds in a quiet, clear-media eye at 14 months is tumour. Strabismus is the second commonest way it is noticed.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Coats disease	It also gives leukocoria in a young child with a normal front of the eye.	Coats shows yellow exudate and telangiectasia, not a chalky mass with seeds.
A	Stage 5 ROP	A total detachment behind the lens does look white through the pupil.	Born at term with no oxygen therapy, so the risk period never existed.

2 • E Same-day referral to a specialist centre

Delay costs the eye and, once tumour leaves the globe, the child. Diagnosis and staging happen at the specialist centre, and no needle enters an eye suspected of holding retinoblastoma.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Needle biopsy	Histology is how most other paediatric tumours are confirmed first.	Perforating the globe seeds tumour outside it and changes survival.
A	Scan, review in 2 weeks	Imaging first looks orderly, and 2 weeks feels prompt in clinic terms.	Growth over 2 weeks can turn an eye-sparing case into an orbital one.

3 • A, E, F EUA of both eyes, MRI, RB1 testing

Both eyes are examined under anaesthesia because disease is often bilateral. MRI stages the nerve and pineal region without radiation, and RB1 testing drives screening of siblings.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	CT of the head	Calcification is the classic sign and CT shows it best.	It irradiates a child who may carry a germline RB1 mutation.
C, D	Biopsy, marrow	Tissue and marrow are how paediatric tumours are usually staged.	Biopsy seeds the orbit, and marrow is for proven extraocular disease.

4 • D Refer today; if she is not brought, escalate

The duty is a same-day referral made by the clinician who holds the suspicion, with that suspicion named in writing. A child not brought to the appointment is a safeguarding matter.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Letter via the parents	It keeps the family informed and involves their usual doctor.	It hands the timing of a sight and life threatening referral to others.
C	Safeguarding first	Involving safeguarding early looks like the careful, thorough route.	Nothing delays the ophthalmic referral. Safeguarding follows non-attendance.

5 • A Germline RB1 is possible in unilateral disease

Laterality does not settle germline status, so testing is offered in unilateral disease too. A germline result changes screening of this child, of siblings and of any future children.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Unilateral rules it out	Bilateral disease is the taught marker of heritable retinoblastoma.	A germline mutation can still produce a single unilateral tumour.
C	Recessive inheritance	Two hits at one gene sounds like recessive transmission.	Transmission is dominant with high penetrance. The second hit is somatic.

IF-THEN RULES FROM THIS CASE

If a child has leukocoria or new strabismus with no red reflex, refer the same day to a retinoblastoma service. If the child is not brought, escalate rather than discharge.

CASE S2-23 • RETINA, CHOROID, VITREOUS • 5 ITEMS

A haze in the vision beside an old scar

A 28-year-old woman has 10 days of floaters and blur OD, without pain. Two years ago co-trimoxazole gave her a rash and fever. BVA 20/60 OD, 20/20 OS. OD: 2+ vitreous cells, a fluffy white retinal focus 1.5 disc diameters superotemporal to the fovea, its edge touching a pigmented atrophic scar, with haze in the vitreous above it. Fine keratic precipitates, 1+ cells, IOP 14. OS normal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Toxocara granuloma with dense localised vitritis
- B Acute retinal necrosis from varicella zoster virus
- C Toxoplasma retinochoroiditis at the edge of a scar
- D Cytomegalovirus retinitis in occult immunosuppression
- E Tuberculous focal choroiditis with vitreous spillover

ANSWER _____ CONFIDENCE L • M • H

2 Which three belong in the initial plan?

TREATMENT • SELECT 3

- A Systemic antiparasitic treatment started this week
- B Oral steroid added 48 hours after the antiparasitic
- C Toxoplasma IgG and IgM to support the diagnosis
- D Oral steroid first, antiparasitic if it fails
- E Intravitreal foscarnet while serology is awaited
- F Topical steroid alone for the anterior reaction

ANSWER _____ CONFIDENCE L • M • H

3 Which regimen fits her drug history best?

TREATMENT

- A Oral aciclovir 800 mg five times daily
- B Oral prednisolone 40 mg daily on its own
- C Pyrimethamine, clindamycin and folinic acid
- D Pyrimethamine, sulfadiazine and folinic acid
- E Oral co-trimoxazole twice daily for six weeks

ANSWER _____ CONFIDENCE L • M • H

4 Why does the new focus sit at the edge of the old scar?

BASIC SCIENCE

- A Glial tissue obstructs flow in the border arterioles
- B Cysts at the scar margin rupture and release parasites
- C Immune complexes settle in previously damaged retina
- D New vessels grow from the scar and leak into retina
- E Fresh infection is seeded there by the choroidal flow

ANSWER _____ CONFIDENCE L • M • H

5 At four weeks, which finding most justifies continuing treatment?

TREATMENT

- A A fluffy lesion border with vitreous haze
- B A pigmented border with a clearing vitreous
- C Acuity 20/30 with a residual central floater
- D A high IgG titre on the repeat sample
- E A few fine keratic precipitates remaining

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-23 • why

1 • C Toxoplasma reactivation at a scar

A fluffy white focus of retinitis touching an old pigmented scar, with vitritis over it and a mild anterior reaction in a healthy young woman, is reactivation at the border of an old lesion.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Acute retinal necrosis	Both give white retinitis with heavy vitritis and an aching eye.	That spreads circumferentially in days and occludes arterioles.
D	CMV retinitis	It also produces a white retinal lesion with haemorrhage in it.	It needs deep immunosuppression and provokes little vitritis.

2 • A, B, C Antiparasitic, then steroid, with serology

A lesion 1.5 disc diameters from the fovea with vitritis is treated systemically. Steroid follows the antiparasitic by about two days, because steroid alone lets the organism multiply.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Steroid first	Vision is being lost to inflammation, which steroid settles fast.	Unopposed steroid enlarges the lesion. Cover the organism first.
E, F	Foscarnet, topical only	One covers a virus, the other treats the reaction you can see.	The organism is not viral, and drops do not reach the retina.

3 • C Pyrimethamine with clindamycin and folinic acid

She reacted to co-trimoxazole with rash and fever, so sulfonamides are out. Clindamycin substitutes for sulfadiazine, and folinic acid protects the marrow from pyrimethamine.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Sulfadiazine regimen	It is the standard combination and the one most often quoted.	Sulfadiazine is a sulfonamide, and she reacted to one already.
E	Co-trimoxazole	It is a convenient oral alternative with real evidence behind it.	It contains the sulfonamide that gave her rash and fever.

4 • B Cysts at the scar margin rupture

Bradyzoites persist in cysts in and around the healed lesion. When a cyst ruptures, parasites invade the adjacent retina, which is why recurrence is satellite to the scar.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Immune complexes	Recurrence looks immunological, and steroid does help the eye.	The lesion is an infection with live parasites, not a deposit.
E	Fresh infection	New exposure from cat litter or raw meat feels like the trigger.	Reactivation is endogenous, from cysts already in that retina.

5 • A A fluffy border with active vitritis

Activity is judged at the lesion border and in the vitreous. A soft edge with haze means live parasites, whereas pigmentation and a clearing vitreous are the signs treatment can stop.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Pigmented border	Any remaining abnormality feels like a reason to keep treating.	Pigment at the edge is the healed state, and the endpoint.
D	High IgG titre	A high titre reads like a measure of how active the disease is.	IgG marks past exposure and stays raised for life.

IF-THEN RULES FROM THIS CASE

If a focal white retinitis touches a pigmented scar with vitritis over it, treat toxoplasma reactivation. If steroid is used, it follows the antiparasitic and never leads it.

CASE S2-25 • RETINA, CHOROID, VITREOUS • 5 ITEMS

The upper field that went and did not come back

A 44-year-old man has had type 1 diabetes for 27 years, HbA1c 96 mmol/mol, blood pressure 168/98, and has not attended for laser in three years. He drives a taxi. BVA 20/200 OD, 20/40 OS. OD: fibrovascular fronds at the disc and along the superior arcade, a taut hyaline membrane, a shallow concave detachment tented over the temporal macula with striae, no break seen, inferior vitreous blood.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Tractional retinal detachment involving the macula
- B Rhegmatogenous detachment from a superior break
- C Combined tractional and rhegmatogenous detachment
- D Diabetic macular oedema with severe thickening
- E Vitreous haemorrhage over a taut hyaline membrane

ANSWER _____ CONFIDENCE L • M • H

2 Which single feature separates this from a rhegmatogenous detachment?

DIAGNOSIS

- A Acuity of 20/200 in that eye at presentation
- B Retinal striae running across the macula
- C No afferent pupillary defect in that eye
- D A concave surface tented to fibrovascular fronds
- E Vitreous haemorrhage filling the inferior vitreous gel

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate management?

TREATMENT

- A Glucose and blood pressure optimised before referral
- B Observation while the inferior vitreous blood clears
- C Vitrectomy with delamination of membranes and endolaser
- D Completion of panretinal photocoagulation, review in three months
- E Intravitreal anti-VEGF alone, review in four weeks

ANSWER _____ CONFIDENCE L • M • H

4 What is the main risk of anti-VEGF injection alone in this eye?

TREATMENT

- A Fibrovascular contraction that worsens the traction
- B Rapid progression of the inferior vitreous haemorrhage
- C Endophthalmitis from the injection itself
- D A sustained rise in intraocular pressure
- E Worsening of his systemic hypertension

ANSWER _____ CONFIDENCE L • M • H

5 He drives a taxi. Which two duties follow from today's findings?

LAW AND ETHICS • SELECT 2

- A Asking his employer to confirm his fitness
- B Telling him the duty to notify the authority is his
- C Recording the acuities and the advice given
- D Notifying the licensing authority yourself this week
- E Telling him he may drive until his operation

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-25 · why

1 · A Tractional detachment, macula involved

A shallow concave elevation tented to fibrovascular fronds, with striae across the macula and no break, is traction. Twenty-seven years of poor control and absent laser explain the fronds.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Combined detachment	Long traction does tear retina, and that would change surgery.	No break is seen, and the surface stays concave, not convex.
B	Rhegmatogenous detachment	Blood and floaters with a field defect is the classic route in.	A break-driven detachment is convex, mobile, and reaches the ora.

2 · D A concave surface pulled to fronds

Traction pulls retina towards the vitreous, so the surface is concave and tethered where the fronds insert. Blood, poor acuity, striae and normal pupils occur in either kind.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Vitreous haemorrhage	Blood accompanies both a retinal tear and a bleeding frond.	It says the vitreous is full of blood, not why retina is off.
B, C	Striae, normal pupils	Both are findings in front of you and both feel diagnostic.	Striae follow any macular distortion, and pupils are often normal.

3 · C Vitrectomy with delamination and endolaser

The macula is detached by traction, and only surgery relieves it. Laser, injection and metabolic control all matter, but none of them lifts a taut membrane off the fovea.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Finish the laser	Laser is overdue and does treat the ischaemia driving all this.	Laser cannot relieve traction, and three months loses the macula.
A	Optimise glucose first	An HbA1c of 96 and 168/98 are genuinely part of his problem.	They change his ten-year risk, not the fovea detached today.

4 · A Fibrous contraction, more traction

Blocking VEGF makes fronds involute and fibrose, and fibrous tissue shortens. On a macula already tented, that tightens the pull. It is given days before surgery, not instead of it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Endophthalmitis	It is the risk quoted at every injection consent.	It is rare, and unrelated to the traction question asked here.
B	More haemorrhage	The eye already has blood, and injections do enter the vitreous.	Anti-VEGF reduces bleeding from fronds rather than provoking it.

5 · B, C Tell him he must notify, and record it

Vocational standards are stricter, and this eye has lost acuity and field, so fitness is decided by the licensing authority. The duty to declare is his, and the advice is documented.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Notify them yourself	A taxi carries passengers, so acting directly feels protective.	The duty is his. You disclose only if he drives on despite advice.
A	Tell his employer	The employer controls whether he works, so it seems practical.	It breaches confidence, and the employer has no assessing role.

IF-THEN RULES FROM THIS CASE

If a detachment is concave and tented to fibrovascular tissue with no break, it is traction and it needs vitrectomy. Incomplete laser does not change that order.

CASE S2-26 • RETINA, CHOROID, VITREOUS • 5 ITEMS

An aching eye that lost its field in five days

A 57-year-old man on no medication has five days of floaters, ache and falling vision OS. He had shingles of the trunk two years ago. BVA 20/200 OS, 20/20 OD. OS: 3+ vitreous cells, keratic precipitates, IOP 30. Confluent creamy peripheral necrosis nasally and temporally, with scalloped borders spreading circumferentially, sheathed occluded arterioles, an oedematous disc, macula spared. OD normal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Behcet occlusive retinal vasculitis with infiltrate
- B Progressive outer retinal necrosis without vitritis
- C Cytomegalovirus retinitis with a mild vitreous reaction
- D Acute retinal necrosis from varicella zoster virus
- E Toxoplasma retinochoroiditis with dense vitritis

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the first 24 hours?

TREATMENT • SELECT 3

- A Oral steroid started before the antiviral
- B Barrier laser at the necrotic border today
- C Intravitreal ganciclovir instead of systemic cover
- D Intravenous aciclovir at induction dose
- E Aqueous or vitreous sample for viral PCR
- F Examination of the fellow eye with a warning

ANSWER _____ CONFIDENCE L • M • H

3 How should the pressure of 30 mmHg and the anterior cells be treated?

TREATMENT

- A A prostaglandin analogue as first-line control
- B Observation until the antiviral takes effect
- C Oral acetazolamide 250 mg four times daily alone
- D Topical steroid with an aqueous suppressant
- E Oral steroid 48 hours before the antiviral starts

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism explains the risk to the fellow eye?

BASIC SCIENCE

- A Viraemia seeding both choroids from the blood
- B Reactivation in both trigeminal ganglia together
- C Antibody against a shared retinal antigen
- D Spread of virus along the visual pathway
- E Immune complexes settling in both retinas at once

ANSWER _____ CONFIDENCE L • M • H

5 Which change at three weeks most needs surgical review?

TREATMENT

- A Pressure settling to 16 on one topical agent
- B PCR confirming varicella zoster virus in the tap
- C A new field defect with a peripheral break
- D A necrotic border that consolidates and pigments
- E Vitritis persisting with acuity steady at 20/80

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-26 • why

1 • D Acute retinal necrosis, zoster

Confluent peripheral necrosis with scalloped borders spreading circumferentially, dense vitritis and occluded arterioles over five days, in a well man with previous shingles, is zoster.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	CMV retinitis	Both give white retina with vessel change in an older adult.	It needs immunosuppression, spreads slowly, and barely inflames.
E	Toxoplasma retinitis	Heavy vitritis over a white focus is exactly its picture.	Its lesion is single, satellite to a scar, and not arteritic.

2 • D, E, F IV antiviral, PCR sample, check the other eye

Necrosis advances hour by hour, so systemic antiviral starts today, and the tap is taken as it begins. The fellow eye is at real risk and has to be examined and the risk explained.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Steroid first	The eye is full of cells, and steroid settles that quickly.	Steroid ahead of antiviral cover lets the necrosis spread faster.
B, C	Laser now, intravitreal only	One walls off detachment, the other hits the virus hard.	Laser waits for a quieter eye, and this virus needs systemic cover.

3 • D Topical steroid with aqueous suppression

Once antiviral cover is running, topical steroid is safe and treats the trabeculitis raising the pressure, with an aqueous suppressant while it settles. Prostaglandins worsen uveitis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Steroid before antiviral	Steroid does control the inflammation that is blurring vision.	In a necrotising retina it must follow cover, never precede it.
B	Observe the pressure	The pressure may fall as the inflammation is treated.	30 mmHg with an oedematous disc is not a pressure to watch.

4 • D Virus travels the visual pathway

Untreated, about a third of these eyes are followed by the fellow eye, because virus passes through the chiasm and central connections. Systemic antiviral is what lowers that risk.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Immune complexes	Bilateral disease sounds autoimmune rather than infective.	This is lytic infection, and antiviral therapy prevents it.
B	Both ganglia at once	Latent zoster does sit in sensory ganglia on both sides.	Ganglia supply skin. Retinal spread follows the visual pathway.

5 • C A break in necrotic retina with a field defect

Necrotic retina tears as the vitreous contracts, and detachment follows in most untreated eyes. A new field defect with a break is a vitreoretinal problem, not a medical one.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Border pigmentation	Any change at the border reads as progression at first glance.	Pigment at the border is healing, and the point of treatment.
B	PCR result	It confirms the diagnosis, which feels like a decision point.	It validates the drug already running. Nothing needs surgery.

IF-THEN RULES FROM THIS CASE

If peripheral necrosis spreads circumferentially with dense vitritis and arteritis, start systemic antiviral the same day and examine the fellow eye. If retina tears, refer for surgery.

CASE S2-27 • RETINA, CHOROID, VITREOUS • 5 ITEMS

The dark lesion that grew between two photographs

A 63-year-old woman with blue irides reports three months of a shadow in her nasal field OD. BVA 20/25 OD. A dome-shaped pigmented choroidal lesion temporal to the right disc, with orange pigment on its surface and subretinal fluid at its edge, no drusen and no halo. Ultrasound: 9.2 mm base, 3.6 mm thickness, low internal reflectivity, choroidal excavation. A photograph two years ago recorded a 7.1 mm base.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Hypertrophy of the retinal pigment epithelium
- B Choroidal naevus with surface drusen and a halo
- C Choroidal melanoma with subretinal fluid
- D Choroidal metastasis from an occult primary
- E Circumscribed choroidal haemangioma

ANSWER _____ CONFIDENCE L • M • H

2 Which three features carry the greatest risk of malignancy?

DIAGNOSIS • SELECT 3

- A A position temporal to the disc
- B Acuity of 20/25 in that eye
- C Thickness of 3.6 mm on ultrasound
- D Orange pigment on the surface
- E Subretinal fluid at the edge
- F Blue irides and fair skin

ANSWER _____ CONFIDENCE L • M • H

3 What is the most appropriate next step?

TREATMENT

- A Photographs repeated in six months for comparison
- B Laser to the area of subretinal fluid
- C Enucleation arranged at the local unit
- D Systemic staging first, then ophthalmic review
- E Referral to an ocular oncology service this week

ANSWER _____ CONFIDENCE L • M • H

4 After oncology assessment, which plan best fits a tumour of this size?

TREATMENT

- A Systemic chemotherapy before local treatment
- B Plaque brachytherapy, with liver surveillance afterwards
- C Enucleation, because subretinal fluid is present
- D Transpupillary thermotherapy as sole treatment
- E Observation with three-monthly photographs

ANSWER _____ CONFIDENCE L • M • H

5 Which route of spread explains where this tumour metastasises?

BASIC SCIENCE

- A Haematogenous, as the uvea has no lymphatics
- B Lymphatic, through the conjunctival channels
- C Direct extension along the optic nerve sheath
- D Seeding of the vitreous and then the meshwork
- E Vortex veins draining to the regional nodes

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-27 • why

1 • C Choroidal melanoma

A dome-shaped pigmented lesion 3.6 mm thick with orange pigment, subretinal fluid, low internal reflectivity and excavation, whose base has grown from 7.1 to 9.2 mm, is malignant.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Choroidal naevus	Most pigmented choroidal lesions are naevi and stay harmless.	A naevus is flat, carries drusen and a halo, and does not grow.
E	Choroidal haemangioma	It is dome-shaped and leaks fluid under the retina as well.	It is orange-red rather than pigmented, and reflects sound highly.

2 • C, D, E Thickness, orange pigment, fluid

Thickness above two millimetres, orange lipofuscin and subretinal fluid each predict growth in a small pigmented lesion. Iris colour is a risk for the disease, not for this lesion.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	Blue irides, fair skin	Light pigmentation is the population risk factor for melanoma.	It says who develops one, not whether this lesion is growing.
A, B	Position, good acuity	Both feel relevant: near the disc, and vision still preserved.	Neither predicts growth. Good acuity is usual in thick tumours.

3 • E Ocular oncology referral this week

Diagnosis, treatment choice and eye-conserving options belong to a specialist service, and growth is already documented. Staging is arranged there, alongside the local treatment plan.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Repeat photographs	Serial imaging is the correct approach to a suspicious naevus.	Growth is documented already. The interval decision is behind her.
C	Enucleation locally	It removes the tumour definitively in a single operation.	Most tumours this size keep the eye with plaque radiotherapy.

4 • B Plaque brachytherapy with liver follow-up

A medium tumour of 3.6 mm thickness is treated with plaque radiotherapy, which keeps the eye and matches enucleation for survival. Surveillance afterwards is aimed at the liver.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Enucleation for fluid	Fluid signals activity, which sounds like a reason to remove it.	Fluid is common and settles after radiotherapy. Size decides.
D	Thermotherapy alone	It is eye-sparing and avoids a radiation plaque altogether.	Recurrence is high in a 3.6 mm tumour treated by heat alone.

5 • A Haematogenous, the uvea has no lymphatics

The uveal tract has no lymphatic drainage, so cells leave in the blood and lodge first in the liver. That is why surveillance is hepatic and why nodes are not sampled.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Lymphatic spread	Most solid tumours reach nodes first, so it is the default.	The uvea has no lymphatics, and nodes are not the first site.
C	Optic nerve sheath	Direct extension occurs and looks anatomically plausible.	It is rare, and it does not explain hepatic metastasis.

IF-THEN RULES FROM THIS CASE

If a pigmented choroidal lesion is over two millimetres thick, carries orange pigment or leaks fluid, refer it as a melanoma. Documented growth removes the option of watching.

CASE S1-06 • SYSTEMIC HEALTH • 5 ITEMS

The rash that stopped at the middle of the forehead

A 68-year-old woman taking methotrexate has four days of right brow pain, then a vesicular rash over the forehead and upper lid that stops at the midline, with lesions on the tip of the nose. BVA 20/30 OD. Pseudodendritic epithelial plaques, reduced corneal sensation, 1+ cells, a few pigmented keratic precipitates, sectoral iris atrophy. IOP 15 OS, 26 OD. Discs normal.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Primary herpes simplex with a periocular rash
- B Herpes zoster ophthalmicus with anterior uveitis
- C Impetiginised contact dermatitis of the lids
- D Bacterial preseptal cellulitis with lid vesicles
- E Eczema herpeticum with secondary keratitis

ANSWER _____ CONFIDENCE L • M • H

2 Which antiviral treatment is most appropriate?

TREATMENT

- A Intravenous aciclovir for seven days as an inpatient
- B Oral valaciclovir 500 mg twice daily for five days
- C Oral aciclovir 800 mg five times daily, seven days
- D Oral aciclovir 400 mg five times daily for five days
- E Topical ganciclovir 0.15% gel five times daily

ANSWER _____ CONFIDENCE L • M • H

3 How should the pressure of 26 mmHg in that eye be managed?

TREATMENT

- A Oral acetazolamide 250 mg four times daily
- B Topical pilocarpine 2% four times daily
- C Topical steroid for the uveitis plus a beta-blocker
- D Topical prostaglandin analogue as first-line control
- E Observe the pressure until the rash has resolved

ANSWER _____ CONFIDENCE L • M • H

4 Which three actions belong in her care over the next month?

TREATMENT • SELECT 3

- A Stop the methotrexate permanently to prevent recurrence
- B Prescribe a topical antibiotic for the crusting lesions
- C Arrange varicella serology to confirm the diagnosis
- D Discuss zoster vaccination once this episode settles
- E Review at intervals for a year for late stromal disease
- F Tell her rheumatology team about the acute infection

ANSWER _____ CONFIDENCE L • M • H

5 She asks whether she can work in a nursery while the rash is active. What is the correct advice?

LAW AND ETHICS

- A Return when the rash is covered by a dressing
- B No restriction, since zoster is not transmissible
- C Stay away for 21 days from the rash appearing
- D Stay away until every lesion has crusted over
- E Return once oral antiviral therapy has started

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-06 • why

1 • B Herpes zoster ophthalmicus with uveitis

A painful vesicular rash confined to one dermatome, respecting the midline and involving the nose tip, with sectoral iris atrophy, pigmented precipitates and a raised pressure, is zoster.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Primary herpes simplex	Vesicles and pseudodendrites overlap, and both are herpesviruses.	Simplex respects no dermatome and does not atrophy the iris in sectors.
D	Preseptal cellulitis	A hot swollen lid in an immunosuppressed patient reads bacterial.	Dermatomal vesicles with intraocular signs are not preseptal disease.

2 • C Oral aciclovir 800 mg five times daily

Zoster needs the higher oral dose five times daily, started as early as possible, to cut ocular complications and neuralgia. Simplex doses undertreat it and topical gel never reaches the skin.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Aciclovir 400 mg	It is the familiar dose carried over from simplex keratitis.	Varicella zoster is less sensitive and needs twice that dose.
A	Intravenous therapy	She is immunosuppressed, so parenteral cover feels safer.	Reserve it for dissemination or encephalitis. She has neither.

3 • C Treat the trabeculitis, add a beta-blocker

The rise here is trabeculitis, so treating the inflammation lowers it, with aqueous suppression added while it settles. Prostaglandins and miotics aggravate inflammation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Prostaglandin analogue	It is the strongest single agent and the usual first choice.	Prostaglandins can worsen uveitis and are avoided while it is active.
B	Pilocarpine	Miosis and better outflow both sound useful in a pressure spike.	Miotics increase inflammation and pull the iris into synechiae.

4 • D, E, F Vaccine discussion, long review, inform the team

Zoster eye disease relapses for years, so review runs long and vaccination is discussed after recovery. Her immunosuppression is a decision for the team that prescribes it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Stop methotrexate	Immunosuppression allowed reactivation, so stopping looks protective.	Her joint disease needs it. Any change belongs to the prescriber.
C	Varicella serology	Laboratory proof feels rigorous before a year of follow-up.	A dermatomal vesicular rash is the diagnosis. Serology changes nothing.

5 • D Away from the nursery until lesions crust

Vesicle fluid transmits varicella to non-immune contacts, and infectivity ends when the lesions crust. Treatment shortens the illness, and a dressing does not remove the duty to stay away.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Back once treated	Antivirals reduce viral shedding, which sounds like safety.	Shedding continues until crusting. Treatment does not end exclusion.
B	No restriction	Zoster needs contact, unlike chickenpox spread through the air.	Vesicle fluid still gives varicella to non-immune children.

IF-THEN RULES FROM THIS CASE

If a dermatomal rash reaches the tip of the nose, examine the eye and start high-dose oral antiviral that day. If the pressure rises, treat the trabeculitis rather than adding a prostaglandin.

CASE S1-16 • SYSTEMIC HEALTH • 5 ITEMS

A drug the rheumatologist started five years ago

A 46-year-old woman with systemic lupus erythematosus takes hydroxychloroquine 400 mg daily, started five years ago. Weight 61 kg, eGFR 88, no tamoxifen, no other ocular medication. Reading is unchanged. BVA 20/20 both eyes. Maculae show no pigment disturbance and no ring of depigmentation. Automated 10-2 fields are full, spectral domain OCT shows an intact outer retina, and no baseline record exists.

1 What does the current dose require of the practitioner?

TREATMENT

- A Advise stopping the drug today, pending further macular imaging
- B Halve the dose in clinic now and repeat the fields in a year
- C Ask that the dose be recalculated on ideal rather than actual weight
- D Continue as prescribed, since retina, fields and acuity are normal
- E Report to the prescriber that the dose exceeds 5 mg/kg and needs review

ANSWER _____ CONFIDENCE L • M • H

2 Which two tests form the core of annual screening here?

BASIC SCIENCE • SELECT 2

- A Fundus autofluorescence of the pole
- B Colour vision with pseudoisochromatic plates
- C Weekly Amsler chart testing at home
- D Spectral domain OCT of the macula
- E Automated 10-2 threshold visual fields

ANSWER _____ CONFIDENCE L • M • H

3 Which pattern would indicate early toxicity?

DIAGNOSIS

- A Loss of the ellipsoid zone directly beneath the foveal centre
- B Cystoid spaces in the inner nuclear layer around the fovea
- C Drusenoid elevation of the pigment epithelium temporal to fixation
- D Choroidal thickening beneath the central macula on enhanced imaging
- E Parafoveal outer retinal thinning with the foveal centre intact

ANSWER _____ CONFIDENCE L • M • H

4 Screening at year seven shows parafoveal ellipsoid loss and a 10-2 ring defect. What next?

TREATMENT

- A Advise halving the dose and continue annual screening as before
- B Add autofluorescence and electroretinography before any advice
- C Report to the prescriber this week so the drug can be withdrawn
- D Repeat the fields and OCT in six months before advising a change
- E Tell the patient to stop the drug today and inform the prescriber later

ANSWER _____ CONFIDENCE L • M • H

5 What does the duty of care require of the screening record?

LAW AND ETHICS

- A The prescriber's dose calculation copied into the record each year
- B A written note that continuing the drug is at the patient's own risk
- C Annual results kept in the practice record and released only on written request
- D Baseline images and fields on file, with every result sent to the prescriber
- E A consent form for screening signed and filed before each annual visit

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-16 • why

1 • E Report the dose above 5 mg/kg

At 61 kg, 400 mg daily is 6.6 mg/kg of actual body weight, above the 5 mg/kg threshold that raises retinopathy risk sharply. Dosing belongs to the prescriber, so the finding is reported with its reason.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Continue as prescribed	Every test today is normal, so there seems to be nothing to act on.	Risk is set by cumulative dose. Normal screening now does not license the dose.
C	Use ideal body weight	Ideal weight dosing was the old rule and it is still widely quoted.	Actual body weight replaced it, because it predicts toxicity better.

2 • D, E SD-OCT with 10-2 automated fields

Screening pairs one structural and one functional test, because either can show the parafoveal change first. Autofluorescence and multifocal electroretinography are adjuncts, not the core.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Autofluorescence	It shows the parafoveal ring beautifully once damage is established.	It is an adjunct, and it changes late compared with OCT and fields.
B, C	Colour plates, Amsler	Both are cheap, quick, and test central function of a sort.	Neither is sensitive to parafoveal loss, which spares the very centre early.

3 • E Parafoveal thinning, fovea spared

Toxicity starts two to six degrees from fixation, thinning the outer nuclear layer and ellipsoid zone in a ring while the centre stays normal. The fundus still looks unremarkable at that stage.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Subfoveal ellipsoid loss	The bull's eye name suggests the centre is the target of the damage.	The centre is spared until late. Central loss points to another disease.
C	Drusenoid elevation	It sits parafoveally, which matches where toxicity is looked for.	Drusenoid change is age-related deposition, not drug-related thinning.

4 • C Report this week for withdrawal

Structure and function agree, which is definite toxicity, and damage continues after the drug stops. Withdrawal is the prescriber's decision and it is needed within days, not months.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Patient stops it today	It is fast and it puts the patient in control of her own risk.	Abrupt withdrawal without the prescriber risks a lupus flare.
D	Repeat in six months	Confirming a screening finding before acting is usually good practice.	Two agreeing tests already confirm it, and six months adds permanent loss.

5 • D Baseline on file, results to the prescriber

Screening detects change, which needs a documented starting point, and only the prescriber can act on the result. No baseline exists here, so establishing one is part of this visit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Kept, released on request	Records are confidential, so waiting to be asked sounds correct.	Shared care obliges active communication to the prescriber, not storage.
B	Note of patient risk	It documents that risk was discussed, which feels protective.	It transfers a clinical duty to the patient. The duty is to inform the prescriber.

IF-THEN RULES FROM THIS CASE

If the daily dose exceeds 5 mg/kg of actual weight, report it whatever the retina looks like. If structure and function agree on parafoveal loss, that is toxicity and the drug must stop.

CASE S1-25 • SYSTEMIC HEALTH • 5 ITEMS

The lid that sits lower by the evening

A 34-year-old woman has six weeks of a right lid that sits lower by the evening and intermittent vertical double vision that changes direction from day to day. No pain. BVA 20/20 each. Right ptosis 3 mm at rest, 4.5 mm after 45 seconds of sustained upgaze, with an upshoot of the lid on refixation. Ptosis 1 mm after a two-minute ice pack. Pupils equal and brisk. Orthophoric at one measurement, 6 Δ left hypertropia at the next.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Aponeurotic ptosis with a decompensating phoria
- B Chronic progressive external ophthalmoplegia
- C Partial right third nerve palsy, pupil sparing
- D Thyroid eye disease with an early restrictive myopathy
- E Ocular myasthenia gravis, no bulbar signs

ANSWER _____ CONFIDENCE L • M • H

2 Which three investigations are appropriate now?

TREATMENT • SELECT 3

- A Repetitive nerve stimulation or single-fibre EMG
- B MRI of the orbits with fat suppression
- C ESR, CRP and temporal artery ultrasound
- D CT angiography of the circle of Willis
- E Acetylcholine receptor antibody assay
- F CT of the chest to look for a thymoma

ANSWER _____ CONFIDENCE L • M • H

3 Which action matters most before she leaves today?

TREATMENT

- A Arrange occlusion of the right eye until review in six months
- B Start a trial of pyridostigmine and review in four weeks
- C Safety-net for breathing and swallowing, with urgent routes
- D Dispense press-on prism for the vertical deviation now
- E Order a ptosis crutch to be fitted at the next visit

ANSWER _____ CONFIDENCE L • M • H

4 Which mechanism explains the fatigue on sustained upgaze?

BASIC SCIENCE

- A Segmental demyelination of the superior division of the third nerve
- B Depletion of acetylcholinesterase in the synaptic cleft
- C Antibodies block postsynaptic acetylcholine receptors
- D Antibodies attack presynaptic voltage-gated calcium channels
- E Deletion of mitochondrial DNA in extraocular muscle fibres

ANSWER _____ CONFIDENCE L • M • H

5 Which single finding would argue hardest against this diagnosis?

DIAGNOSIS

- A A hypertropia that measures differently on two days
- B Improvement of the ptosis after a two-minute ice pack
- C Normal deep tendon reflexes throughout the four limbs
- D A fixed right pupil of 6 mm with a poor light reaction
- E Weakness of orbicularis oculi on forced lid closure

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S1-25 • why

1 • E Ocular myasthenia gravis

Variable ptosis that worsens on sustained upgaze, a lid twitch on refixation, improvement after ice and a hypertropia that changes size are junctional fatigue. The pupils are always spared.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Aponeurotic ptosis	Ptosis worse by evening is what patients with lid laxity describe.	Aponeurotic ptosis does not fatigue on upgaze or lift after an ice pack.
B	External ophthalmoplegia	It gives painless ptosis with motility loss and no pupil signs.	It is slowly progressive and symmetrical, and it does not vary hourly.

2 • A, E, F AChR antibodies, chest CT, nerve stimulation

Antibody testing confirms most cases, electrophysiology catches the seronegative ones, and a thymoma changes management in about one in ten. Orbital and vascular imaging answer a different question.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Orbital MRI	Ptosis with a variable vertical deviation looks orbital.	Muscles and orbit are normal here; the fault is at the neuromuscular junction.
D	CT angiography	Ptosis with diplopia raises the aneurysm reflex.	The pupils are equal and brisk and the signs fatigue with effort.

3 • C Safety-netting for bulbar and respiratory signs

About fifteen percent of ocular cases generalise within two years, and a crisis is a respiratory emergency. She needs to know which symptoms mean the emergency department rather than the clinic.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Press-on prism now	The diplopia is the complaint and prism is the optical fix.	A deviation that changes within the day cannot be prismsed reliably.
B	Start pyridostigmine	It is the right drug class for the disease.	Prescribing sits with the neurologist, and it is not today's priority.

4 • C Postsynaptic receptor blockade

Receptor loss lowers the endplate potential, so repeated release fails to reach threshold as vesicle stores run down. Rest and cooling restore transmission, which is why ice lifts the lid.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	Calcium channel antibody	It is a real antibody-mediated junction disease with weakness.	That is Lambert-Eaton: strength improves with use and reflexes are lost.
E	Mitochondrial deletion	It is the mechanism of the disease this most resembles on inspection.	Mitochondrial ptosis is fixed and progressive, never fatiguable.

5 • D A fixed, poorly reactive pupil

The iris sphincter is not affected in myasthenia, so pupil involvement forces a different diagnosis. Orbicularis weakness and day-to-day variability are typical rather than exclusionary.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	Orbicularis weakness	Extra weakness sounds like a bigger and different disease.	Orbicularis involvement is common in ocular myasthenia and supports it.
C	Normal reflexes	Normal reflexes seem to rule out a neuromuscular disorder.	Reflexes are normal in myasthenia and lost in Lambert-Eaton.

IF-THEN RULES FROM THIS CASE

If ptosis varies within the hour, fatigues on upgaze or lifts after ice, test for myasthenia and image the chest. If the pupil is involved, it is not myasthenia.

CASE S2-31 • SYSTEMIC HEALTH • 5 ITEMS

Vision that dropped like a blind for four minutes

A 71-year-old man has had three episodes of painless loss of vision OD in 10 days, each a grey curtain falling from above, lasting about four minutes, with full recovery. He smokes 20 a day. BP 168/94. No jaw claudication, no scalp tenderness, ESR 12 mm/h. BVA 20/25 both eyes, pupils equal with no RAPD. A bright refractile plaque lies at an arteriolar bifurcation OD. A right carotid bruit is audible.

1 What is the most likely diagnosis?

DIAGNOSIS

- A Transient obscurations from raised intracranial pressure
- B Giant cell arteritis with transient visual loss
- C Retinal transient ischaemic attack from carotid embolism
- D Retinal migraine with recurrent monocular aura
- E Ocular ischaemic syndrome from carotid occlusion

ANSWER _____ CONFIDENCE L • M • H

2 Which three steps belong in the first 24 hours?

TREATMENT • SELECT 3

- A Same-day referral to a transient ischaemic attack clinic
- B Carotid duplex booked as an outpatient in four weeks
- C Oral prednisolone 60 mg daily while the ESR is repeated
- D A second antihypertensive today, with review in a month
- E Same-day carotid duplex ultrasound of both sides
- F Aspirin 300 mg today, then a maintenance antiplatelet

ANSWER _____ CONFIDENCE L • M • H

3 Duplex shows 78% stenosis of the right internal carotid. What is the most appropriate management?

TREATMENT

- A Antiplatelet and statin alone, with duplex repeated in six months
- B Endarterectomy within two weeks, with antiplatelet and a statin
- C Dual antiplatelet therapy for 90 days, then reassessment
- D Carotid stenting deferred until a further episode occurs
- E Warfarin to an INR of 2 to 3, with no surgical referral

ANSWER _____ CONFIDENCE L • M • H

4 Which source best explains the refractile plaque seen OD?

BASIC SCIENCE

- A Lipid deposit from untreated hypertriglyceridaemia
- B Cholesterol embolus from an ulcerated carotid plaque
- C Calcific embolus from a stenosed aortic valve
- D Platelet-fibrin embolus from atrial fibrillation
- E Talc embolus from intravenous drug use

ANSWER _____ CONFIDENCE L • M • H

5 He asks whether he may keep driving to work. What is the correct advice and record?

LAW AND ETHICS

- A Advise him to stop driving now, notify the authority, and record it
- B Advise him to stop driving only once the surgery has been booked
- C Tell him the practice will notify the licensing authority on his behalf
- D Leave driving to the vascular clinic and make no entry now
- E No restriction, since acuity is 20/25 and the field is full

ANSWER _____ CONFIDENCE L • M • H

TARGET 6 MIN Started _____ Finished _____ Flagged _____

Case S2-31 • why

1 • C Retinal TIA, carotid embolic source

Painless monocular loss lasting minutes with full recovery, repeated three times, with a refractile plaque at a bifurcation and an ipsilateral bruit, is embolic retinal ischaemia from the carotid.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	Giant cell arteritis	It is the transient loss that must never be missed, and he is the right age for it.	ESR 12 with no jaw claudication, scalp tenderness or systemic symptoms.
D	Retinal migraine	Brief stereotyped episodes with complete recovery fit its description well.	First episodes at 71, with a visible plaque and a bruit, are embolic.

2 • A, E, F Same-day duplex, aspirin, TIA clinic

Stroke risk peaks in the days after a retinal TIA, so imaging, antiplatelet cover and specialist assessment all happen today. The benefit of surgery falls sharply after the first two weeks.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Oral prednisolone	Covering for arteritis feels like the cautious move in a man of 71.	ESR 12 with no cranial symptoms, and steroid does nothing to an embolus.
B, D	Duplex in four weeks, BP	Both are correct actions at the tempo of a routine vascular review.	The window for endarterectomy is measured in days. Four weeks wastes it.

3 • B Endarterectomy within two weeks

Symptomatic stenosis of 70 to 99% carries a high early stroke risk, and the gain from surgery is largest inside two weeks of the last event. Antiplatelet and statin run alongside, not instead.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	Medical therapy alone	Drugs carry no operative risk and the eye looks normal today.	At 78% and symptomatic, the stroke risk exceeds the risk of operating.
E	Warfarin	Anticoagulation sounds stronger than an antiplatelet for stopping emboli.	This is platelet debris from plaque, not thrombus from a fibrillating atrium.

4 • B Cholesterol embolus from the carotid plaque

A bright yellow refractile fleck lodged at a bifurcation is cholesterol shed from an ulcerated plaque. The bruit places the source in the ipsilateral carotid rather than the heart.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Calcific, valvular	Cardiac emboli are the other classic cause of transient monocular loss.	Calcific emboli are chalky white and dull, and they sit more proximally.
D	Platelet-fibrin embolus	Fibrillation is common at 71 and does throw emboli to the retina.	Platelet emboli are dull grey and pass within minutes. This one persists.

5 • A Stop driving now, he notifies, advice recorded

Three events in 10 days make him unfit to drive from today, and the duty to notify the licensing authority rests with the licence holder. The advice and his response are entered at this visit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	Practice notifies for him	It feels like the protective act, and the risk to other road users is real.	Notification is his duty. Disclosure follows only if he drives on regardless.
E	No restriction	He sees 20/25 and the eye is entirely normal between the attacks.	The hazard is a sudden loss at the wheel, not his acuity in the chair.

IF-THEN RULES FROM THIS CASE

If monocular loss lasted minutes and recovered fully, call it a retinal TIA and image the carotids the same day. If symptomatic stenosis reaches 70%, refer for surgery within two weeks.

PART 10

Reference

The answer key, an eighteen-day schedule keyed to real case numbers, a cut-line recall deck, and the sources with their link status recorded.

ANSWER KEY

Letters were generated to a distribution, not chosen by an author. The audit that proves it is on page 4, printed from the same build.

RECALL DECK

Cut on the dashed lines. Ten seconds per prompt, answered aloud. Misses go back into tomorrow's pile.

EIGHTEEN DAYS

New cases on the left, recall on the right. If you are short of time, cut new cases and keep the recall column.

SOURCES

Every claim carries a state: verified, unconfirmed, or removed. Nothing in this manual is certified by its own author.

Answer key

Letters are generated to the distribution audited on page 4, not chosen by whoever wrote the item. Guessing one letter throughout scores close to a fifth of the paper, which is what it should score. Multiple letters in a cell mean an all-or-none item: every one, or no mark.

Session 1

CASE	1	2	3	4	5	6
S1-01	B	A, D, F	E	A	A	
S1-02	C	D, E, F	A	B	D	
S1-03	C	B	A	B	D, E	
S1-04	A	B, C, D	C	A	D	
S1-05	A	D	A, F	D	A	
S1-06	B	C	C	D, E, F	D	
S1-07	C	D	B	E	A, E	
S1-08	D	D	A, B	A	B	
S1-09	B	C, D, E	A	E	B	
S1-10	E	B	A, B, C	E	A	
S1-11	E	C	A, B, F	A	E	
S1-12	B	A, B, F	E	B	E	
S1-13	C	B	A, B	A	D	
S1-14	B	E	C, D, E	C	A	
S1-15	B	A, E	B	D	D	
S1-16	E	D, E	E	C	D	
S1-17	B	E	A, B, F	E	E	
S1-18	C	E	A, B	B	E	
S1-19	D	A, B	D	D	E	
S1-20	C	A	D	C, D	E	
S1-21	B	C, D, E	B	B	D	
S1-22	C	B, C, D	A	B	C	
S1-23	A	C	A, B, F	C	D	
S1-24	E	A, B, F	A	C	C	
S1-25	E	A, E, F	C	C	D	
S1-26	B	D, E, F	A	C	A	
S1-27	C	A	B, C, D	C	B	
S1-28	E	B, C, D	D	D	D	
S1-29	E	A	A	B	D, E, F	
S1-30	C	C	A, E, F	C	D	
S1-31	C	D	D, E	D	D	
S1-32	E	D, E, F	D	E	A	
S1-33	C	D	A, B	B	A	
S1-34	A	A	E	D, E	D	
S1-35	C	C	A, B, F	B	E	

Session 2

CASE	1	2	3	4	5	6
S2-01	E	A, B, E	A	A	D	
S2-02	D	D	C, D, E	E	E	
S2-03	E	C	B, C, D	B	C	
S2-04	B	D	A, B, C	B	A	
S2-05	E	C	C	A	B, C	
S2-06	E	E	A, E, F	B	E	
S2-07	A	B	D	E	A, E	
S2-08	A	C	C, D, E	A	C	
S2-09	E	D	B	B	A, B, F	
S2-10	A	B, C, D	B	C	B	
S2-11	E	A, B, C	C	C	D	
S2-12	B	E	A, E, F	D	A	
S2-13	D	A	E	C, D, E	A	
S2-14	B	D	A	B, C, D	D	
S2-15	E	C	A	A, B, C	A	
S2-16	D	B, C, D	E	B	D	
S2-17	A	A, B, C	E	D	B	
S2-18	D	C	C	D, E, F	E	
S2-19	D	C	D, E, F	E	A	
S2-20	E	D, E, F	D	D	E	
S2-21	B	C, D, E	D	C	B	
S2-22	B	B, C, D	A	E	D	
S2-23	C	A, B, C	C	B	A	
S2-24	D	C	A, E, F	E	C	
S2-25	A	D	C	A	B, C	
S2-26	D	D, E, F	D	D	C	
S2-27	C	C, D, E	E	B	A	
S2-28	C	B	C	C	D, E	
S2-29	C	B	A, B, F	D	B	
S2-30	D	E	E	D, E, F	C	
S2-31	C	A, E, F	B	B	A	
S2-32	B	D, E, F	C	E	C	
S2-33	B	C, D, E	A	D	E	
S2-34	A	B, C, D	A	B	A	
S2-35	B	A, B	A	B	D	

Eighteen days

Both sessions, eighteen days, generated from the case numbers in this book. Cases are answered cold, once. Everything after that is recall, not rereading. The right-hand column is the part people skip and the part that moves the score.

DAY	NEW	WORK	RECALL
Day 1	S1-01 to S1-05	Cases cold, keys the same evening	-
Day 2	S1-06 to S1-10	Cases cold, keys the same evening	S1-01 to S1-05, missed items
Day 3	S1-11 to S1-15	Cases cold, keys the same evening	S1-01 to S1-10, missed items
Day 4	S1-16 to S1-20	Cases cold, keys the same evening	S1-06 to S1-15, missed items
Day 5	S1-21 to S1-25	Cases cold, keys the same evening	S1-11 to S1-20, missed items
Day 6	S1-26 to S1-30	Cases cold, keys the same evening	S1-16 to S1-25, missed items
Day 7	S1-31 to S1-35	Cases cold, keys the same evening	S1-21 to S1-30, missed items
Day 8	Score Session 1	By item type, not just the total	The two weakest columns, worked through
Day 9	S2-01 to S2-05	Cases cold, keys the same evening	-
Day 10	S2-06 to S2-10	Cases cold, keys the same evening	S2-01 to S2-05, missed items
Day 11	S2-11 to S2-15	Cases cold, keys the same evening	S2-01 to S2-10, missed items
Day 12	S2-16 to S2-20	Cases cold, keys the same evening	S2-06 to S2-15, missed items
Day 13	S2-21 to S2-25	Cases cold, keys the same evening	S2-11 to S2-20, missed items
Day 14	S2-26 to S2-30	Cases cold, keys the same evening	S2-16 to S2-25, missed items
Day 15	S2-31 to S2-35	Cases cold, keys the same evening	S2-21 to S2-30, missed items
Day 16	Score Session 2	By item type, not just the total	The two weakest columns, worked through
Day 17	Repair	Every item missed with high confidence	Rewrite each as one if-then rule
Day 18	Timed run	Session 1 again, 175 items in 3.5 hours	Compare to the first pass by column
Day 19	Timed run	Session 2 again, under the same conditions	Score by item type
Day 20	Consolidate	Emergency gate and competing pairs, aloud	Formulas drilled, then stop

READ THE TWO SCORE SHEETS SIDE BY SIDE

Compare them by item-type column, not by total. A total that rose while the treatment column fell is a warning, not progress, because treatment carries the largest share of the blueprint and decides TMOD separately.

IF YOU HAVE FEWER THAN EIGHTEEN DAYS

Keep the recall column and cut new cases, not the reverse. Ten cases reviewed to the point of recall beat thirty-five read once. With four days: the emergency gate, the competing pairs, and one timed half session.

Cut-line recall deck

Cut on the dashed lines. Ten seconds per prompt, answered aloud.

Misses go back into tomorrow's pile.

Painful red eye in a contact lens wearer Microbial keratitis until the slit lamp says otherwise. Measure it, culture if central, treat intensively, review inside 24 hours. Never patch.	Pain out of proportion, water exposure, poor antibiotic response Acanthamoeba. Culture, PCR, or confocal, repeated if suspicion persists. Ring infiltrate is late, not required. Steroid makes it worse.
Lid swelling plus painful restricted motility Orbital cellulitis. Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ENT. Colour and motility are the gate.	Over 50, profound loss, chalky pallid disc oedema Arteritic AION. ESR, CRP, platelets, and immediate high-dose systemic steroid. A normal ESR does not clear it.
Sudden painless profound monocular loss Retinal artery occlusion, an acute stroke syndrome. Stroke pathway now, embolic source workup, GCA screen by age. No office massage.	Flashes, shower of floaters, pigment in the vitreous Retinal break until the periphery has been examined with indentation. A normal central OCT proves nothing.

Deck, continued

New distortion with subretinal fluid in an older patient Neovascular AMD until proven otherwise. Urgent retina referral for anti-VEGF. Good acuity is a reason to treat quickly, not to wait.	Bilateral myopic shift with uniformly shallow chambers on a new drug Ciliochoroidal effusion, commonly topiramate. Stop the drug with the prescriber, suppress aqueous, cycloplege. Not pupillary block: no iridotomy, no miotics.
Acute ptosis and miosis after neck pain Painful Horner: carotid dissection until imaged. Emergency head and neck vascular imaging. Do not wait to finish localisation drops.	Recurrent chalazion at the same site with lash loss Sebaceous carcinoma. Full-thickness biopsy with conjunctival mapping, examine nodes. Not a third curettage.
Multiple-response item Count the requested number, judge each option alone, choose patient-specific over generally true. Partial sets score zero.	High-confidence miss Top of the error log. Name the false rule, write the replacement as one if-then sentence, retest tomorrow from a changed stem.

FINAL 48 HOURS

Red flags, drug contraindications, formulas, image algorithms, and logistics only. No new content. Between the two sessions, do not autopsy Session 1: eat, hydrate, reset, and start the second session on pacing.

Sources and status

Official NBEO documents control scope and weighting. Everything below was checked on 15 August 2026, and link status is recorded so a reader can tell what was actually opened.

Official

STATUS	DOCUMENT
READ	NBEO. Part II PAM/TMOD exam information. optometry.org/exams/part-ii-pam-tmod . Source for item count, case size, session structure, timing, TMOD breakout, and delivery.
RESOLVES	NBEO. Part II Content Matrix, PDF dated May 2026. Link returns 200; content not machine-read for this edition, so domain ranges remain unconfirmed.
RESOLVES	NBEO. Part II PAM Content Outline and TMOD Content Outline, PDFs dated May 2026.
RESOLVES	NBEO. PAM Sample Cases and Test Day Instructions, PDFs dated May 2026.
DEAD	Pearson VUE tutorial deep link. Did not resolve; NBEO states the tutorial is being updated. Reach it from the exam page.

Clinical anchors

Mac Grory B, et al. Management of central retinal artery occlusion. *Stroke* 2021;52:e282-e294. · Marmor MF, et al. Recommendations on screening for chloroquine and hydroxychloroquine retinopathy, 2016 revision. *Ophthalmology* 2016;123:1386-1394. · Herpetic Eye Disease Study Group. *N Engl J Med* 1998;339:300-306. · Beck RW, et al. Corticosteroids in acute optic neuritis. *N Engl J Med* 1992;326:581-588. · AREDS2 Research Group. *JAMA* 2013;309:2005-2015. · CDC. Sexually transmitted infections treatment guidelines, 2021. · American Diabetes Association. Standards of care in diabetes.

Learning method

Dunlosky J, et al. Improving students' learning with effective learning techniques. *Psychol Sci Public Interest* 2013;14:4-58. · Roediger HL, Karpicke JD. Test-enhanced learning. *Psychol Sci* 2006;17:249-255.

KNOWN LIMITS OF THIS EDITION

Domain ranges and item-type bands are transcribed, not confirmed against the current matrix. Both sessions are complete: 70 cases and 350 items, every one with a generated answer letter and a distractor table. Clinical anchors are exam-oriented simplifications: verify labelling, dose, interactions, and local law before treating a patient.

ORIGINALITY AND INDEPENDENCE

Every case, option, rationale, table, and diagram here is original. Nothing reproduces NBEO secure items or copyrighted case images. This is an independent resource, not endorsed by NBEO, and no resource can guarantee a passing score.

How this was made

Recorded because the previous edition's defects were invisible to its reader, and because a claim you cannot check is worth less than a method you can.

Build

Content	Cases are structured data. Each item carries its options with the correct one flagged in place, an explanation, and a distractor table naming what makes each wrong option tempting and the finding that rules it out.
Answer letters	Generated after authoring, to an even distribution with no run longer than three and every letter live in both sessions. Options are rotated rather than shuffled, so ordered options stay ordered.
Layout	Every page is a fixed letter-sized box. A cost model estimates the height of each block and packs content into sheets, and a headless browser then measures the real render and fails the build if any page is clipped.
Numbering	Page numbers, the table of contents, both case indexes and the printed answer key are all generated after layout, so they cannot drift away from the content.
Print	Printed through the browser's own engine at letter size with the document owning its page geometry. The exported file is then checked for page count, page box and embedded fonts.

Typography

Display	Instrument Serif, for the page titles only.
Text	Spectral at 10.5 point, down to 8 point in the distractor tables and nowhere smaller. Chosen because it has static weights: the print engine silently substitutes a fallback for variable-axis fonts, which loses the body face in the exported file without any warning on screen.
Labels and numbers	IBM Plex Mono, for anything that has to be scanned rather than read.
One deliberate exception	The Greek delta in prism notation is set in a fallback face, in both weights, because Spectral has no Greek coverage. That was established by exporting the file with the character removed and comparing the embedded fonts. The alternative was to write prism notation incorrectly.
Colour	Petrol keys structure and navigation, red keys danger and prohibition, ochre keys a caution or a rule to memorise. Red is never used decoratively, so a red strip on a page always means the same thing.

Known limits of this edition

Unconfirmed figures	Per-domain item ranges and item-type percentage bands are transcribed from the prior edition. The official Content Matrix resolves but was not machine-read for this build. Both are marked unconfirmed wherever they appear.
Clinical simplification	Drug figures are exam anchors. Verify labelling, dose, interactions and local law before treating anybody.
Independence	Every case, option, rationale, table and diagram is original. Nothing reproduces secure NBEO items or copyrighted case images. This is an independent resource, not endorsed by NBEO, and no resource can guarantee a passing score.

Built 15 August 2026. 178 pages, 70 cases, 350 items.